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OFFICIAL PROGRAMME BOOK

# MTERMS 2026

10th Malaysian Tissue Engineering and  
Regenerative Medicine Scientific Conference



7 – 8 September 2026



Concorde Hotel, Shah Alam, Malaysia

## From Discovery to Therapy:

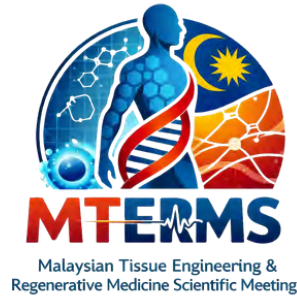
Advances in Tissue Engineering  
and Regenerative Medicine

RESEARCH  
INNOVATION  
COLLABORATION  
IMPACT



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# MTERMS 2026

## 10th Malaysian Tissue Engineering and Regenerative Medicine Scientific Conference

7-8 September 2026  
Concorde Hotel, Shah Alam

### Organised by



**TISSUE ENGINEERING AND  
REGENERATIVE MEDICINE  
SOCIETY OF MALAYSIA**

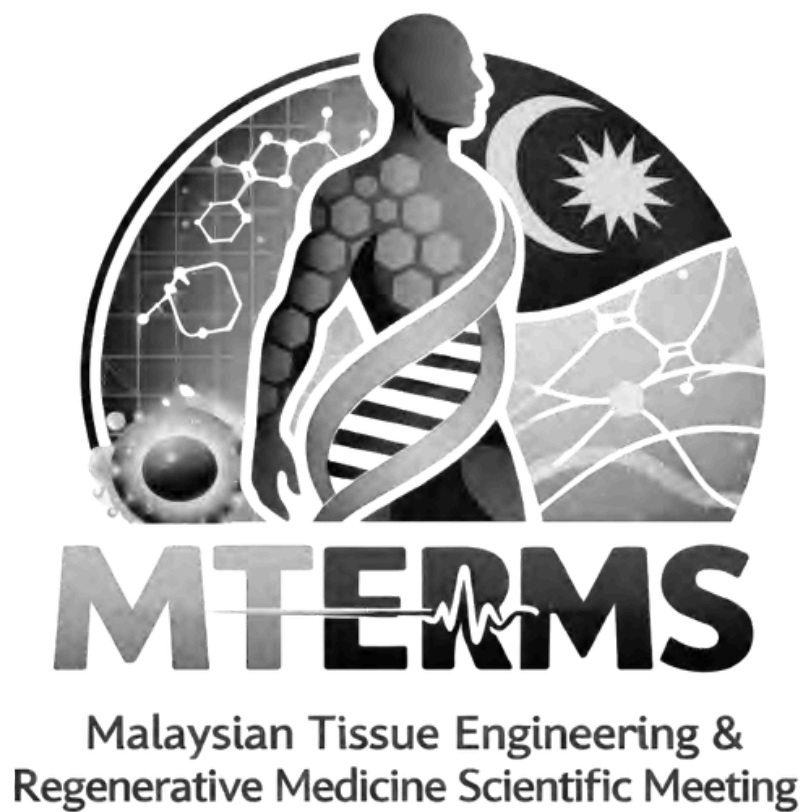


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**UNIVERSITI  
TEKNOLOGI  
MARA**

**Fakulti  
Pergigian**

## FORWARDS

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## FOREWORD

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**Assalamualaikum and a very good day.**

It is with great pleasure and pride that I welcome our distinguished speakers, delegates, researchers, clinicians, industry partners, students and guests to the 10th Malaysian Tissue Engineering & Regenerative Medicine Scientific Meeting (MTERMS 2026).

This year's theme, "From Discovery to Therapy: Advances in Tissue Engineering and Regenerative Medicine," captures both the remarkable progress of our field and the responsibility before us. Scientific discovery is the foundation of innovation, but its greatest impact is realised when knowledge is translated into safe, effective and accessible therapies that improve patients' lives.

Tissue engineering and regenerative medicine are advancing at an extraordinary pace. Developments in stem cell technologies, biomaterials, biofabrication, extracellular vesicles, cell and gene therapies, and other emerging approaches continue to redefine the possibilities of regenerative healthcare. Translating these advances from the laboratory to clinical practice, however, requires strong partnerships across disciplines and sectors, together with rigorous science, responsible innovation and a shared commitment to patient benefit.

MTERMS has long provided a platform where these connections can flourish. As we celebrate its 10th edition, we also recognise how far our scientific community has progressed and look ahead to the opportunities that lie before us. I hope MTERMS 2026 will inspire meaningful scientific exchange, foster new collaborations and provide our young investigators with the confidence and opportunities to become the next generation of leaders in regenerative medicine.

TESMA remains committed to advancing excellence in tissue engineering and regenerative medicine, strengthening collaboration among academia, healthcare and industry, and enhancing Malaysia's contribution and visibility within the regional and global regenerative medicine community.

I extend my sincere appreciation to the MTERMS 2026 Organising Committee, our co-organisers and collaborating institutions, distinguished speakers, sponsors, partners, exhibitors and volunteers whose dedication has made this meeting possible. I also thank every participant for being part of this important scientific gathering.

May MTERMS 2026 be a meeting that not only showcases discoveries, but also builds the partnerships that transform discovery into therapy.

I wish you an inspiring, productive and memorable MTERMS 2026.

**Warmest regards,**

**Professor Dr. Norshariza Binti Nordin**

President, TESMA

2025–2027



## DEAN FACULTY OF DENTISTRY UiTM

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**Assalamualaikum Warahmatullahi Wabarakatuh,  
Salam Sejahtera and Warm Greetings.**

On behalf of the Faculty of Dentistry, Universiti Teknologi MARA (UiTM), I am delighted that our faculty is part of the host institution for this significant scientific gathering, which continues to serve as a premier platform for advancing knowledge and collaboration in tissue engineering and regenerative medicine.

The theme of this year's meeting, **"From Discovery to Therapy: Advances in Tissue Engineering and Regenerative Medicine,"** aptly reflects the remarkable journey of scientific innovation—from fundamental discoveries in the laboratory to translational research and clinical applications that ultimately enhance patient care and improve quality of life. As regenerative medicine continues to evolve, interdisciplinary collaboration remains the driving force behind transformative healthcare solutions.

In dentistry and craniofacial sciences, regenerative approaches have become increasingly important in addressing complex clinical challenges, including bone regeneration, periodontal tissue repair, biomaterial innovation, and tissue reconstruction. The scientific contributions compiled in this abstract book exemplify the dedication, creativity, and commitment of researchers, clinicians, engineers, and industry partners in advancing these frontiers. Together, these works showcase not only scientific excellence but also the collective aspiration to translate research discoveries into meaningful therapeutic outcomes.

The conference celebrates the enthusiasm and achievements of students and early-career researchers, whose contributions represent the future of regenerative medicine. Their active participation in presenting new ideas, engaging in scientific discourse, and building professional networks reflects the vibrant and sustainable research ecosystem that we continue to cultivate in Malaysia and beyond.

I hope that the abstracts presented in this volume will inspire meaningful discussions, stimulate new collaborations, and spark innovative ideas that contribute to future breakthroughs in tissue engineering and regenerative medicine. May this publication serve as a valuable reference and testament to our shared commitment to advancing regenerative healthcare through research, innovation, and collaboration.

I wish all participants every success and an enriching scientific experience throughout MTERMS 2026.

**Thank you.**

**Assoc. Prof. Dr. Mas Suryalis Ahmad**  
Dean  
Faculty of Dentistry  
Universiti Teknologi MARA (UiTM)

## CHAIRPERSON MTERMS 2026

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It is my great honour and privilege to welcome you to the Malaysian Tissue Engineering & Regenerative Medicine Symposium (MTERMS) 2026.

MTERMS has always been more than a scientific meeting. It is a platform where ideas are challenged, knowledge is shared, collaborations are formed, and discoveries begin their journey towards meaningful applications. The theme and scientific scope of MTERMS 2026 reflect how rapidly our field is evolving. Today, regenerative medicine sits at the intersection of multiple disciplines. Advances in stem cell science, biomaterials, tissue engineering, organoids, 3D bioprinting, gene and cell engineering, immunomodulation, extracellular vesicles, smart biomaterials and biofabrication.

Within these pages are the ideas, experiments, findings and aspirations of researchers working across different disciplines, from fundamental mechanisms of stem cells and cell-material interactions to sophisticated scaffold technologies, regenerative strategies and translational applications. Equally important to MTERMS 2026 is our commitment to developing the next generation of scientists. Through the Student and Young Investigator Symposium (SYIS), we provide emerging researchers with a platform not merely to present their findings, but to question, discuss, connect and become part of a wider scientific community

As we celebrate scientific advancement, we must also recognise that innovation carries responsibility. The future of regenerative medicine will depend not only on what we are capable of developing, but also on how effectively, ethically and responsibly those discoveries can be translated. Issues surrounding clinical translation, regulatory science, biomanufacturing, sustainability, ethics and commercialisation therefore form an essential part of the conversations we seek to advance through MTERMS 2026.

On behalf of the MTERMS 2026 Organising Committee, I extend my sincere appreciation to our distinguished speakers, scientific and organising committees, reviewers, presenters, delegates, collaborators, sponsors and partners. Your expertise, dedication and support have made this symposium possible.

To every researcher whose work appears in these pages, thank you for sharing your science with us. May the discussions initiated at MTERMS 2026 develop into new collaborations, stronger networks and, ultimately, innovations with meaningful impact on science, healthcare and society.

**Associate Professor Ts Dr. Nur Aliana Hidayah Mohamed**

Chairperson The Malaysian Tissue Engineering and Regenerative Medicine  
Scientific Meeting (MTERMS)

## Tissue Engineering and Regenerative Medicine Society of Malaysia - The History

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The Tissue Engineering Society of Malaysia (TESMA) was founded in 2004, during a formative period when tissue engineering was emerging in Malaysia as a distinct multidisciplinary field. Its establishment followed the first Malaysian conference dedicated to biomaterials and tissue engineering, the International Conference on Biomaterials and Tissue Engineering (IC-BITE 2004), held in Kuala Lumpur that same year. These developments brought together scientists, clinicians, engineers and industry representatives who recognised the need for a national platform to advance research, education and clinical translation in the field.

TESMA was formally registered on 10 November 2004. Professor Emerita Dato' Dr Ruszymah Hj. Idrus, widely recognised as a pioneer of tissue engineering and regenerative medicine in Malaysia, served as the Society's founder and first President. Following their postdoctoral training in the United States, she and her husband, Dr Aminuddin Saim, an otorhinolaryngologist, established and expanded the Tissue Engineering Research Group at Universiti Kebangsaan Malaysia Medical Centre in 1999.

This pioneering research group subsequently developed into the Tissue Engineering Centre in 2008 and later evolved into the Centre for Tissue Engineering and Regenerative Medicine and the Department of Tissue Engineering and Regenerative Medicine under the Faculty of Medicine, UKM, in 2020 and 2024, respectively. Working with multidisciplinary collaborators, the group contributed significantly to the development of Malaysia's capabilities in cell culture, biomaterials, engineered skin, cartilage, bone, cornea, peripheral nerve and respiratory epithelium, as well as clinical-grade cell and tissue processing.

As the discipline expanded beyond the engineering of replacement tissues to encompass broader regenerative strategies such as cell and gene therapy, smart biomaterials, cell-free biologics and small molecules the Society adopted the name Tissue Engineering and Regenerative Medicine Society of Malaysia in 2010, while retaining the acronym TESMA. It also became affiliated with the Tissue Engineering and Regenerative Medicine International Society (TERMIS). TESMA's mission has remained focused on advancing science and technology, promoting education and research, encouraging ethical professional practice, and strengthening collaboration among academia, healthcare institutions, government agencies and industry.

In 2012, TESMA launched its official scientific journal, *Regenerative Research*. This open-access, peer-reviewed, semi-annual online journal provides a platform for the dissemination of research in tissue engineering and regenerative medicine in Malaysia and throughout the region.

TESMA's flagship activity is the Malaysian Tissue Engineering and Regenerative Medicine Scientific Meeting (MTERMS). Originating from the national scientific gathering in 2004 and subsequently developing into a biennial meeting, MTERMS has provided a sustained forum for keynote lectures, symposia, oral and poster presentations, regulatory discussions, and dedicated section for students and young investigators. It has connected Malaysian researchers with regional and international leaders while promoting collaboration among universities, research institutes, regulatory agencies and industry partners.



## Tissue Engineering and Regenerative Medicine Society of Malaysia - The History



Today, TESMA remains Malaysia's principal professional society dedicated to tissue engineering and regenerative medicine. As the community gathers for the 10th MTERMS, let us continue to build upon the vision of its founders and successive leadership committees by strengthening multidisciplinary collaboration, nurturing future generations of researchers, and supporting the translation of scientific discoveries into safe, ethical and clinically meaningful regenerative solutions.



An early photograph of participants at the 3rd Malaysian Tissue Engineering & Regenerative Medicine Scientific Meeting at Universiti Putra Malaysia in 2010. Professor Emerita Dato' Dr Ruszymah Hj. Idrus, who spearheaded Malaysia's pioneering tissue-engineering community, is pictured fifth from the left in the front row.

Sources: UKM Department of Tissue Engineering & Regenerative Medicine, website: <https://www.ukm.my/dtermfper>; Radzi et al., "A Review of Malaysian Experience in Tissue Engineering Research and Development" (2018); TESMA website: <https://tesma.org.my>

**10<sup>th</sup> Malaysian Tissue Engineering and Regenerative Medicine  
Scientific Conference (MTERMS) 2026**

**Venue: Concorde Hotel**

**Date: 7-8 September 2026**

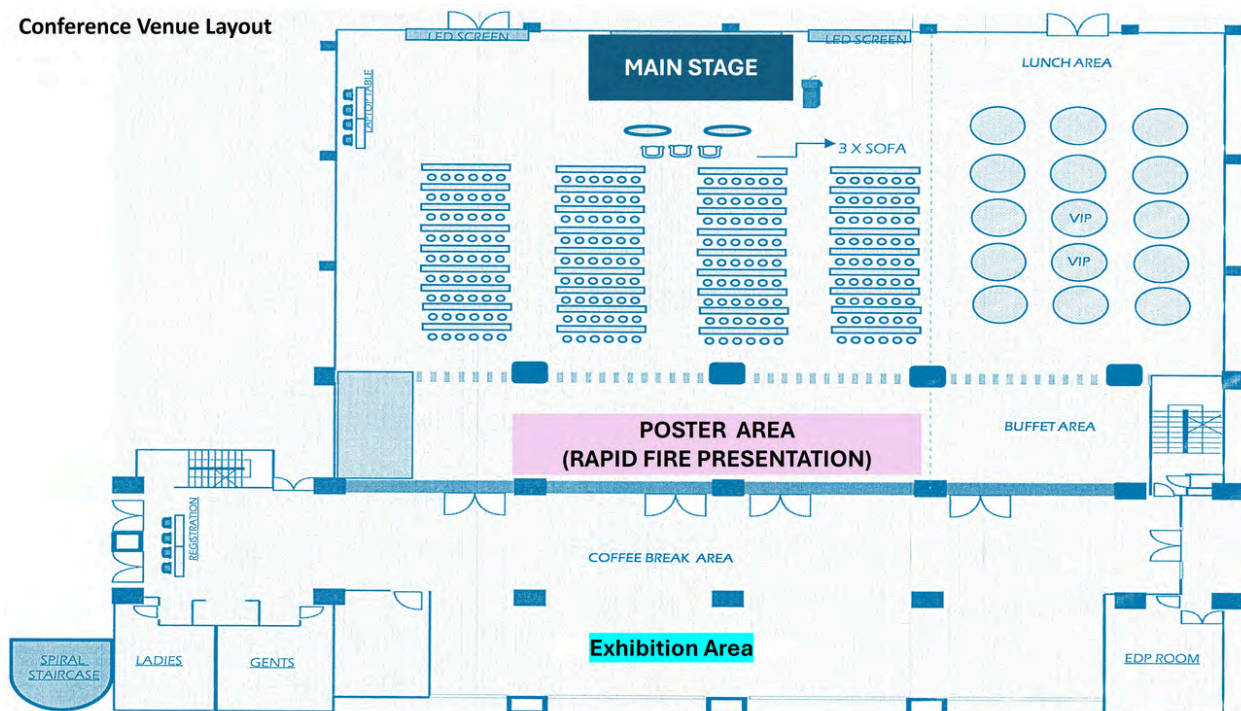
| 7 September 2026, Monday (Day 1) |                                                                                                                                                                                    |                                                                                                      |                                                                                  |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Time                             | Event                                                                                                                                                                              |                                                                                                      |                                                                                  |
| 07:30 – 08:30                    | Registration                                                                                                                                                                       |                                                                                                      |                                                                                  |
|                                  | Venue: Grand Ballroom                                                                                                                                                              |                                                                                                      |                                                                                  |
| 08:30 – 09:00                    | <b>OPENING CEREMONY</b><br>Prof. Datuk Ts. Dr. Shahrin bin Sahib @ Sahibuddin, FASc., VC UiTM                                                                                      |                                                                                                      |                                                                                  |
| 09:00 – 10:00                    | <b>OPENING KEYNOTE</b><br>Prof. Emeritus John Mason<br>Title: Stem Cell-derived organoids as tools to understand brain development and diseases                                    |                                                                                                      |                                                                                  |
| 10:00 – 10:30                    | <b>PLENARY 1</b><br>Dr. Chua Kien Hui<br>Title: From Fundamental Discovery in Regenerative Medicine to Commercial Impact                                                           |                                                                                                      |                                                                                  |
| 10:30 – 11:00                    | Tea Break<br>Exhibition (Exhibition Hall) & e-Poster Viewing (Poster Area)                                                                                                         |                                                                                                      |                                                                                  |
|                                  | <b>SYMPOSIUM</b>                                                                                                                                                                   |                                                                                                      |                                                                                  |
| 11:00 – 12:15                    | <b>Grand Ballroom</b><br><b>SP1:</b><br>Functional Biomaterials for Regenerative Medicine                                                                                          | <b>Grand Patio</b><br><b>SP2:</b><br>Cell-Free Therapies: Secretome, Exosomes, and Conditioned Media | <b>Patio1</b><br><b>SP3:</b><br>iPSC: Innovations Toward Clinical Translation    |
| 12:15 – 12:45                    | <b>INDUSTRIAL TALK 1 (Grand Ballroom)</b><br>Mr Sean Ng<br>Founder and Managing Director, Ming Medical Sdn. Bhd.<br>Title: <a href="#">Advance Regenerative Medicine Solutions</a> |                                                                                                      |                                                                                  |
| 12:45 – 14:00                    | Lunch<br>Exhibition (Exhibition Hall) & e-Poster Rapid Fire Presentation (Poster Area)                                                                                             |                                                                                                      |                                                                                  |
|                                  | <b>ORAL PRESENTATION (OP)</b>                                                                                                                                                      |                                                                                                      |                                                                                  |
| 14:00 – 15:00                    | <b>Grand Ballroom</b><br><b>Theme 1:</b><br>Stem Cell Innovations and Clinical Applications                                                                                        | <b>Grand Patio</b><br><b>Theme 2:</b><br>Biomaterials and Tissue Scaffolds                           | <b>Patio1</b><br><b>Theme 3:</b><br>Extracellular Vesicle and Disease Remodeling |
| 15:00 – 15:10                    | Tea Break<br>Exhibition (Exhibition Hall) & e-Poster Viewing (Poster Area)                                                                                                         |                                                                                                      |                                                                                  |
|                                  | <b>Student and Young Investigator Symposium (SYIS)</b>                                                                                                                             |                                                                                                      |                                                                                  |
| 15:10 – 16:30                    | <b>Grand Ballroom</b>                                                                                                                                                              | <b>Grand Patio</b>                                                                                   | <b>Patio1</b>                                                                    |

|               |                                                                                                                                                                                                                                                                                                                                                                       |                                                    |                                                                                            |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------|
|               | <b>Theme 1:</b><br>Stem Cell Innovations and<br>Clinical Applications                                                                                                                                                                                                                                                                                                 | <b>Theme 2:</b><br>Hydrogels and<br>Biomaterials   | <b>Theme 3:</b><br>Innovations in<br>Bioscaffolds for Tissue<br>Engineering                |
|               | <b>Venue: Grand Ballroom</b>                                                                                                                                                                                                                                                                                                                                          |                                                    |                                                                                            |
| 16:30 – 17:00 | <b>PLENARY 2</b><br>Prof. Ika Dewi Ana<br>Title: Preventing biofilm formation and reducing persistent infections associated with indwelling scaffolds                                                                                                                                                                                                                 |                                                    |                                                                                            |
| 17:00 – 18:00 | <b>KEYNOTE TALK</b><br>Prof. Amin Tamadon<br>Title: From Discovery to Therapy: Building a National ATMP and Mesenchymal Stromal Cell-Derived Extracellular Vesicle Platform in Kazakhstan—Opportunities for Malaysia–Kazakhstan Collaboration                                                                                                                         |                                                    |                                                                                            |
| 18:00         | <b>Adjourn Day 1</b>                                                                                                                                                                                                                                                                                                                                                  |                                                    |                                                                                            |
|               |                                                                                                                                                                                                                                                                                                                                                                       |                                                    |                                                                                            |
|               | <b>8 SEPTEMBER 2026, Tuesday (Day 2)</b>                                                                                                                                                                                                                                                                                                                              |                                                    |                                                                                            |
| 08:30 – 09:00 | Registration                                                                                                                                                                                                                                                                                                                                                          |                                                    |                                                                                            |
|               | <b>Venue: Grand Ballroom</b>                                                                                                                                                                                                                                                                                                                                          |                                                    |                                                                                            |
| 09.00– 09:30  | <b>PLENARY 3</b><br>Prof. Kyung-Soon Park<br>Title: 3D-Printed Patient-Specific Implants for Reconstruction of Massive Acetabular Defects in Revision Total Hip Arthroplasty                                                                                                                                                                                          |                                                    |                                                                                            |
| 09:30 – 10.45 | <b>Distinguished Expert Forum</b><br><b>Theme:</b> Bridging Discovery to Delivery: Translating Regenerative Science into Clinical and Commercial Reality<br><b>Panel:</b> Prof. Dr. John Mason, Prof. Dr. Bassem Sadek, Dr. Chua Kien Hui, Prof. Amin Tamadon, Prof. Dr. Ika Dewi Ana and Prof. Dr. Kyung-Soon Park<br><b>Moderator:</b> Assoc. Prof. Dr. Ng Min Hwei |                                                    |                                                                                            |
| 10:45-11:00   | Tea Break<br>Exhibition (Exhibition Hall) & e-Poster Viewing (Poster Area)                                                                                                                                                                                                                                                                                            |                                                    |                                                                                            |
|               | <b>SYMPOSIUM</b>                                                                                                                                                                                                                                                                                                                                                      |                                                    |                                                                                            |
| 11:00 – 12.15 | <b>Grand Ballroom</b><br><b>SP4:</b><br>Translational Biofabrication in Regenerative Medicine: Bridging Cells, Biomaterials, and Clinical Applications                                                                                                                                                                                                                | <b>Grand Patio</b><br><b>SP5:</b><br>TERM Research | <b>Patio1</b><br><b>SP6:</b><br>Vascular Regenerative and Reparative Medicine - A Frontier |
| 12:15 – 13:15 | <b>22nd TESMA Annual General Meeting (AGM) 2026</b><br>Venue: Grand Ballroom                                                                                                                                                                                                                                                                                          |                                                    |                                                                                            |
| 13:15 – 14:00 | Lunch<br>Exhibition (Exhibition Hall) & e-Poster Viewing (Poster Area)                                                                                                                                                                                                                                                                                                |                                                    |                                                                                            |
|               | <b>SYMPOSIUM</b>                                                                                                                                                                                                                                                                                                                                                      |                                                    |                                                                                            |

## SCIENTIFIC PROGRAM

|               |                                                                                                                                                                              |                                                                                   |                                                                                                     |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| 14:00-15:15   | <b>Grand Ballroom</b><br><b>SP7:</b><br>Nerve & Skin<br>Regeneration                                                                                                         | <b>Grand Patio</b><br><b>SP8:</b><br>Smart Technology for<br>Medical Applications | <b>Patio1</b><br><b>SP9:</b><br>Innovation to Clinical<br>Translation in Emerging<br>Cell Therapies |
| 15:15-15:45   | Tea Break<br>Exhibition (Exhibition Hall) & e-Poster Viewing (Poster Area)                                                                                                   |                                                                                   |                                                                                                     |
|               | <b>Venue: Grand Ballroom</b>                                                                                                                                                 |                                                                                   |                                                                                                     |
| 15:45 – 16:45 | <b>CLOSING KEYNOTE</b><br>Prof. Dr. Bassem Sadek<br>Title: Nanoparticle-Enabled Biomaterials: Translating Nanoengineered Platforms<br>from Discovery to Regenerative Therapy |                                                                                   |                                                                                                     |
| 16:45 – 17:15 | <b>CLOSING AND AWARD GIVING CEREMONY</b>                                                                                                                                     |                                                                                   |                                                                                                     |
| 17:15         | <b>END</b>                                                                                                                                                                   |                                                                                   |                                                                                                     |

## Conference Venue Layout





## **SPEAKERS ABSTRACTS**

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## Stem Cell-derived Organoids as Tools to Understand Brain Development and Disease

John O. Mason, David J. Price and Wai Kit Chan

*Institute for Neuroscience and Cardiovascular Research, University of Edinburgh, UK*

**Abstract:** Understanding the molecular mechanisms that control embryonic development of the brain is important for identifying the origins of neurodevelopment disorders, such as autism. Until recently, mice were commonly used as models to investigate mechanisms of brain development. While this has given us a good understanding of many aspects of how the brain develops embryonically, there are significant differences between mouse and human brain development. The advent of cerebral organoids - small tissue rudiments that closely resemble human brain tissue which are grown in culture from stem cells - opened up the possibility to study human brain-like tissue directly. Much of our work has focussed on the PAX6 gene, a key regulator of multiple aspects of forebrain development. Humans with heterozygous mutations in PAX6 display mild structural abnormalities of the brain and some show autism-like symptoms. We have generated human iPSC lines carrying homozygous mutations in PAX6. Cerebral organoids grown from these cells show a significant increase in the number of inhibitory neurons, altering the balance between excitatory and inhibitory cells, a mechanism widely believed to underlie some autism spectrum disorders. We found that the additional inhibitory neurons most likely arise through alteration to intercellular signalling pathways, notably the pleiotrophin pathway. We are currently exploring the phenotypes of PAX6 $\pm$  human organoids, which more closely resemble the human patients.

**Biography:** Professor John O. Mason, BSc, PhD is an emeritus professor of molecular neural development. John Mason's long association with the University of Edinburgh began in 1979 when he enrolled in the University's BSc Molecular Biology programme. After graduation in 1983, he moved to Cambridge for his PhD, working at the MRC's renowned Laboratory of Molecular Biology where he first developed his life-long interest in understanding the control of gene expression. After completing his PhD in 1987, he was awarded a prestigious fellowship from the European Molecular Biology Organisation to support a postdoctoral position in the laboratory of Nobel Laureate Harold Varmus at the University of California at San Francisco, where he first became interested in understanding embryonic development. He returned to the University of Edinburgh in 1991, initially at the Centre for Genome Research in the College of Science and Engineering and then to a lectureship in Physiology in the College of Medicine and Veterinary Medicine in 1996, rising to the award of a personal chair in Molecular Neural Development in 2020. John's research on the molecular mechanisms that control embryonic development of the brain has been continuously funded for over 30 years including substantial grants from the BBSRC, MRC, Wellcome Trust, Simons Initiative for the Developing Brain and others, totalling in excess of £10m. Using genetically modified animal and human stem cell based models, his research has made a major contribution to our understanding of how the brain develops during embryogenesis and how mutations in specific genes involved in controlling brain development can lead to neurodevelopmental disorders such as autism. He has shared resources generated in his lab including novel transgenic strains with many research teams around the world. His work has led to the publication of more than 70 primary research papers and several well-cited review articles. He has given invited talks on his research in the UK, Europe, USA and Asia and has served as a member of multiple grant panels in the UK, USA and Netherlands. He was elected a fellow of the Royal Society of Biology in 2019. John has taken a leading role in the teaching of genetics and molecular and cell biology topics within Biomedical Sciences for almost 30 years, including the creation of a successful new honours programme in Anatomy and Development. To accompany one of his undergraduate courses, he co-authored an award winning textbook on developmental neuroscience, which ran to two English language editions (2011 and 2017) and a simplified Chinese translation (2020).



## From Discovery to Therapy: Building a National ATMP and Mesenchymal Stromal Cell-Derived Extracellular Vesicle Platform in Kazakhstan—Opportunities for Malaysia–Kazakhstan Collaboration

Professor Amin Tamadon, DVM, PhD

*Department of Natural Sciences, West Kazakhstan Marat Ospanov Medical University, Aktobe, Republic of Kazakhstan*

**Abstract:** Advanced therapy medicinal products (ATMPs) are reshaping regenerative medicine, yet the clinical translation of mesenchymal stromal cell-derived extracellular vesicles (MSC-EVs) remains constrained by variability in cell sources and manufacturing, incomplete potency assays, scale-up challenges, quality-system requirements, and evolving regulatory expectations. This keynote presents the development of a national-scale translational platform in Kazakhstan centred on West Kazakhstan Marat Ospanov Medical University and its developing ATMP Center. The programme integrates umbilical cord MSC sourcing and controlled culture, EV isolation and purification, physicochemical and biological characterization, formulation, stability and potency assessment, preclinical efficacy and safety studies, regulatory documentation, and early clinical planning. Supported by a major nationally funded exosome-therapy programme, the platform is being used to investigate EV-based approaches for reproductive disorders, wound repair, regenerative dentistry, oncology-related models, and other tissue-regeneration indications. It also supports an early-phase clinical programme evaluating the dermal safety and tolerability of a topical EV-containing liquid in healthy adults. The presentation will discuss practical lessons in distinguishing research-grade from clinical-grade evidence; defining critical quality attributes and release criteria; selecting scalable purification methods; developing mechanism-linked potency assays; controlling sterility, endotoxin, and mycoplasma risks; establishing GMP-compatible documentation; and designing a staged pathway from preclinical proof-of-concept to first-in-human evaluation. Beyond scientific development, the keynote proposes a Malaysia–Kazakhstan collaboration framework that connects Malaysia's strengths in tissue engineering, biomaterials, biomanufacturing, and commercialization with Kazakhstan's emerging ATMP infrastructure and Central Asian research and clinical networks. Priority opportunities include joint preclinical validation, biomaterial-based EV delivery systems, multicentre early-phase studies, researcher training, regulatory and quality-system exchange, joint grant applications, technology transfer, and university–industry partnerships. The overarching objective is to demonstrate how integrated institutional platforms and international cooperation can convert promising EV discoveries into reproducible, quality-controlled, and clinically relevant therapeutic products.

**Keywords:** Advanced therapy medicinal products, mesenchymal stromal cells, extracellular vesicles, exosomes, regenerative medicine, clinical translation, Malaysia–Kazakhstan collaboration

**Biography:** Professor Amin Tamadon, PhD, is a Full Professor and Principal Investigator of Regenerative Medicine at West Kazakhstan Marat Ospanov Medical University, Kazakhstan. Since 2010, his research has focused on mesenchymal stromal cells, extracellular vesicles and exosomes, tissue engineering, reproductive regenerative medicine, biomaterials, and translational therapy development. He earned his PhD from Shiraz University and completed postdoctoral fellowships at Seoul National University, Republic of Korea, and Fudan University, China. With nearly 300 scholarly works recorded in ORCID and more than 230 Scopus-indexed documents, he has contributed extensively to cell-based and cell-free regenerative therapies. He is a senior scientific leader in Kazakhstan's nationally funded exosome and ATMP development programme and contributes to the establishment of the Advanced Therapy Medicinal Products Center at his university, integrating EV manufacturing, characterization, quality control, preclinical evaluation, regulatory preparation, and early-phase clinical translation. He also serves as Editor-in-Chief of the West Kazakhstan Regenerative Medicine Journal and promotes international partnerships in regenerative medicine, technology transfer, researcher training, and Malaysia–Kazakhstan scientific collaboration.



## From Fundamental Discovery in Regenerative Medicine to Commercial Impact

Chua Kien Hui  
*Supergenics Berhad*

**Abstract:** Over the past two decades, regenerative medicine has evolved from exploratory laboratory science into a translational discipline with tangible clinical and commercial outcomes. Early discoveries in stem cell biology and tissue regeneration provided essential insights into cellular plasticity, micro-environmental signalling and scaffold-based tissue repair. However, bridging the gap between laboratory findings and therapeutic application requires more than scientific rigor; it demands interdisciplinary integration, regulatory compliance, intellectual property protection and scalable manufacturing under cGMP facility. Drawing from experience in academic mentorship, industry consultancy and technology commercialization; this sharing will outline key translational milestones including patent development, GMP process design, regulatory engagement and industry collaboration. Critical challenges in commercialization of regenerative medicine such as production standardization, reproducibility of product quality and efficacy, ethical governance, investor confidence and market education are areas need to carefully consider and addressed. Emphasis will be placed on building translational ecosystems that integrate academia, clinicians, regulators and industry partners to accelerate safe and effective therapy deployment. As regenerative medicine enters an era of advanced cell therapy, biomimetic engineering, and precision regenerative strategies; the future lies not only in scientific discovery but in sustainable innovation models. The ultimate impact of regenerative medicine will depend on our ability to transform knowledge into accessible, regulated and commercially viable therapies that improve patient outcomes.

**Biography:** Professor Dr. Chua Kien Hui currently holds several leadership roles, including Executive Director (R&D) at Supergenics Berhad, Chief Technology Officer at Glyness Industries Sdn. Bhd., and Medical Science Advisor at Wellspring Solution Sdn Bhd. His previous administrative experience includes serving as the Head of the Physiology Department for the Faculty of Medicine at Universiti Kebangsaan Malaysia (UKM) and as the Deputy Executive Director (Performance and Rating) for the Strategy Centre at UKM. Dr. Chua completed his Bachelor of Biomedical Sciences (HONS) at UKM in 1997, followed by a Master of Biotechnology at the University of Malaya (UM) in 1999, and a PhD in Physiology specializing in Tissue Engineering at UKM in 2005. His academic and research contributions are extensive, comprising 140 articles, two books, and two book chapters. These publications have earned 4,729 citations on Google Scholar, resulting in an H-index of 38. Dr. Chua has also demonstrated a strong commitment to education through the supervision of 20 PhD candidates, 15 Master's students, and 25 undergraduate project theses. Furthermore, his expertise has led to 32 engagements as an invited speaker and participation in five panel committees. His innovative work is reflected in nine patents, two trademarks, and two copyrights. Dr. Chua has received various accolades, such as the Excellent Service Award from the UKM Medical Faculty in 2008 and 2010, and the Excellent Academic and Research Award in 2009. He was also part of the group that received the Excellent Research Group Award at UKM in 2011 and 2012. In 2018, he was honored with the Anugerah Bitara & Perkhidmatan UKM for product commercialization and recognized as Tokoh Penyelidik. More recently, in 2019, he was named an Expert of the Research Institute for the Guangdong Bird's Nest Industry Association. Professionally, he has been a member of the Tissue Engineering and Regenerative Medicine International Society and the Malaysian Society of Pharmacology and Physiology since 2005, and a life member and committee member of the Malaysian Society of Tissue Engineering since 2009.





## From bench to bedside: Preventing biofilm formation and reducing persistent infections associated with indwelling scaffolds

Ika Dewi Ana

*Department of Dental Biomedical Sciences, Faculty of Dentistry, Universitas Gadjah Mada, Yogyakarta 55281, Indonesia*

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*Center of Excellence for Synthetic Extracellular Matrix and Biomedical Adjuvant Industry, Universitas Gadjah Mada, Yogyakarta 55281, Indonesia*

**Abstract:** Antimicrobial resistance (AMR) represents a critical global health challenge, particularly in hospital-acquired infections (HAIs) associated with medical implants and indwelling scaffolds. These infections are frequently driven by multidrug-resistant (MDR) pathogens capable of forming biofilms, which complicate clinical management, increase morbidity and mortality, and substantially elevate healthcare costs especially in low-resource settings. Persistent biofilm-associated infections contribute to prolonged hospitalization, repeated surgical interventions, and poor therapeutic outcomes. As conventional antibiotic-based strategies continue to lose efficacy, innovative and integrative approaches are urgently needed. In this presentation, we highlight recent advances from our laboratory in the development of carbonate hydroxyapatite (CHA)-based synthetic extracellular matrices with intrinsic antimicrobial properties. Particular emphasis is placed on strategies to prevent biofilm formation within scaffold structures, as well as on the translational challenges encountered in bridging laboratory findings to clinical applications. We also present bioresource-based technologies designed to enhance biocompatibility, durability, and therapeutic efficacy in tissue engineering and regenerative medicine. Special attention is given to plant-derived materials, including plant-derived exosome-like nanovesicles (PDENs), as multifunctional bioactive agents. These platforms aim to target a broad spectrum of MDR pathogens, modulate inflammation through macrophage polarization, minimize cytotoxic effects, and reduce the risk of exacerbating antimicrobial resistance.

**Keywords:** Indwelling scaffold, implant, extracellular vesicle (EV), antimicrobial resistance, regenerative therapy

**Biography:** Professor Ika Dewi Ana, DDS., PhD, is a distinguished dentist and biomedical scientist specializing in calcium phosphate chemistry, tissue engineering, and drug delivery systems. Her research encompasses the development of personalized and AI-assisted regenerative therapies, as well as high-impact projects like the MicroLung Project and the BioTUNE Consortium, which focus on lung-on-a-chip models and medical implant properties. Beyond her scientific expertise, she is highly trained in innovation management, having completed specialized coursework at Ashridge Business School in the UK. She has successfully transitioned from laboratory research to commercial impact, supported by 21 patents and 10 authored book chapters. She has successfully commercialized four major innovations in Indonesia including Carbonate Apatite (CHA) bone grafts and haemostatic sponges and is currently valorizing new technologies such as CHA-based vaccine adjuvants. She has held significant leadership roles, including serving as a commissioner for a medical device start-up and as a Council Member for TERMIS-AP. Since 2022, she has led the Research Collaboration Centre for Biomedical Scaffolds, which was designated in 2025 as a national Centre of Excellence (PUI-PT) for extracellular matrix and adjuvants. Her contributions extend to national policy and professional standards, where she has been instrumental in formulating Indonesian National Standards (SNI) for medical devices and conducting clinical trials. She has received prestigious recognition for her work, including the Habibie Prize in 2022, the WIPO National Award for Inventors in 2023, the Asia Best Innovation Award from the Hitachi Global Foundation in 2024 and a STEM Senior Scientist and a Gold Medal Academic Leader in Health and Medical Sciences in 2025 by the Indonesian government.



## 3D-Printed Patient-Specific Implants for Reconstruction of Massive Acetabular Defects in Revision Total Hip Arthroplasty

Kyung-Soon Park

*Chonnam National University Hwasun Hospital, Department of Orthopedic Surgery, South Korea*

**Abstract:** Reconstruction of massive acetabular bone defects in revision total hip arthroplasty (THA) remains a major surgical challenge. Conventional techniques—including jumbo cups, structural allografts, trabecular metal augments, cages, and cup-in-cage constructs—often have limitations in restoring anatomy and achieving stable fixation in cases with severe bone loss such as Paprosky type III defects or pelvic discontinuity. Recent advances in additive manufacturing have enabled the development of patient-specific 3D-printed implants designed to match individual pelvic anatomy and defect morphology. This study presents our experience with 3D-printed patient-specific implants for complex acetabular reconstruction in revision THA. Preoperative planning was performed using high-resolution computed tomography with thin slices to generate a three-dimensional model of the pelvis. A patient-specific implant was designed using dedicated software, allowing customization of iliac, ischial, and pubic flanges to achieve optimal fixation and restoration of hip center. A physical 3D-printed bone model was created for surgical simulation. The final implant was manufactured using electron beam melting technology to produce a porous titanium structure that promotes biological fixation. The patient-specific implants provided stable initial fixation through multiple flanges and screw fixation to the ilium, ischium, and pubis. The porous structure of the implant facilitated biological ingrowth without additional surface treatment. In the presented cases of multiple acetabular revisions with severe bone defects, satisfactory restoration of hip biomechanics and stable fixation were achieved. 3D-printed patient-specific implants represent a promising option for reconstruction of massive acetabular bone defects in revision THA. By allowing precise preoperative planning, anatomical customization, and enhanced biological fixation, this technology may improve stability and functional outcomes in highly complex revision cases. Further studies with larger cohorts and longer follow-up are required to confirm long-term implant survivorship.

**Biography:** Kyung-Soon Park, M.D., Ph.D., is a Professor at Chonnam National University Medical College and serves as a Department Head at Chonnam National University Hwasun Hospital. He completed his undergraduate medical education in 2003 and later earned a Ph.D. in 2011 from Chonnam National University Medical School. He achieved his board certification from the Korean Board of Orthopedic Surgery in February 2008. Dr. Park's leadership experience extends to his role as the Director of the Planning and Research department at Chonnam National University Bitgoeul Hospital. A highly engaged member of the medical community, Dr. Park is affiliated with several professional organizations, including the Korean Orthopaedic Association, the Korean Hip Society, and the Korean Society for Biomaterials. He has actively contributed to these organizations, notably serving on scientific committees. His clinical and research excellence is highlighted by multiple Academic Awards received from these societies between 2007 and 2025. Furthermore, Dr. Park is an accomplished researcher with a portfolio of 100 international publications. His extensive research primarily explores advancements in total hip arthroplasty, bone tissue regeneration, and complex orthopedic fracture management.



# Advance Regenerative Medicine Solutions

Sean Ng See Nguan  
*Ming Medical Sdn. Bhd.*

**Abstract:** Regenerative medicine is rapidly transforming the landscape of modern healthcare by moving beyond conventional symptom management towards tissue repair, regeneration and restoration of biological function. This presentation highlights emerging regenerative medicine solutions based on stem cells, extracellular vesicles (exosomes), secretomes and immune cell therapies, with a focus on their translational potential across a broad range of medical applications. The presentation explores the biological properties of stem cell-derived secretomes and exosomes, including their role in cellular signalling, tissue repair, angiogenesis and regeneration. The potential applications discussed include orthopaedics, neurological and neurodegenerative disorders, diabetes and metabolic diseases, cardiovascular and vascular regeneration, wound healing, ocular conditions, ageing and longevity, and immune-based therapies. Examples of regenerative approaches involving stem cells, exosomes and NK cells demonstrate the expanding potential of cell- and cell-free therapies in addressing complex and chronic diseases. The presentation also addresses the importance of advancing regenerative medicine from laboratory research towards clinical translation through standardised manufacturing, GLP studies, first-in-human studies, clinical trials and cGMP infrastructure. With continued advances in biotechnology and precision medicine, regenerative therapies have the potential to reshape healthcare by providing innovative, targeted and biologically driven solutions for tissue repair and restoration of function.

**Biography:** Sean Ng See Nguan is the Founder and Managing Director of Ming Medical Sdn. Bhd., a company specialising in cell and regenerative medicine therapy. Since establishing Ming Medical in 2014, he has been actively involved in the development and promotion of regenerative medicine, providing scientific and strategic consultation to healthcare practices in Malaysia and internationally, particularly across the Middle East and Southeast Asia. His areas of interest include stem cell therapy using mesenchymal stem cells (MSCs), exosomes, natural killer (NK) cells, and longevity medicine. He has also been involved as an industrial partner and co-investigator in local and international research projects, contributing to scientific publications and research collaborations. Prior to entering the regenerative medicine field, Sean built an extensive career in the technology industry, including founding and managing iTopia, serving as Managing Director of Informix Software (IBM Database), and holding various positions with IBM Malaysia. He began his professional career as a Clinical Research Officer at Imperial College London. Sean holds a BSc in Clinical Biochemistry from the University of Liverpool, UK. Over the years, he has been invited to speak at numerous international conferences, medical forums and regenerative medicine events across Asia, the Middle East and Europe. His combination of scientific training, technology leadership and entrepreneurial experience has positioned him at the intersection of healthcare, biotechnology and regenerative medicine, with a strong focus on translating emerging scientific discoveries into real-world healthcare applications.

## INVITED SPEAKER ABSTRACTS

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# Therapeutic Effects of SHED-conditioned Medium on Traumatic Buccal Mucosa Ulcers

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**Abstract:** Introduction: Traumatic ulcers are among the most common injuries, affecting as many as 95% of people. This study evaluates a nanoemulgel containing spent medium from stem cells derived from human exfoliated deciduous teeth (SHED-SM) for its effects on NIH-3T3 cell viability and healing of buccal mucosal traumatic ulcers. Materials and methods: SHED-SM cytotoxicity was tested on NIH-3T3 fibroblasts using the MTT assay. Thirty male Wistar rats aged 2–3 months were used to create traumatic ulcer models by excising the right buccal mucosa with a 3 mm punch biopsy. Animals were split into two groups of 15: treatment (SHED-SM) and control. The gel was applied once daily for four days. Assessments occurred on days 1, 3, 5, 7, and 14. Tissue sections were stained with Hematoxylin–Eosin and by immunohistochemistry to quantify neutrophils and to measure COX-2, VEGF, and Col-1 expression. Results: Statistical analysis (one-way ANOVA) indicated significant changes in the MTT viability assay, neutrophil infiltration, and COX-2, Col-1, and VEGF expression. All these measures were elevated in the SHED-SM–treated animals relative to the control group. Conclusion: SHED-SM appears to promote faster recovery of traumatic buccal mucosal wounds.

**Keywords:** Nanoemulgel, Spent medium, Stem Cells from Human Exfoliated Deciduous Teeth (SHED), Traumatic ulcers, Wound healing

# Development of Carbon Fiber and Zinc Oxide Reinforced Polyamide 6 Filaments for Craniofacial Reconstruction

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**Abstract:** Objectives: This study aimed to develop multifunctional polyamide 6 (PA6) composite filaments reinforced with carbon fiber (CF) and zinc oxide (ZnO) for fused deposition modelling (FDM) applications. It was hypothesized that the incorporation of CF and ZnO would enhance the mechanical performance and antibacterial properties of PA6 while maintaining suitable physical characteristics for filament fabrication. Materials and Methods: PA6 composite filaments containing CF and ZnO were fabricated through melt compounding followed by filament extrusion. The developed filaments were characterized to evaluate their mechanical, physical, antibacterial, and morphological properties. Mechanical testing was conducted to assess the reinforcing effect of CF, while physical characterization examined surface roughness. Antibacterial performance was evaluated against selected bacterial strains, and morphological analysis was performed to examine filler dispersion and interfacial interaction within the PA6 matrix. Results: The incorporation of CF improved the mechanical properties of the PA6 filaments through enhanced stiffness and reinforcement effects. Surface roughness increased with filler addition, indicating changes in the physical characteristics of the composite filaments. ZnO imparted antibacterial activity against the selected bacterial strains due to its inherent antimicrobial behaviour. Morphological observations confirmed acceptable dispersion of CF and ZnO within the PA6 matrix, indicating good interfacial interaction between the fillers and polymer. Conclusions: The combined incorporation of CF and ZnO successfully produced multifunctional PA6 composite filaments with enhanced mechanical performance and antibacterial properties. These findings demonstrate the potential of the developed composite filaments as advanced materials for FDM 3D printing applications requiring both structural integrity and antimicrobial functionality.

**Keywords:** Polyamide 6, Filament, 3D Printing

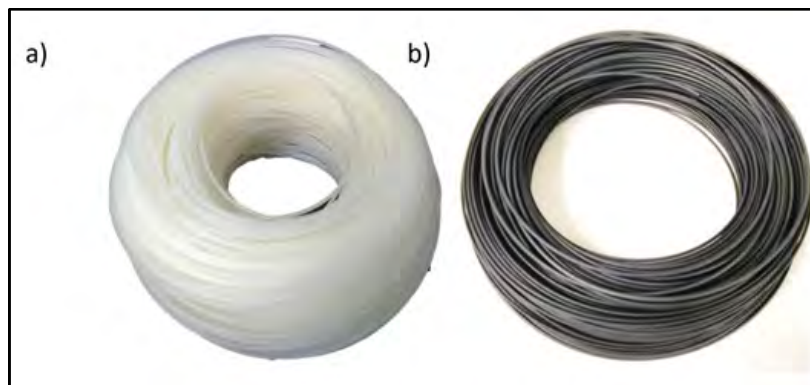


Figure 1: Fabricated filament feedstock (a) Unfilled PA 6, (b) CF and ZnO filled PA 6

# Engineering Functional Hydroxyapatite Coatings with Plant-Derived Polyphenols for Bone Regeneration and Orthopaedic Implants

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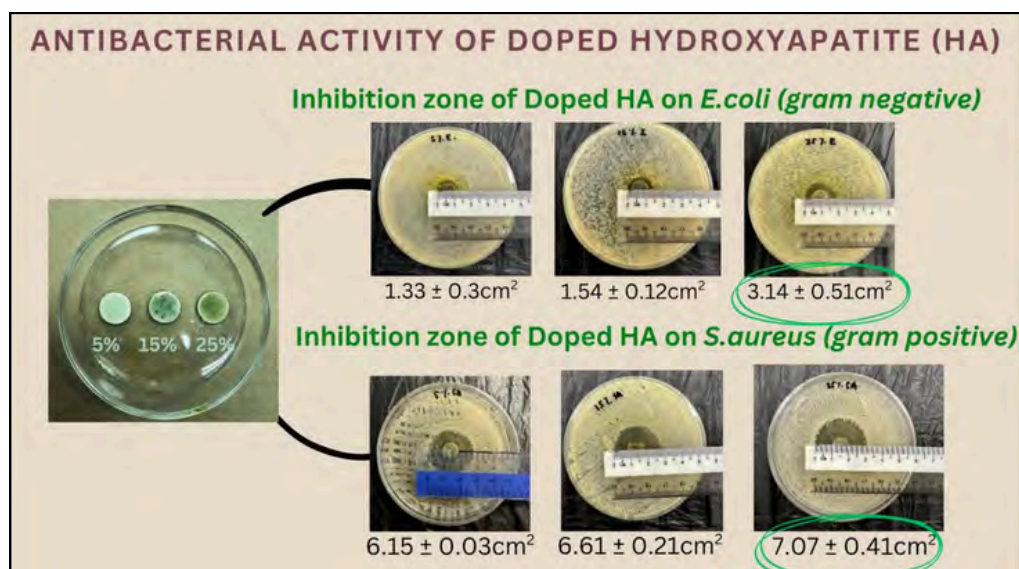
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**Abstract:** Objectives: Hydroxyapatite (HA) is widely used as an orthopedic implant coating because of its excellent biocompatibility and osteoconductivity. However, its clinical performance is limited by poor antibacterial activity and low drug-loading capacity, increasing the risk of implant-associated infections. This study investigates the potential of *Euphorbia tirucalli* L. extract as a natural polyphenol-rich additive to enhance the biological and physicochemical properties of HA coatings. Materials and Methods: The extract was prepared using different solvents, with aqueous ethanol producing the highest total phenolic content ( $2.501 \pm 0.50$  mg GAE/g) and total flavonoid content ( $1.307 \pm 0.11$  mg QE/g). High-performance liquid chromatography identified gallic acid as the major bioactive compound ( $557.9 \pm 8.3$  mg/g). The extract was incorporated into HA through agitation, filtration, drying, and grinding. Successful incorporation was confirmed by X-ray diffraction, Fourier-transform infrared spectroscopy, and field-emission scanning electron microscopy. Antibacterial evaluation showed that HA containing 25% extract exhibited the largest inhibition zones against *Staphylococcus aureus* ( $7.07 \pm 0.41$  cm<sup>2</sup>) and *Escherichia coli* ( $3.14 \pm 0.51$  cm<sup>2</sup>). Cytocompatibility assessed using the MTT assay demonstrated that the same formulation achieved  $120.0 \pm 5.3\%$  cell viability after seven days, exceeding the ISO 10993-5 threshold for non-cytotoxic materials. Drug release studies revealed a sustained release profile, with the 5% extract formulation achieving the highest cumulative release (85.9% after 1440 min), following First-Order kinetics ( $R^2 > 0.95$ ). Conclusions: These findings demonstrate that E. tirucalli-doped HA is a promising multifunctional coating that combines antibacterial activity, excellent biocompatibility, and controlled drug release, providing a sustainable alternative for orthopedic implant surface engineering.

**Keywords:** Hydroxyapatite, *Euphorbia tirucalli*, Orthopaedic implant, Polyphenols, Biocompatibility



# A Machine Learning Liver-On-A-Chip System For Safer Drug Formulation

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**Abstract:** Drug metabolism critically influences therapeutic efficacy, toxicity, and drug–drug interactions, highlighting the need for predictive preclinical models. We developed a whole-tissue ex vivo liver-on-a-chip platform that recapitulates hepatic drug metabolism and enables high-throughput assessment of hepatotoxicity. Using acetaminophen (APAP) as a model compound, we characterized metabolic pathways and identified strategies to reduce toxic metabolite formation. Integration of this platform with machine learning–guided screening enabled the rapid identification of rutin and eugenol as novel functional excipients that significantly mitigated APAP-induced hepatotoxicity in vivo. We further engineered a controlled-release oral formulation incorporating these protective excipients, providing a scalable strategy for safer drug development and formulation design.

**Keywords:** Liver-on-a-chip, High-throughput screening, Drug metabolism, Machine learning, Drug formulation

# Natural Herb–Primed Neural Stem Cell-Derived Small Extracellular Vesicles Enhance Cross-Species Neurogenesis

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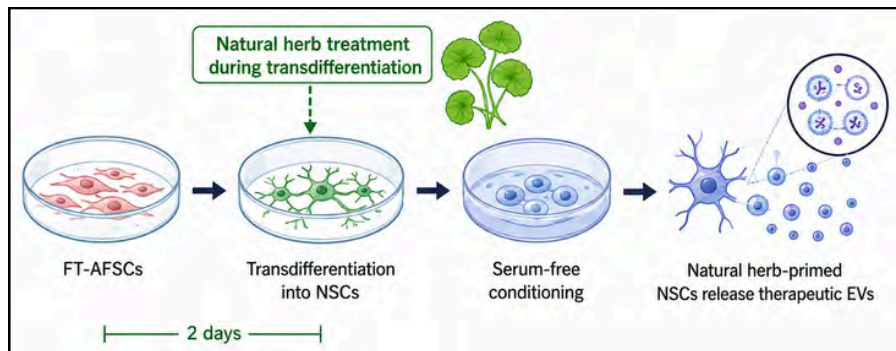
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**Abstract:** Neurodegenerative diseases (NDs) are characterised by progressive neuronal loss and the limited regenerative capacity of the central nervous system (CNS), highlighting the need for effective cell-free strategies to support neuronal repair. Objective: This study investigated whether priming full-term amniotic fluid stem cells (FT-AFSCs) with a natural herb during transdifferentiation into neural stem cells (NSCs) could generate NSC-derived small extracellular vesicles (sEVs) with enhanced neurogenic activity, with particular emphasis on their cross-species efficacy. Methods: Rat FT-AFSCs undergoing transdifferentiation into NSCs were primed with a natural herb extract for two days, followed by two days of serum-free conditioning for sEV collection. Small extracellular vesicles were isolated by sequential ultracentrifugation and characterised for particle size and distribution using nanoparticle tracking analysis, ultrastructure by transmission electron microscopy, and bioactive cargo, including insulin-like growth factors, by ELISA. Their functional activity was assessed in heterologous mouse embryonic stem cells (ESCs) undergoing neural differentiation by evaluating neural lineage marker expression, neuronal maturation, synaptic development, and neurite outgrowth. Results: Natural herb priming enhanced the neural commitment of FT-AFSCs, as demonstrated by increased Nestin and Sox1 expression and enhanced neurosphere formation, indicating improved acquisition and maintenance of an NSC-like phenotype. Small extracellular vesicles subsequently derived from these primed NSCs exhibited neurogenic activity across species. Treatment of mouse ESCs promoted neuronal differentiation and maturation, with increased expression of  $\beta$ III-tubulin (TUJ1), microtubule-associated protein 2 (MAP2), synaptophysin (SYP), and choline acetyltransferase (ChAT), accompanied by enhanced neurite outgrowth. Conclusion: Priming FT-AFSCs with a natural herb during neural transdifferentiation represents a biologically driven strategy for generating functionally enhanced NSC-derived sEVs. Their ability to promote neuronal differentiation, maturation, synaptic development, and neurite extension across species supports their potential as a scalable, cell-free platform for neurotherapeutic development in neurodegenerative diseases.

**Keywords:** Acellular therapy, Stem Cells, Exosome, Natural herb extract





# 3D Bioprinted Genipin-Crosslinked Gelatin Hydrogel with MSC-sEVs for Wound Healing

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**Abstract:** Objectives: This study aimed to develop a 3D-bioprinted gelatin–genipin hydrogel incorporating umbilical cord mesenchymal stem cell-derived small extracellular vesicles (UC-MSC-sEVs) to enhance stability, delivery, and wound healing efficacy. Materials and Methods: Four hydrogel formulations were prepared: non-crosslinked gelatin (8% and 10%) and genipin-crosslinked variants (8% and 10%). Physicochemical and mechanical properties were evaluated. Cells and vesicles were isolated and characterised using established international guidelines. In vitro assays assessed cell compatibility, fibroblast viability, and wound closure. In vivo evaluation was performed using a mouse full-thickness wound model with three groups: saline, hydrogel alone, and sEV-loaded hydrogel. Results: The 10% crosslinked hydrogel showed optimal performance, including rapid gelation, high mechanical strength, suitable swelling, controlled degradation, and appropriate moisture transmission. sEV incorporation significantly improved fibroblast viability without toxicity. In vivo, treated wounds demonstrated accelerated closure, complete epithelialisation by day 14, enhanced collagen deposition, and increased formation of new blood vessels. Conclusions: The developed 3D-bioprinted genipin-crosslinked gelatin hydrogel effectively improves sEV delivery and promotes wound healing. This platform shows strong potential for regenerative medicine applications and warrants further optimisation and mechanistic investigation.

**Keywords:** 3D bioprinting, Extracellular vesicles, Hydrogel, Regenerative medicine, Wound healing

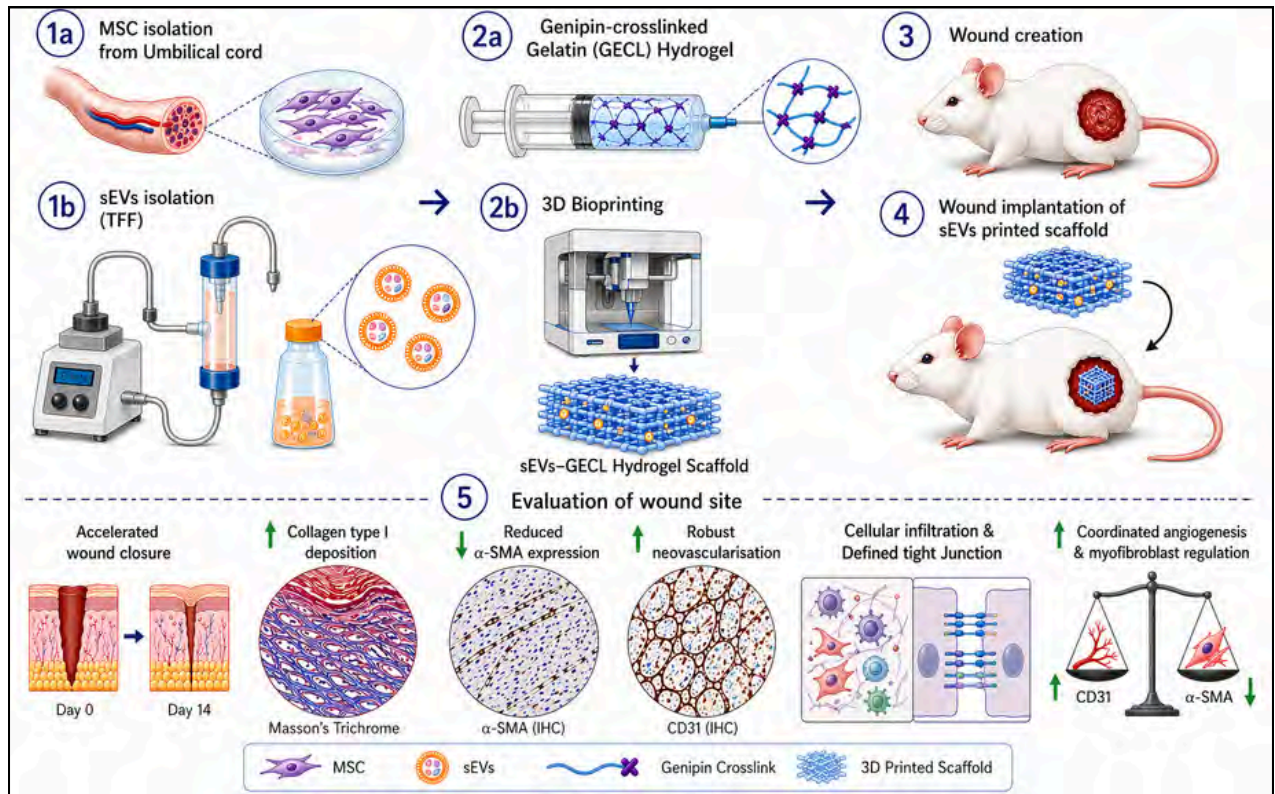


Figure 1: Development and in vivo evaluation of small extracellular vesicle (sEV)-functionalised genipin-crosslinked gelatin (GECL) hydrogel scaffolds for full-thickness skin wound healing. (1a) Mesenchymal stem cells (MSCs) were isolated from human umbilical cord tissue. (1b) sEVs were isolated using tangential flow filtration (TFF). (2a) sEVs were incorporated into the GECL hydrogel prior to printing. (2b) fabricated into sEV-GECL porous scaffolds via extrusion-based 3D bioprinting. (3) Full-thickness dorsal skin wounds were created in a murine model, followed by (4) implantation of the sEV-GECL scaffold at the wound site. (5) Wound healing outcomes were evaluated through macroscopic and histological analyses.

# Therapeutic potential of mesenchymal stem cell-derived extracellular vesicles in skin diseases

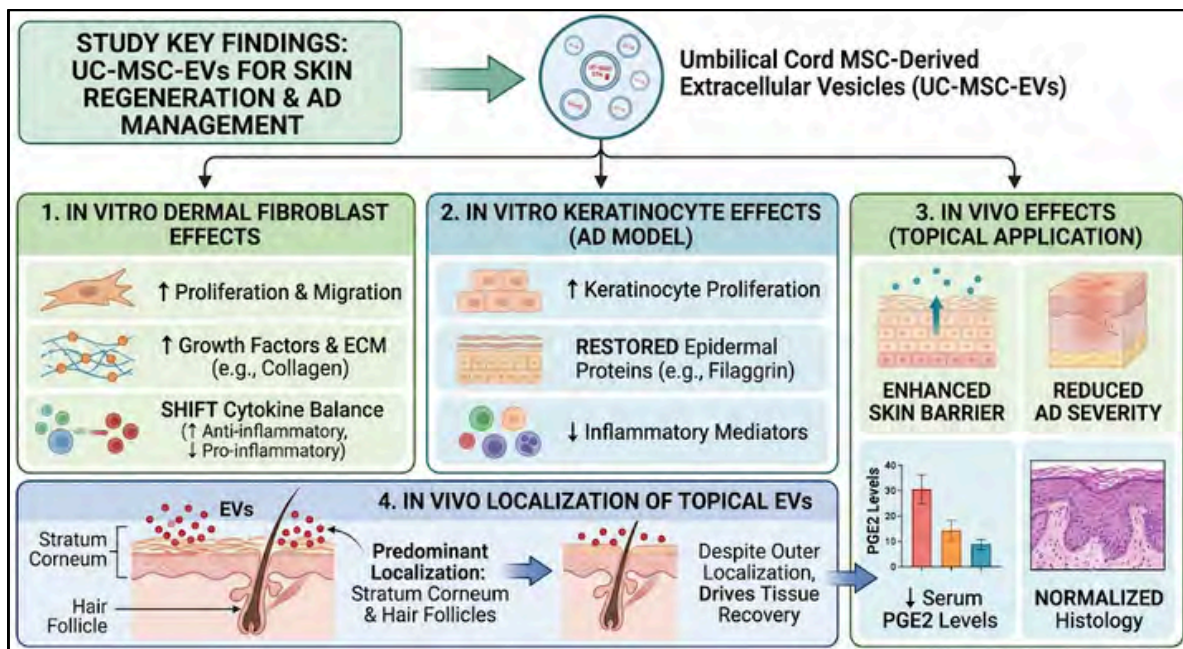
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**Abstract:** Objectives: Mesenchymal stem cell-derived extracellular vesicles (MSC-EVs) have gained considerable attention as cell-free therapeutic agents for skin diseases due to their potential to promote wound healing, reduce inflammation, and minimize scarring. In this study, we investigated the effects of umbilical cord MSC-EVs (UC-MSC-EVs) on skin regeneration. Materials and Methods: Experiments were performed to investigate the ability of UC-MSC-EVs to modulate dermal fibroblast proliferation, migration, cytokine secretion, growth factor production, and extracellular matrix (ECM) gene expression, as well as their potential to support keratinocyte proliferation, enhance skin barrier repair, and reduce inflammation in both in vitro and in vivo models of atopic dermatitis (AD). Results: UC-MSC-EVs were found to promote dermal fibroblast proliferation, viability, and migration; reduce the secretion of pro-inflammatory cytokines while enhancing the release of anti-inflammatory cytokines; and increase the production of growth factors and ECM components. In a keratinocyte model of AD, UC-MSC-EVs enhanced keratinocyte proliferation, restored the expression of epidermal structural proteins, and reduced the production of inflammatory mediators. In vivo, topical application of UC-MSC-EVs improved skin barrier function, reduced AD severity, and normalized histological features and serum prostaglandin E2 (PGE2) levels. Notably, UC-MSC-EVs were efficiently internalized by skin cells in vitro, but following topical application in vivo, they were predominantly localized within the stratum corneum and hair follicles. Conclusion: These findings highlight the multifaceted therapeutic potential of UC-MSC-EVs in skin repair and AD management, offering a promising novel cell-free therapeutic approach for the treatment of skin disorders.

**Keywords:** Cosmeceutical, Eczema, Exosome, Mesenchymal stromal cells, Wound healing





# Extracellular Vesicle Therapy: Promise and Challenges

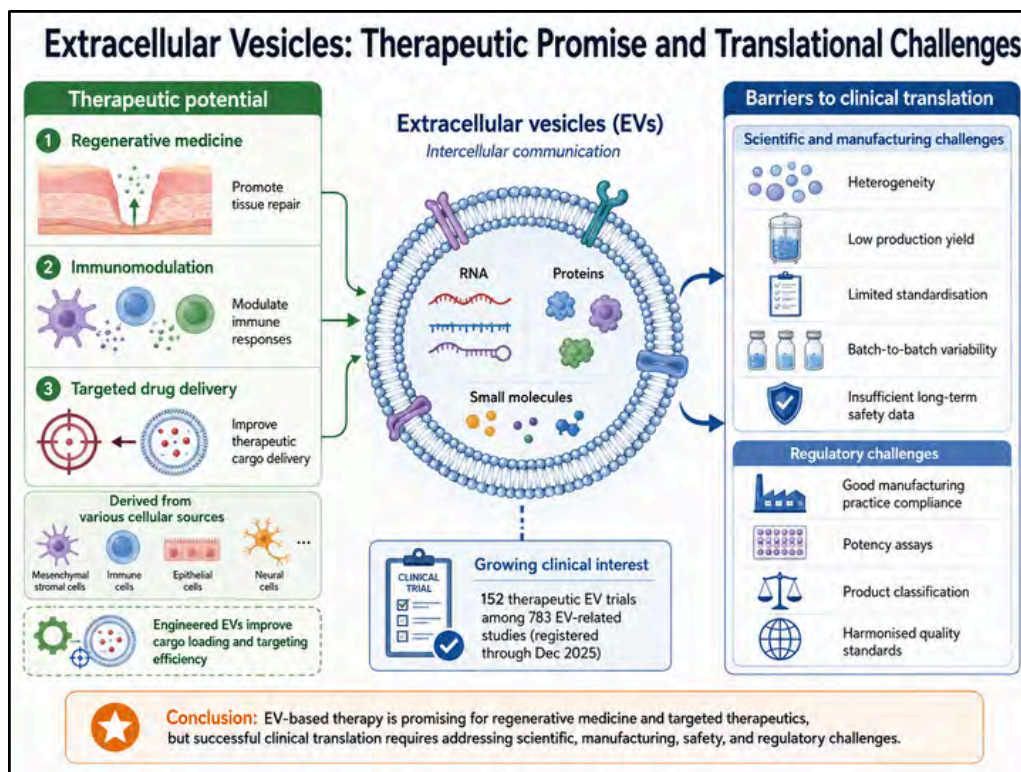
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**Abstract:** Objectives: Extracellular vesicles are nanosized lipid-bilayer particles involved in intercellular communication. Due to their regenerative, immunomodulatory, and drug-delivery capabilities, they have emerged as promising cell-free therapeutics for various disorders. This presentation aims to discuss their current therapeutic potential and the key translational and regulatory limitations impeding clinical implementation. The working hypothesis is that extracellular vesicles have substantial therapeutic potential, but successful translation requires major scientific, manufacturing, safety, and regulatory challenges to be addressed. Materials and Methods: Recent preclinical, translational, clinical-trial, manufacturing, and regulatory evidence on extracellular vesicle-based therapeutics was reviewed, focusing on regenerative medicine, immunomodulation, targeted drug delivery, commercial development, and clinical translation. Results: Extracellular vesicles derived from various cellular sources demonstrated potential to modulate immune responses, promote tissue repair, and improve therapeutic cargo delivery. Engineered extracellular vesicles further improved cargo delivery and targeting efficiency. A recent analysis of ClinicalTrials.gov records registered through December 2025 identified 152 therapeutic extracellular vesicle trials among 783 extracellular vesicle-related studies, demonstrating growing clinical interest. However, challenges remain, including vesicle heterogeneity, low production yield, limited standardisation, batch-to-batch variability, and insufficient long-term safety data. Regulatory concerns involving good manufacturing practice compliance, potency assays, product classification, and harmonised quality standards also remain major barriers. Conclusions: Extracellular vesicle-based therapy represents a promising strategy for regenerative medicine and targeted therapeutics. Addressing current scientific, manufacturing, and regulatory challenges is essential to facilitate successful clinical translation.

**Keywords:** Clinical translation, Extracellular vesicles, Immunomodulation, Regenerative medicine, Regulatory science



# Extracellular Vesicles-Derived MicroRNAs Signatures in Osteoarthritis: Comparative Profiling of Synovial Fluid and Systemic Circulation

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**Abstract:** Small extracellular vesicles (EV), also known as exosomes, are involved in cell-cell communication in normal physiological conditions and diseased states i.e., osteoarthritis (OA). EV-microRNAs (EV-miRNAs) have been identified and reported to play important roles in OA. This study aims to identify differentially expressed (DE) EV-miRNAs in OA plasma (PB) and synovial fluid (SF) compared to non-OA. We hypothesized that specific EV-miRNAs are differentially expressed (DE) in both PB and SF of individuals with OA compared to non-OA controls. The PB and SF samples were collected from consented arthroplasty and arthroscopy subjects and used for EV-derived RNA isolation. Then, 50 ng of EV-derived total RNA was used for small RNA sequencing analysis. Bioinformatics analysis was conducted to identify the DE miRNAs in OA plasma and SF. A total of 1929 EV-miRNAs were identified in the SF and PB samples; of which nine SF miRNAs were significantly differentially expressed, and 19 PB miRNAs were differentially expressed. The putative target mRNA of these differentially expressed miRNAs and their downstream pathways were analyzed. One of the identified miRNAs, miR 31-5p, is involved in chondrocyte survival and bone regeneration, is downregulated in both OA SF and PB samples, which corresponds to previous studies that investigated either SF or PB. In conclusion, DE EV-miRNA were identified in OA SF and PB. The identified DE EV-miRNAs provide potential molecular targets for improving OA diagnosis and developing personalized therapeutic strategies.

**Keywords:** Musculoskeletal sciences, Osteoarthritis, Extracellular vesicles, Orthopaedics



# Therapeutic potential of mesenchymal stem cell-derived extracellular vesicles in skin diseases

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**Abstract:** Down syndrome (DS), caused by trisomy 21, is the most common chromosomal disorder associated with intellectual disability and early-onset neurodegeneration. Despite extensive research, the molecular mechanisms underlying abnormal neurodevelopment and neural dysfunction in DS remain incompletely understood. Increasing evidence suggests that dysregulation of key neurodevelopmental pathways contributes to impaired neurogenesis, altered gliogenesis, synaptic dysfunction, and cognitive deficits in DS. This study aimed to investigate the mechanistic basis of neurodevelopmental dysregulation using human induced pluripotent stem cell (hiPSC)-derived neural models. Patient-derived and isogenic control hiPSCs were differentiated into neural progenitor cells, neurons, astrocytes, and cerebral organoids to model early human brain development. Molecular, cellular, and functional analyses were performed to evaluate neurodevelopmental alterations associated with DS. Transcriptomic and protein expression analyses revealed significant dysregulation of pathways associated with neurogenesis, gliogenesis, inflammation, oxidative stress, and synaptic development. Particular emphasis was placed on the role of RE1-silencing transcription factor (REST), a critical regulator of neuronal gene expression and neural differentiation. DS-derived neural models demonstrated aberrant REST-associated signalling accompanied by a neurogenic-to-gliogenic shift, characterised by reduced neuronal differentiation and increased astrocytic lineage commitment. Functional characterisation further demonstrated impaired neuronal maturation, altered neural network activity, and reduced synaptic connectivity in DS neurons and cerebral organoids. Pharmacological modulation of dysregulated signalling pathways partially restored neuronal differentiation and improved functional neuronal phenotypes, highlighting the therapeutic potential of targeting developmental signalling abnormalities in DS. Collectively, these findings provide mechanistic insights into the cellular and molecular basis of neurodevelopmental dysregulation in Down syndrome. Human hiPSC-derived neural models offer a powerful translational platform for elucidating disease mechanisms and identifying potential therapeutic strategies to improve neurodevelopmental outcomes and cognitive function in individuals with DS.

**Keywords:** Cerebral organoids, Down syndrome, Intellectual disability, Induced pluripotent stem cell, REST

# Generation of Multiple Cell Types Derived from hiPSC for Multi-tissue Organoids

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**Abstract:** Human induced pluripotent stem cells (hiPSCs) provide a platform that can be used for disease modelling and developing a better understanding of developmental biology. Traditional two-dimensional (2D) co-culture of different cell types derived from hiPSCs is the simplest platform for modelling multicellular tissue similar to an in vitro model, however, these 2D co-cultures are not always physiologically relevant and fail to reproduce tissue complexity, especially in terms of 3D architecture. To bridge this gap, we require robust protocols to generate different cell types and combined them as 3D organoids for more accurate and human-specific disease modelling and other applications are desperately needed. To address this issue, we have optimised a chemically defined, xeno-free differentiation platform that can generate macrophages (MΦ), dendritic cells (DCs), fibroblasts (FBs), and vascular endothelial cells (VECs) from the same starting population of human induced pluripotent stem cells (hiPSCs) with a minimal seeding density by inducing mesoderm differentiation over 2 days and then changing growth factor composition which can drive the mesoderm towards FBs, VECs, MΦ, and DCs. We have validated the resulting cell types via mRNA-seq as well as using cell type specific functional assays including polarisation of MΦ into M1 and M2 cells assay, FBs activation into MyoFBs using TGF-β and ECs tube formation assay. Furthermore, to demonstrate how our approach works, we combined these non-parenchymal cells (MΦ, DCs, FB, and VECs) with parenchymal cells (PCs) representative of liver hepatocytes which were generated using our established protocol in chemically defined and xeno-free medium to generate multi-tissue organoids. In summary, we generated isogenic MΦ, DCs, FBs and VECs in a chemically defined, xeno free, single cell differentiation protocol that have gene expression and functional profiles similar to their primary equivalents, which could be combined with PCs from different tissues including heart, lung, kidney, gut and liver.

**Keywords:** Dendritic cells, Endothelial cells, Fibroblasts, hiPSCs, Macrophages

# Computational Approach to Accelerate Culture Media Optimization for Cell Therapy

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**Abstract:** Culture media optimization plays a vital role in bioprocess development. Achieving an optimal formulation for the culture media is crucial as it enables maximum cell growth and subsequently yields the highest possible cell density. The current workflow involves experimental Design of Experiments (DoE) to determine the optimal culture media formulation. A paradigm shift is underway in the optimization of culture media, wherein a modelling approach can be employed to accelerate culture media optimization. In this talk, I will introduce a computational approach involving genome- scale metabolic modelling and model-guided DoE approach, and how this workflow can help the industry, particularly for cell therapy, to accelerate culture media design and optimization.

**Keywords:** Culture media, Computational media design, Metabolic flux model

# From Skin to Synapse: Building Malaysia's iPSC-Brain Organoid Platform for Rare Disease Discovery at IMR

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**Abstract:** Advances in induced pluripotent stem cell (iPSC) technology have enabled the development of human-relevant models for studying neurological and rare diseases. At the Institute for Medical Research (IMR), we are developing a local iPSC-based research pipeline to support the generation of new approach methodologies (NAMs) for disease modelling, functional studies, and future translational applications. A key emphasis of this initiative is the establishment of sustainable workflows and reproducible processes that can support long-term stem cell research capacity, biobanking, and collaborative rare disease studies within the local setting. This presentation will provide an overview of our ongoing efforts to establish an integrated workflow that begins with patient skin collection and primary fibroblast culture, followed by cellular reprogramming into iPSCs and differentiation into neural cell types and three-dimensional (3D) brain organoids. The platform incorporates facilities for stem cell culture, neural differentiation, organoid generation, imaging, and molecular characterisation, supporting both foundational model development and downstream disease investigations. In parallel, we are also exploring alternative cellular engineering strategies, including direct reprogramming approaches, to develop more efficient and accessible human neural models. A major focus of our work has been the establishment of reproducible and scalable protocols for neural modelling. Alongside cerebral organoids, we are also developing two-dimensional (2D) neural systems comprising of neurons, astrocytes and Schwann cells, as well as exploring scaffold-supported culture approaches to improve cellular organisation and tissue architecture. These models are intended to provide more physiologically relevant alternatives to conventional experimental systems. Using this platform, we have begun applying patient-derived models to rare neurological disorders, including mitochondrial disease such as Leigh syndrome. Early work has demonstrated successful generation of patient-specific neural cultures and brain organoids, with observable differences in growth and cellular morphology compared to controls. Collectively, these efforts represent an important step toward building locally relevant human cellular models and strengthening Malaysia's capacity in stem cell and rare disease research.

**Keywords:** Brain organoids, Human induced pluripotent stem cells, Neurological disease, New approach methodologies (NAMs), Rare disease

# In Vivo Evaluation of a 3D-Bioprinted Gelatin-Hydroxyapatite Scaffold for Periodontal Tissue Regeneration in a Rat Model

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**Abstract:** Periodontal regeneration remains challenging due to the complex architecture of periodontal tissues and limitations of conventional regenerative materials. Three-dimensional (3D) bioprinting offers the potential to fabricate customised scaffolds with controlled architecture for periodontal regeneration. Objective: This study evaluated the physicochemical, biological, and regenerative properties of a 3D-bioprinted gelatin-hydroxyapatite (GelHA) scaffold. Methods: Gelatin (8%) containing 1%, 3%, or 5% hydroxyapatite (HA) and 0.1% genipin was fabricated using extrusion-based 3D bioprinting. Scaffold printability, morphology, porosity, biodegradation, wettability, chemical composition, and mechanical properties were evaluated. Human periodontal ligament stem cells (hPDLSCs) were used to assess cell viability and proliferation. Based on these findings, the GelHA5 formulation was selected for in vivo evaluation. Standardised fenestration defects were surgically created at the mandibular first molar region of 24 male Sprague-Dawley rats, which were allocated to untreated, HA, 3D-bioprinted gelatin, and 3D-bioprinted GelHA groups. After eight weeks, regeneration was evaluated using micro-computed tomography and histological staining. Results: GelHA5 demonstrated favourable printability, porosity, mechanical properties, and biodegradation, with hPDLSC viability exceeding 80% and enhanced proliferation over seven days. In vivo, GelHA5 showed the highest mean bone mineral density, while the HA group demonstrated the highest mean bone volume fraction, although differences were not statistically significant ( $p>0.05$ ). Histologically, GelHA5 demonstrated organised periodontal ligament fibres, mineral deposition along dentin, new bone formation, and angiogenic features. Periodontal ligament fibre angulation was comparable to native tissue ( $p>0.05$ ). Conclusion: The 3D-bioprinted GelHA5 scaffold demonstrated favourable biomaterial and biological properties and showed potential to support periodontal tissue regeneration in a rat defect model.

**Keywords:** 3D Bioprinting, Animal model, Periodontal regeneration, Tissue engineering



# 3D-Bioprinted GelHA Scaffold for Periodontal Regeneration

## Experimental Workflow and Key Findings Flowchart

### STUDY OBJECTIVE

Evaluate the physicochemical, biological, and regenerative properties of a 3D-bioprinted gelatin-hydroxyapatite (GelHA) scaffold for periodontal tissue regeneration.

### PHASE 1: IN VITRO FORMULATION & SCREENING

#### Fabrication via Extrusion-Based 3D Bioprinting

8% Gelatin scaffold containing 1%, 3%, or 5% Hydroxyapatite (HA) and 0.1% Genipin crosslinker

#### Characterization & Evaluation

- Printability, morphology, porosity, biodegradation, wettability, chemical composition, mechanics
- Cell viability and proliferation assessed using human periodontal ligament stem cells (hPDLSCs)

Selection Criteria

### GELHA5 SELECTION FOR IN VIVO EVALUATION

#### Optimized Formulation Chosen Based On:

- Favorable printability, porosity, mechanical properties, and biodegradation rate
- High biocompatibility (hPDLSC cell viability > 80% and enhanced proliferation over 7 days)

### PHASE 2: IN VIVO EVALUATION (8-WEEK RAT MODEL)

**Surgical Model:** Standardized fenestration defects in mandibular 1st molar of 24 male SD rats  
**4 Treatment Groups:** Untreated | Hydroxyapatite (HA) | 3D gelatin | 3D GelHA (GelHA5)  
**Evaluation Methods:** Micro-computed tomography (micro-CT) and Histological staining at 8 weeks

### MICRO-CT FINDINGS

#### Bone Mineral Density (BMD)

- GelHA5: Highest mean BMD of all groups

#### Bone Volume Fraction (BVF)

- HA group: Highest mean BVF

\*Note: Group differences not statistically significant ( $p > 0.05$ )\*

### HISTOLOGICAL FINDINGS (GELHA5)

- PDL alignment: Highly organized fibers
- PDL angulation: Comparable to native tissue ( $p > 0.05$ )
- Mineralization: Mineral deposition along dentin
- Bone formation: New bone formation observed
- Blood vessels: Prominent angiogenic features

### STUDY CONCLUSION

The 3D-bioprinted GelHA5 scaffold demonstrated favorable biomaterial and biological properties and showed excellent potential to support periodontal tissue regeneration in-vivo.

# Angiogenic signaling molecules in periodontal regeneration: A scoping review

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**Abstract:** Introduction: Angiogenesis is essential for periodontal regeneration because newly formed vessels provide oxygen, nutrients, and regenerative cells to healing defects. This scoping review evaluated angiogenic signaling molecules, scaffold components, vascularization patterns, and assessment methods reported in animal models of scaffold-assisted periodontal regeneration. Methods: MEDLINE/PubMed, EBSCOhost, and Web of Science were searched from inception to 2025. Animal studies using scaffolds containing biomaterials, cells, and/or growth factors for periodontal defect regeneration were included when angiogenesis was assessed by histology, histomorphometry, immunohistochemistry, radiography, or endothelial-cell assays. Findings from the 12 included studies were narratively synthesized according to scaffold composition, signaling mediators, and angiogenesis outcomes. Results: Angiogenesis was observed across included studies as vascularized regenerated periodontal ligament, blood-vessel ingrowth within scaffolds and material-tissue interfaces, and differences in vessel size or density. Cell- and growth-factor-loaded scaffolds produced the most prominent angiogenic responses. Reported mediators and pathways included VEGF, CD31, BMP9-Smad1/5/8, FGF-2/Wnt- $\beta$ -catenin, PI3K/Akt-ERK1/2, and exosomal miR-21-NOTCH1/DLL4 signaling. Exosome-functionalized silk fibroin/collagen/nano-hydroxyapatite scaffolds increased endothelial tube formation and CD31-positive neovascular structures, while VEGF immunoreactivity was detected in new bone and perivascular regions. Osteogenesis generally followed angiogenesis and was increased or comparable with controls. However, most studies relied on descriptive histology, with limited use of endothelial-specific markers or in vitro confirmation. Conclusion: Angiogenic signaling molecules enhance scaffold-mediated periodontal regeneration, particularly when combined with stem cells, exosomes, or growth factors. Standardized evaluation of early and late angiogenic markers is required to clarify temporal mechanisms and guide scaffold design.

**Keywords:** Angiogenesis, Bone scaffolds, Periodontal regeneration, Signaling molecules, Tissue engineering

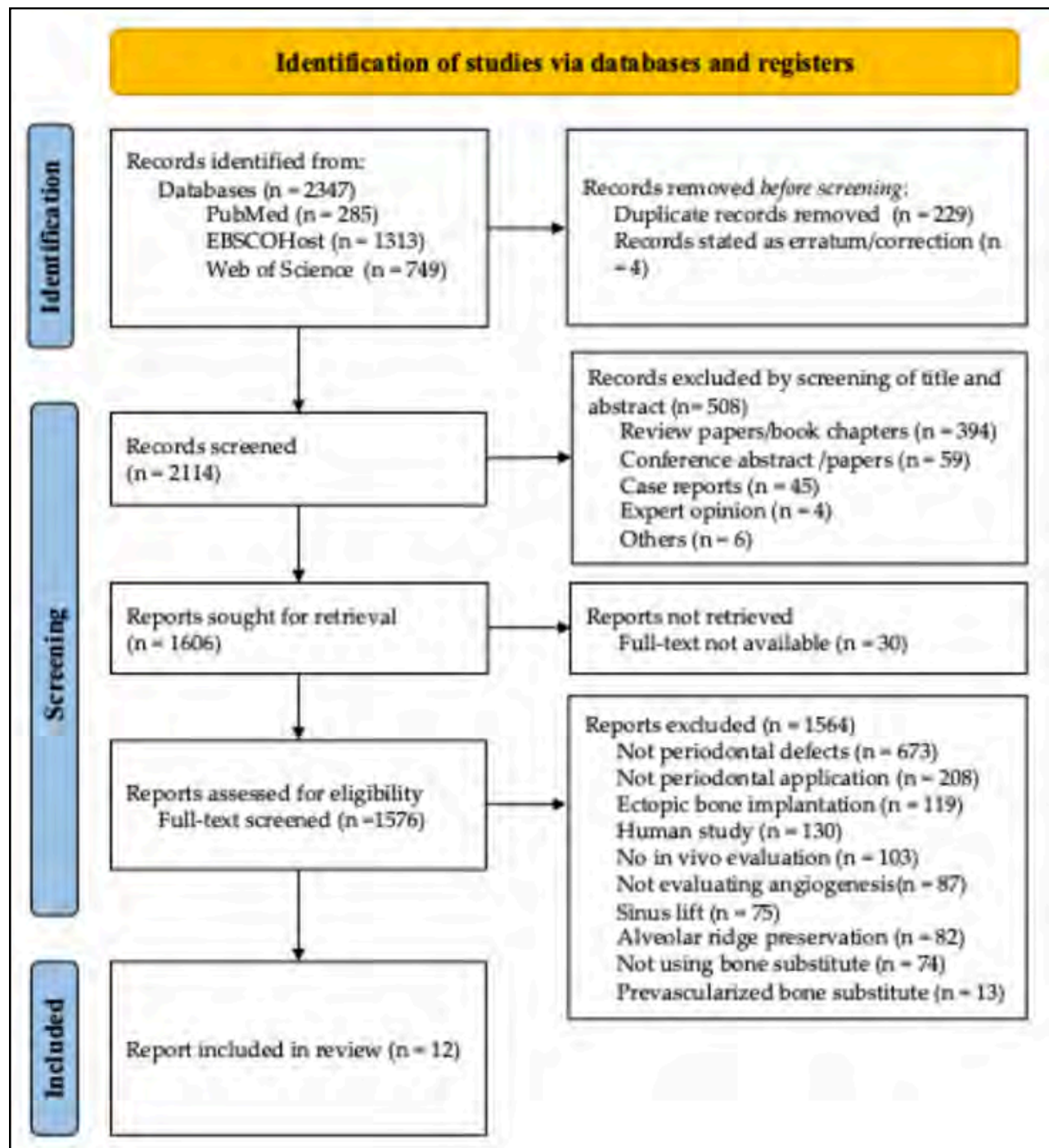


Figure 1: PRISMA flow diagram of study selection

# Engineering a Bioactive Lyophilised Platelet-Rich Fibrin-Based Scaffold for Craniofacial Bone Regeneration

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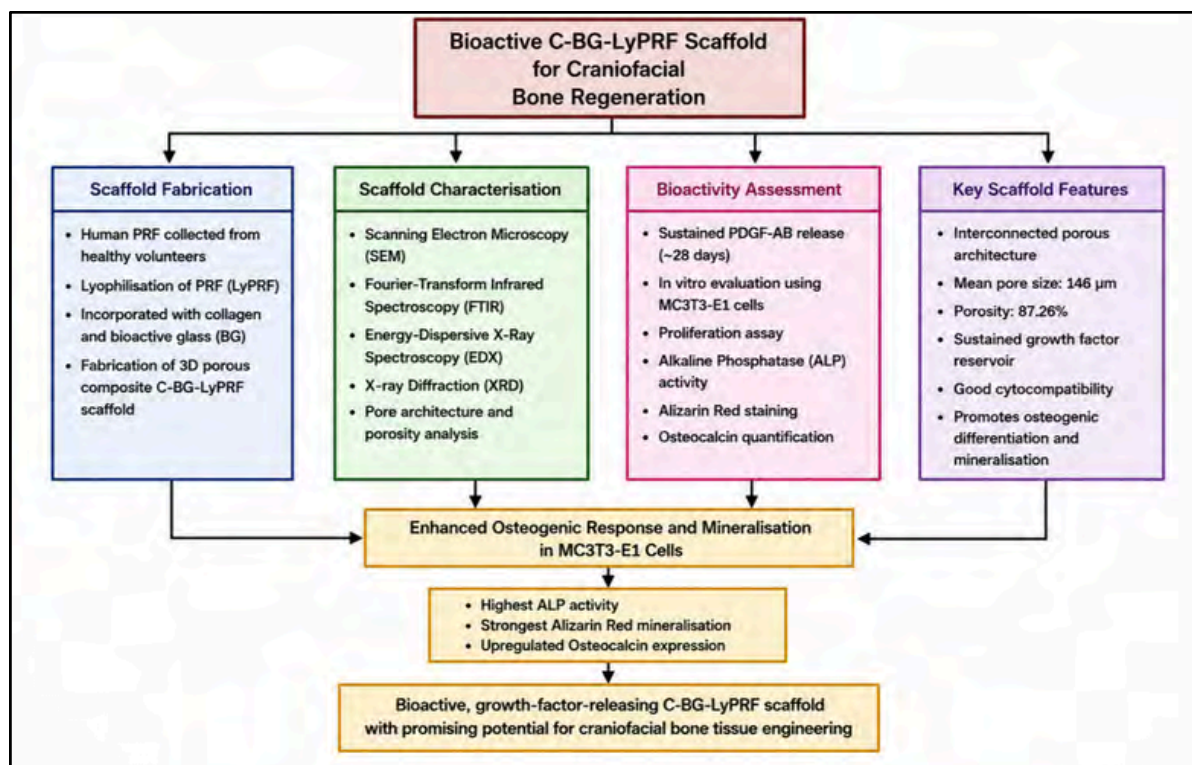
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**Abstract:** Craniofacial bone defects arising from congenital anomalies, trauma, and tumour surgery remain challenging to reconstruct. Platelet-rich fibrin (PRF) is an autologous source of bioactive growth factors; however, its limited stability and storage potential restrict its wider clinical application. This study developed and characterised a collagen-bioactive glass-lyophilised PRF (C-BG-LyPRF) scaffold for craniofacial bone regeneration. Human PRF was prepared from healthy volunteers, lyophilised, and incorporated with collagen and BG to fabricate a three-dimensional porous scaffold. Scaffold architecture and chemical characteristics were evaluated using Scanning Electron Microscopy, Fourier-Transform Infrared Spectroscopy, Energy-Dispersive X-Ray Spectroscopy, and X-ray diffraction. PDGF-AB release and osteogenic potential were assessed using MC3T3-E1 cells through proliferation, alkaline phosphatase activity, Alizarin Red staining, and osteocalcin quantification. The C-BG-LyPRF scaffold exhibited an interconnected porous architecture with a mean pore size of 146  $\mu\text{m}$  and porosity of 87.26%. Sustained PDGF-AB release was observed for approximately 28 days, supporting its potential as a growth-factor reservoir. The scaffold demonstrated good cytocompatibility and promoted osteogenic activity in MC3T3-E1 cells. Notably, the composite scaffold exhibited the strongest mineralisation response among the tested groups, supported by enhanced alkaline phosphatase activity and Alizarin Red staining, while osteocalcin expression further supported osteogenic differentiation. Collectively, these findings demonstrate that integrating LyPRF with C-BG creates a bioactive, growth-factor-releasing scaffold with promising potential for craniofacial bone tissue engineering.

**Keywords:** Lyophilised platelet-rich fibrin, Bioactive glass, Craniofacial bone regeneration, Tissue engineering, Bioactive scaffold



# Temperature-Tuned Fusion of Simvastatin-Loaded PLGA/TiO<sub>2</sub> Microparticle Scaffolds for Bone Regeneration

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**Abstract:** Introduction: Critical-sized bone defects require scaffolds that balance interconnected porosity with sufficient mechanical stability for cell infiltration and tissue support. Objectives: This study evaluated whether titanium dioxide nanotubes (TNTs) and controlled low-temperature fusion could strengthen simvastatin-loaded poly(lactic-co-glycolic acid) (PLGA) microparticle scaffolds without eliminating porosity. We hypothesised that incorporating TNTs and fusing near the PLGA glass-transition temperature would enhance interparticle bonding and compressive strength. Methods: Porous PLGA microparticles, with or without simvastatin and TNTs, were fabricated by double-emulsion solvent evaporation and fused into cylindrical scaffolds at 37, 40 or 50 °C. Morphology, porosity, wettability, thermal behaviour, chemical structure, crystalline phase and compressive strength were evaluated. Results: Compressive strength increased with fusion temperature. Between 37 and 50 °C, strength increased from 0.012 to 0.189 MPa for PLGA, 0.024 to 0.208 MPa for simvastatin/PLGA, 0.063 to 0.218 MPa for PLGA/TNTs and 0.067 to 0.240 MPa for simvastatin/PLGA/TNTs; differences between 37 and 50 °C were significant ( $p < 0.05$ ). The 50 °C simvastatin/PLGA/TNT scaffold retained approximately 38-40% porosity. Differential scanning calorimetry identified glass-transition temperatures of 49.9-54.0 °C, while X-ray diffraction confirmed anatase TNTs. Conclusion: Fusion at 50 °C produced the strongest scaffold among the investigated formulations while preserving a porous structure. Combining nanotube reinforcement, simvastatin loading and controlled fusion provides a promising platform for subsequent osteogenic and antibacterial evaluation in oral and maxillofacial bone regeneration.

**Keywords:** Bone regeneration, PLGA, Porous scaffold, Simvastatin, Titanium dioxide nanotubes

**Table 1. Compressive strength (MPa) of PLGA-based scaffolds at different fusion temperatures.**

| Formulation           | 37 °C | 40 °C | 50 °C |
|-----------------------|-------|-------|-------|
| PLGA                  | 0.012 | 0.044 | 0.189 |
| Simvastatin/PLGA      | 0.024 | 0.053 | 0.208 |
| PLGA/TNTs             | 0.063 | 0.068 | 0.218 |
| Simvastatin/PLGA/TNTs | 0.067 | 0.085 | 0.240 |

Values are mean compressive strengths; 37 versus 50 °C,  $p < 0.05$ .



# Advanced 3d-Printed Ocular Implants for Long-Acting Glaucoma Therapy

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**Abstract:** Glaucoma remains one of the leading causes of irreversible blindness worldwide, necessitating the development of long-acting ocular drug delivery systems capable of improving therapeutic efficacy and patient adherence. This presentation will discuss the engineering and characterization of 3D-printed timolol maleate (TM)-loaded ocular implants fabricated via vat-photopolymerization technology for sustained glaucoma treatment. The study explored the use of polyethylene glycol diacrylate (PEGDA)-based hybrid polymer matrices combined with poly(lactic-co-glycolic acid) (PLGA) or polylactic acid/polyurethane acrylate (PLA/PUA) to fabricate customizable subconjunctival implants with tunable drug-release behavior. The implants were successfully produced with different surface area-to-volume (SA/V) ratios, enabling modulation of TM release kinetics over a prolonged period of up to 180 days. Comprehensive physicochemical characterization demonstrated successful photocrosslinking and uniform drug incorporation without significant chemical interactions between TM and the polymer matrices. Differential scanning calorimetry and thermogravimetric analyses confirmed the thermal stability of the developed systems, while scanning electron microscopy revealed distinct surface morphologies depending on polymer composition and drug loading. In vitro release studies demonstrated that TM release was strongly influenced by implant geometry and polymer composition, with PEGDA and PEGDA-PLA/PUA systems exhibiting biphasic release profiles, whereas PEGDA-PLGA implants demonstrated triphasic release behavior associated with PLGA bulk erosion. Drug release predominantly followed a diffusion-controlled mechanism consistent with the Korsmeyer–Peppas model. Furthermore, the developed implants showed favorable in vitro biocompatibility in fibroblast cells, with cell viability exceeding 80% in both direct and indirect assays. Overall, this work highlights the potential of vat-photopolymerization 3D printing as a versatile platform for the development of customizable long-acting ocular implants for glaucoma therapy and demonstrates the promise of advanced manufacturing technologies in next-generation ophthalmic drug delivery systems.

**Keywords:** 3D Printing, ocular implant, Timolol maleate, Glaucoma, Sustained drug delivery

# Engineering Fibrotic Microenvironments for Tissue Regeneration Using MicroRNA-Loaded Chitosan Nanocomplexes

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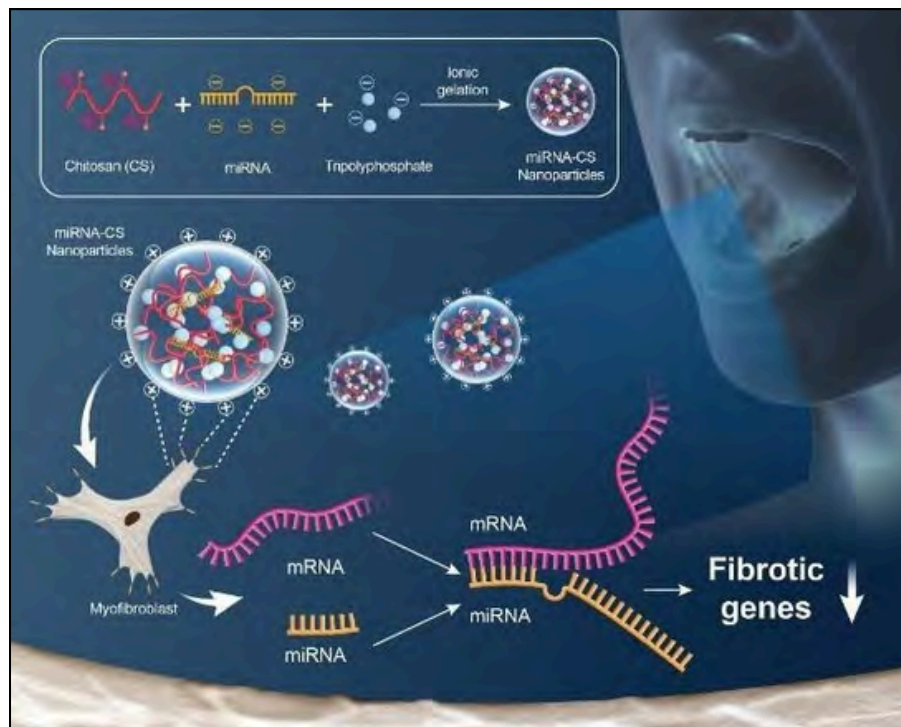
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**Abstract:** Objectives: Fibrosis is characterized by persistent activation of the TGF- $\beta$ /Smad signaling pathway, leading to myofibroblast differentiation and excessive extracellular matrix (ECM) deposition in diseases such as oral submucous fibrosis (OSF), idiopathic pulmonary fibrosis (IPF), and liver fibrosis. Dysregulated microRNAs (miRNAs) have emerged as promising therapeutic targets, but their clinical application is limited by poor in vivo stability and off-target effects. Materials and Methods: We developed chitosan nanocomplexes (NPs) with intrinsic anti-fibrotic properties as a delivery platform for anti-fibrotic miRNAs. Results: Chitosan NPs were prepared by ionic gelation and exhibited a particle size of  $121.9 \pm 0.1$  nm, a polydispersity index of  $0.162 \pm 0.004$ , and a zeta potential of  $+22.4 \pm 0.5$  mV. In an arecoline-induced OSF model, human oral submucosal fibroblasts were co-transfected with miR-29a-3p, miR-196a-3p, and miR-509-5p using chitosan NPs. Treatment significantly reduced the expression of fibrosis-related genes, while suppressing Smad2/3 phosphorylation and type I collagen protein expression. Chitosan NP-mediated miRNA delivery also significantly inhibited fibroblast migration. In vivo, administration of miRNA-loaded chitosan NPs attenuated ECM deposition in OSF mice and markedly alleviated fibrosis in bleomycin-induced pulmonary fibrosis. Conclusions: Chitosan NP-mediated delivery of anti-fibrotic miRNAs effectively remodels fibrotic microenvironments by suppressing TGF- $\beta$ /Smad signaling and ECM accumulation, thereby promoting tissue regeneration for fibrotic diseases.

**Keywords:** Fibrosis, TGF- $\beta$ /Smad pathway, microRNA, Chitosan nanocomplex



# HDAC11–STAT3 axis drives infection-induced endothelial inflammation and dysfunction in diabetic iPS-Ecs

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**Abstract:** Diabetes mellitus is a metabolic disease characterised by hyperglycaemia and is associated with various complications, including macrovascular and microvascular disorders. People with diabetes are more susceptible to infections, particularly bacterial infections. *Escherichia coli* (*E. coli*) is one of the most common bacteria contributing to diabetic complications. Endothelial cells (ECs), which form the innermost layer of blood and lymphatic vessels, possess the ability to recognise pathogens through pattern recognition receptors (PRRs) such as Toll-like receptors (TLRs), NOD-like receptors (NLRs), and RIG-I-like receptors (RLRs). The interaction between TLR2/4 and *E. coli* lipopolysaccharides (LPS) or outer membrane proteins (OMPs) triggers the amplification of inflammatory signalling cascades, thereby initiating a series of immune responses. This study aimed to elucidate the mechanisms underlying the exacerbation of inflammatory responses in diabetic conditions. We utilised EC-derived induced pluripotent stem cells (iPS-ECs) obtained from diabetic donors. These cells were exposed to *E. coli* for 2 hours and subsequently monitored for up to 48 hours post-infection. Our findings demonstrated that infected iPS-ECs derived from diabetic donors exhibited significantly higher inflammatory signatures compared with infected non-diabetic cells and uninfected diabetic controls. RNA sequencing (RNA-Seq) analysis revealed a clear distinction in signalling pathway activation between non-diabetic infected ECs (NDE) and diabetic infected ECs (DBE). Notably, the IL6/JAK/STAT3 signalling pathway was among the most enriched pathways in DBE compared with DB cells and was strongly associated with inflammatory response hallmark. Furthermore, our study identified histone deacetylase 11 (HDAC11) as a key factor involved in exacerbating inflammatory responses through its interaction with STAT3. To determine whether HDAC11 regulates STAT3, gene manipulation experiments were performed using both shRNA-mediated knockdown and pharmacological inhibition of HDAC11. Inhibition of HDAC11 significantly reduced the expression of both STAT3 and phosphorylated STAT3 at tyrosine 705 (pSTAT3 Y705). Immunoprecipitation assays were conducted to further investigate the interaction between HDAC11 and STAT3. The results demonstrated that HDAC11 interacts with both STAT3 and pSTAT3. In conclusion, our study demonstrated that HDAC11 interacts with STAT3 and pSTAT3 in iPS-ECs derived from diabetic donors in the presence of *E. coli*. This interaction contributes to the sustained inflammatory response observed under diabetic infectious conditions.

**Keywords:** HDAC11, STAT3, Inflammation, *E. coli* infection, Diabetes

# Engineering artificial tissue through innovative 3D cell culture model

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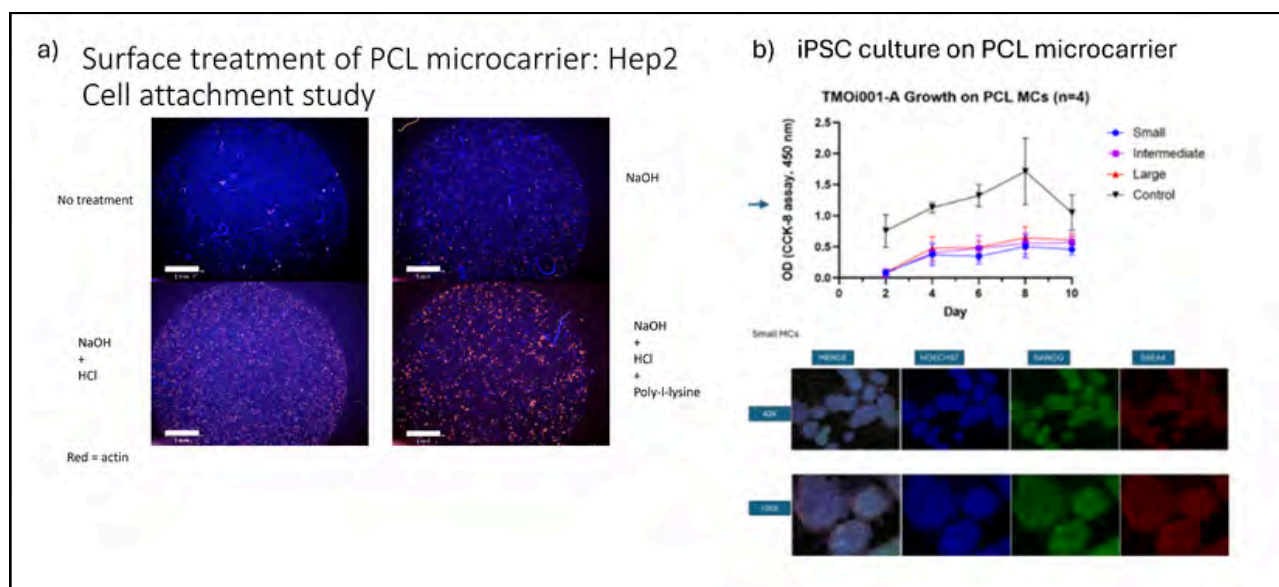
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**Abstract:** For many decades, drug discovery endeavors have relied on two-dimensional (2D) culture methods of eukaryotic cells on flat substrates made from glass or polystyrene. The lack of relevant architectural features is the major disadvantage which may lead to differing biological responses than in the native tissue, hence a poor predictive model for the clinical outcome. Consequently, the failure rate of drug discovery remains high with 90% of drugs failing in the clinical trial phase. Three-dimensional (3D) cell culture systems are seeing a rapid rate of development, principally driven by the need for conducting studies in better predictive model systems toward improving the success rate of drug discovery. Our works explore the application of tissue engineering techniques to generate “artificial tissue in a dish” models through 3D cell culture systems. To establish such models, we utilized innovative biofabrication methods such as electrospinning and the emulsion-solidification technique to fabricate 3D scaffolds as the architectural foundation for our experimental 3D cell culture systems. For instance, we experimented with aligned electrospun polycaprolactone (PCL) microfibrillar scaffolds to establish a 3D cell culture system for peripheral nerve tissue. Through neuronal-Schwann cell co-culture in the scaffolds, we have demonstrated that the 3D cell culture system can mimic the native tissue by recapitulating the anisotropic organization of the peripheral nerve tissue. We also experimented with PCL microcarriers as a potential scaffold for 3D cell culture systems with promising results. Our preliminary study on PCL microcarriers using Hep-2 cell line indicates that these scaffolds can support cell attachment and growth. Our studies support the use of electrospun PCL microfibrillar scaffolds and PCL microcarriers as scaffolds for 3D cell culture systems. Such scaffolds will be valuable for the future of drug discovery and development.

**Keywords:** 3D cell culture, Tissue Engineering, Electrospinning, Microcarrier, Peripheral nerve engineering



# Precision Bone Regeneration in Implant Dentistry

Wei Jen Chang

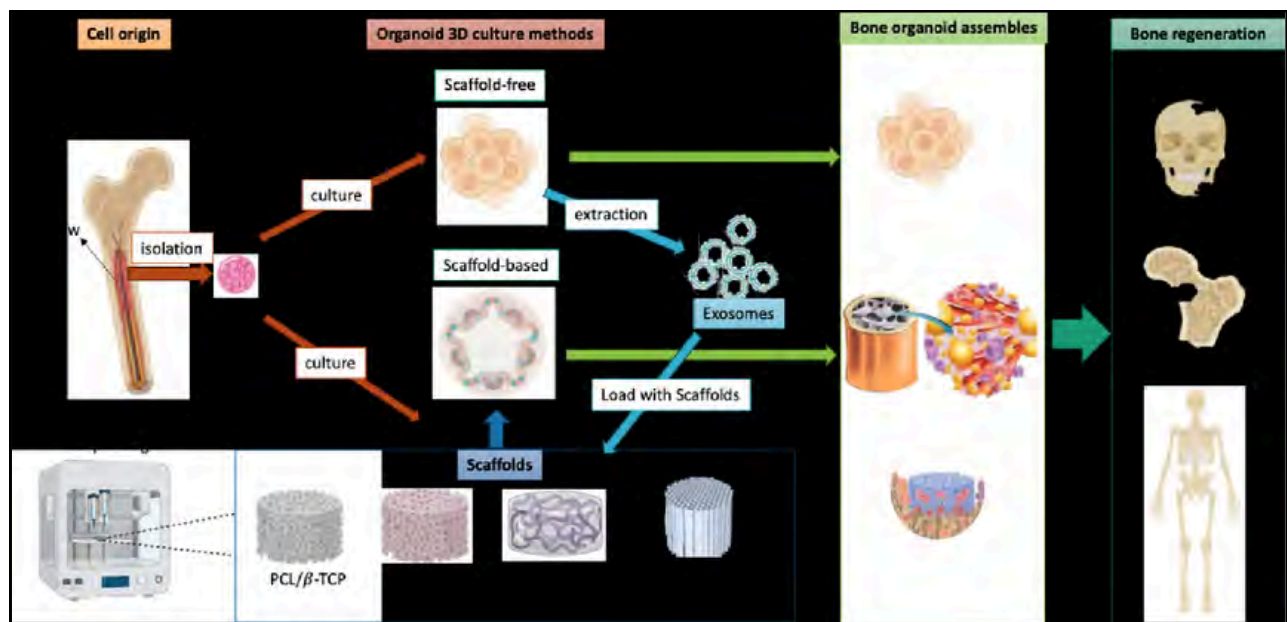
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**Abstract:** Regeneration of alveolar bone remains a fundamental challenge in implant dentistry, particularly in cases involving critical-sized defects, compromised bone quality, or impaired healing capacity. Recent advances in biomaterials, biofabrication, and regenerative biology have shifted therapeutic strategies from passive structural augmentation toward biologically instructive approaches that actively regulate tissue repair. This current progress in the development of bioengineered platforms for implant-related bone regeneration are the integration of three-dimensional (3D)-printed scaffolds, functional hydrogels, and bioactive therapeutics. Collectively, these developments demonstrate how the convergence of biomaterials science, tissue engineering, and regenerative biology is enabling precision regenerative therapies for implant dentistry.

**Keywords:** 3D Printing, Hydrogel, Tissue engineering, Dental implant





# Therapeutic Angiogenesis

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**Abstract:** In Japan, the number of patients with peripheral arterial disease (PAD) is estimated to be approximately 3 million. It is estimated that 10-20% of patients with PAD experience disease progression, and approximately 2% ultimately develop CLI, requiring major limb amputation. Nearly 10,000 patients annually undergo limb amputation because they do not respond adequately to conventional medical and surgical therapies. Historically, such advanced cases were considered avenues for their treatment. In 2000, Japan pioneered the use of autologous bone marrow-derived mononuclear cell implantation in patients with severe PAD who had no treatment options other than amputation. This approach provided the first clinical evidence worldwide demonstrating the efficacy and safety of cell-based therapeutic angiogenesis. Advances in understanding of vascular development and differentiation have clarified that neovascularization occurs not only through cells from preexisting vessels, but also through vasculogenesis, which involves the mobilization, differentiation, and proliferation of endothelial progenitor cells derived from the bone marrow or circulating blood. Recently, therapeutic strategies, including gene therapy using angiogenic growth factors, cell transplantation, and physical modalities such as extracorporeal shock wave therapy and low-intensity pulsed ultrasound, have been applied clinically at multiple institutions, with accumulating evidence supporting their efficacy. In this presentation, I summarize the therapeutic effects of vascular regenerative therapies for CLI, discuss long term outcomes, and address current limitations and future perspectives, with particular emphasis on their underlying mechanisms of actions.

**Key words:** Regenerative medicine, Therapeutic angiogenesis, Critical limb ischemia

# State-of-the-art Cell Processing Center at Hiroshima University Hospital for Translational Regenerative Medicine

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**Abstract:** Regenerative medicine is attracting attention as a promising treatment strategy for patients with intractable diseases or those who have no effective treatment options. In Japan, the implementation of regenerative medicine is regulated under “the Act on the Safety of Regenerative Medicine”, which mandates strict standards for cell processing and clinical applications. To accelerate translational research and clinical application in regenerative medicine, a new Cell Processing Center (CPC) will be established at Hiroshima University Hospital in September 2026. The CPC supports the development and manufacturing of cell-based therapeutic products for investigator-initiated clinical research and advanced regenerative medicine programs. The establishment of this CPC represents an important step toward expanding access to safe and effective regenerative medicine and strengthening international cooperation in translational medicine research. This presentation will introduce the newly established CPC facility and discuss its future plans.

**Keywords:** Regenerative medicine, Cell Processing Center, Translational research



**HIROSHIMA UNIVERSITY  
HOSPITAL**  
Hiroshima, Japan

## Cell Processing Center

Pioneering advanced technologies and stringent quality management  
to realize regenerative medicine and contribute to the future of healthcare.



### Advanced Aseptic Environment and Equipment

|                                                                                                                                                                                                                    |                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p><b>Cleanroom Environment</b></p> <p>Maintained high levels of cleanliness to ensure a controlled cleanroom environment.</p> |  <p><b>Safety Cabinets (Class II A2/B2)</b></p> <p>Equipped with safety cabinets that provide a secure environment for cell culture and processing.</p> |  <p><b>Cell Culture and Processing Equipment</b></p> <p>Introduction of advanced equipment to support high-quality cell culture and processing.</p> |  <p><b>Quality and Environmental Monitoring</b></p> <p>Continuous monitoring of temperature, humidity, differential pressure, particle counts, and more, 24/7.</p> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Potential of proteomics in determining therapeutic molecular characteristics for tissue regeneration

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**Abstract:** Objectives: To investigate the molecular properties of bone marrow mononuclear cells (BM-MNC) and peripheral blood mononuclear cells (PB-MNC) used in cell therapies for chronic wounds. We hypothesized that distinct proteomic signatures exist between these cell sources, which may influence therapeutic mechanisms and patient outcomes. Materials and Methods: We performed a proteomic analysis on BM-MNC and PB-MNC samples from eight chronic wound patients receiving autologous cell implants. Proteins were extracted from the cells, processed using high-sensitivity mass spectrometry, and compared against a human protein database to identify differential protein expression and associated biological pathways. Results: Five of the eight patients achieved more than 50% wound size reduction within six months. Principal component analysis of the protein profiles revealed a primary separation between the cell sources, accounting for 27.4% of the variance. Proteomic profiling identified 16 significantly differentially expressed proteins ( $p < 0.05$ ) between the two cell populations. PB-MNCs exhibited significant upregulation in metabolic and immune-associated proteins, reflecting an activated, repair-oriented state, whereas BM-MNCs upregulated structural proteins indicative of a quiescent, resting state. Conclusions: PB-MNCs present a viable, functionally active cell therapy for chronic wounds, possessing distinct molecular properties compared to BM-MNCs. Assessing protein factors prior to implantation could help identify likely responders and non-responders, thereby advancing precision medicine and improving treatment efficacy in wound care.

**Keywords:** Cell therapy, Chronic wound, Mononuclear cells, Precision medicine, Proteomics

**Table 1.** Summary of Key Clinical and Proteomic Findings

| Parameter                        | BM-MNC Group               | PB-MNC Group                |
|----------------------------------|----------------------------|-----------------------------|
| Number of Patients               | 4                          | 4                           |
| >50% Wound Healing (at 6 months) | 2 patients                 | 3 patients                  |
| Associated Cellular Phenotype    | Quiescent / Niche-resident | Activated / Repair-oriented |
| Numbers of Proteins Identified   | 25                         | 90                          |

# Vascular stabilization as a strategy to prevent bone loss: insights from LIPUS under unloading Dental Tissue Engineering in the 21st Century

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**Abstract:** Objectives: Mechanical unloading during spaceflight causes rapid skeletal deterioration, yet practical non-pharmacological countermeasures remain limited. We investigated whether daily low-intensity pulsed ultrasound (LIPUS) attenuates bone loss and preserves osteovascular integrity in a hindlimb unloading (HLU) mouse model of simulated microgravity. Materials and Methods: Six-month-old male mice were assigned to control, HLU, or HLU plus LIPUS groups for 14 days. Bone microarchitecture was assessed by micro-computed tomography. Osteoclast and osteoblast parameters were evaluated by TRAP staining and osteocalcin immunostaining, respectively. Periosteal vascular architecture was analyzed by Endomucin and  $\alpha$ SMA immunostaining. To evaluate local hypoxic status, mice received intraperitoneal injection of Hypoxyprobe 1 h before sacrifice. Serum inflammatory cytokines were also analyzed. Results: HLU markedly reduced trabecular and cortical bone parameters and disrupted periosteal vascular architecture. Daily LIPUS significantly preserved trabecular bone volume fraction, trabecular thickness and number, cortical thickness, and cortical bone mineral density, although these protective effects remained partial compared with non-unloaded controls. Histological analyses demonstrated that LIPUS suppressed HLU-induced osteoclast activation and partially restored osteocalcin-positive osteoblast parameters. In parallel, LIPUS significantly increased Endomucin-positive and  $\alpha$ SMA-positive vascular loop formation and enlarged vascular loop size in the periosteal region. Importantly, vascular parameters positively correlated with trabecular bone volume fraction. Furthermore, HLU-induced periosteal hypoxia and elevations in systemic inflammatory cytokines, including IL-1 $\beta$ , TNF- $\alpha$ , and IL-6, were significantly attenuated by LIPUS treatment. Conclusion: Daily LIPUS mitigates unloading-induced skeletal deterioration while preserving periosteal vascular architecture and suppressing hypoxic and inflammatory responses under simulated microgravity.

**Keywords:** Low-intensity pulsed ultrasound, Osteovascular integrity, Bioactive stimulus

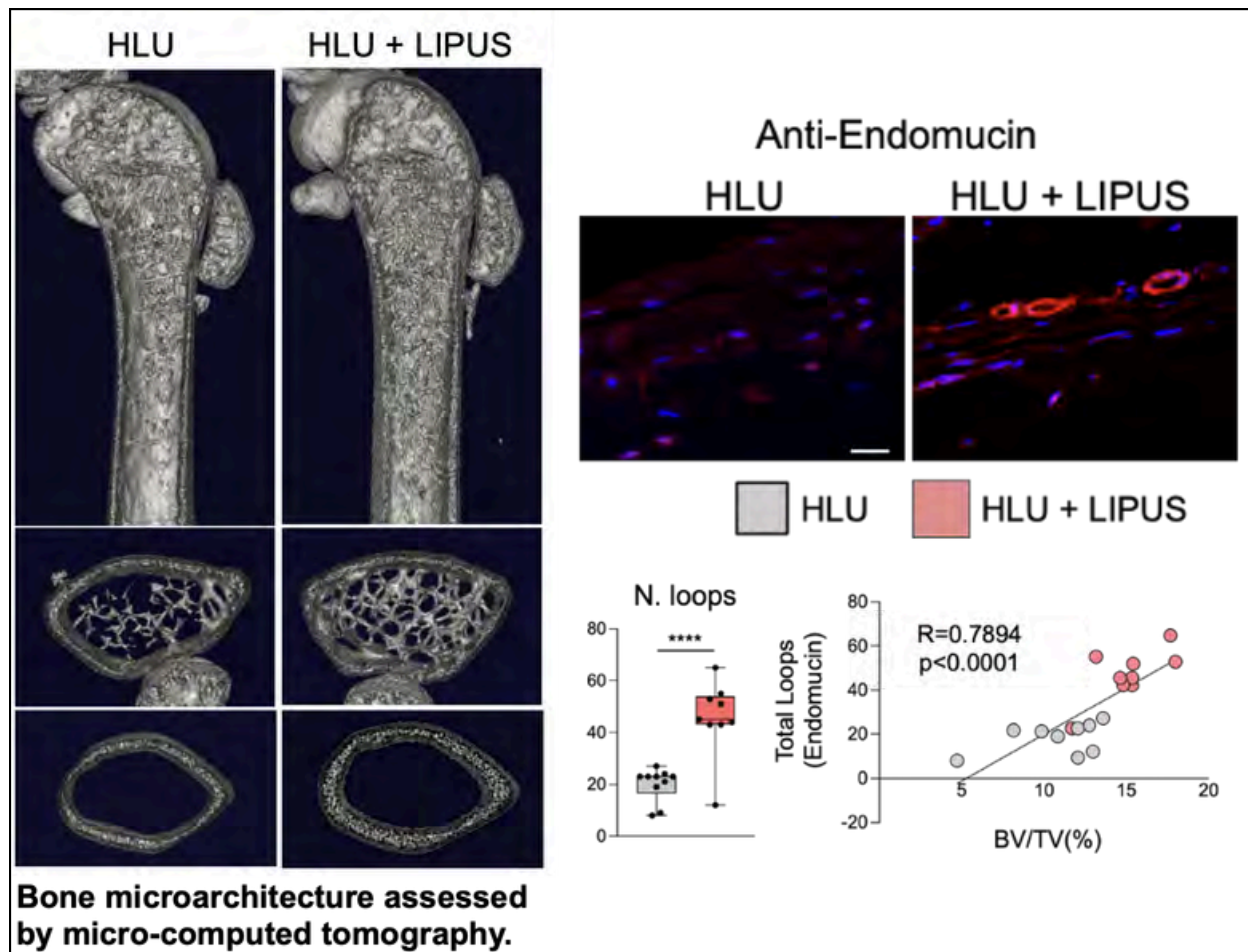


Figure: Vascular stabilization as a strategy to prevent bone loss with low-intensity pulsed ultrasound (LIPUS)



# Paradigm Shift for Vascular Regeneration in Regenerative Medicine

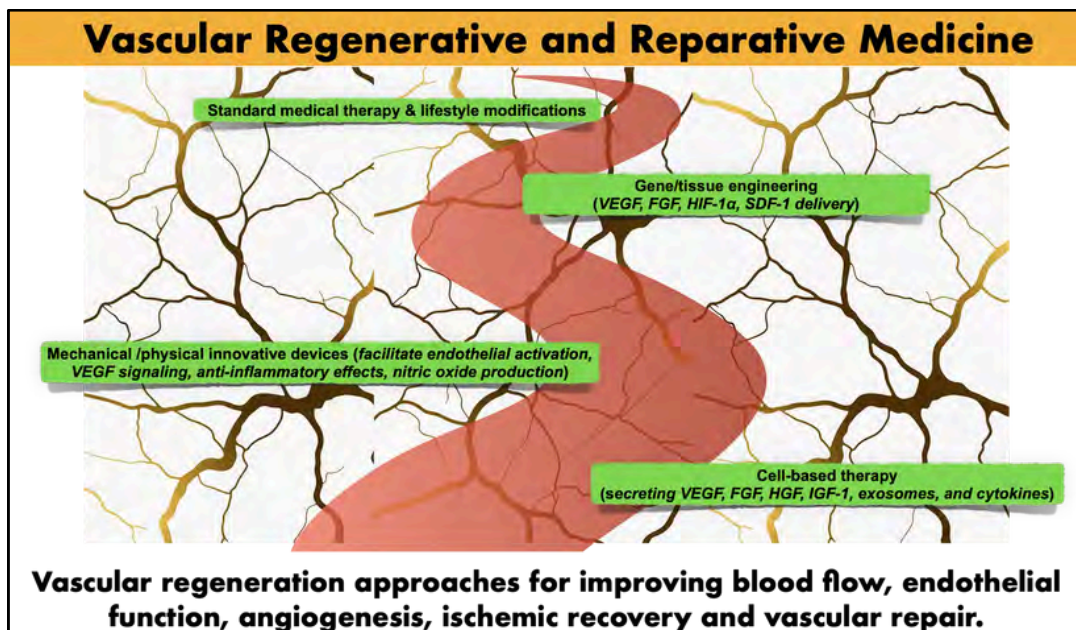
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**Abstract:** Various potential pathways are being explored for therapeutic options beyond present limitations in regenerative medicine. Vascular-associated microenvironment with tissue through oxygen and nutrient delivery, end-organ metabolite exchange and hormonal signaling influence regulation of systemic blood flow perfusion and endothelial function. Tissue dysfunction may begin with microvascular impairment and vice versa. The tissue and micro-vasculature should be considered as a functional unit, where dysfunction of one may induce or exacerbate the function of its partner. Impaired endogenous vascular regenerative capacity, has been linked to metabolic, trauma or radiation-induced and age-related diseases, particularly in adverse cardiovascular outcomes. Advances in understanding of vascular development and differentiation have clarified that neovascularization occurs not only through cells from preexisting vessels, but also through vasculogenesis, which involves the mobilization, differentiation, and proliferation of endothelial progenitor cells derived from the bone marrow or circulating blood. These mechanisms can be utilized to treat intractable vascular diseases and vascular-related conditions. Recent research strategies include RNA therapy, transplantations of cells from different sources, catheter-based regenerative agents, organoids or tissue engineering and mechanical modalities such as photobiomodulation, extracorporeal shock wave therapy and low-intensity pulsed ultrasound. Responders and non-responders in clinical trials need to be recognized to ascertain factors differentiating the outcomes. Vascular regenerative and reparative medicine is a frontier to be explored towards holistic approach in providing therapeutic options through personalized treatments and hybrid mechanical-biological solutions. Vascular regeneration strategies can be integrated to advance standard care in medicine to improve cost-effective clinical outcomes and patients' quality of life.

**Keywords:** Vascular regeneration, Paradigm shift, Advancements



# Transcriptome Analysis of Ischemic Stroke Recovery Induced by Neural Stem Cell Preconditioned with Baicalein-Enriched Fraction of *Oroxylum indicum*

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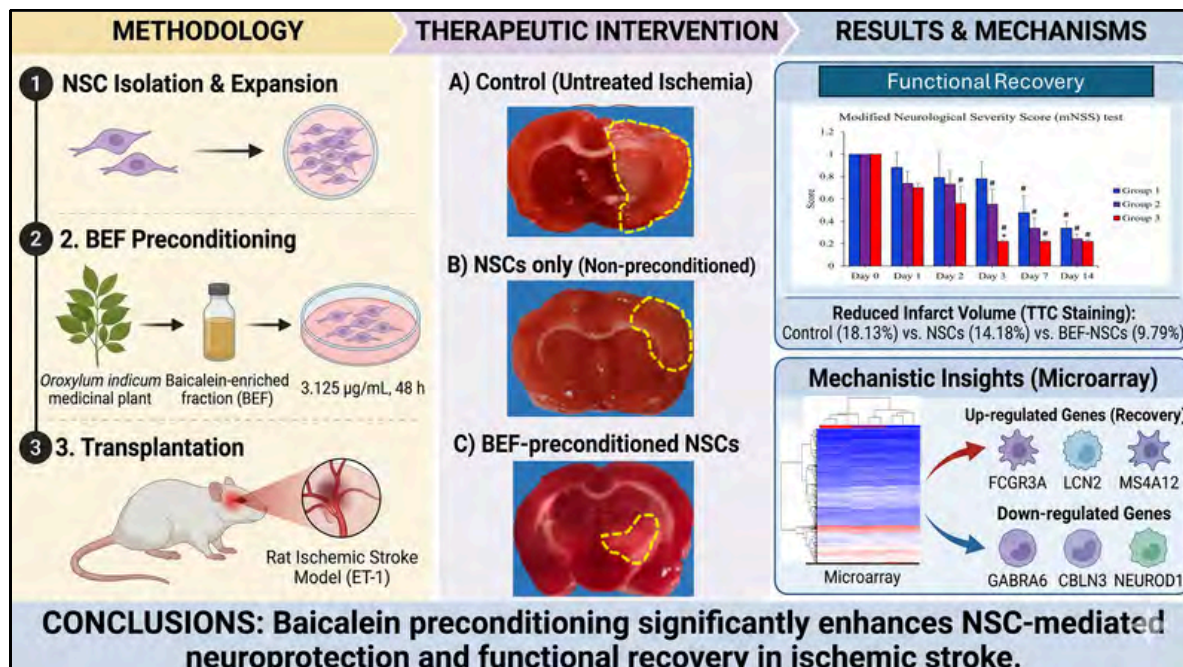
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**Abstract:** Ischemic stroke remains a leading cause of mortality and long-term disability worldwide. Neural stem cell (NSC) transplantation has emerged as a promising regenerative strategy; however, its therapeutic efficacy is limited by the hostile post-ischemic microenvironment, which reduces graft survival. Objectives: This study investigated the therapeutic potential of NSCs preconditioned with baicalein-enriched fraction (BEF) extracted from *Oroxylum indicum* medicinal plant for ischemic stroke. Methods: In vitro-expanded NSCs were preconditioned with BEF (3.125 µg/mL, 48 h) prior to transplantation into an endothelin-1 (ET-1)-induced ischemic stroke rat model. Neurological outcomes were evaluated over 14 days. Gene expression profiling was performed using microarray analysis to identify transcriptomic changes associated with BEF-preconditioned NSC treatment. Results: Rats treated with BEF-preconditioned NSC (Group 3) showed significant functional improvement compared to non-preconditioned NSC (Group 2) and untreated group (Group 1). TTC staining demonstrated a significant reduction in infarct volume in the BEF-preconditioned group ( $9.79 \pm 2.63\%$ ) compared to non-preconditioned NSCs ( $14.18 \pm 0.45\%$ ) and controls ( $18.13 \pm 2.16\%$ ). Furthermore, microarray-based transcriptomic profiling identified differential gene expression associated with recovery, including downregulation of GABRA6, CBLN3, and NEUROD1, and upregulation of FCGR3A, LCN2, and MS4A12. Conclusions: Baicalein preconditioning significantly enhances NSC-mediated neuroprotection and functional recovery in ischemic stroke, highlighting its potential as a novel strategy to improve stem cell-based therapy efficacy.

**Keywords:** Ischemic stroke, Neural stem cells, Baicalein, *Oroxylum indicum*, Neuroprotection



# Development of an In Vitro Parkinson's Disease Model Using Differentiated SH-SY5Y Cells for Phoenixin Characterization

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**Abstract:** Parkinson's disease (PD) is characterized by progressive degeneration of dopaminergic neurons in the substantia nigra. Differentiated SH-SY5Y cells are widely used as an in vitro PD model, while exposure to 1-methyl-4-phenylpyridinium (MPP<sup>+</sup>) induces mitochondrial dysfunction, oxidative stress, and neuronal injury. Phoenixin (PNX) is an endogenous peptide naturally present in the body and is expressed in several tissues, including the central nervous system. As a pleiotropic neuropeptide, PNX has been implicated in neuroinflammatory regulation and cellular stress responses, with emerging evidence suggesting its potential neuroprotective role. This study aimed to establish and characterize an MPP<sup>+</sup>-induced SH-SY5Y cellular model suitable for investigating the neuroprotective potential of PNX. SH-SY5Y cells were initially differentiated with retinoic acid to promote a neuronal-like phenotype, followed by exposure to varying concentrations of MPP<sup>+</sup> to induce mitochondrial-associated cellular injury and identify an appropriate experimental concentration for subsequent neuroprotective studies. The endogenous expression and cellular localization of PNX were subsequently evaluated using ELISA and immunocytochemistry, respectively. The established SH-SY5Y PD model enabled the detection and characterization of endogenous PNX, with PNX expression successfully measured and its cellular localization predominantly observed within the cytoplasm. These results establish a reproducible MPP<sup>+</sup>-induced SH-SY5Y model for studying PNX in the context of Parkinson's disease and confirm its endogenous presence and cytoplasmic localization in neuronal-like SH-SY5Y cells. This model provides a foundation for further investigation into the functional role and underlying mechanisms of PNX in mitochondrial dysfunction, oxidative stress, and neuronal survival in PD.

**Keywords:** Parkinson's disease, SH-SY5Y, Phoenixin, in vitro model, Neuropeptide

# Development of In Vitro Neurodegenerative Models Using Stem Cell-Derived Neural Cells for Natural Product Screening

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**Abstract:** Neurodegenerative diseases (ND), including Alzheimer's disease and Parkinson's disease, are characterised by progressive neuronal dysfunction and loss, with oxidative stress contributing significantly to their progression. Natural products are increasingly investigated for their nutritional and neuroprotective potential. However, their therapeutic screening is often limited by conventional in vitro models, which may not adequately reflect the complexity and diversity of neuronal populations. Objectives: This study aimed to develop in vitro ND models using neural cells derived from the 46C mouse embryonic stem cell and the R3 rat amniotic fluid-derived stem cell, owing to their capacity to differentiate into neural lineages. Methods: The 46C cells were differentiated using the 4-/4+ embryoid body protocol with all-trans retinoic acid, whereas R3 cells underwent adherent monolayer differentiation protocol. The derived neural cells were exposed to hydrogen peroxide (H<sub>2</sub>O<sub>2</sub>) for 24 h in a dose-dependent manner and assessed for morphology, cell viability, and intracellular ROS accumulation. For natural product screening, H<sub>2</sub>O<sub>2</sub>-injured neural cells were treated with *Centella asiatica* (CA) extract (1 and 10 µg/mL) for 48 h, followed by evaluation of treatment responses. Results: H<sub>2</sub>O<sub>2</sub> significantly reduced cell viability in 46C-derived neural cells treated with 1500 µM (53.3 ± 2.4%) and in R3-derived neural cells treated with 2000 µM (66.8 ± 8.3%), with marked neurite damage and increased intracellular ROS. Furthermore, CA extract improved cell viability and promoted neurite outgrowth in H<sub>2</sub>O<sub>2</sub>-injured neural cells. Conclusions: The 46C- and R3-derived neural cells may provide a promising in vitro model for studying oxidative stress-related neurodegeneration and for screening potential therapeutic agents.

**Keywords:** Neurodegenerative diseases, Mouse embryonic stem cell, Rat amniotic fluid-derived stem cell, in vitro models, Natural product



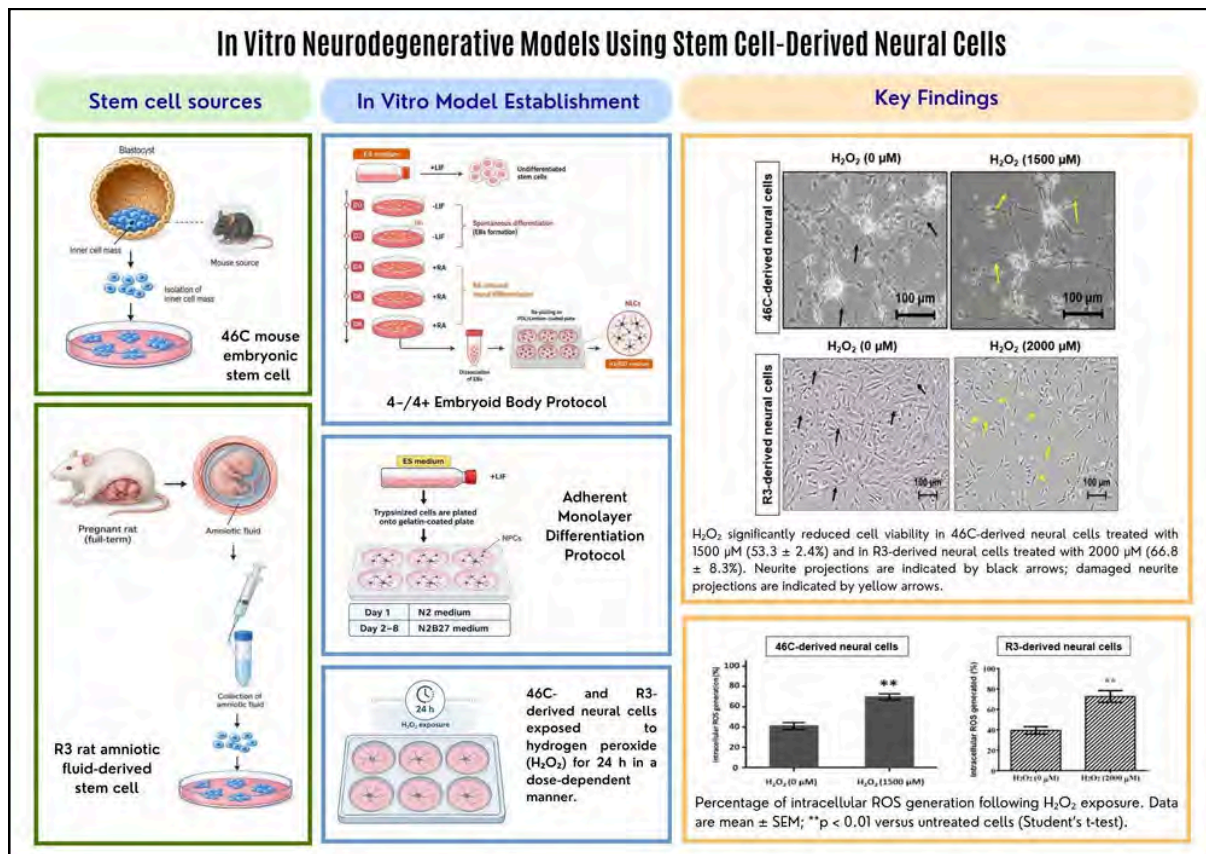


Figure: Key findings of this study

# Evaluation of Hyaluronic Acid-Based 3D Printable Hydrogel as a Bioink for Skin Tissue Engineering

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**Abstract:** Skin injuries can result in acute or chronic wounds that require effective therapeutic strategies to promote tissue regeneration. Bioactive polymer-based three-dimensional (3D) bioprinting hydrogels have emerged as promising biomaterials for wound healing applications. Objectives: This study evaluated the physicochemical, biological, and wound-healing potential of gelatin (GEL) and hyaluronic acid (HA)-based bioinks for tissue regeneration in vitro and in vivo. Materials and Methods: Genipin was selected as a natural crosslinker due to its compatibility and ability to improve mechanical properties. The physicochemical and biological characteristics of the hydrogels were assessed, with a focus on scaffold performance in human primary fibroblast cultures. Hydrogels were printed and crosslinked using 0.1 wt% genipin. Results: Crosslinked hydrogels exhibited excellent water retention (>1000%) and increased surface hydrophobicity following HA incorporation. The GEL-HA hydrogels maintained high biocompatibility, supporting over 85% cell viability. The incorporation of TQ significantly enhanced Ki-67 and collagen type-I (Col-I) expression, indicating increased cell proliferation and extracellular matrix production. In vivo, the GEL-HA-TQ hydrogel accelerated wound closure, promoted re-epithelialization and collagen deposition, reduced scar formation, and achieved approximately 94.7% wound closure by day 15. No systemic toxicity was observed, as liver, kidney, and pancreas function markers remained within normal ranges. Conclusion: GEL-HA hydrogels crosslinked with genipin, particularly when incorporated with TQ, exhibit excellent physicochemical, biocompatible, and wound-healing properties, making them strong candidates for bioinks in chronic wound treatment.

**Keywords:** Printed hydrogel, Bioink, 3D-bioprinting, Diabetic C57BL/6 mice, Wound healing

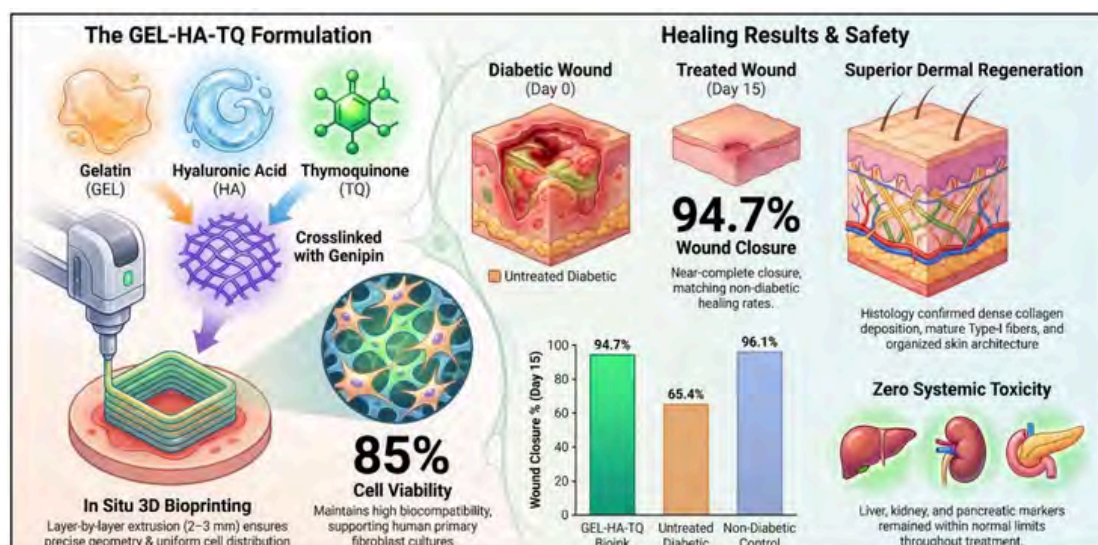


Figure 1. This infographic highlights a breakthrough 3D-bioprinted platform represents a major step forward in personalizing and optimizing therapeutic strategies for chronic diabetic wound



# A Non-Invasive Microwave Sensor for Cerebrospinal Fluid Detection in a Layered Head Phantom

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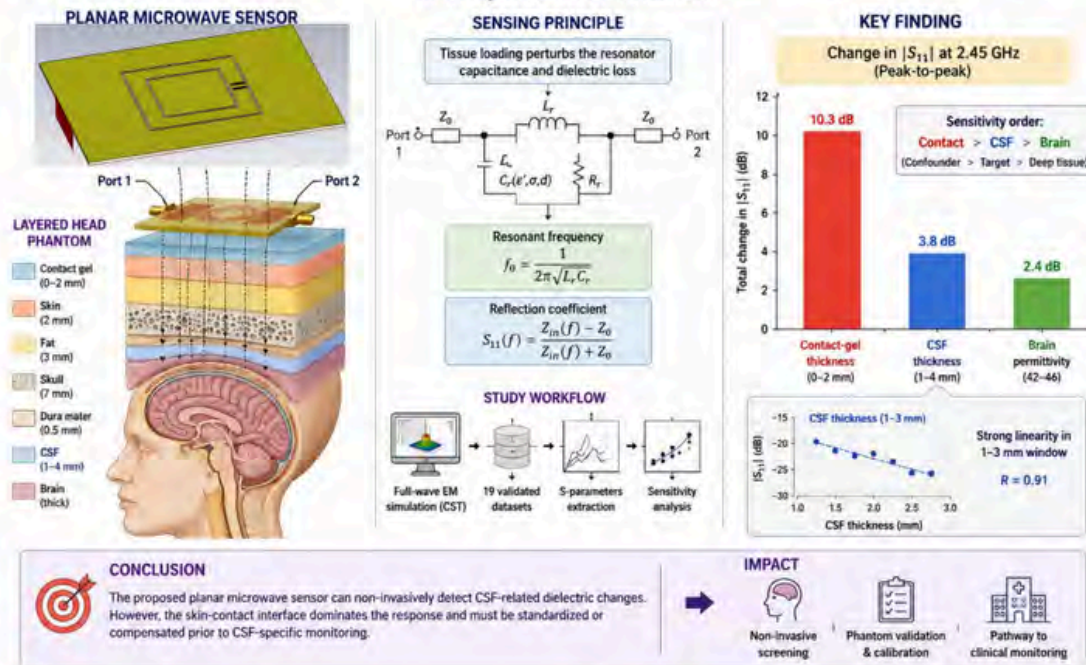
**Abstract:** Objectives: Changes in cerebrospinal fluid (CSF) and head-tissue composition occur in several neurological and post-surgical conditions, but monitoring commonly depends on imaging or invasive probes. This study evaluates a compact non-invasive microwave sensor for detecting dielectric changes in a layered head phantom, with the hypothesis that its reflection response tracks CSF thickness while also revealing contact-interface interference. Materials and Methods: A two-port planar sensor integrating a complementary split-ring resonator and interdigital capacitive loading was designed on a low-loss microwave laminate and operated near 2.45 GHz. Full-wave electromagnetic simulations were performed using a seven-layer phantom comprising contact gel, skin, fat, skull, dura, CSF and brain. Nineteen validated datasets were analysed by independently varying CSF thickness (1–4 mm), contact-gel thickness (0–2 mm) and brain relative permittivity (42–46). Results: The sensor remained well matched across all datasets, with S11 below –10 dB and S21 approximately –2.7 to –3.1 dB. At 2.45 GHz, S11 changed by 3.8 dB with CSF thickness, 10.3 dB with contact-gel thickness and 2.4 dB with brain permittivity, producing the sensitivity order contact interface > CSF > brain dielectric variation (Table 1). Within the 1–3 mm CSF range, the response showed strong linearity ( $r = 0.91$ ). Conclusions: The proposed planar microwave sensor can detect CSF-related dielectric changes non-invasively, but contact conditions dominate the response and must be standardised or compensated before CSF-specific monitoring.

**Keywords:** Cerebrospinal fluid, Contact interface, Head phantom, Microwave sensor, S-parameters

Table 1. Key simulated  $S_{11}$  sensitivity at 2.45 GHz.

| Varied factor               | Range  | Total $S_{11}$ change |
|-----------------------------|--------|-----------------------|
| Contact-gel thickness       | 0–2 mm | 10.3 dB               |
| CSF thickness               | 1–4 mm | 3.8 dB                |
| Brain relative permittivity | 42–46  | 2.4 dB                |

### A Non-Invasive Microwave Sensor for Cerebrospinal Fluid Detection in a Layered Head Phantom



This graphical abstract summarizes the development and simulation validation of a non-invasive microwave sensor for detecting CSF-related dielectric changes. The left panel illustrates the planar two-port resonant sensor placed over a multilayer head phantom. The central panel explains the sensing mechanism, where tissue loading alters the resonator capacitance, dielectric loss, resonance behavior, and reflection coefficient. The right panel presents the key quantitative finding: contact-gel thickness produced the largest  $S_{11}$  change at 2.45 GHz, followed by CSF thickness and brain permittivity. The figure emphasizes that the proposed sensor is promising for CSF-sensitive monitoring, but reliable use requires standardization or compensation of the skin-contact interface.

# From Beam Steering to Body Sensing: Advanced Lens and Planar Antenna Technologies for Next-Generation Biomedical Applications

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**Abstract:** The convergence of biomedical sciences and microwave engineering is creating new opportunities for non-invasive healthcare, tissue characterization, and smart regenerative medicine. In particular, antenna technologies have emerged as key enablers for next-generation biomedical sensing, diagnosis, monitoring, and therapeutic systems. This work highlights recent advances in dielectric lens and planar antenna technologies and their potential applications in biomedical engineering and regenerative medicine. The presentation begins by discussing the interaction between electromagnetic waves and biological tissues, where variations in dielectric properties provide unique signatures for tissue characterization and disease detection. Recent developments in shaped dielectric lens antennas capable of producing high-gain, wide-angle, and multibeam radiation patterns will be presented. Advanced lens shaping techniques based on have enabled compact and lightweight lens structures with excellent beam-steering performance, making them attractive for biomedical imaging, microwave diagnostics, and high-resolution sensing applications. The talk will also explore planar antenna technologies, including flexible, wearable, and textile-based antennas for continuous health monitoring and wireless body area networks (WBAN). These platforms support unobtrusive monitoring of physiological parameters and facilitate real-time healthcare delivery. Emerging applications in microwave imaging, RFID-enabled healthcare, and non-destructive characterization of biomaterials and engineered tissues will be discussed. Finally, future directions involving the integration of artificial intelligence, Internet of Medical Things (IoMT), and microwave sensing technologies will be highlighted, demonstrating how advanced antenna systems can contribute towards personalized, intelligent, and patient-centric healthcare solutions for tissue engineering and regenerative medicine.

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Key words. Biomedical Sensing, Dielectric Lens Antenna, Microwave Imaging, Planar Antenna, Wearable Antenna.

# High Directive Microwave Sensor

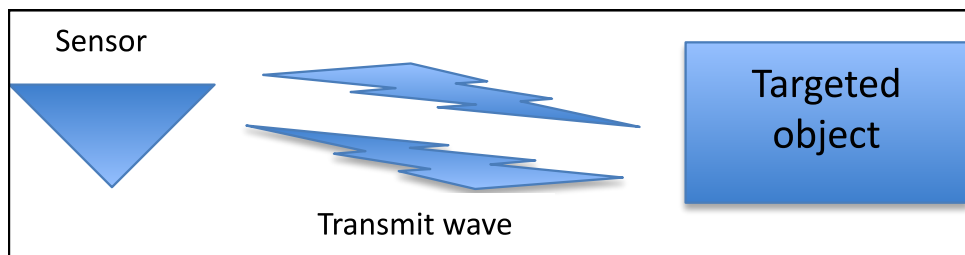
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**Abstract:** This work presents the design and performance evaluation of a high-directivity microwave sensor for biomedical applications. The sensor consists of a modified microstrip radiating structure integrated with a directive enhancement technique to improve impedance matching, gain, and directivity over the desired operating frequency range. A metasurface-based configuration is employed to manipulate the electromagnetic field distribution, resulting in improved energy concentration and sensing capability. The sensor is designed and optimized using a full-wave electromagnetic simulator, and its performance is evaluated in terms of reflection coefficient ( $S_{11}$ ), bandwidth, gain, directivity, radiation pattern, and electric field distribution. The simulated results demonstrate that the proposed sensor achieves excellent impedance matching with a wide operating bandwidth while providing significantly higher directivity compared with the conventional sensor configuration. The concentrated electric field enables greater penetration and sensitivity for detecting variations in the dielectric properties of biological tissues. This characteristic makes the proposed sensor suitable for biomedical sensing applications. Furthermore, the enhanced directivity reduces the influence of surrounding tissues, thereby improving measurement accuracy and reliability. The proposed sensor offers a compact, low-cost, and non-ionizing solution for microwave biomedical diagnostics and monitoring. Owing to its improved radiation characteristics and sensing performance, the developed sensor has strong potential for integration into biomedical systems.

**Keywords:** Microwave sensor, Directive sensor



# Advances in AI-driven Microwave Technologies for Skin Wound Classification and Monitoring

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**Abstract:** Objectives: Non-invasive microwave sensing techniques are increasingly important in biomedical applications because they enable tissue assessment and wound monitoring without causing discomfort, tissue damage, or risk of infection. This study investigates the classification of wound severity level using microwave sensing integrated with artificial intelligent (AI) deep learning technique. Materials and Methods: The experimental setup comprised a microwave sensor operating at 1–3 GHz, a mini-Vector Network Analyzer (VNA), microcomputer for signal processing and tissue phantoms designed to simulate skin, fat, and muscle. The dielectric characteristics of different tissue types and wound conditions were measured using the reflection coefficient (S11) across multiple frequency sweep. Actual rats (Sprague-dawley) then were used for real data measurement. From analysis, the S11 measurement data show some pattern of frequency shifting and different magnitude level for different type of wound normal, mild, and severe. Then acquired frequency-domain data were used and transformed into scalogram images for training purposes using deep learning convolutional neural network (CNN) to classify tissue materials and wound severity levels. Results: The model designed starts from 1-layer convolutional layer to 4-layer convolution which gives accuracy results improvement from 33% to 99.63%. The unseen data of 360 images from each class were fed into the trained model, only 4 images returned into false class. The trained model was subsequently deployed on a Raspberry Pi to demonstrate its real-time implementation. Conclusions: These findings indicate that integrating microwave sensing with AI deep learning provides an early promising non-invasive method for tissue monitoring, wound assessment, and biomedical diagnostics.

**Keywords:** Signal processing, Wound classification, Deep learning, Convolutional neural network

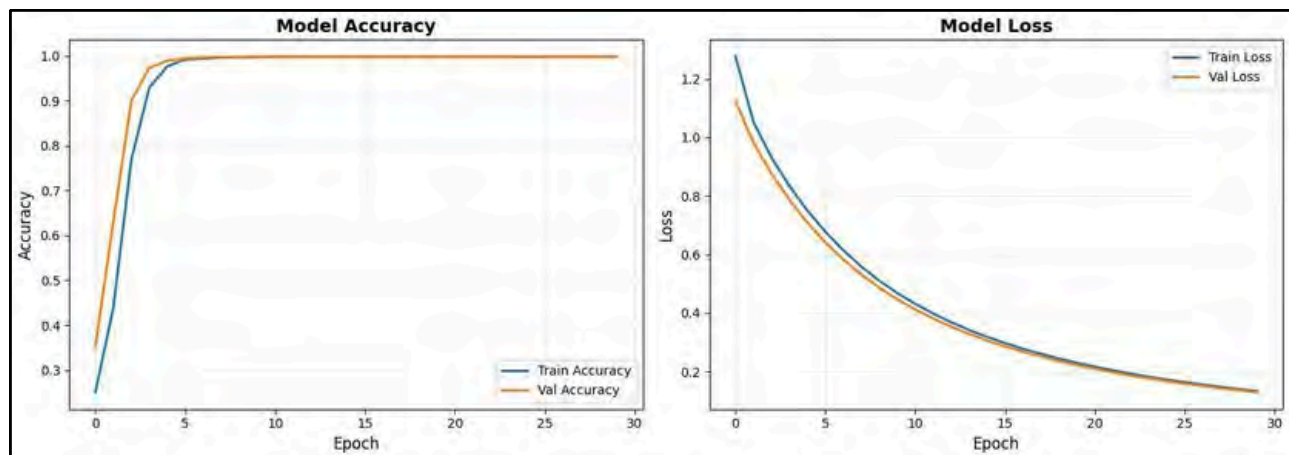


Figure 1. Training model



# Characterization of the Dielectric Properties of Normal Mouse Tissues and Organs

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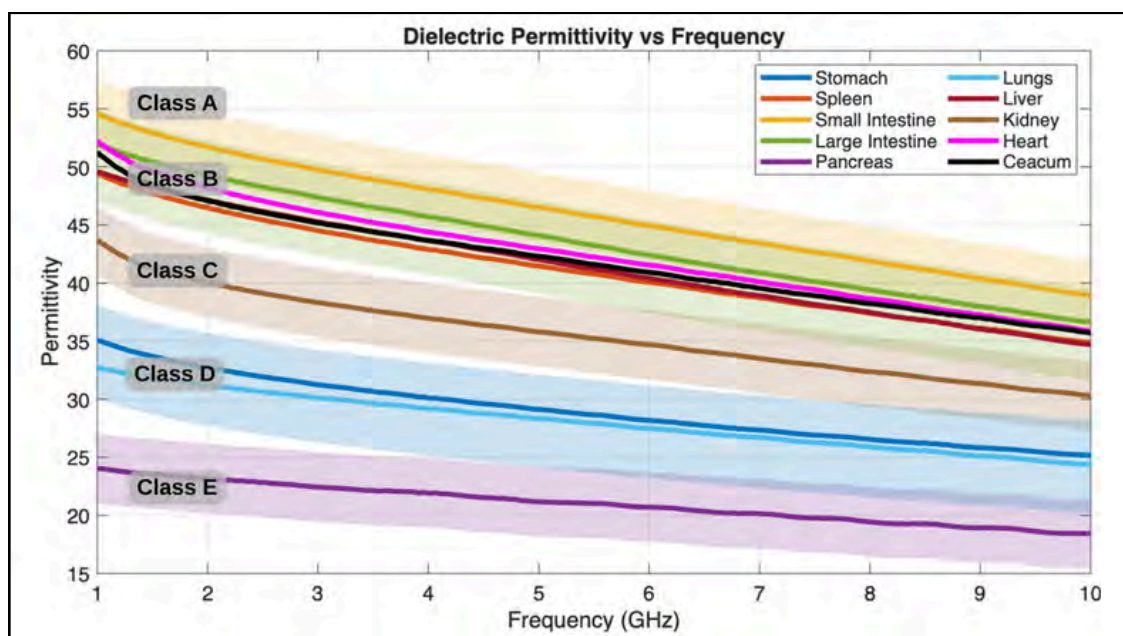
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**Abstract:** Objectives: Dielectric properties of biological tissue are important for developing and testing microwave-based medical devices such as microwave imaging, hyperthermia and implantable antennas. Such data are well known for human tissue but limited for mice, which are commonly used to test these devices before human trials. This work aimed to measure and compare the electrical properties of normal mouse organs across the 1 to 10 GHz frequency range. Materials and Methods: Ten organs, including stomach, spleen, small intestine, large intestine, pancreas, lungs, liver, kidney, heart, and cecum, were collected from three mice. Three repeated readings were taken per organ to check measurement consistency, and probe calibration using standard short open load and reference liquid procedures yielded a measurement uncertainty of  $\pm 5\%$ . Results: Permittivity decreased consistently with frequency across all organs, consistent with expected tissue dispersion behavior. Tissues were divided into five dielectric classes based on their spectral magnitude and trend: Class A showed the highest readings (small intestine) ~55 to 39 dB, Class B formed a middle-upper band (spleen, liver, large intestine, heart, cecum) ~51 to 35 dB, Class C followed (kidney) ~44 to 30 dB, Class D formed a lower band (stomach, lungs) ~35 to 25 dB, and Class E showed the lowest readings (pancreas) ~24 to 18 dB. Conclusions: Mouse organs show consistent, organ-specific electrical behavior across microwave frequencies, likely reflecting differences in water content and structure. This reference data can support preclinical microwave device design and computer modeling of small-animal tissue.

**Keywords:** Dielectric properties, Mouse organs, Permittivity, Preclinical model, Tissue characterization



# From NCEs to CGTPs: A CRO Perspective on Regulatory Challenges in Emerging Therapies

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**Abstract:** Background: Cell and gene therapy products (CGTPs) raise regulatory questions that extend beyond conventional product classification. While established pharmaceutical and biologic frameworks provide essential principles for evaluating quality, safety and efficacy, their application becomes less straightforward when living cells constitute the therapeutic product. CRO Perspective: Experience across conventional medicines, combination products and biologics demonstrates that product complexity is not new. Within mature regulatory pathways, concepts such as identity, dose, potency, stability and comparability have recognisable evidentiary expectations (Table 1). For CGTPs, the same concepts must address living, variable and process-dependent products. Product regulation may also intersect with requirements governing manufacturing, human materials, healthcare facilities, professional practice, distribution, traceability and medical devices. Consequently, a single therapy may involve several regulatory interfaces before it can be manufactured, supplied and administered. Regulatory Challenge: These difficulties are not confined to one national authority. Established International Council for Harmonisation guidelines provide useful foundations, but their original scopes do not always encompass living-cell products. CGTP-specific harmonisation is developing, while regulators and developers continue to interpret existing principles alongside emerging scientific evidence. Conclusion: The transition from new chemical entities to CGTPs does not require established regulatory expertise to be discarded. It requires that expertise to expand across disciplines and adapt to products whose characteristics challenge conventional development assumptions. CROs can support this transition by converting scientific uncertainty into coordinated, product-specific and executable development controls.

**Keywords:** Cell and gene therapy products, Contract research organisation, Emerging therapies, Regulatory framework, Regulatory science

| Table 1. Application of Established Regulatory Principles to CGTP-Specific Challenges |                               |
|---------------------------------------------------------------------------------------|-------------------------------|
| Established principle                                                                 | CGTP-specific consideration   |
| Identity                                                                              | Living-product heterogeneity  |
| Comparability                                                                         | Process-dependent attributes  |
| Safety                                                                                | Persistence and delayed risks |

# Enhancing MSC Therapeutic Potential Through Hypoxia and IFN- $\gamma$ Priming for Mitochondrial Disease

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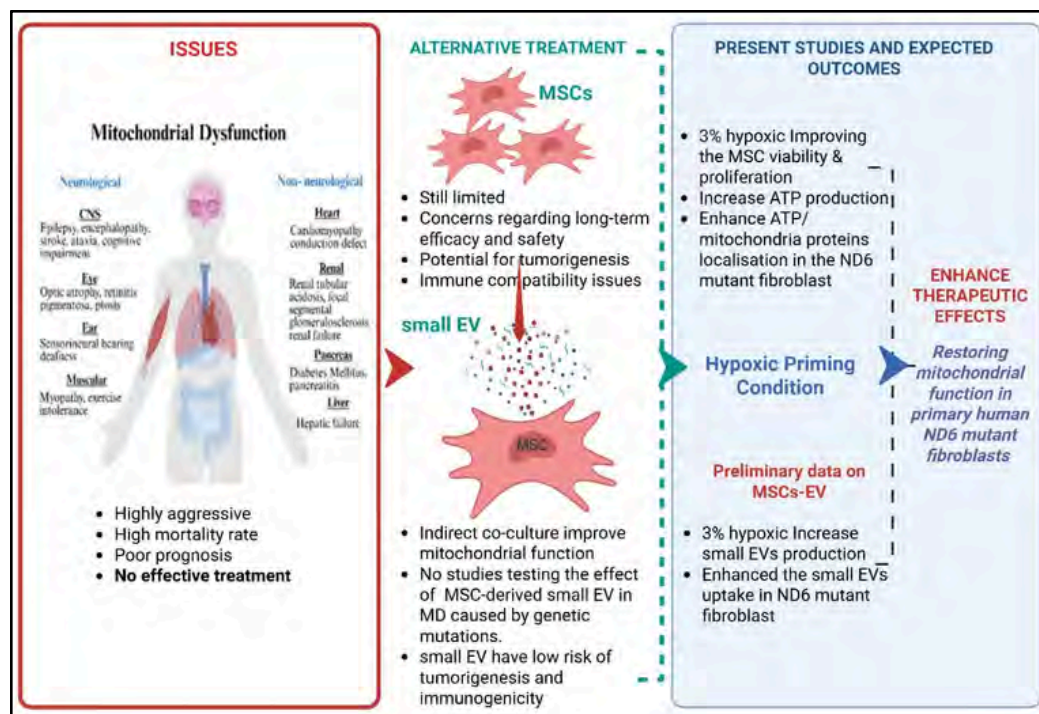
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**Abstract:** Mitochondrial diseases caused by pathogenic mitochondrial DNA (mtDNA) mutations impair oxidative phosphorylation and ATP production, with limited effective treatment options. This study investigated whether hypoxia and interferon-gamma (IFN- $\gamma$ ) priming could enhance the therapeutic potential of mesenchymal stem cells (MSCs) for restoring mitochondrial function in ND6-mutant fibroblasts. Human primary MSCs from three donors were cultured under normoxic conditions, 3% hypoxia, 100 ng/mL IFN- $\gamma$ , or combined hypoxia and IFN- $\gamma$  priming. Primed MSCs were subsequently indirectly co-cultured with patient-derived ND6-mutant fibroblasts. MSC proliferation and ATP production were assessed, while mitochondrial transfer was evaluated using MitoTracker Green co-localization. Combined hypoxia and IFN- $\gamma$  priming significantly enhanced MSC proliferation ( $p = 0.0018$ ), whereas hypoxia alone and combined priming significantly increased MSC ATP production (both  $p < 0.001$ ). Mitochondrial transfer from MSCs to ND6-mutant fibroblasts was confirmed by MitoTracker Green co-localization. Indirect co-culture with primed MSCs increased ATP levels in ND6-mutant fibroblasts by approximately 40–60%. Preliminary findings further demonstrated uptake of MSC-derived small extracellular vesicles (sEVs) by ND6-mutant fibroblasts, suggesting that extracellular vesicle-mediated mechanisms may also contribute to the observed therapeutic effects. Collectively, these findings demonstrate that hypoxic and IFN- $\gamma$  priming can enhance key therapeutic properties of MSCs and promote ATP restoration in ND6-mutant fibroblasts, supporting the potential of primed MSC-based approaches as a novel therapeutic strategy for mitochondrial diseases.

**Keywords:** MSC, ATP, Hypoxia, Mitochondrial disease, IFN- $\gamma$ , Small Evs



# Clinical translational research: UKM GMP Experience

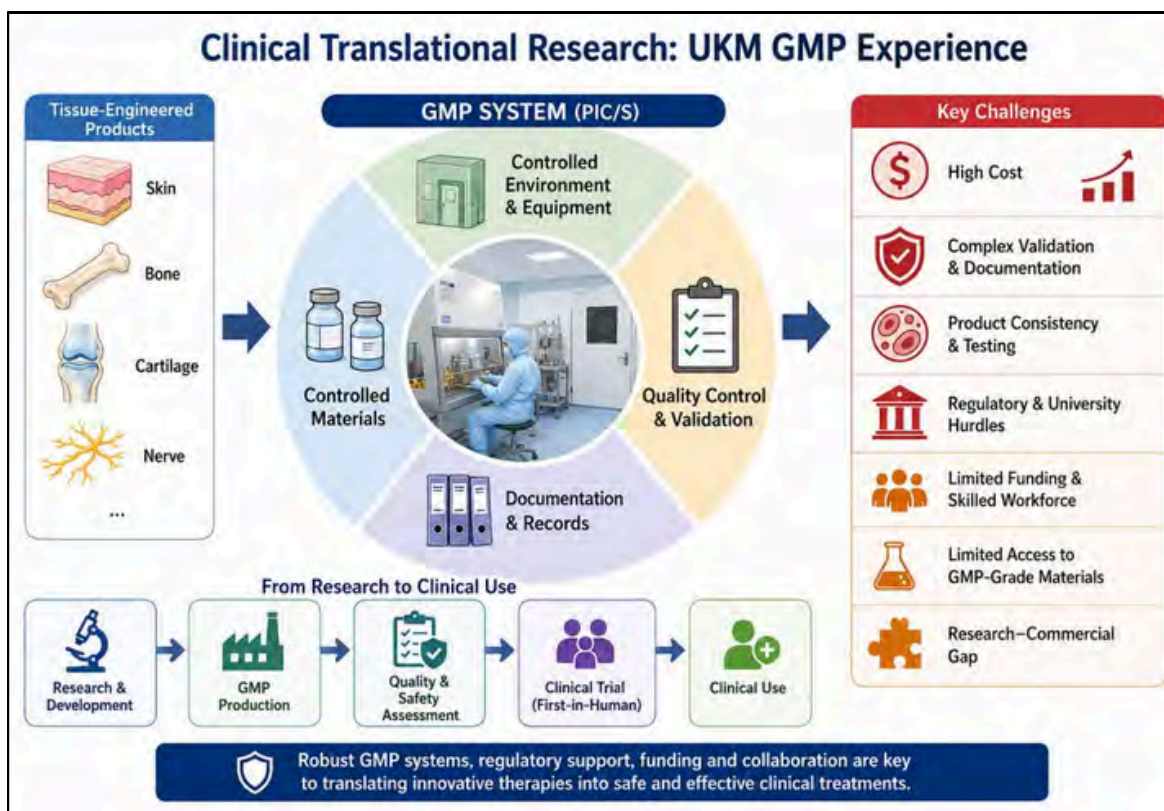
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**Abstract:** Universiti Kebangsaan Malaysia has developed numerous tissue-engineered products involving skin, bone, cartilage, nerve, and other tissues with strong potential for clinical translation. However, the transition from laboratory research to clinical application requires strict compliance with Good Manufacturing Practice (GMP) standards and regulatory requirements. This study describes UKM's experience in implementing GMP systems for clinical translational research and highlights the major challenges encountered. GMP activities were conducted according to PIC/S guidelines, emphasizing controlled cleanroom environments, certified equipment, validated clinical-grade reagents, quality control procedures, and comprehensive documentation. Critical equipment underwent IQ, OQ, and PQ certification, while production processes included media fill simulations, dry runs, cleaning validation, and aseptic competency assessments. Significant challenges include high annual maintenance costs of GMP facilities, extensive validation requirements, difficulties in standardizing cell characterization and batch consistency, and limited availability of validated assays for sterility, identity, viability, potency, and safety testing. Additional barriers include regulatory hurdles for first-in-human clinical trials, limited funding support, shortage of skilled personnel, restricted access to GMP-grade materials, and a persistent gap between research and commercialization. Strengthening regulatory support, funding, and technical expertise is essential to accelerate clinical translation of regenerative medicine products in Malaysia.

**Keywords:** Good Manufacturing Practice (GMP), Clinical translation, Validation



# Engineering MSC 2.0: An Integrated Platform for Next-Generation MSC Therapeutics

Woo Jun Yung

**Abstract:** Mesenchymal stem cells (MSCs) possess remarkable functional plasticity, with the capacity to exert both pro- and anti-inflammatory effects in response to their surrounding environment. This “yin-yang” biology underpins their therapeutic potential but can also contribute to variable and context-dependent outcomes. At AGeM Bio, we developed a platform that uses transient, non-viral engineering to programme MSCs with defined therapeutic functions. Rather than relying solely on their native responses, MSC 2.0 is designed to tip the balance towards a disease-appropriate pro- or anti-inflammatory state. This presentation will introduce the MSC 2.0 concept and illustrate how the same cellular platform can be adapted for distinct therapeutic applications. Regenerative applications will be briefly discussed, including the use of modified MSCs to promote angiogenesis, regulate inflammation and support tissue repair in non-healing wounds. The main focus will be AGeM-102, our lead oncology candidate for recurrent high-grade glioma, and the mechanisms supporting its development. These studies explore how engineered MSCs can deliver localised anti-tumour activity while inducing a transient pro-inflammatory state within the tumour microenvironment. Together, this work demonstrates how the intrinsic yin-yang properties of MSCs can be deliberately directed to address diseases with fundamentally different biological needs.

**Biography:** Dr Woo Jun Yung is Co-Founder and Chief Scientific Officer of AGeM Bio, a Singapore-based biotechnology company developing engineered mesenchymal stem cell therapeutics. He leads the company's scientific strategy and research and development activities, spanning platform development, preclinical validation, manufacturing translation and pipeline expansion. He received his PhD in Biochemistry from the National University of Singapore, where his research focused on MSC engineering and cell-based therapeutics. Dr Woo was a key contributor to the development of AGeM Bio's platform technologies and its lead candidate, AGeM-102, a locoregional chemo-immunotherapeutic for recurrent high-grade glioma that is advancing towards clinical evaluation. His work focuses on translating programmable MSCs into practical therapeutic products across oncology and regenerative medicine by integrating mechanistic biology with scalable manufacturing and clinical development.



# Can Robots Help the Human Body Heal? Intelligent Sensing, AI and Automation for Regenerative Medicine

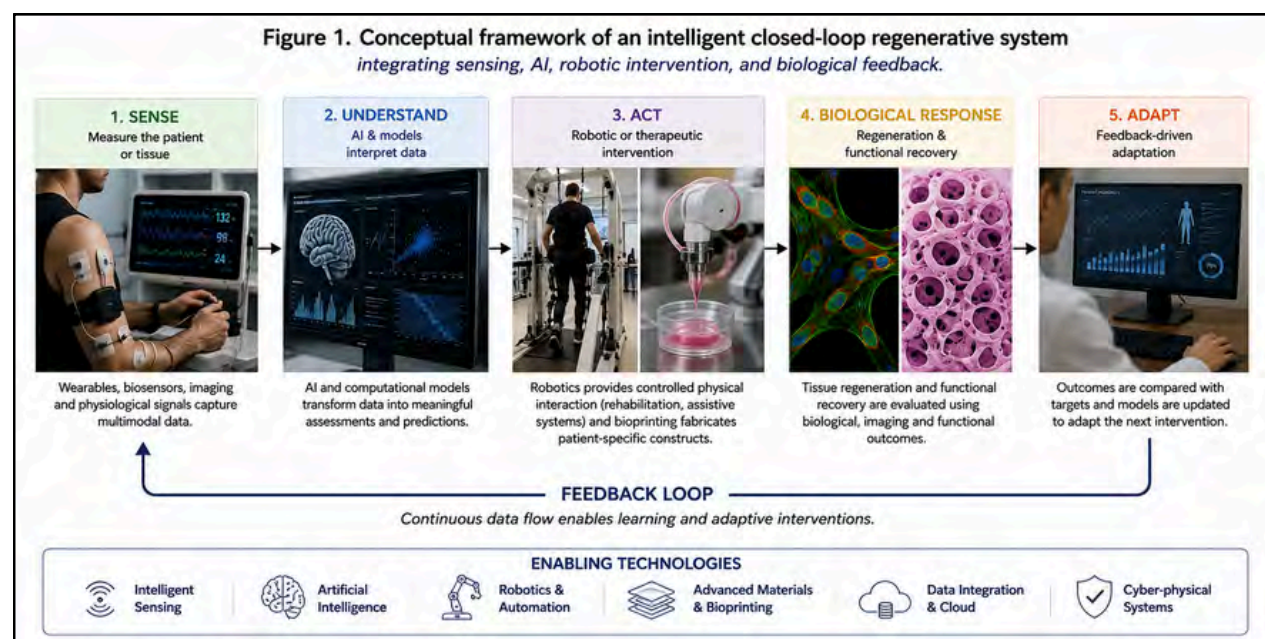
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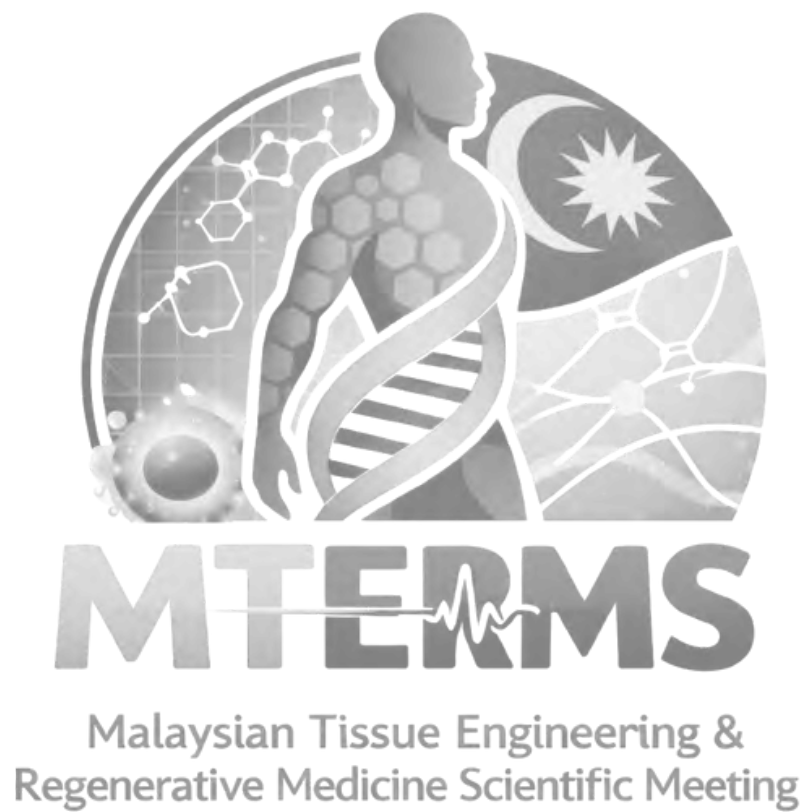
**Abstract:** Regenerative medicine is advancing rapidly through developments in biomaterials, tissue engineering, bioprinting, and personalised therapies. At the same time, robotics, intelligent sensing, artificial intelligence (AI), and automation are transforming how complex systems are measured, controlled, and adapted. The convergence of these fields presents new opportunities to support the transition from biological discovery to intelligent and personalised therapeutic solutions. This plenary explores regenerative medicine from an automation and robotics perspective through a simple framework: sense, understand, act, and adapt. Intelligent and wearable sensors can provide quantitative information on physiological and functional states, while AI and computational models can transform multimodal data into meaningful assessments and predictions. Robotics can subsequently provide controlled and repeatable physical interaction through rehabilitation and assistive systems, while robotic bioprinting and automated biomanufacturing extend these principles towards the fabrication of living constructs. The presentation further considers the emerging concept of closed-loop regenerative systems, in which sensing, AI, robotic intervention, and biological or functional feedback are continuously integrated to enable adaptive and potentially patient-specific interventions. Emerging developments in robotic in-situ bioprinting, digital twins, and autonomous biomanufacturing illustrate how engineering and biological systems are becoming increasingly interconnected. Finally, key challenges involving biological variability, sensing reliability, AI trustworthiness, robotic safety, clinical validation, and regulation are discussed. This convergence offers an exciting interdisciplinary pathway towards more measurable, adaptive, and personalised regenerative healthcare.

**Keywords:** Regenerative Medicine, Robotics, Intelligent Sensing, Artificial Intelligence, Bioprinting



## ORAL PRESENTER ABSTRACTS

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# Characterisation of Skeletal Muscle Stem Cell Responses During Dystrophic Muscle Regeneration Using the mdx Mouse Model

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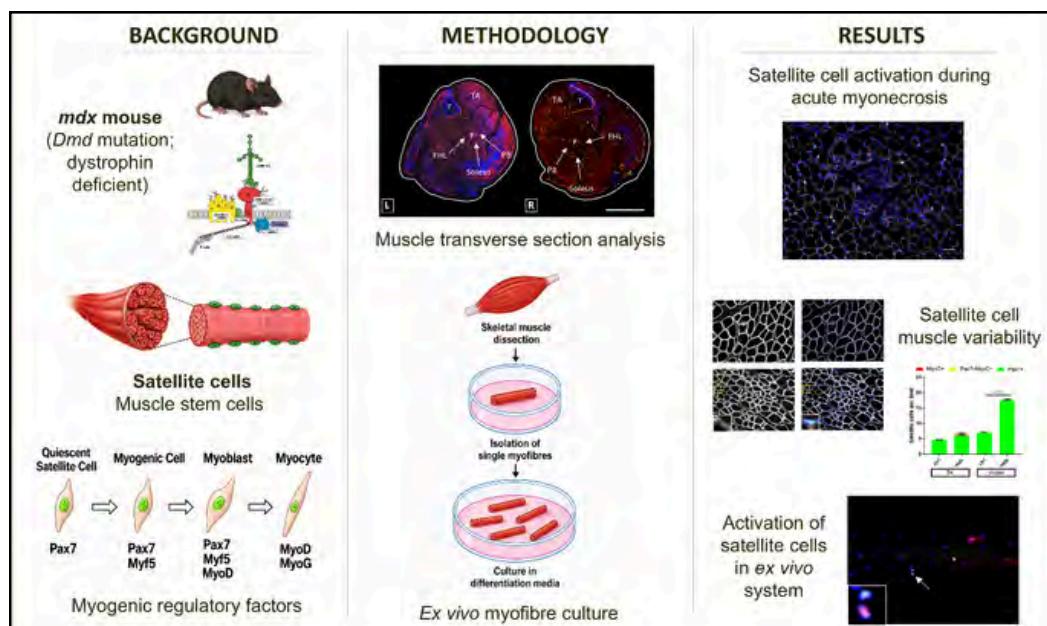
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**Introduction:** The mdx mouse is a murine model of Duchenne Muscular Dystrophy (DMD), harbouring mutations in the Dmd gene that result in dystrophin deficiency and progressive muscle degeneration. Unlike human DMD, mdx mice exhibit preserved regenerative capacity, suggesting active satellite cell-mediated myofibre regeneration during disease progression. **Objectives:** To evaluate endogenous regenerative response and satellite cell dynamics in dystrophic skeletal muscle of juvenile and adult mdx mouse. **Methods:** Satellite cell activity was evaluated in vivo using hindlimb muscle sections from juvenile (3–4 weeks) and adult (12 weeks) mdx mice, and ex vivo using isolated myofibre cultures from 8-week-old mdx and control mice. To evaluate satellite cell activity and myogenic progression, immunofluorescence analysis was performed using Pax7 and myogenic regulatory factor markers (Myf5, MyoD, and MyoG). **Results:** Early juvenile mdx muscles demonstrated marked satellite cell activation and active myofibre regeneration, with increased Pax7<sup>+</sup>/MyoD<sup>+</sup> and MyoG<sup>+</sup> cells within degenerating regions. Significantly elevated satellite cell numbers were observed in the tibialis anterior, flexor hallucis longus, and soleus muscles compared to controls, with the soleus showing prominent MyoG expression and centrally nucleated fibres. In adult mdx mice, muscle-specific variation persisted, with the soleus maintaining higher satellite cell counts, while tibialis anterior muscles demonstrated larger fibre calibre. Ex vivo myofibre culture of mdx satellite cells showed early activation and apicobasal division within 24 hours, with comparable progression through myogenic stages relative to controls. **Conclusion:** mdx satellite cells retain substantial regenerative and self-renewal capacity despite ongoing dystrophic degeneration, with distinct muscle-specific variation in myogenic activity.

**Keywords:** Satellite cell, Dystrophic, mdx, Muscle Regeneration



# PRP as a Xeno-Free Alternative for Maintaining Stemness in 3D UC-MSC Culture

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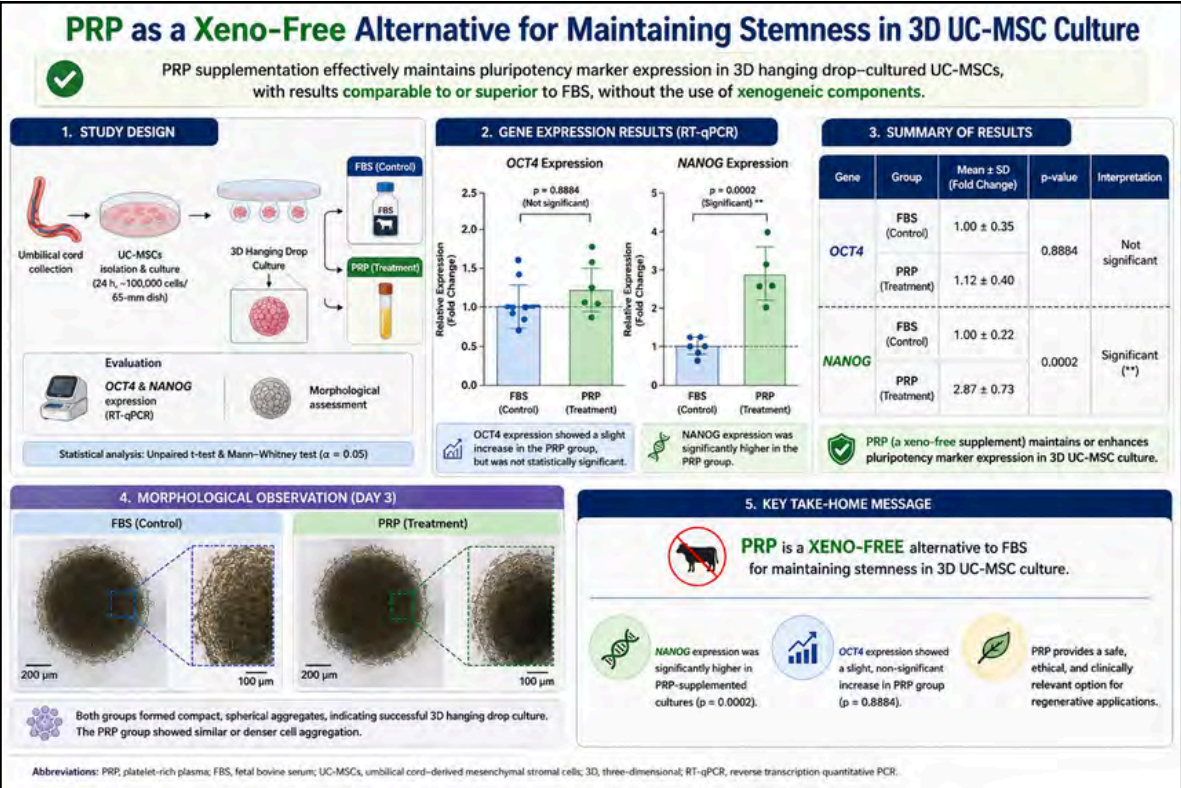
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Umbilical cord-derived mesenchymal stromal cells (UC-MSCs) possess significant potential for regenerative therapy due to their multipotent and self-renewal properties. Three-dimensional (3D) hanging drop culture better mimics the in vivo microenvironment and may enhance pluripotency-related gene expression. However, fetal bovine serum (FBS), commonly used in cell culture, carries ethical and immunogenic concerns due to xenogeneic components. **Objectives:** This study aimed to compare the effects of platelet-rich plasma (PRP) and FBS supplementation on OCT4 and NANOG gene expression in 3D-cultured UC-MSCs. **Methods:** UC-MSCs were cultured in vitro in a 3D hanging drop system and divided into FBS-supplemented (control) and PRP-supplemented (treatment) groups. Gene expression levels of OCT4 and NANOG were analysed using RT-qPCR. Morphological assessment and statistical analysis using an unpaired t-test and Mann–Whitney test were performed. **Results:** PRP supplementation maintained pluripotency marker expression comparable to or better than FBS. OCT4 expression showed a slight but non-significant increase in the PRP group ( $p = 0.8884$ ), whereas NANOG expression was significantly higher in the PRP group ( $p = 0.0002$ ). Both groups successfully formed compact spheroids under 3D culture conditions. **Conclusions:** PRP is a promising xeno-free alternative to FBS for maintaining stemness-related gene expression in 3D UC-MSC cultures.

**Keywords:** 3D Culture, Platelet-Rich Plasma, Stem Cells, Xeno-Free





# ***Engineering ACE2- and IL-37-Expressing Mesenchymal Stem Cells for Dual Targeting of SARS-CoV-2 and Acute Respiratory Distress Syndrome (ARDS) in vitro***

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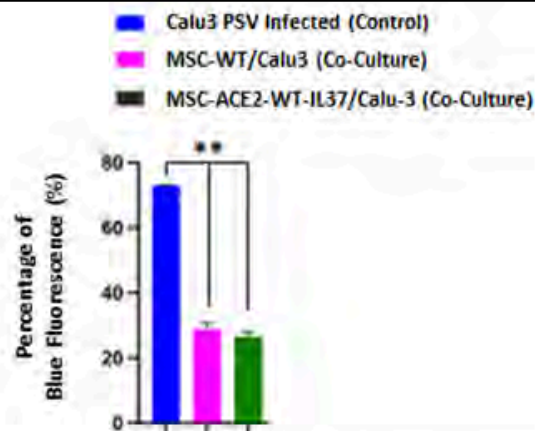
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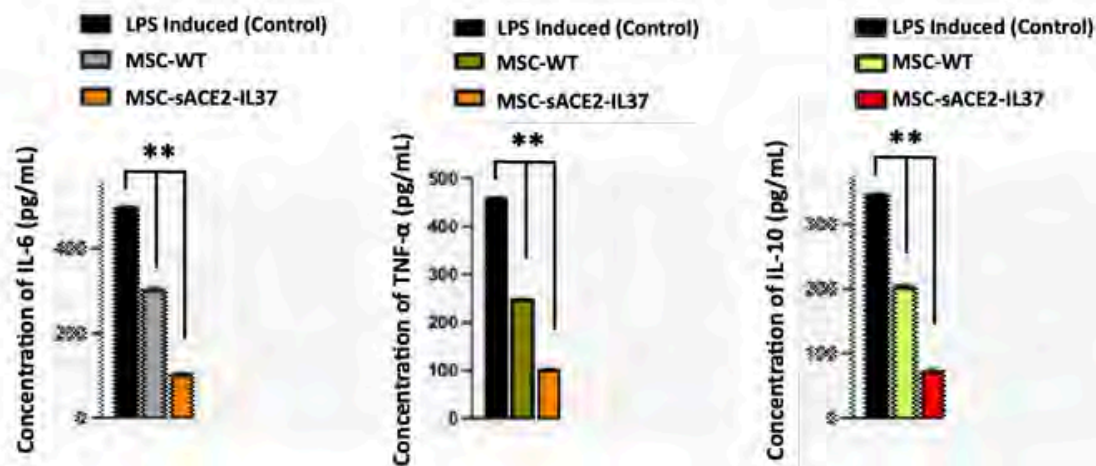
**Introduction:** Severe COVID-19 frequently progresses to ARDS (acute respiratory distress syndrome), a life-threatening lung condition driven by hyperinflammation and massive tissue damage. Mesenchymal stem cells (MSCs) represent a promising therapeutic strategy for COVID-19 and ARDS due to their innate immunomodulatory properties. **Methodology:** This study explores pre-clinical in vitro model to study the therapeutic potential of genetically modified MSCs expressing ACE2 (Angiotensin-Converting Enzyme 2) and IL-37 (Interleukin-37) by employing pseudo-typed SARS-CoV-2 expressing blue fluorescence and LPS-induced macrophages as models for COVID-19 and ARDS. **Results and Discussion:**

The ACE-2 and IL-37 expressing MSCs (MSC-ACE2-IL37) generated in this study sustained their mesodermal lineage differentiation and maintained stable expression of both proteins for up to 15 days. Co-culture of the modified MSC (MSC-ACE2-IL37) with Calu-3 cells resulted in a significant reduction in the infectivity of pseudo-typed SARS-CoV-2 to the Calu-3 cells, demonstrating an effective viral neutralization (Figure 1). The MSC-ACE2-IL37 co-cultured with LPS-treated Calu3 cells or macrophages was able to significantly reduce the secretion of inflammatory cytokines (IL-6, TNF- $\alpha$ , and IL-10) in both Calu3 (Figure 2) and macrophages indicating that the modified MSCs were effective in reducing inflammation. Furthermore, we found that MSC-ACE2-IL37 effectively reduced macrophage viability in the XTT assay, suggesting that these modified MSCs exert an anti-inflammatory effect by deactivating macrophages. **Conclusion:** Collectively, these findings highlight the potential of engineered MSCs as therapeutic platforms against infectious diseases, particularly COVID-19 and its associated inflammatory symptoms. Further translational studies are warranted to evaluate their safety and efficacy in pre-clinical animal models.





**Figure 1: Flow cytometry analysis of neutralization of pseudo-typed SARS-CoV-2 infection in Calu-3 cells by MSC-ACE2-IL37 in a co-culture assay.** Significant reduction in the percentage of blue fluorescence in Calu-3 cells when co-cultured with either MSC-WT (Wild Type) or MSC-ACE2-IL37 (\*\* $p < 0.001$ ).



**Figure 2: Treatment of LPS-induced-Calul3 by MSC-ACE2-IL37 in a co-culture assay.** Significant reduction in the secretion of IL6, TNF-α and IL-10 was noticed in LPS-treated Calu3 cells as analyzed by ELISA (\*\* $p < 0.001$ ).

**Keywords:** Mesenchymal Stem Cell (MSC), ACE2, IL37, COVID-19, acute respiratory distress syndrome (ARDS).

# An Factor X-Enriched Gastric Stromal Clone Promotes Gastric Stem Cell Maintenance

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**Introduction:** Gastric stem cells are regulated by multiple signals from surrounding stromal populations, but the contribution of distinct gastric stromal clones remains incompletely understood. This study investigated whether there is a novel factor-enriched gastric stromal clone that supports gastric stem-cell maintenance.

**Methods:** Stromal cells were isolated from adult mouse stomach, immortalised, and established as clonal cell lines. Gastric stromal clone 6 (C6) was characterised by immunofluorescence using established stromal markers. The effects of C6-conditioned medium on gastric organoids were assessed by measuring organoid perimeter and analysing gene expression using quantitative reverse-transcription polymerase chain reaction (qRT-PCR). One-colour microarray profiling revealed that C6 was enriched in factor X family genes while exhibiting low expression of factor Z family genes. The biological activities of these candidate factors were subsequently validated using recombinant factor X and factor Z. **Results:** C6-conditioned medium increased gastric organoid size and expression of the proliferation and stem-cell markers *Ki67* and *Sox9*. It reduced expression of the pit-cell marker *Muc5ac* and parietal-cell marker *Atp4b*, while the chief-cell marker *Pgc* was not significantly altered. Transcriptomic analysis of C6 showed enrichment of factor X-family genes, together with low factor Z-family expression. Recombinant factor X reproduced the growth-promoting phenotype by increasing organoid size, *Ki67* and *Sox9* and suppressing *Muc5ac* and *Atp4b*. Meanwhile, the treatment with recombinant factor Z did not significantly increase organoid size. **Conclusion:** A novel factor X-enriched gastric stromal clone promotes gastric stem-cell proliferation and maintenance while restraining pit- and parietal-cell differentiation. These findings identify factor X-family signaling as a potential stromal niche mechanism governing gastric epithelial homeostasis.

**Keywords:** Cell signaling; Gastric organoid; Gastric stem cell; Stem cell niche; Stromal cell

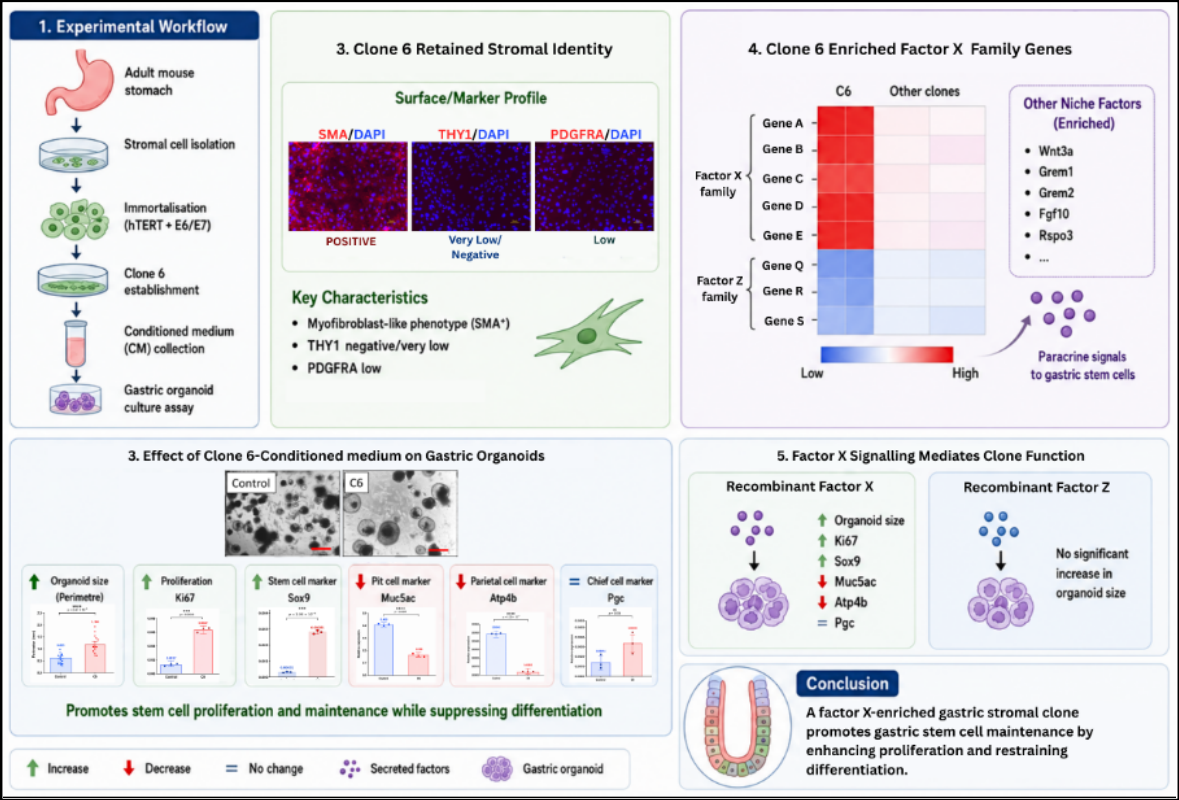


Figure 1: An factor X-enriched gastric stromal clone promotes gastric stem-cell proliferation and maintenance while restraining epithelial differentiation.

# Building Malaysia's Patient-Specific iPSC-Derived Cardiomyocyte Platform for Precision Cardio-Oncology

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Doxorubicin is a highly effective chemotherapeutic agent for lymphoma and other malignancies but is associated with irreversible cardiotoxicity and long-term cardiac dysfunction. Understanding patient-specific susceptibility requires physiologically relevant human models, as conventional cell lines lack individual genetic diversity and animal models often fail to accurately predict human cardiac responses. Human induced pluripotent stem cell-derived cardiomyocytes (hiPSC-CMs) provide a promising platform for disease modelling and cardiotoxicity assessment. However, inter-individual variability, genetic stability, and differentiation efficiency remain significant challenges, and no Malaysian patient-specific hiPSC-CM platform for cancer research has been reported. This study aimed to establish and characterise Malaysian patient-specific hiPSC lines and evaluate the reproducibility of an in-house cardiac differentiation protocol across lymphoma patients and healthy donors. Peripheral blood mononuclear cells (PBMCs) were collected with informed consent and ethical approval, then reprogrammed using non-integrating episomal vectors under feeder-free, antibiotic-free conditions. Established hiPSC lines were characterised by morphology, pluripotency marker expression, genomic stability, and trilineage differentiation before cardiac differentiation using a chemically defined protocol. Cardiomyocyte identity was confirmed by the expression of cardiac markers such as alpha-actinin and troponin-T and the ability of cells to beat rhythmically at the end stage of differentiation. Patient-specific hiPSC lines were successfully generated from both lymphoma patients and healthy controls. All lines displayed typical pluripotent morphology, maintained genomic integrity, expressed key pluripotency markers, and differentiated efficiently into spontaneously beating cardiomyocytes despite donor-specific genetic backgrounds. This study establishes the first Malaysian patient-specific hiPSC-CM platform for cardio-oncology research, providing a robust foundation for investigating doxorubicin-induced cardiotoxicity, identifying predictive biomarkers, and advancing precision medicine and future cardioprotective strategies.

**Keywords:** Human induced pluripotent stem cells (hiPSCs), Cardiotoxicity, Lymphoma, Cardiomyocytes, Patient-specific disease modelling

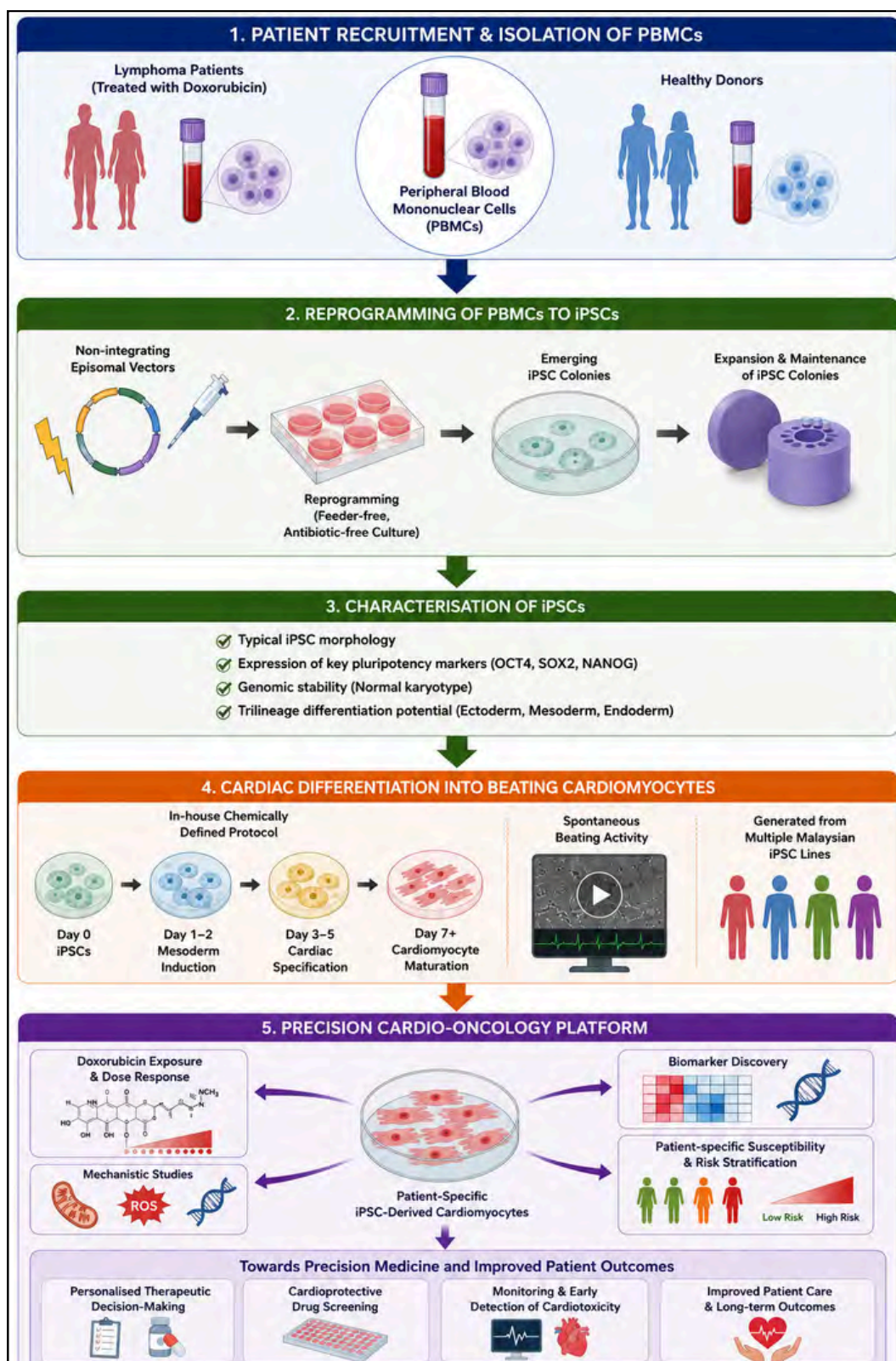


Figure 1: Graphical overview of the establishment of the Malaysian patient-specific hiPSC-derived cardiomyocyte platform for precision cardio-oncology.



# Chemical-Thermal Crosslinking of Gelatine Membranes for Tunable Degradation and Enhanced HaCaT Cell Adhesion and Migration

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Due to high biodegradability, gelatine membranes readily dissolve upon moisture exposure. **Objectives:** The present study aimed to develop gelatin membranes with controlled degradation through a chemical-thermal crosslinking approach. **Methods:** Gelatine was first crosslinked with 0, 0.5, 1 wt% glutaraldehyde (GTA), followed by microwave heating at 180, 300 and 450 W to induce thermal crosslinking. The effects of GTA concentration and microwave power on surface morphology, functional groups, and degradation profiles were investigated. The in vitro efficacy of the crosslinked membranes in promoting cell adhesion and migration was further evaluated using adhesion and wound scratch assays on HaCaT cells, respectively. **Results:** Surface characterization revealed that membrane roughness increased with GTA concentration and microwave power. FTIR analysis confirmed the formation of aldimine linkages between free amine groups in gelatine and GTA. Notably, membranes crosslinked with 1 wt% GTA showed the slowest degradation rate, retaining about 80% of their mass after 7 days of incubation in PBS, whereas those crosslinked with 0.5 wt% GTA and uncrosslinked gelatine degraded completely. Cell adhesion assay using fluorescein diacetate staining confirmed successful adhesion of HaCaT cells on all crosslinked membranes. Membranes dried at 450 W exhibited the highest cell adhesion due to their increased surface roughness. In the wound scratch assay, gelatine crosslinked with 0.5 wt% GTA and microwave dried at 450 W exhibited the highest healing rate as evidenced by the smallest wound closure area, indicating profound in vitro efficacy. **Conclusions:** These findings suggest that the chemical-thermally crosslinked gelatine membranes show potential as wound healing materials.

**Keywords:** Gelatine, Crosslinking, Microwave, Wound healing

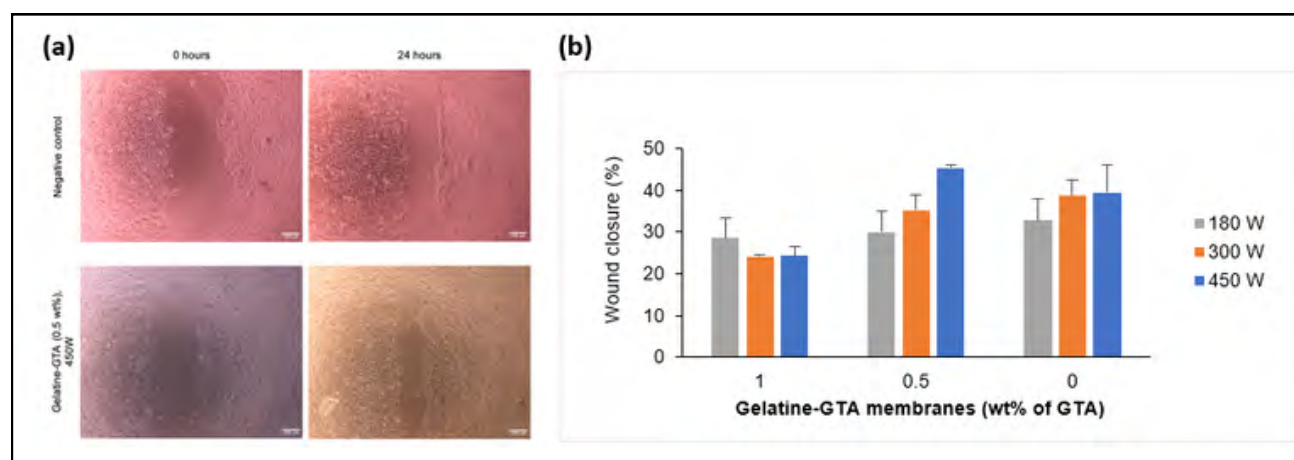


Figure 1: Effect of gelatine membranes crosslinked with GTA and microwave dried at different powers on an in vitro wound healing of HaCaT cells. (a) Representative micrographs of the wounds before and after an incubation of 24 h for control and gelatine membrane crosslinked with 0.5 wt% GTA and microwave dried at 450 W. (b) Wound closure after 24 h for cells treated with crosslinked gelatine membranes at 180, 300 and 450 W. Values represent means  $\pm$  SD (n = 3).

# Polymeric nanoparticulate system for the co-delivery of curcumin and 5-Fluorouracil for the potential colorectal cancer treatment

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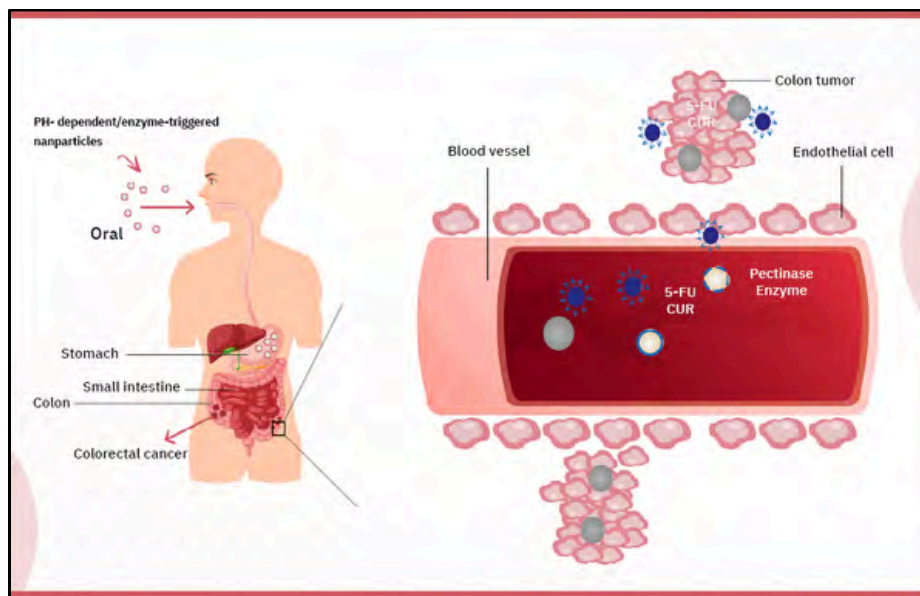
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Colorectal cancer (CRC) is the third leading cancer diagnosis in the world and one of the foremost causes of cancer-related fatalities. However, current treatments possess high resistance and debilitating side effects. Objectives: This study aimed to formulate and optimize co-loaded CUR and 5-FU chitosan pectinate nanoparticles for colonic delivery. Methods: The nanoparticles were formulated using the ionotropic gelation method and were characterized in terms of particle size, zeta potential, encapsulation efficiency, and release profiles. For simultaneous quantification of curcumin and 5-FU, a validated HPLC-PDA analysis method was developed. Results: The physicochemical characteristics of the optimized CUR-5-FU co-loaded nanoparticles were  $381.3 \pm 0.9$  nm, 0.397, and  $+36.0 \pm 0.9$  mV for the particle size, pDI, and zeta potential, respectively. The increase in particle size for the encapsulated nanoparticles showed statistically significant results ( $p < 0.01$ ). SEM analysis illustrated spherical morphology. The surface of the particles appeared smooth without cracks, which suggests sufficient crosslinking. The obtained encapsulation efficiency (EE%) values were  $68\% \pm 1.2\%$  and  $43\% \pm 0.85\%$  for curcumin and 5-FU, respectively. During the simulated gastrointestinal drug release studies, negligible release of both drugs was obtained in the simulated gastric fluid (SGF). Nevertheless, bursts of the drugs were released ( $p < 0.01$ ) within one hour of placing the nanoparticulate system in the simulated colonic fluid (SCF). The drugs continued to release until 72.1% and 82.9% of the encapsulated curcumin and 5-FU, respectively. Finally, stability studies at 4 and 25 °C showed that the nanoparticles remained within the clinically acceptable range, confirming the high physical stability of the system. Conclusion: CUR-5-FU co-loaded chitosan pectinate nanoparticles demonstrated an established targeting and simultaneous delivery of drugs for the potential synergistic anti-colorectal cancer effects.

**Keywords:** Curcumin, 5-Fluorouracil, Colorectal cancer, Nanoparticles



## ***Achatina fulica* mucus inhibits premature aging of human skin fibroblasts by decreasing miR-23a expression**

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**Background:** The extremely high ultraviolet (UV) index in Indonesia causes premature aging. *Achatina fulica* mucus contains components that can prevent premature aging. **Objective:** To examine changes caused by UVB exposure in miR-23a, p53 mRNA and HAS2 mRNA expressions in prematurely aged human skin fibroblasts treated with *A. fulica* mucus. **Methods:** *A. fulica* mucus was collected from 50 *A. fulica* fish stimulated with 5-10V electrical stress for 30-60 seconds and then freeze-dried. Human skin fibroblasts were divided into 6 groups: Group 1 (normal control) which is fibroblasts in complete medium, Group 2 (negative control) which is fibroblasts + UVB 100mJ/cm<sup>2</sup> in complete medium, group 3 (positive control) consists of fibroblasts + UVB 100mJ/cm<sup>2</sup> + PRP in complete medium, group 4 (AF 3) is fibroblasts + UVB 100mJ/cm<sup>2</sup> + AF 3.9µg/ml in complete medium, group 5 (AF 15) is fibroblasts + 100mJ/cm<sup>2</sup> + AF 15.625µg/ml in complete medium and group 6 (AF 62) is fibroblasts + UVB 100mJ/cm<sup>2</sup> + AF 62.5µg/ml in complete medium. After treatment, each group was measured for miR-23a-3p, p53 mRNA, and HAS2 expression by qPCR. Data obtained from the observations were tested using one-way ANOVA and post hoc LSD tests. **Results:** The group given *A. fulica* mucus showed significantly lower p53 and miR-23a mRNA expression ( $p < 0.05$ ) by 5-fold each, while HAS2 mRNA expression was significantly higher ( $p < 0.05$ ) by 5-fold compared to the UVB group. **Conclusion:** *A. fulica* mucus prevents premature aging of human skin fibroblasts exposed to UVB by inhibiting the expression of p53 and miR-23a mRNA, and increasing HAS2 mRNA.

**Keywords:** *Achatina fulica*, UVB, Fibroblast, p53, miR-23a, HAS2

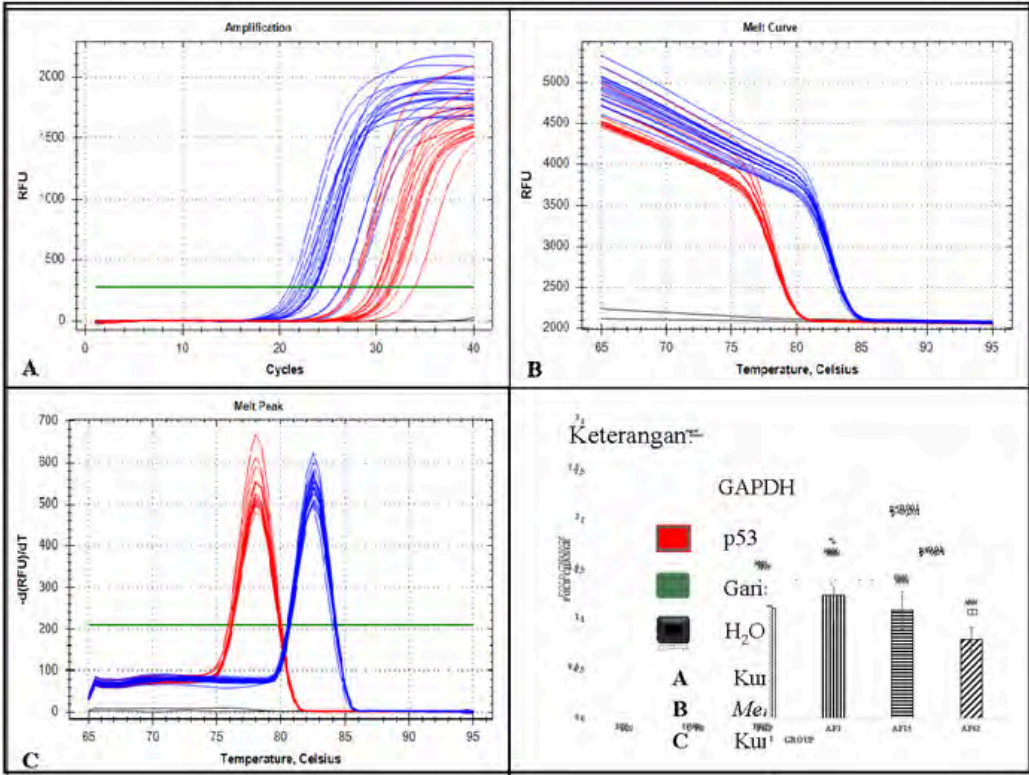


Figure 1: Amplification curve, melt curve, melt peak and p53/GAPDH mRNA expression

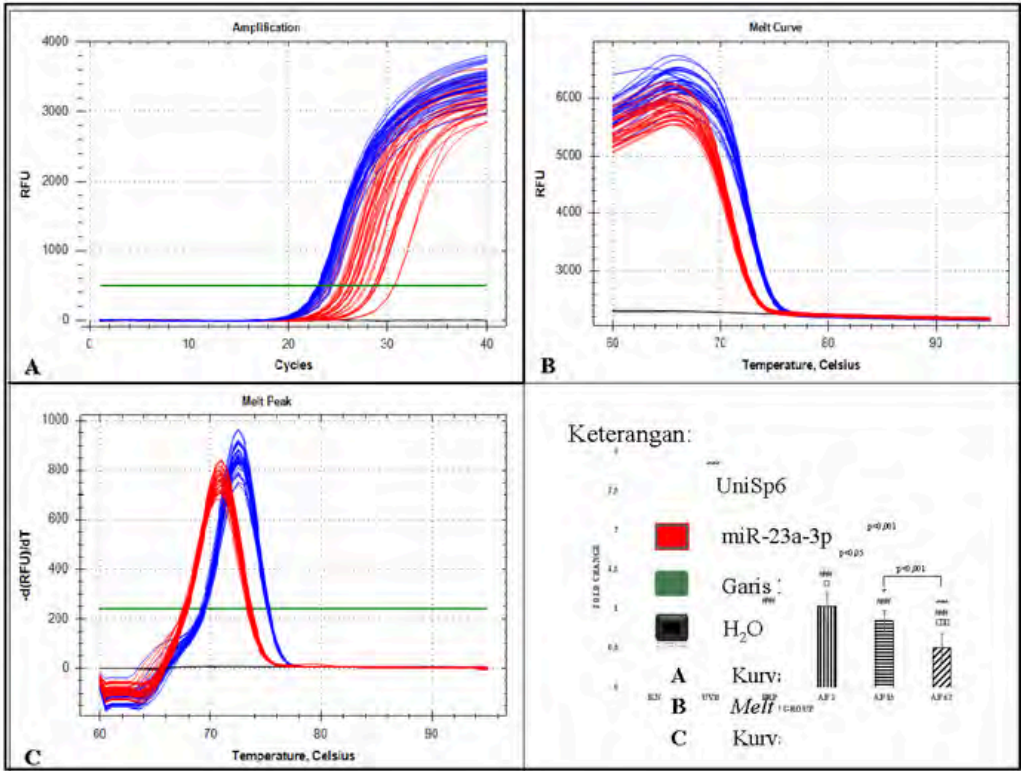


Figure 2: Amplification curve, melt curve, melt peak and miR-23a/UniSp6 expression

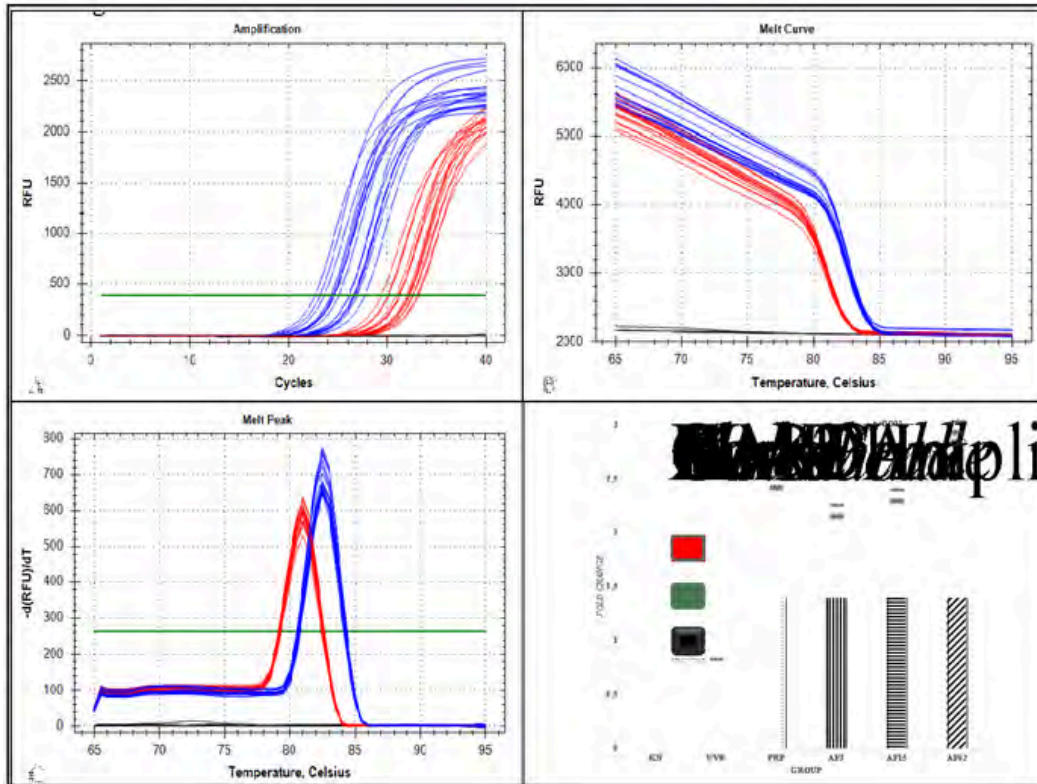


Figure 3: Amplification curve, melt curve, melt peak and HAS2/GAPDH mRNA expression

KN is group of fibroblasts in complete medium;

UVB is group of fibroblasts + UVB 100mJ/cm<sup>2</sup> in complete medium;

PRP is group of fibroblasts + UVB 100mJ/cm<sup>2</sup> + PRP in complete medium;

AF3 is group of fibroblasts + UVB 100mJ/cm<sup>2</sup> + AF 3.9µg/ml in complete medium;

AF 15 is group of fibroblasts + 100mJ/cm<sup>2</sup> + AF 15.625µg/ml in complete medium;

AF 62 is group of fibroblasts + UVB 100mJ/cm<sup>2</sup> + AF 62.5µg/ml in complete medium.

\* $p < 0,05$  vs KN, \*\* $p < 0,01$  vs KN, \*\*\* $p < 0,001$  vs KN.

# $p < 0,05$  vs UVB, ## $p < 0,01$  vs UVB, ### $p < 0,001$  vs UVB.

§  $p < 0,05$  vs PRP, §§ $p < 0,01$  vs PRP, §§§ $p < 0,001$  vs PRP.



# The Effects of Dual Growth Factor and Bisphosphonate-Loaded Hydroxyapatite(HA) Scaffold in Rat Critical-Sized Femoral Defect Model

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**Introduction:** Bone tissue regeneration is a rapidly evolving field focused on the development of biocompatible scaffolds capable of supporting the repair of damaged osseous tissue. These scaffolds provide a structural framework that promotes cell growth, osteogenic differentiation, and matrix mineralization. The incorporation of osteogenic growth factors and anti-resorptive agents within hydroxyapatite (HA) based scaffolds has emerged as a promising strategy to enhance the regenerative microenvironment and improve bone healing outcomes. **Objective:** This study aimed to investigate the in-vivo efficacy of a multi-component hydroxyapatite (HA) scaffold functionalized with alendronate (Aln) and loaded with vascular endothelial growth factor (VEGF) and bone morphogenetic protein-2 (BMP-2) for the repair of critical-sized osseous defects. **Methods:** An engineered HA-VEGF+BMP-2/Aln composite was implanted into a critical-sized rat femoral defect model. Regenerative performance was quantified via histological analysis, immunohistochemical (IHC) profiling of the osteoprotegerin (OPG)/receptor activator of nuclear factor kappa-B ligand (RANKL) pathway, and Western blot analysis of bone sialoprotein II (BSPII) expression. **Results:** Histological evaluation demonstrated that the Aln-functionalized scaffolds significantly enhanced architectural organization and tissue continuity while reducing trabecular gaps compared to HA-only controls. IHC analysis revealed a distinct shift in the OPG/RANKL expression ratio, indicating that the localized delivery of Aln effectively modulated the bone remodeling microenvironment. Furthermore, Western blot data confirmed a significant upregulation of BSPII expression, indicating accelerated extracellular matrix maturation and advanced osteogenic differentiation. **Conclusion:** The localized integration of Aln with VEGF and BMP-2 within the HA scaffold promoted angiogenesis, osteogenesis, and regulated bone remodeling, highlighting its potential for critical-sized bone defect repair.

**Keywords:** Hydroxyapatite scaffold, Alendronate, Bone regeneration, VEGF/BMP-2, Critical-sized bone defect

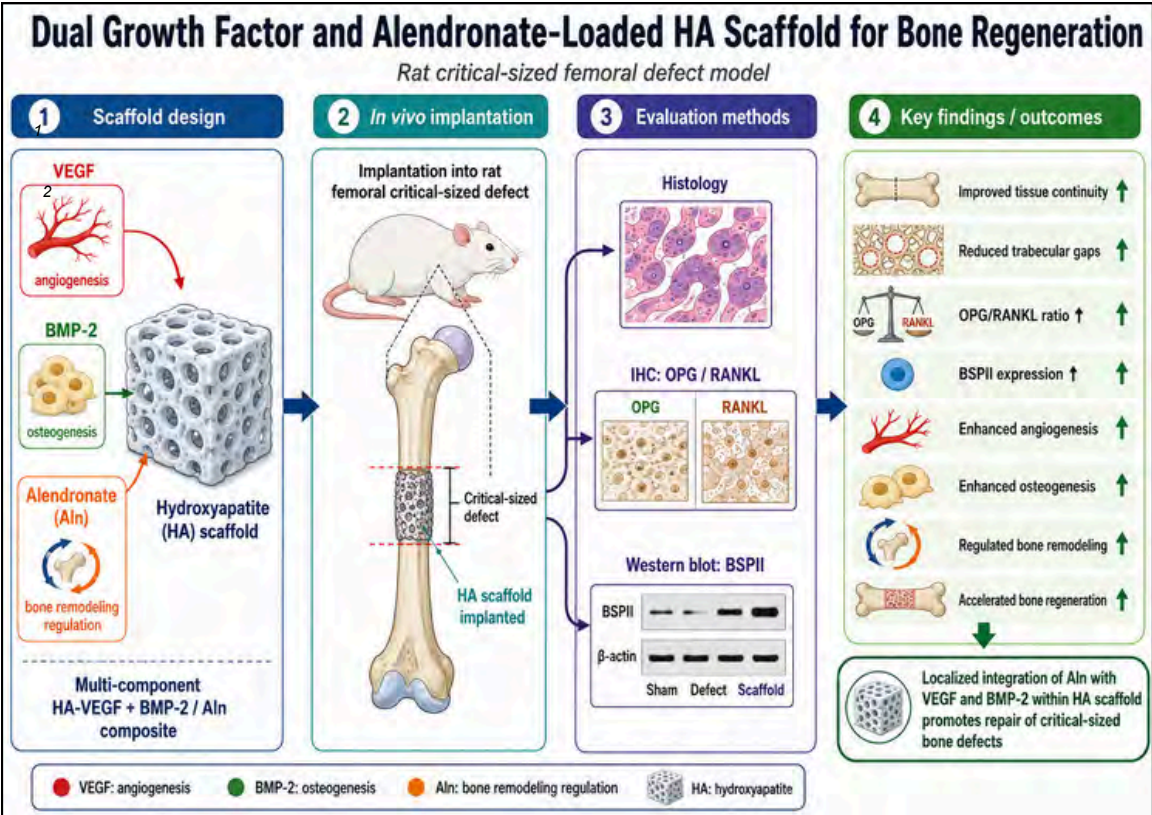


Figure 1: Study design and summary of the physiochemical and biological performance of the formulated scaffolds

# Human Platelet-Derived Extracellular Vesicles Attenuate Alveolar Bone Loss in Ligature-Induced Periodontitis

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Periodontitis causes progressive destruction of periodontal tissues and alveolar bone loss. Objectives: This study aimed to evaluate the therapeutic potential of human platelet-derived extracellular vesicles (hPLT-EVs) compared with platelet-rich plasma (PRP) on alveolar bone resorption and inflammation in a rat periodontitis model. Methods: hPLT-EVs isolated from expired platelet concentrates via tangential flow filtration were characterised by nanoparticle track analysis, transmission electron microscopy, and Western blotting (CD63+, TSG101+, CD41+). Periodontitis was induced in male Wistar rats via 4-0 silk ligatures around maxillary second molars for 14 days across five groups: SHAM, negative control, PRP, platelet-rich plasma (PRP), EV30 (30 X 106 particles) and EV40 (40 x 106 particles). Treatments were administered via palatal injection every 7 days over 28 days. Left maxillary gingival tissues underwent multiplex cytokine analysis, while H&E-stained right maxillary bone sections were evaluated for cemento-enamel junction to alveolar bone crest (CEJ-ABC) distance. Results: Characterisation confirmed intact small extracellular vesicles of platelet origin with a mean diameter of 131–157 nm. Multiplex cytokine analysis revealed distinct immunomodulatory signatures, with EV40 significantly enhancing anti-inflammatory cytokine expression (IL-4, IL-10) compared to PRP and EV30. Histological analysis demonstrated that EV40 significantly reduced CEJ-ABC distance compared to the negative control, indicating a reduction in alveolar bone loss. EV groups displayed a consistent trend toward greater bone preservation compared to PRP. Conclusions: Palatal administration of hPLT-EVs (40 x 106 particles) offers a promising cell-free therapy by modulating local inflammation and reducing alveolar bone destruction in experimental periodontitis.

**Keywords:** Extracellular vesicles, Platelet-rich plasma, Periodontitis, Alveolar bone loss, Immunomodulation

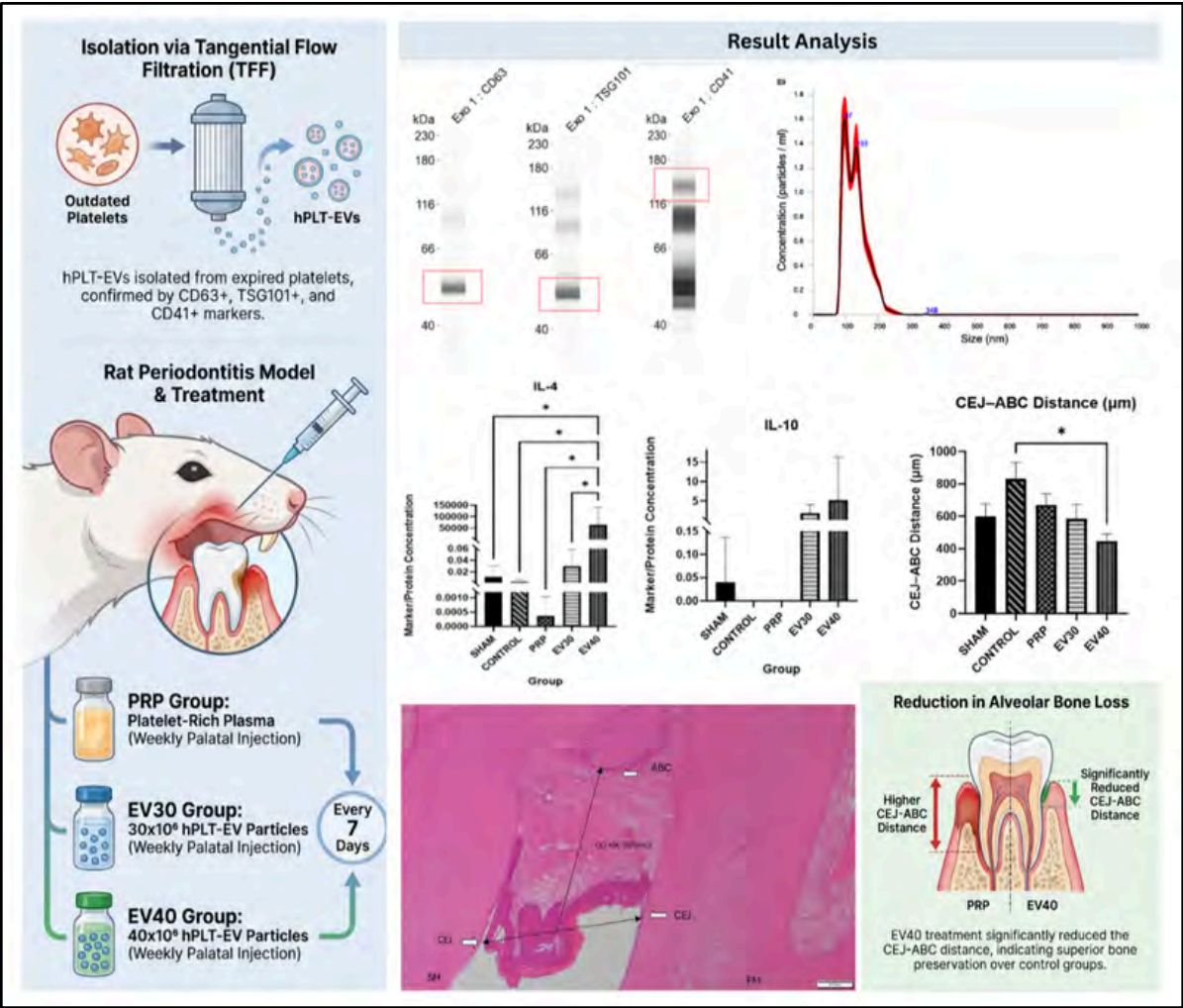


Figure 1: Research Findings of the Study

# Therapeutic Effects of MSC-Derived Exosomes in the Chronic Phase of Renal Ischemia–Reperfusion Injury

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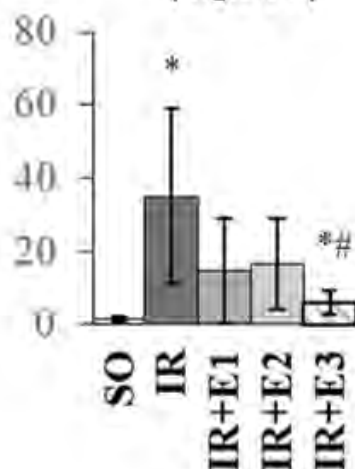
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Chronic kidney disease (CKD) often develops as a consequence of maladaptive or incomplete repair following acute kidney injury (AKI). Persistent inflammation, oxidative stress, and progressive fibrosis drive the transition from AKI to CKD, creating a need for regenerative therapies that promote adaptive kidney repair. Mesenchymal stem cell-derived exosomes represent a promising cell-free therapeutic approach capable of delivering bioactive molecules that modulate inflammation and attenuate fibrosis. However, its efficacy in promoting renal recovery during the chronic phase following ischemia-reperfusion (IR) injury is not yet fully understood. Objectives: This study aimed to evaluate the effects of Human Wharton's Jelly mesenchymal stem cell (hWJMSC)-derived exosomes in a rat model of chronic-phase renal IR injury. Methods: Rats were treated with exosomes via tail vein injection one day after ischemia–reperfusion surgery at three different doses ( $8.5 \times 10^3$ ,  $8.5 \times 10^4$  and  $8.5 \times 10^5$  particle per animal). All animals were euthanized 21 days after surgery. Therapeutic efficacy was evaluated by assessing renal function, histopathological changes, and the expression of genes associated with renal injury and fibrosis. Results: The high-dose exosome-treated group exhibited significantly lower urinary albumin excretion, proteinuria score and fibrotic area fraction than the IR group. In addition, the high-dose group showed significantly lower COL1A1 mRNA expression and significantly higher E-cadherin expression than the IR group. Conclusions: Exosomes HWJMSC demonstrated renoprotective effects by improving renal function, attenuating renal fibrosis, and preserving epithelial integrity in a chronic phase of renal IR model.

**Keywords:** Exosomes, Mesenchymal Stem Cells, Renal fibrosis, Ischemia-reperfusion injury

**Urinary Albumin Excretion  
(mg/24 h)**





# Longevity: elongating the telomere through exosomes

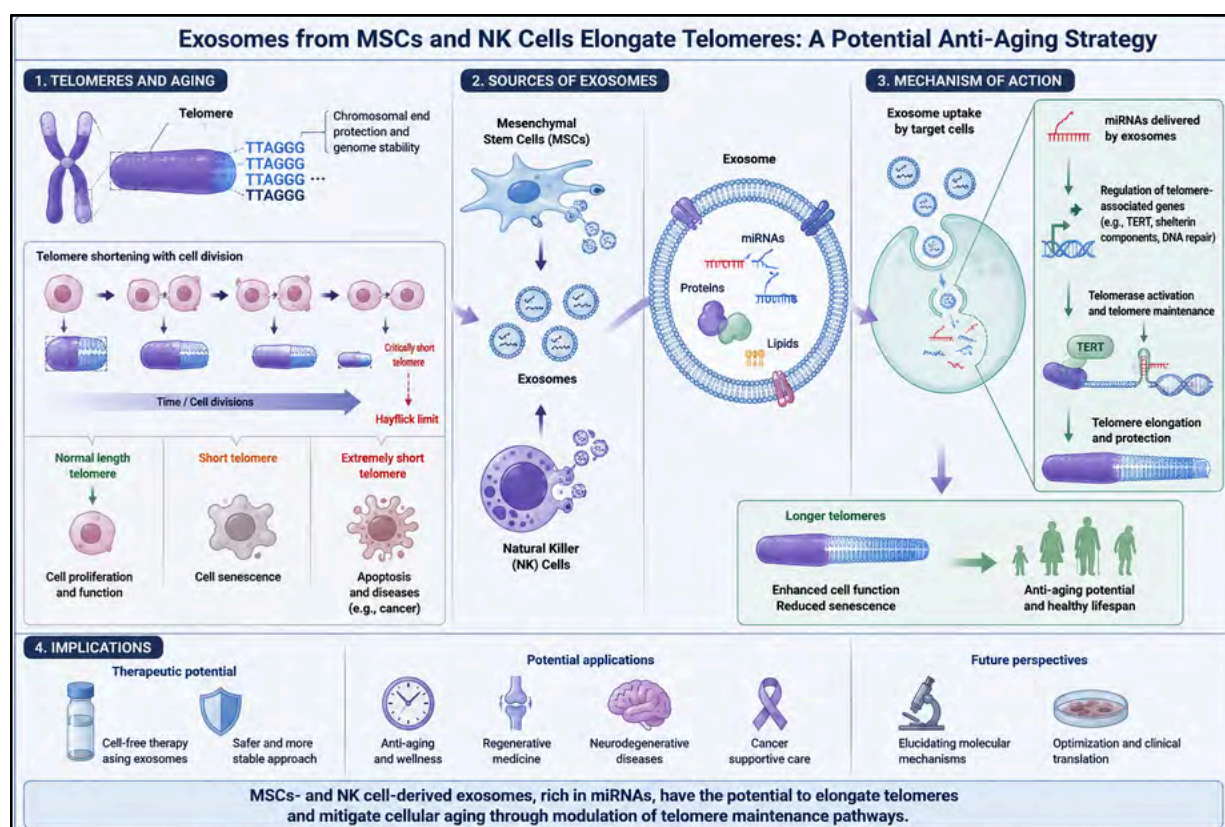
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**Introduction:** Telomere shortening is a hallmark of cellular ageing and contributes to senescence and reduced regenerative capacity. Exosomes derived from mesenchymal stem cells (MSCs) and natural killer (NK) cells have emerged as promising cell-free therapeutic agents due to their regenerative and immunomodulatory properties. This study investigated the effects of MSC- and NK-derived exosomes on telomere length in human fibroblasts. **Methods:** Human fibroblasts were treated with MSC- and NK-derived exosomes under in vitro conditions. Telomere length was measured before and after treatment and compared with untreated controls. Changes in telomere length across different cell passages were also evaluated to assess the relationship between cellular ageing and exosome treatment. **Results:** Both MSC- and NK-derived exosomes increased telomere length compared with untreated controls, with NK-derived exosomes demonstrating a greater effect ( $14.6 \pm 1.1$  kb) than MSC-derived exosomes ( $12.0 \pm 0.9$  kb) and controls ( $11.1 \pm 0.8$  kb). Following exosome treatment, telomere length increased from 10.7 kb to 16.7 kb in treated fibroblasts. Progressive telomere shortening was observed with increasing cell passage number, consistent with replicative ageing. **Conclusion:** MSC- and NK-derived exosomes demonstrate the potential to preserve or restore telomere length in cultured fibroblasts, suggesting a promising cell-free approach for regenerative medicine and healthy ageing. However, maintaining optimal telomere length is crucial, as excessive elongation may increase the risk of genomic instability and oncogenesis. Further mechanistic studies and in vivo investigations are warranted to establish the safety and therapeutic efficacy of exosome-based interventions.

**Keywords:** Longevity, Telomeres, Exosomes, Therapy



# Banana Peel-Based Cookies Exhibit Antihyperglycemic and Pancreatic Protective Properties in Diabetic Mice Induced by Streptozotocin: A Comprehensive Analysis of Sensory Acceptability and Histopathological Findings

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**Objectives:** Diabetes mellitus involves persistent hyperglycaemia and progressive pancreatic  $\beta$ -cell damage. This study evaluated the antihyperglycaemic and pancreatic protective effects of cookies formulated from three banana peel varieties (kepok, ambon, and raja) in streptozotocin (STZ)-induced diabetic mice, alongside their sensory acceptability. **Methods:** Male BALB/c mice were divided into six groups: healthy control, diabetic control, positive control, and three treatment groups receiving kepok, ambon, or raja banana peel cookies for two weeks. Blood glucose was measured before and after treatment. Pancreatic tissue was examined histologically using haematoxylin and eosin staining with semi-quantitative scoring, and sensory acceptability was assessed using a hedonic test. **Results:** All treatment groups showed significantly lower blood glucose than the diabetic control ( $p<0.05$ ), with the kepok group achieving the lowest final level ( $118.2\pm27.4$  mg/dL), closest to the healthy control ( $92.3\pm11.9$  mg/dL). Pancreatic histopathology scores in treatment groups (0.36–0.40) were comparable to the healthy control (0.24), indicating preserved  $\beta$ -cell integrity, compared with markedly higher scores in the diabetic control (5.60) and positive control (3.24). Sensory evaluation showed all formulations were acceptable, with raja cookies achieving the highest overall acceptability score ( $7.61\pm1.72$ ,  $p=0.031$ ). **Conclusions:** Banana peel cookies exhibit antihyperglycaemic and pancreatic protective properties while maintaining favourable sensory acceptance, supporting their potential as a functional food for diabetes management. Further studies are needed to confirm the underlying mechanisms and clinical applicability.

**Keywords:** Antihyperglycaemic activity, Banana peel cookies, Functional food, Pancreatic histopathology, Streptozotocin-induced diabetes

Table 1. Summary of Key Findings on Blood Glucose, Pancreatic Histopathology, and Sensory Acceptability

| Group            | Blood GlucoseDay 14 (mg/dL) | Pancreatic HistopathologyScore | Overall Acceptability(1–10) |
|------------------|-----------------------------|--------------------------------|-----------------------------|
| Healthy control  | 92.3 ± 11.9                 | 0.24 ± 0.43                    | –                           |
| Diabetic control | 420.6 ± 112.4               | 5.60 ± 0.82                    | –                           |
| Positive control | 334.0 ± 126.1               | 3.24 ± 1.41                    | –                           |
| Kepok cookies    | 118.2 ± 27.4                | 0.36 ± 0.49                    | 7.19 ± 1.86                 |
| Ambon cookies    | 140.2 ± 33.1                | 0.40 ± 0.50                    | 7.17 ± 2.02                 |
| Raja cookies     | 131.2 ± 23.0                | 0.40 ± 0.50                    | 7.61 ± 1.72                 |

Note: All comparisons between diabetic control and treatment/ healthy control groups were significant ( $p<0.05$ ). Overall acceptability differed significantly among cookie formulations ( $p=0.031$ ), with raja cookies scoring highest.

# Advances in human brain organoids for Alzheimer's disease modelling

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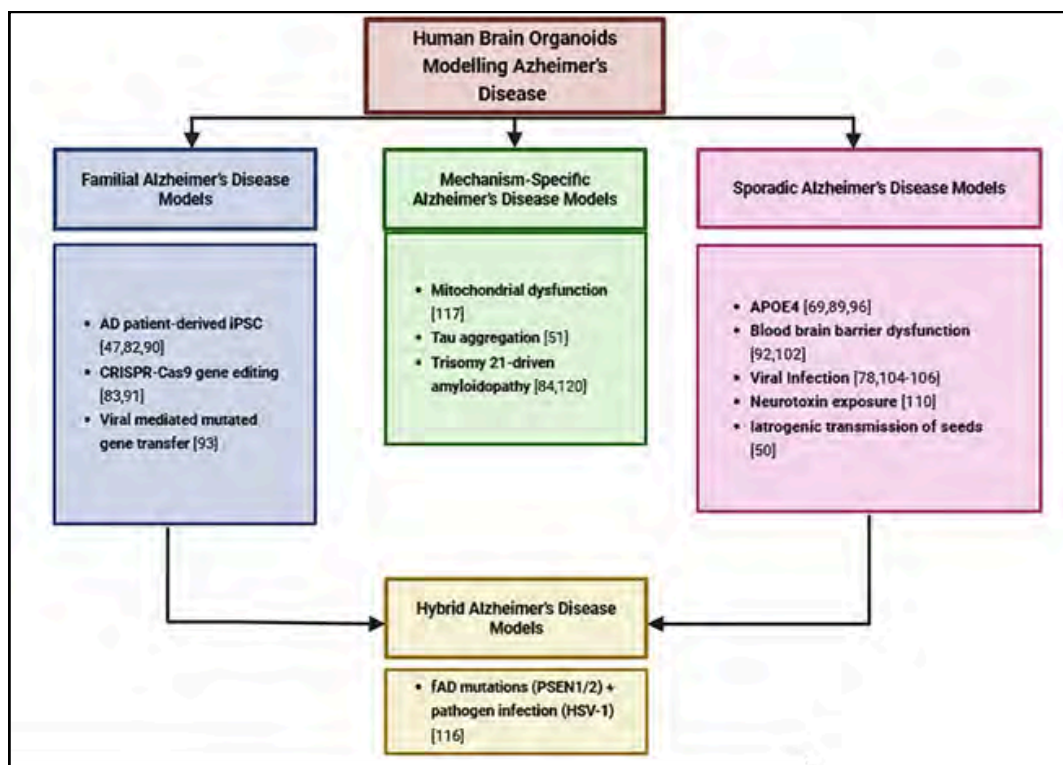
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Alzheimer's disease (AD) is a critical neurodegenerative disorder with no current effective disease-modifying therapy. Establishing a translational relevant preclinical AD model is considered as the bottleneck in AD drug development. Human brain organoid (hBO) is a suitable candidate to model AD in the context of human physiology. In the recent years, hBO platforms have evolved to incorporate robust population of microglia, astrocytes and vascular endothelial cells. Concurrently, the adoption of innovative modelling strategies has expanded the realm of AD modelling in hBOs, extending from familial AD to sporadic AD and beyond. This review discusses recent advances in hBO designs for modelling AD, highlighting emerging strategies to model different facets of AD and evaluates the current extent to which AD phenotypes are recapitulated in hBOs systems. HBO models exhibit varying levels of recapitulation in AD phenotypes, with degrees ranging from strong, moderate, emerging but limited, weak and variable, and weak but consistent. Collectively, advances in AD modelling using hBOs have partially alleviated limitations related to cellular diversity and complexity, while expanding strategies for modelling distinct subtypes of AD. These advances have also been successful in strengthening several AD phenotypes recapitulation and unravelling emerging AD phenotypes in hBOs. However, early-generation of AD hBO models are lacking cellular and physiological complexity, as well as modelling strategy for different AD subtypes. Future works are still required to enhance the robustness, credibility, and translational utility of AD hBO models in biomedical applications.

**Keywords:** Tissue engineering, Disease modelling, Brain organoid



# Efficiency and Stability of NILV Versus Conventional Lentiviral System Using PGK-GFP and Reprogramming Factor Constructs

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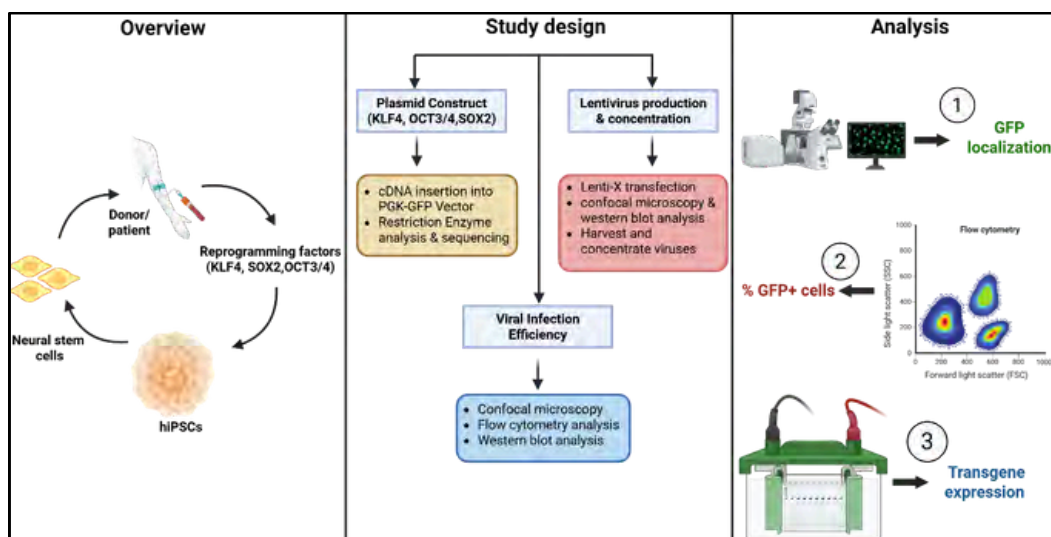
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**Introduction:** Integrase-free lentiviral vectors (NILVs) represent a safer alternative gene-delivery by minimizing the risk of genomic integration, making them suitable for applications involving transient gene expression. This study compares the efficiency, expression profile, and stability of NILV and wild-type lentiviral vector (LV) systems using a PGK-GFP reporter construct. In parallel, plasmids encoding key reprogramming factors KLF4, SOX2, and OCT3/4 were synthesized, validated, and characterized for future cellular reprogramming strategies. **Methods:** PGK-GFP, PGK-OCT3/4-SOX2-GFP, and PGK-KLF4-GFP plasmids were constructed and validated by restriction enzyme digestion and sequence analysis. LV and NILV virus particles were produced using Lenti-X™ Packaging Single Shots® premixed with Xfect® Transfection Reagent. Viral transduction efficiency was evaluated in U937 cells. GFP expression was observed by confocal microscopy and quantified by flow cytometry, while Western blot confirmed the intracellular expression of GFP and the delivered reprogramming factors. **Results:** All plasmid constructs were successfully generated and sequence-verified. In Lenti-X 293T HEK cells, both LV and NILV systems efficiently expressed GFP and reprogramming-factor proteins. In U937 cells, LV-PGK-GFP produced stable and progressively increasing GFP mean fluorescence intensity over time, indicating sustained transgene expression. In contrast, NILV-PGK-GFP expression declined after Day 4, consistent with episomal dilution in dividing cells due to the absence of integrase-mediated genomic integration. **Conclusion:** NILVs supported efficient yet transient gene expression, whereas LVs provided stable, long-term expression. Successful validation of all reprogramming gene constructs highlights their potential for safe, non-integrating gene-delivery applications.

**Keywords:** Transcription factors, Cell reprogramming, Lentiviral vector, Plasmid construction, GFP expression



# Differential Effects of Oxygen Tension on iPSC Function and Exosome Production

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**Introduction:** Induced pluripotent stem cell (iPSC)-derived exosomes are increasingly explored for regenerative applications, yet the effect of oxygen tension on iPSC behaviour and exosome production remains insufficiently defined. **Objective:** This study aimed to evaluate the effects of oxygen tension (21% O<sub>2</sub>, 3% O<sub>2</sub>, and 1% O<sub>2</sub>) on iPSC proliferation and viability, and to assess its impact on exosome production and characteristics using EdU, MTT, and nanoparticle tracking analysis (NTA). **Method:** iPSCs were cultured under defined oxygen conditions prior to conditioned media collection. Cell viability and metabolic activity were assessed using the MTT assay, while DNA synthesis was quantified via EdU incorporation. Exosomes were isolated from conditioned media using differential ultracentrifugation and analysed for particle concentration and size distribution using NTA. **Results:** Oxygen tension influenced iPSC function. Moderate hypoxia (3% O<sub>2</sub>) maintained cell viability and metabolic activity comparable to normoxia, whereas severe hypoxia (1% O<sub>2</sub>) reduced metabolic output, indicating cellular stress. Proliferation was highest under normoxia, but remained sustained under 3% O<sub>2</sub> without significant impairment. NTA analysis demonstrated higher exosome particle concentrations under 3% O<sub>2</sub> ( $1.71 \times 10^{10}$  particles/mL) compared with normoxia ( $6.86 \times 10^9$  particles/mL) and 1% O<sub>2</sub> ( $6.19 \times 10^9$  particles/mL), while particle sizes remained within the expected extracellular vesicle range. Oxygen-dependent differences in particle concentration and size distribution suggest that hypoxic conditions modulate exosome production. **Conclusion:** Moderate hypoxia (3% O<sub>2</sub>) provides a balanced culture environment that preserves iPSC viability and supports stable proliferation while maintaining exosome production, highlighting its potential as an optimized condition for scalable iPSC-derived extracellular vesicle applications.

**Keywords:** Exosomes, Induced pluripotent stem cell (iPSCs), Nanoparticles, Hypoxic



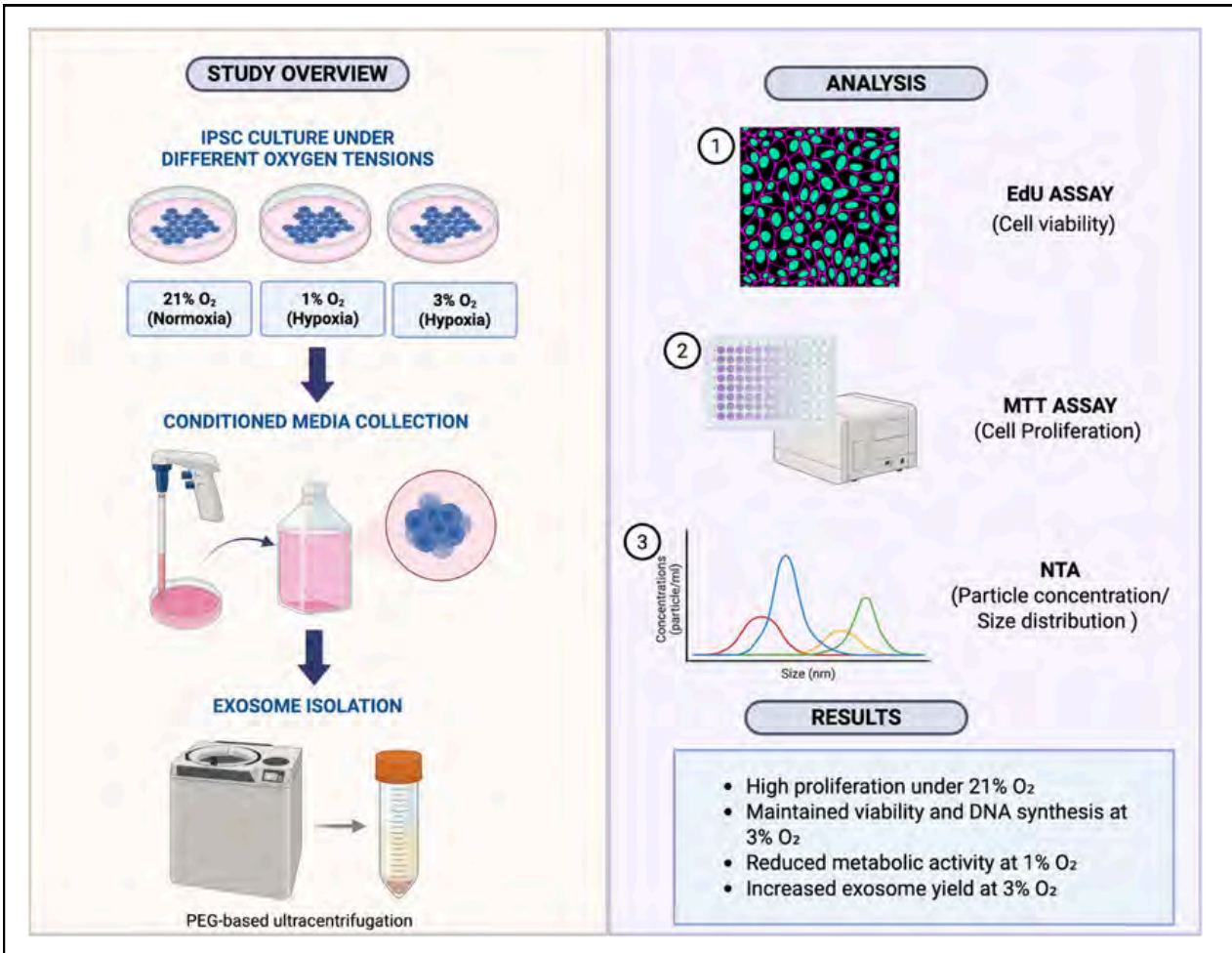


Figure 1: Key findings in this study

# Phoenixin as a Therapeutic Modulator of Neurodegenerative Disease Progression

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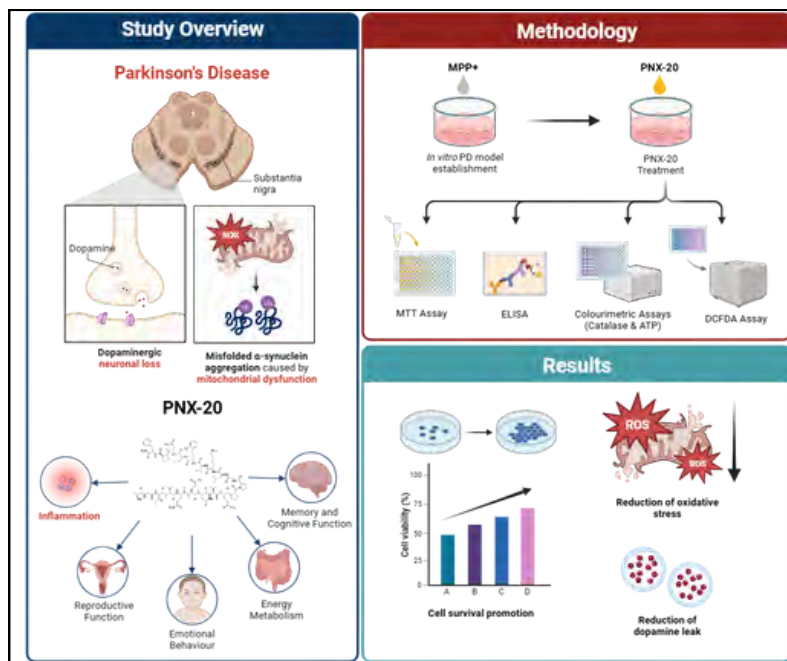
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Parkinsons' disease (PD) is a progressive neurodegenerative disorder associated with dopaminergic neuron loss in the substantia nigra pars compacta (SNpc) of the brain. Recent research has suggested a potential link between neurodegenerative disorders and phoenixin (PNX), a pleiotropic neuropeptide that has demonstrated neuroprotective effects through the promotion of mitochondrial activity and enhancement of neuronal survival. Objectives: This study aims to understand the effect of PNX, specifically PNX-20, on cell survival, dopaminergic content and mitochondrial function in a PD-like *in vitro* model. Methods: The PD-like model was established with differentiated SH-SY5Y cells and MPP+(1-methyl-4-phenylpyridinium)-induced cytotoxicity. Treatment groups were pre-treated with PNX-20 before the addition of cytotoxic agent MPP+, and parameters were evaluated 24 hours after PNX-20 administration. Dose-response of MPP+ and PNX-20 was assessed via MTT assay. ELISA was used to measure the effect of PNX-20 on intracellular dopaminergic content, while mitochondrial function was evaluated via DCFDA Assay, Catalase Assay, as well as ATP Assay. Results: PNX-20 significantly attenuated MPP+-induced cytotoxicity and promoted cell viability up by approximately 10% compared to the untreated control. Mitochondrial function was also significantly improved with the reduction of reactive oxygen species and elevated catalase activity in PNX-20-treated groups. Interestingly, MPP+-induced elevation of intracellular dopaminergic content was significantly reduced by PNX-20. Conclusion: These findings implicate PNX-20 as a potential agent in targeted therapeutics for neurodegenerative diseases including PD through the promotion of cell survival and enhancement of mitochondrial function.

**Keywords:** Parkinsons' Disease, Phoenixin, Neuroprotection, Mitochondrial Function, PNX-20



# Beyond Wharton's Jelly: Defining Optimal Perinatal Mesenchymal Stem Cell (MSC) Sources for Regenerative Medicine

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Mesenchymal stem cells (MSCs) derived from perinatal tissues represent a superior alternative to adult-derived sources due to their high proliferative capacity, low immunogenicity, and ethical accessibility. Wharton's Jelly-derived MSCs (WJMSCs) are currently the established benchmark in regenerative medicine, other perinatal tissues, specifically the amniotic membrane, amniotic fluid, and decidua, remain underutilized and less comprehensively characterized. Objective: This study aimed to conduct a rigorous comparative analysis of MSCs isolated from the human amniotic membrane (hAMSCs), amniotic fluid (hAFMSCs), and decidua (DMSCs) against the WJMSC gold standard to evaluate their potential for clinical scale-up and therapeutic applications. Methods: Using standardized isolation and culture protocols, MSCs from all four sources were assessed through passages P1–P5. Evaluation criteria included adherence to International Society for Cellular Therapy (ISCT) minimal criteria, assessment fibroblast-like morphology, and precise quantification of proliferation. Results: All perinatal sources exhibited the characteristic spindle-shaped morphology and stable plastic adherence required for MSC identification. WJMSCs maintained the fastest growth profile, hAMSCs and DMSCs demonstrated robust proliferation, achieving 80% confluency in ( $10 \pm 2$  days) and ( $12 \pm 2$  days), respectively, significantly outperforming hAFMSCs ( $14 \pm 2$  days;  $p < 0.05$ ). Immunophenotypic characterization confirmed high-purity populations across all sources, with  $>95\%$  expression of positive markers (CD90, CD73, CD105) and  $<2\%$  expression of hematopoietic and immune-associated markers (CD45, CD34, CD11b, CD79a, and HLA-DR). Conclusion: The findings establish hAMSCs and DMSCs as high-quality, viable alternatives to WJMSCs, possessing favorable growth characteristics and stable immunophenotypes, providing a standardized benchmark for these underutilized tissues. This research facilitates the expansion of perinatal stem cells, offering new avenues to optimize cellular therapies in regenerative medicine.

**Keywords:** Perinatal Mesenchymal Stem Cells, Wharton's Jelly-Derived MSCs, Regenerative Medicine; Immunophenotypic Profiling, Clinical Translation

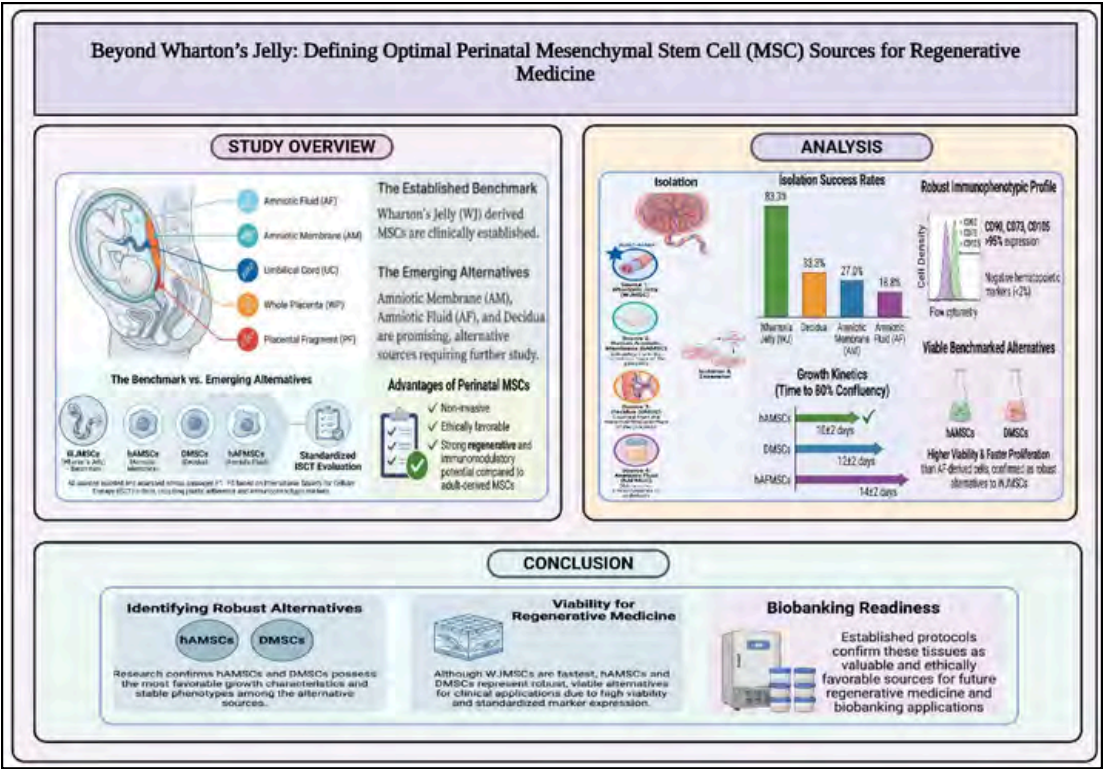


Figure 1: Key finding of the study



# Optimization and Validation of Lentiviral Reprogramming Constructs for Endometriosis Patient-Derived iPSCs

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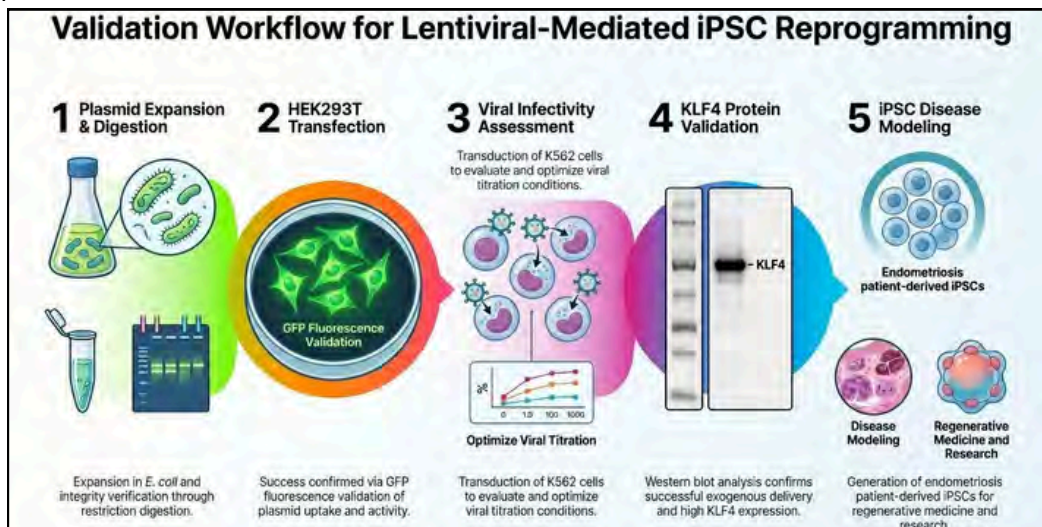
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Induced pluripotent stem cells (iPSCs) provide a promising platform for endometriosis disease modelling and regenerative medicine. Objectives: This study aimed to optimize and validate lentiviral constructs carrying reprogramming factors prior to the generation of endometriosis patient-derived iPSCs. Materials and Methods: Plasmid constructs encoding green fluorescent protein (GFP) and reprogramming factors (KLF4, SOX2 and OCT 3/4) were expanded in ampicillin-resistant *Escherichia coli*, extracted, and validated by restriction digestion to confirm plasmid integrity. Functional validation was performed by transfecting HEK293T producer cells, where GFP fluorescence confirmed successful plasmid delivery and expression. Lentiviral particles generated from producer cells were subsequently used to transduce K562 cells to evaluate viral infectivity and optimize viral titration before application to target fibroblasts. Protein lysates from transduced cells were analysed by western blot to validate transgene expression. Results: Restriction digestion demonstrated high-quality plasmids with expected banding patterns. GFP fluorescence was observed following transfection and transduction, confirming functional viral production. However, flow cytometry detected only approximately 1% GFP-positive cells, indicating suboptimal transduction efficiency and the need for further optimization of parameters including cell preparation, viral stability, and polybrene exposure. Western blot analysis showed consistent housekeeping protein expression and confirmed expression of GFP, KLF4, and SOX2, with KLF4 and SOX2 exhibiting higher expression in transduced cells than in untransduced controls. Detection of SOX2 supports successful delivery of the SOX2/OCT3/4 construct. Conclusion: These findings validate the functionality of the lentiviral reprogramming system while highlighting the need to improve viral transduction efficiency before reprogramming endometriosis-derived fibroblasts into iPSCs. Further optimization will facilitate the establishment of a reliable iPSC disease model.

**Keywords:** Transcription factors, Cell reprogramming, Lentiviral vector, Plasmids construction, GFP expression





# Repeated High-Dose Wharton's Jelly Mesenchymal Stromal Cells Extend Survival and Preserve Locomotion in Advanced Ageing

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Wharton's jelly mesenchymal stromal cells (WJMSCs) have been shown to attenuate chronic inflammation, a key hallmark of ageing. However, evidence regarding their safety and efficacy in oldest-old adults remains limited. Objective: This pilot feasibility study investigated the effects of WJMSCs in advanced ageing model. Methods: 29-month-old male C57BL/6J mice (n=4), approximately equivalent to 80-year-old humans, received three intraperitoneal doses of  $2.5 \times 10^7$  cells/kg at two-week intervals. Untreated age-matched and 13-month-old mice served as negative and baseline controls. Functional, physiological and inflammatory domains were evaluated pre- and post-treatment using a behavioural battery (open field maze, novel object recognition, Morris water maze, accelerated rotarod, four-limb wire hang), Dual-Energy X-ray Absorptiometry (DEXA) and serum cytokine multiplex ELISA. Following euthanasia at two months, multi-organ histopathology assessed in vivo safety of WJMSC administration. Results: Notably, WJMSC-treated mice demonstrated a 50% survival rate, whereas the negative control group reached 100% mortality. At functional level, the significant age-related decline in locomotor activity observed in negative and baseline controls ( $p < 0.05$ ) was attenuated following WJMSC treatment. However, WJMSC intervention did not significantly improve recognition memory, spatial learning, motor coordination, or neuromuscular endurance. DEXA analysis revealed a significant reduction in fat mass accumulation within the WJMSC-treated group ( $p < 0.05$ ), although changes were comparable across groups. No significant differences were observed in body mass index, lean mass, skeletal integrity, or systemic inflammatory profiles. Conclusion: These findings suggest advanced ageing may reach a threshold where late-stage cellular intervention cannot fully restore functional decline. Future studies should investigate middle-age preventive windows for regenerative cell therapies.

**Keywords:** Advanced ageing, Mesenchymal stromal cells, Cognitive function, Physical function, Body composition

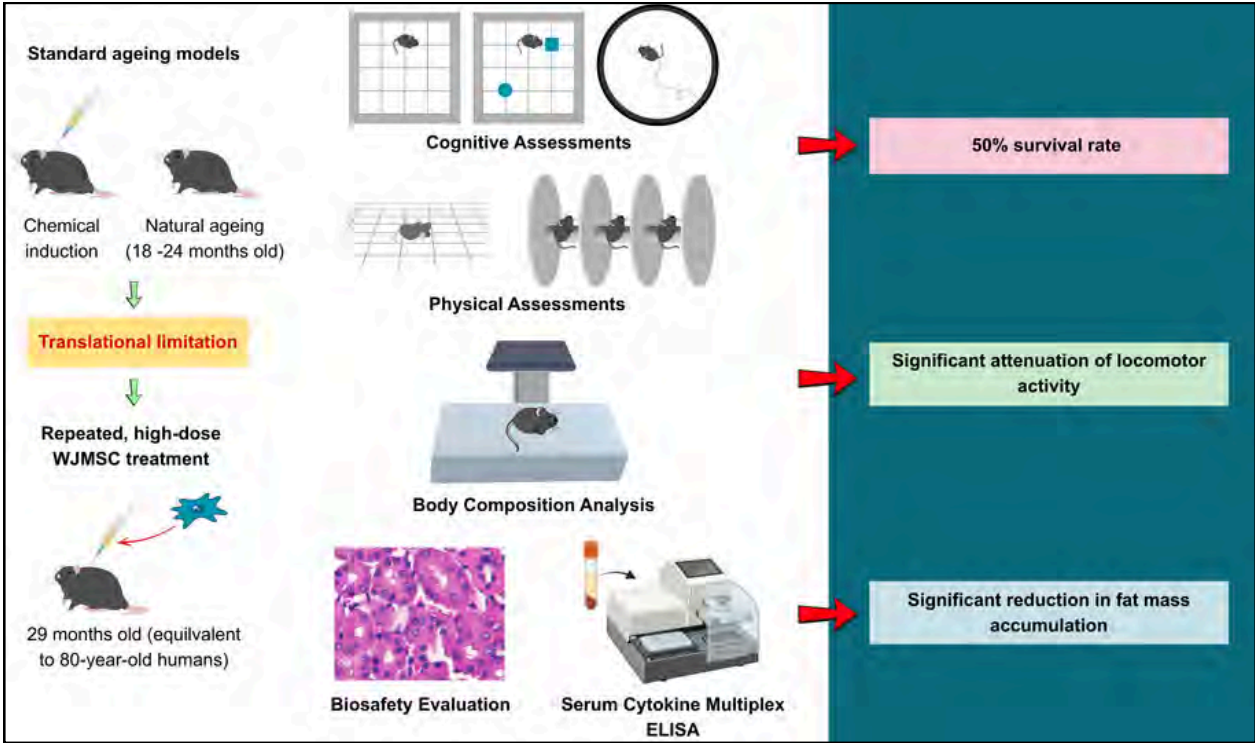


Figure 1: Key findings of the study

# WJ-MSC-Derived Extracellular Vesicles Facilitate Diabetic Wound Repair via Inflammatory Modulation, Angiogenesis, and Matrix Remodeling

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Diabetic chronic wounds remain a significant clinical challenge due to persistent inflammation, impaired vascularisation, and dysregulated tissue remodelling. Extracellular vesicles (EVs) derived from Wharton's Jelly mesenchymal stem cells (WJ-MSCs) have emerged as a promising acellular therapeutic strategy, offering paracrine regenerative benefits without the limitations of cell-based therapies. Objectives: This study aimed to evaluate the therapeutic efficacy of WJ-MSC-derived EVs in modulating inflammation, promoting angiogenesis, and enhancing tissue remodelling in a streptozotocin and high-fat diet-induced Type 2 diabetic rat model. Methods: Male Sprague Dawley rats were induced into diabetic via streptozotocin combined with a high-fat diet. Full-thickness excisional wounds were treated with medium- or high-dose WJ-MSC-derived EVs administered intradermally. Wound contraction was monitored over 14 days. Inflammatory and remodelling markers were quantified via ELISA, while histological and immunohistochemical analyses were performed to assess angiogenesis, collagen deposition, re-epithelialisation. Results: EV-treated groups demonstrated superior wound contraction, with medium-dose achieving 79.6% contraction by Day 10. Pro-inflammatory cytokines were markedly attenuated, with TNF- $\alpha$  and IL-6 reduced by up to 53.4% and 87.2%, respectively, at Day 14. Anti-inflammatory IL-10 peaked at Day 7, indicating timely inflammatory resolution. MMP-9 was reduced by up to 85%, supporting controlled matrix remodelling. Histological evaluation confirmed complete re-epithelialisation, enhanced collagen deposition, and evidence of new blood capillary formation in treated groups. Conclusions: These findings suggest that WJ-MSC-derived EVs hold significant potential as a cell-free therapeutic approach for diabetic chronic wounds, acting through coordinated modulation of inflammation, angiogenesis, and tissue remodelling.

**Keywords:** Extracellular vesicles, Wharton's Jelly, Mesenchymal stem cells, Diabetic wound healing, Tissue remodelling



# 3D-Printed Gelatin Scaffolds Loaded with Kelulut Honey and Asiaticoside for Wound Healing

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The management of diabetic foot ulcers (DFUs) remains a complex clinical challenge, characterised by prolonged inflammation, impaired cellular proliferation, and abnormal tissue remodelling. Conventional treatments, such as total contact casting, successfully relieve pressure but restrict patient mobility and increase susceptibility to bacterial infections. To address the critical need for customised wound coverage, extrusion-based 3D bioprinting offers a promising precision-medicine approach, enabling the fabrication of personalised scaffolds that conform to unique wound topographies. This research aimed to formulate and evaluate novel biomaterial inks comprising genipin-crosslinked gelatin incorporated with natural active compounds: Kelulut honey (KH) or Asiaticoside (ASI). Biomaterial inks were formulated utilising 9% and 10% (w/v) gelatin crosslinked with 0.1% genipin to ensure structural stability and low cytotoxicity. Active compound concentrations were optimised via 2D MTT assays on human dermal fibroblasts (HDFs). Formulations were evaluated for rheological properties and printability. The 3D-printed scaffolds were subsequently assessed for biocompatibility (Live/Dead assay), cell migration (scratch assay), and matrix remodelling (Collagen I and alpha-SMA expression) via immunocytochemistry. 2D MTT assays identified optimal viability at 0.1%–0.5% (v/v) for KH and 0.01%–0.1% (w/v) for ASI, sustaining cell viability over 72 hours. Rheological profiling confirmed shear-thinning behaviour suitable for extrusion. Mechanically, all hydrogels exhibited high resilience (~95–100%) and consistent compressibility (~70–80%). KH yielded highly elastic scaffolds, while ASI produced more compressible matrices. Optimised 3D scaffolds facilitated near-complete wound closure (~100%) by Day 5 and stimulated Collagen I synthesis with minimal  $\alpha$ -SMA expression, indicating high-quality healing. In conclusion, genipin-crosslinked 9% gelatin biomaterial inks loaded with KH or ASI offer excellent printability and suitable mechanical properties. Their ability to promote rapid cell migration and healthy extracellular matrix deposition makes them a highly viable platform for advanced DFU management.

**Keywords:** Biomaterials ink, 3D bioprinting, Kelulut honey, Asiaticoside, Wound healing



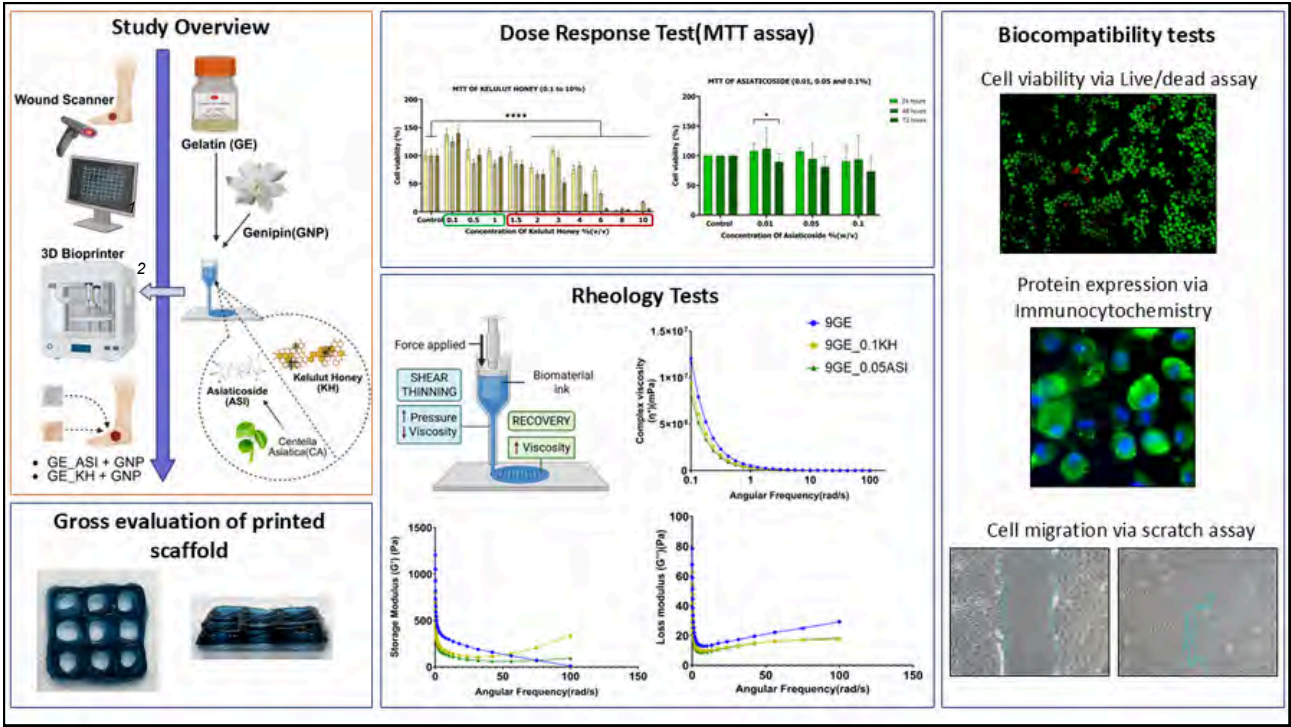


Figure 1: Study design and summary of the physiochemical and biological performance of the formulated scaffolds

# Improving Wound Healing with Curcumin-Infused Topical Hydrogels

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**Objective:** Chronic wounds are slow-healing and prone to infection; existing dressings often lack adaptability, bioactivity, and sufficient moisture control. We developed a topical sodium alginate-polyethylene glycol (SA-PEG) hydrogel loaded with curcumin to combine moisture retention, mimic the ECM, and possess antibacterial activity for irregular wound surfaces. **Methodology:** Sodium alginate (1.5 g) and PEG (1.0 g) were dissolved separately in deionised water, stirred, then combined and divided into 4 mL aliquots. Curcumin (5, 10, 15, or 20 mg) was incorporated in the aliquots and sonicated for 15 minutes before gelation with CaCl<sub>2</sub> (0.8% was optimised). Controls contained no curcumin. Characterisation included, FTIR, SEM, Rheology, Swelling/degradation, Encapsulation Efficiency, Cell assay and antimicrobial testing was done against E. coli, S. aureus and Candida Albicans. **Results:** FTIR confirmed incorporation of curcumin in the SA-PEG network with visible curcumin peaks as seen in Figure 1. SEM revealed porous structures at 0.8% CaCl<sub>2</sub>, which potentially could support cell attachment. Rheology showed shear-thinning in most formulations, with 15–20 mg curcumin gels exhibiting higher viscosity, indicating more sustained release but reduced injectability. Degradation and UV-Vis analysis demonstrated controlled degradation for up to 14 days. The hydrogel recorded an encapsulation efficiency of 86.48% to 98.31%. Biocompatibility testing, conducted through CCK8 assay, demonstrated sustained cell growth and attachment for up to 7 days, confirming the hydrogel's potential biocompatibility as seen in figure 1. Scratch assay indicated successful closure within 24 hours for all experimental groups. **Discussion and Conclusion:** The results confirm that SA-PEG-curcumin hydrogels are structurally stable and biocompatible. FTIR confirmed chemical incorporation, while SEM revealed a porous architecture favorable for cell attachment. Although rheological analysis showed shear-thinning behavior suitable for tube delivery, higher curcumin concentrations increased viscosity, making application more difficult. The scaffolds achieved over 80% encapsulation efficiency and demonstrated sustained degradation for 14 days, indicating prolonged therapeutic potential. Biologically, the hydrogels supported the proliferation of human dermal fibroblasts over 7 days, with scratch assays recording complete wound closure within only 24 hours. Collectively, these findings suggest that SA-PEG-curcumin hydrogels are promising candidates for wound dressings that combine ease of application with extended bioactive release.

**Keywords:** Curcumin hydrogel, Wound healing, Therapeutics, Biomaterials, Tissue engineering

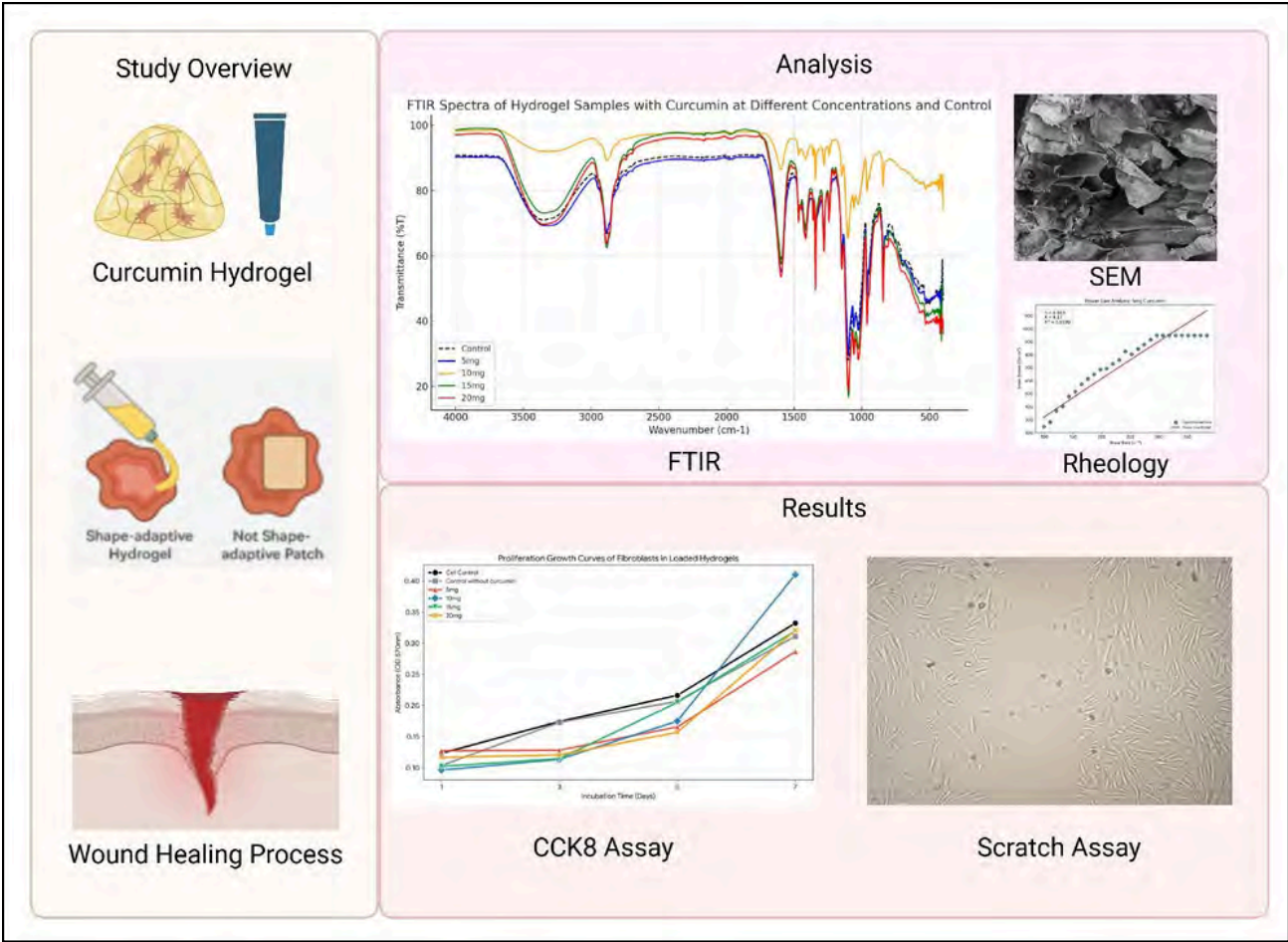


Figure 1: Key findings of the study

# Genipin-Crosslinked Gelatin/PVA Hydrogels for Cartilage Tissue Engineering: Fabrication, Characterization, and Chondrocyte Biocompatibility

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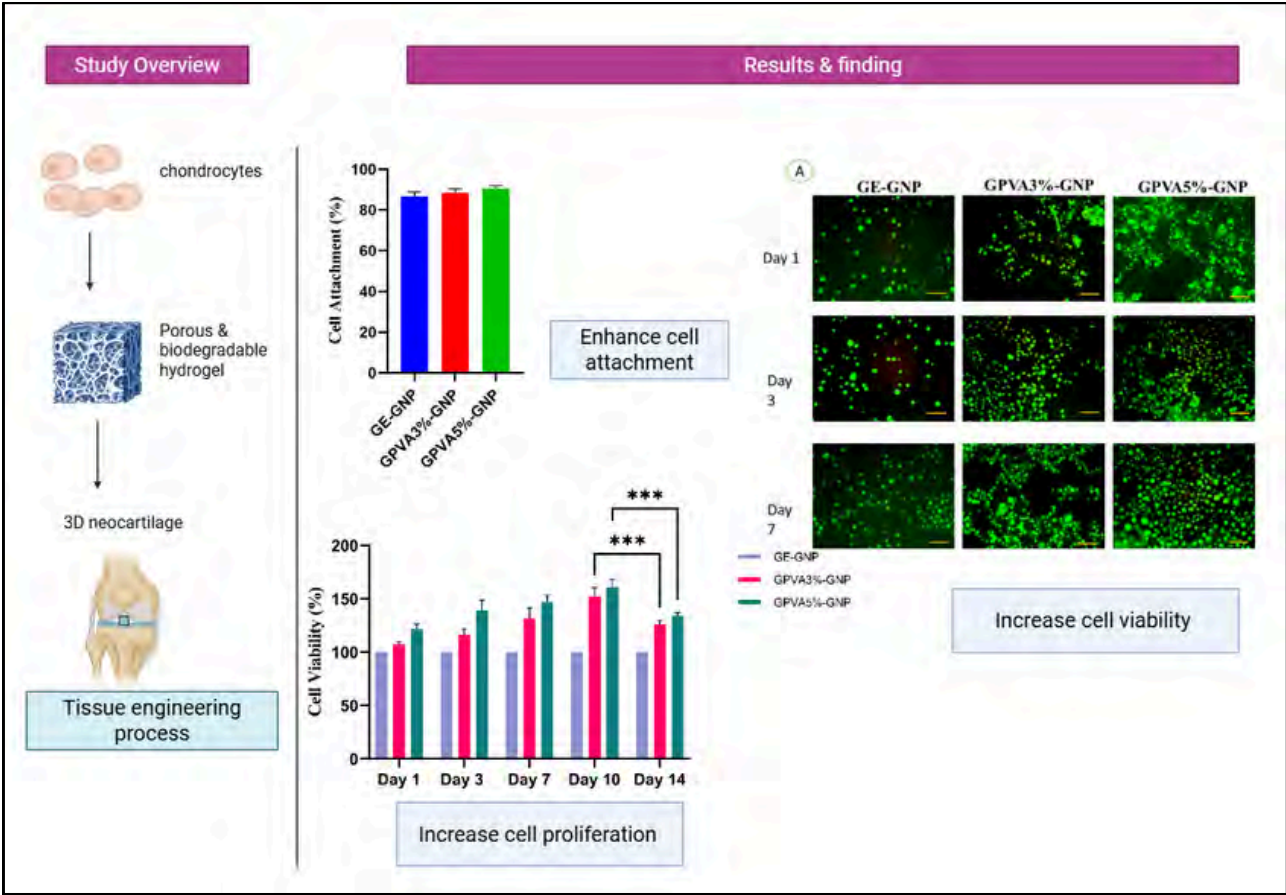
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**Introduction:** Osteoarthritis (OA) presents a significant therapeutic challenge, necessitating innovative biomaterial strategies to provide a stable environment for cartilage regeneration. Hybrid hydrogels combining natural and synthetic polymers offer a promising solution for mimicking the native extracellular matrix. **Objective:** This study developed and characterized a hybrid gelatin (GE)/polyvinyl alcohol (PVA) hydrogel crosslinked with genipin (GNP) to create a biocompatible scaffold with optimized structural properties for tissue engineering. **Methods:** Scaffolds were synthesized by incorporating 6% GE into PVA at concentrations of 3% and 5%, and crosslinked with 0.1% (w/v) GNP. The formulations were evaluated for morphology, swelling behaviour, hydrophilicity, biodegradation, chemical characterization, and mechanical strength. Biocompatibility was assessed using human chondrocytes via MTT assays and Live/Dead imaging. **Results:** The GPVA5%-GNP formulation was identified as optimal, featuring an interconnected porous structure and high thermal stability. Physicochemical analysis revealed an optimal swelling ratio ( $640.43 \pm 22.28\%$ ), improved hydrophilicity ( $44.50 \pm 1.81^\circ$ ), and enhanced compressive strength ( $96.88 \pm 0.72\%$ ). The hydrogel demonstrated controlled biodegradation and excellent cytocompatibility, with cell viability exceeding 95%. Live/Dead imaging confirmed increased viable cell density and supported cell proliferation within the 3D matrix from Day 1 to Day 7. **Conclusion:** The GPVA5%-GNP hydrogel provides a robust, protective mechanical environment that supports chondrocyte survival and growth. With its balanced degradation profile and superior structural integrity, this hybrid biomaterial can serve as a promising localized platform for cartilage regeneration and for managing osteoarthritic environments.

**Keywords:** Gelatin/PVA, Genipin, Osteoarthritis, Hydrogel, Tissue Engineering





# Evaluating a Novel Genipin-Crosslinked Hydrogel for Atopic Dermatitis in In Vitro and In Vivo Models

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Atopic dermatitis (AD) is a skin disorder with complex etiologies and limited long-term treatment options that hinder the skin's natural healing process. Objectives: This study evaluated the therapeutic potential of a novel genipin-crosslinked hydrogel patch (GEHALB) in treating skin barrier function and inflammation in both in vitro and in vivo AD models. Methods: GEHALB patches were fabricated from natural biopolymers crosslinked with 0.1% genipin and loaded with Derma-Clera<sup>TM</sup> (DC) and Ocâline<sup>TM</sup> PF (OC). In vitro AD-like model was established in keratinocytes using a cocktail of cytokines. Cytocompatibility and migration were assessed via MTT, Live/Dead, and scratch assays. In vivo efficacy was evaluated in DNCB-induced BALB/c mice models, monitoring morphology, TEWL, and systemic lymphoid organ weights. Results: An induction concentration of 35 ng/mL was selected for the in vitro model based on optimal cytokine expression and morphological changes. The bioactive-loaded hydrogels demonstrated high cytocompatibility (>90% viability) and keratinocyte migration, reaching  $86.64 \pm 7.83\%$  (GEHALB\_DC) and  $85.56 \pm 8.63\%$  (GEHALB\_OC) within 24 h, outperforming the control. Morphological observations showed superior skin barrier restoration in the hydrogel groups as compared to 1% hydrocortisone, with TEWL returning to healthy levels (<20 g/m<sup>2</sup>/h) by Day 35. Furthermore, treatments reduced spleen and subiliac lymph node weights, indicating suppression of systemic inflammatory responses. Conclusions: These findings suggest that the hydrogel patches offer a promising potential for long-term management of chronic inflammatory skin disorders by supporting both cellular regeneration and systemic immune modulation.

**Keywords:** Atopic dermatitis, Hydrogel, Regenerative medicine, Skin tissue Engineering

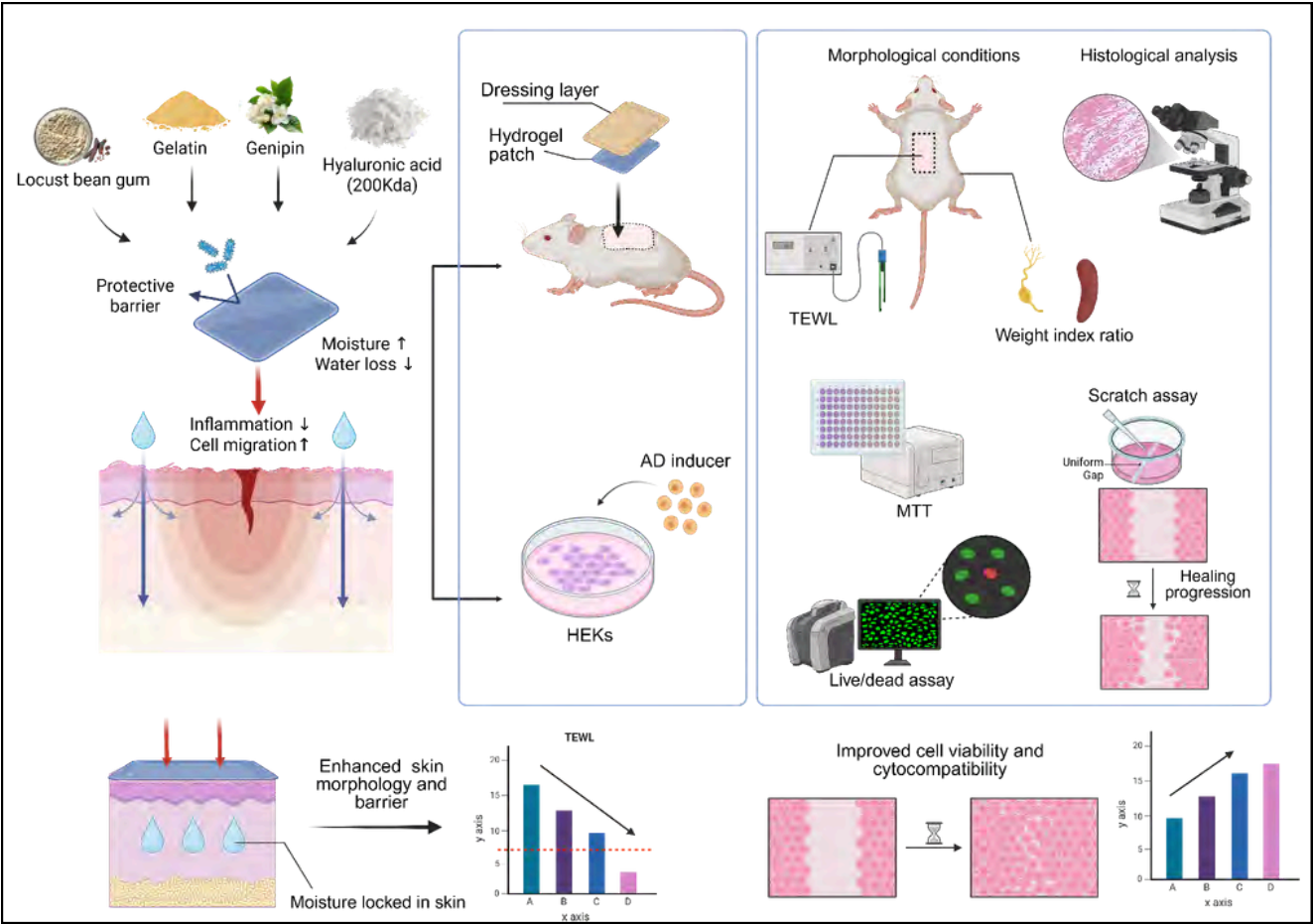


Figure 1: Key findings of the study

# Characterisation of Kelulut Honey-Loaded Injectable Gelatin–PVA-nanoCollagen–Graphene Oxide Hydrogel

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Wound healing is a complex biological process involving haemostasis, inflammation, proliferation, and remodelling to restore skin integrity. Conventional wound treatments often face limitations such as infection risk, frequent dressing changes, and delayed healing. Injectable hydrogels are promising wound dressings due to their biocompatibility, structural stability, moisture-retention ability, and adaptability to irregular wound shapes. Objective: This study aimed to develop and characterise a hybrid gelatin–polyvinyl alcohol (GE–PVA) hydrogel incorporated with nanoCollagen–graphene oxide and Kelulut honey for wound healing applications. Methods: Hybrid hydrogels were prepared using gelatin (6% w/v), polyvinyl alcohol (5% w/v), nanoCollagen–graphene oxide (0.15 mg/mL), and Kelulut honey (1% v/v), followed by genipin crosslinking (0.1% w/v). Four formulations were prepared: GE–PVA, GE–PVA–GNP, GE–PVA–H1–NC, and GE–PVA–H1–NC–GNP for injectability, swelling ratio, contact angle, degradation rate, and water vapour transmission rate (WVTR). FTIR and SEM analyses were conducted to determine chemical interactions and morphology. Results: GE–PVA–H1–NC–GNP demonstrated the lowest extrusion force and enhanced physicochemical properties. GE–PVA–H1–NC and GE–PVA–H1–NC–GNP showed optimal swelling ratios (214.40% and 178.85%) and favourable WVTR values (534–554 g m<sup>-2</sup> h<sup>-1</sup>). Reduced contact angle values indicated improved hydrophilicity. FTIR analysis confirmed enhanced hydrogen bonding, while SEM revealed improved pore interconnectivity and uniform porous structures. Conclusion: The incorporation of nanoCollagen–graphene oxide and Kelulut honey improved the injectability, hydrophilicity, structural integrity, and moisture retention of GE–PVA hydrogels, demonstrating their potential as injectable wound dressing materials.

**Keywords:** Gelatin–PVA hydrogel, Injectable hydrogel, Kelulut honey, nanoCollagen–graphene oxide, Wound healing

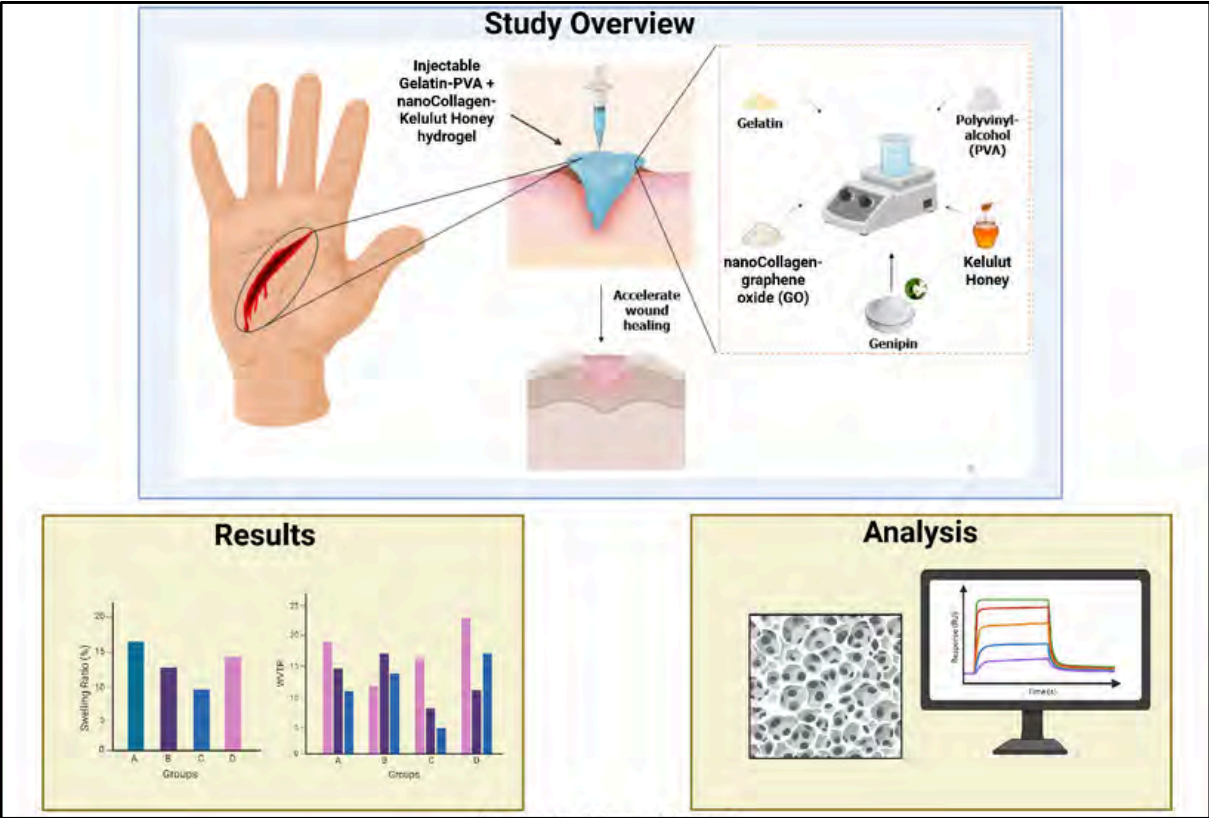


Figure 1: Key findings of the study

# Design of Experiment-Based Optimisation of Hydroxychloroquine-Loaded PLGA Microparticles Fabrication

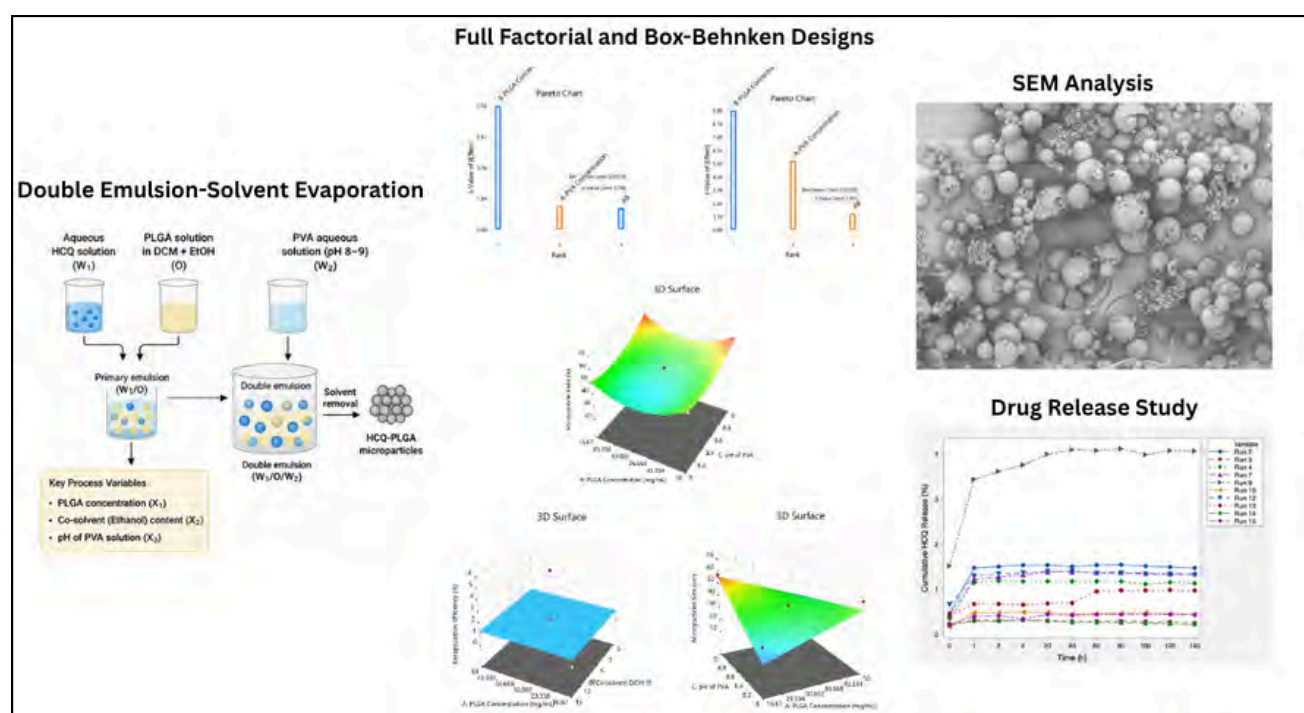
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Poly(lactic-co-glycolic acid) (PLGA) microparticles offer a promising delivery system to improve hydroxychloroquine (HCQ) drug delivery and at the same time, reduce systemic and cardiac exposure. Objective: The goal of this study is to develop an optimized formulation of HCQ incorporated with PLGA microparticles for pulmonary drug delivery. Methods: The Full-Factorial Design (FFD) was introduced to evaluate PVA concentration (%) and PLGA concentration (mg/ml). Then, the Box-Behnken Design (BBD) was used in optimization process and elucidating the main interacting parameters on the synthesis of HCQ-PLGA microparticles, which includes: Concentration of PLGA (mg/ml), additional of co-solvent ethanol (%), and pH of PVA by using response surface methodologies (RSM). Results: According to FFD, PLGA concentration plays major role on the microparticles formulation with high regression coefficient value of >0.600 for all responding variables. However, the BBD model results showed variable responses for each responding variables. The chosen parameters gave high regression coefficient value of 0.8466 for microparticle yield. The statistical analysis suggested by BBD on encapsulation efficiency and particle size of HCQ-PLGA microparticles resulted in a poor correlation between the independent variables and the response within the experimental design space. Despite the results suggested by BBD, HCQ was successfully encapsulated inside the PLGA microparticles and the drug release analyses of the microparticles samples showed controlled release profile. Conclusions: The current study focuses the impact of chemical interaction within the formulation on the HCQ-PLGA microparticles. Further investigation needed to study the impact of the mechanical properties on the formulation.

**Keywords:** Design of Experiment, Hydroxychloroquine, Microparticles, PLGA





# An Ontology-Aware Computational Workflow for Exploratory Mapping of Cardiac Bioscaffold Design for Cardiac Tissue Engineering Application

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**Objective:** In cardiac regenerative therapy, bioscaffolds offer compelling benefits by providing mechanical support, facilitating mass transport, and mediating biochemical cue delivery. However, highly fragmented literature reporting diverse material combinations, fabrication methods, and evaluation conditions obscures recurring design patterns. To bridge this gap, this study presents an ontology-aware computational workflow for the exploratory mapping of a literature-derived dataset comprising 3,396 unique cardiac bioscaffold formulations. **Methodology:** A comprehensive feature matrix was assembled via a multi-agentic pipeline, then strictly audited to eliminate downstream study-modality variables, preventing experimental context from confounding material recipes. Utilizing this purified 169-feature matrix, Uniform Manifold Approximation and Projection (UMAP) was coupled with Hierarchical Density-Based Spatial Clustering (HDBSCAN) to uncover latent structures. For transparency, an explainable AI (XAI) suite combining gradient-boosted tree ensembles (XGBoost) and SHAP feature attribution was deployed to map the multi-dimensional design space. **Results:** The UMAP-HDBSCAN pipeline yielded a robust global landscape partitioning of 46 distinct clusters (Mean Adjusted Rand Index = 0.251), while the XGBoost surrogate achieved a fidelity score of 0.826. Our findings demonstrate that removing study-modality variables eliminates target-leakage shortcuts, shifting the computational focus away from reporting artifacts and toward intrinsic material relationships. Crucially, the architecture resolved distinct, multi-dimensional design-space corridors, sharply isolating prolonged structural epicardial patches from rapid-remodelling, bio-actively enriched hydrogels and electroconductive anisotropic templates. **Discussion and Conclusion:** The results confirm that eliminating study-modality variables successfully controls for target leakage, ensuring the model identifies true structural patterns rather than experimental biases. The identification of 46 distinct clusters demonstrates the framework's capacity to partition complex bio-material data effectively, while the high surrogate fidelity of the XGBoost model verifies the reliability of the SHAP feature attributions. Ultimately, this leakage-controlled framework organizes fragmented literature data, uncovers under-explored design gaps, and generates high-probability candidate regions for future experimental prioritization.

**Keywords:** Cardiac Bioscaffold, Unsupervised Clustering; Explainable Machine Learning; Uniform Manifold Approximation and Projection (UMAP), HDBSCAN, Shapley Additive exPlanations (SHAP)

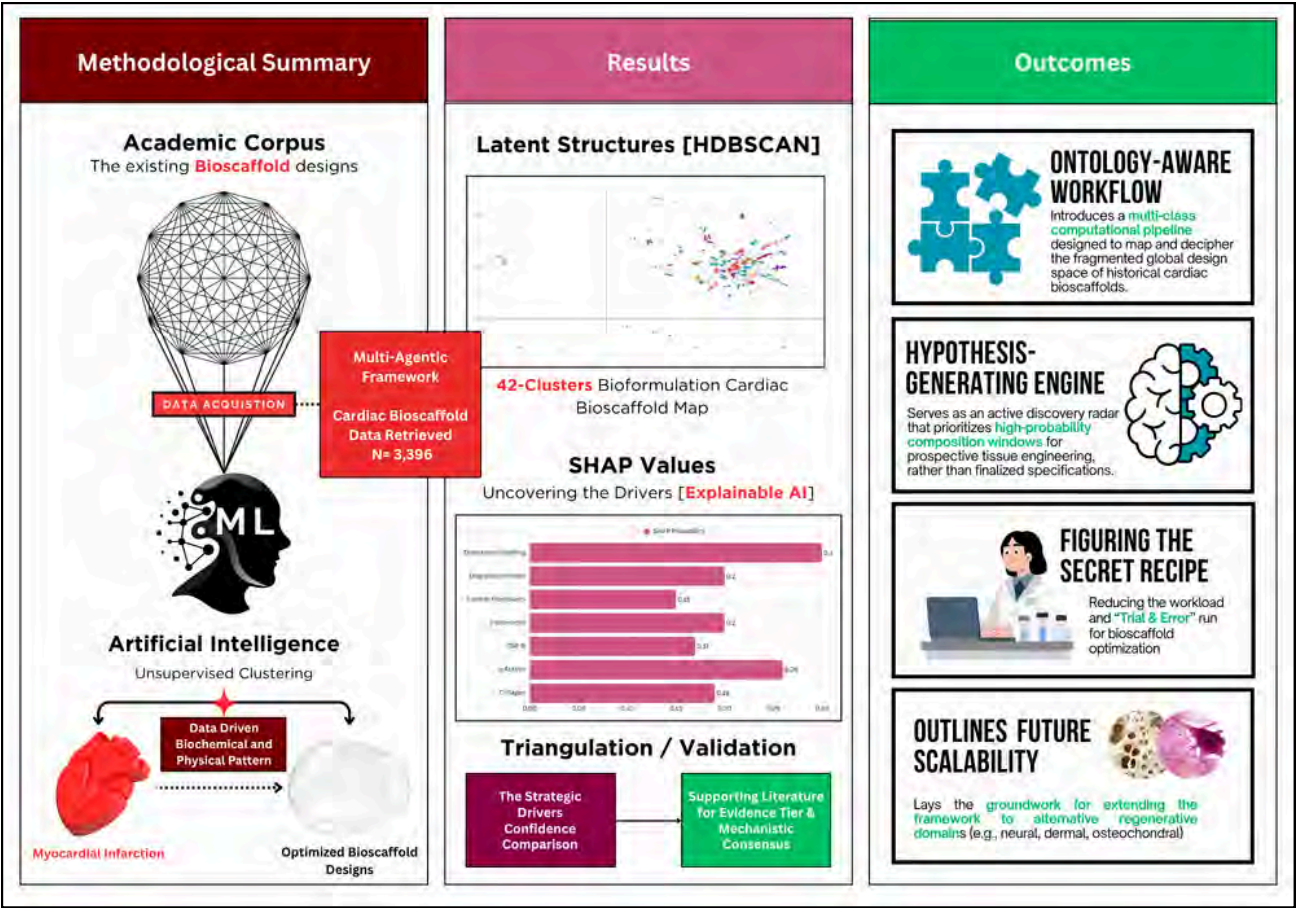


Figure 1: Highlihts of the study

## Sequence Length-Dependent Peptide Effects on Hyperglycaemia-Induced Dermal Fibroblasts

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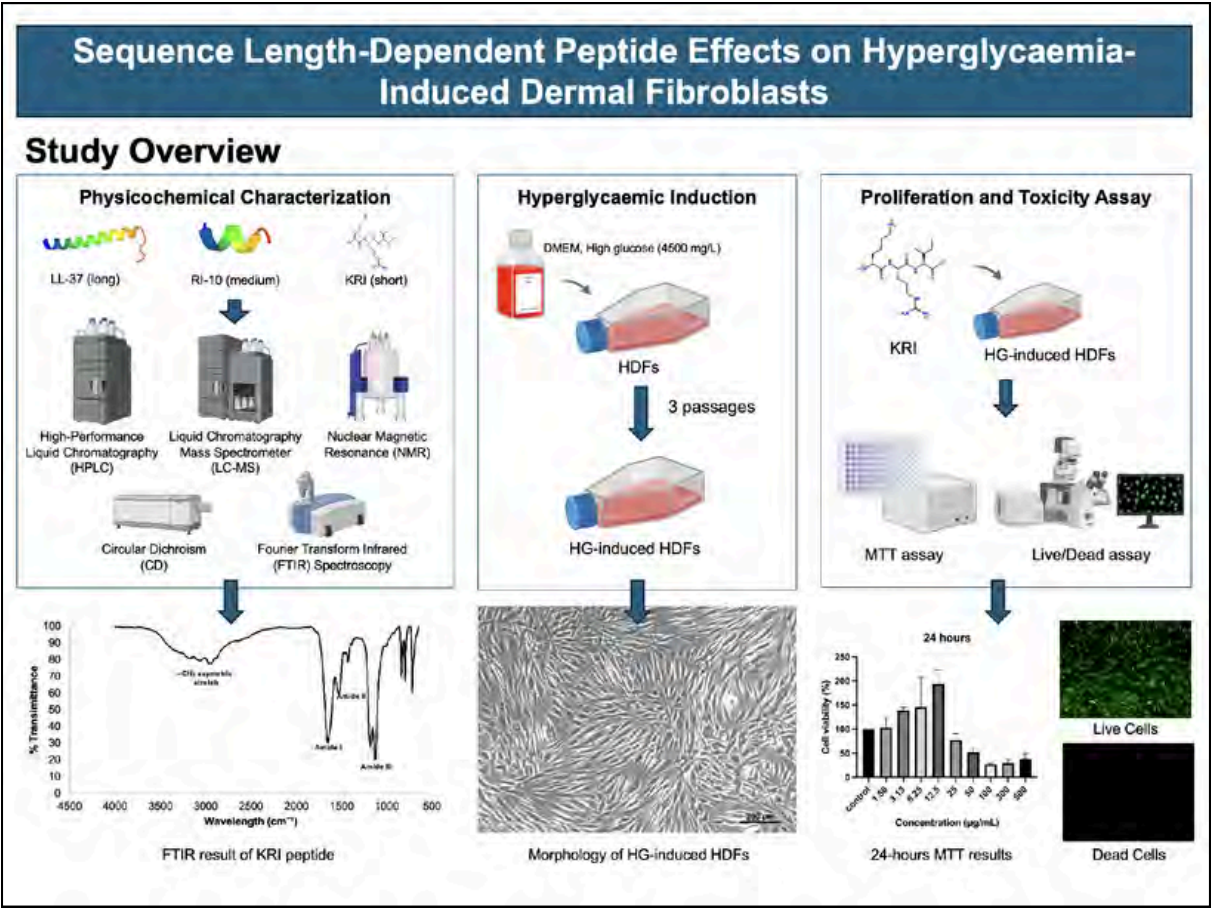
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Malaysia records a high prevalence of diabetes, contributing to chronic hyperglycaemic wounds that characterized by compromised cellular activity, chronic inflammation with high microbial infection risk. To address the cost and clinical translation limitation of current approaches, bioactive peptides have emerged as a promising alternative because of their regenerative, immunomodulatory and antimicrobial properties. However, long and medium peptides face limitations including high enzymatic degradation susceptibility and poor skin penetration. Therefore, short peptides with simpler sizes are developed to enhance stability while minimizing cytotoxicity. Objectives: This research aims to evaluate the physicochemical properties and wound healing efficiency of long (> 10 amino acids), medium (6-10 amino acids) and short peptides (< 5 amino acids) on hyperglycaemia-induced human dermal fibroblasts (HDFs). Methods: The physicochemical properties of these peptides were determined with HPLC, LCMS, NMR, FTIR and CD. Hyperglycaemic conditions were induced in HDFs using high-glucose DMEM for at least three passages. MTT and Live/Dead assays were conducted to assess the proliferative and cytotoxic effects of short peptides on hyperglycaemic HDFs. Results: The purity, molecular weight, amino acid composition, secondary structure and functional groups of the peptides were successfully characterized. Hyperglycaemic induction in HDFs was confirmed through morphological observations and senescence assays. Short peptides enhanced cellular proliferation at low concentrations, while cytotoxic effects were observed above 100 µg/mL. Conclusions: The findings demonstrate that peptide sequence length plays a critical role in determining the physicochemical properties of bioactive peptides. Short peptides promoted proliferation at low concentrations with minimal cytotoxicity, highlighting their potential for wound healing applications.

**Keywords:** Hyperglycaemic wounds, Antimicrobial peptides, Peptide sequence, Physicochemical characteristics



# Evaluation of Cold-Water Fish Skin Gelatin as a Substrate for Enhanced 2D Adhesion Kinetics and 3D Plant-Based Scaffolding of Primary Ovine Fibroblasts

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**Introduction:** Establishing effective cell-substrate interactions is a bottleneck in tissue engineering, particularly for anchorage-dependent cells like fibroblasts. While plant-based scaffolds offer sustainable, vascular-mimetic architectures, their non-mammalian chemistry lacks the necessary cues for fibroblast attachment. This study utilizes fish gelatin as a cost-effective, IgG-free alternative to mammalian matrices to overcome these adhesion barriers. The objective is to evaluate how fish gelatin enhances attachment kinetics in 2D culture and enables cells to adhere to the complex topographical templates of 3D plant-based scaffolds. **Methods:** Primary sheep fibroblasts were cultured on substrates coated with fish skin gelatin at concentrations ranging from 0.1% to 1% w/v. Attachment kinetics were quantified in 2D via MTT assay 2 hours post-seeding. For 3D evaluation, decellularized plant leaves were functionalized with gelatin (1% to 10% w/v) to assess the ability of fibroblasts to adhere to and spread across complex venation of the scaffold. **Results:** In 2D cultures, the 0.1% fish gelatin coating achieved >90% cell adhesion within only 2 hours, compared to ~60% on uncoated surfaces. In 3D trials, a 5% coating enabled robust attachment. Notably, fibroblasts on coated leaves exhibited preferential alignment and healthy, elongated morphology along the leaf venation. In contrast, cells on uncoated controls remained rounded with poor spreading. **Conclusion:** Fish gelatin serves as a highly effective bio-interface that substantially improves primary fibroblast attachment in both 2D and 3D culture settings. These findings highlight the potential use of fish gelatin as a sustainable platform for applications in cultured meat, tissue engineering and regenerative medicine.

**Keywords:** Biomaterial, Fish gelatin, 2D culture, 3D culture, Surface coating



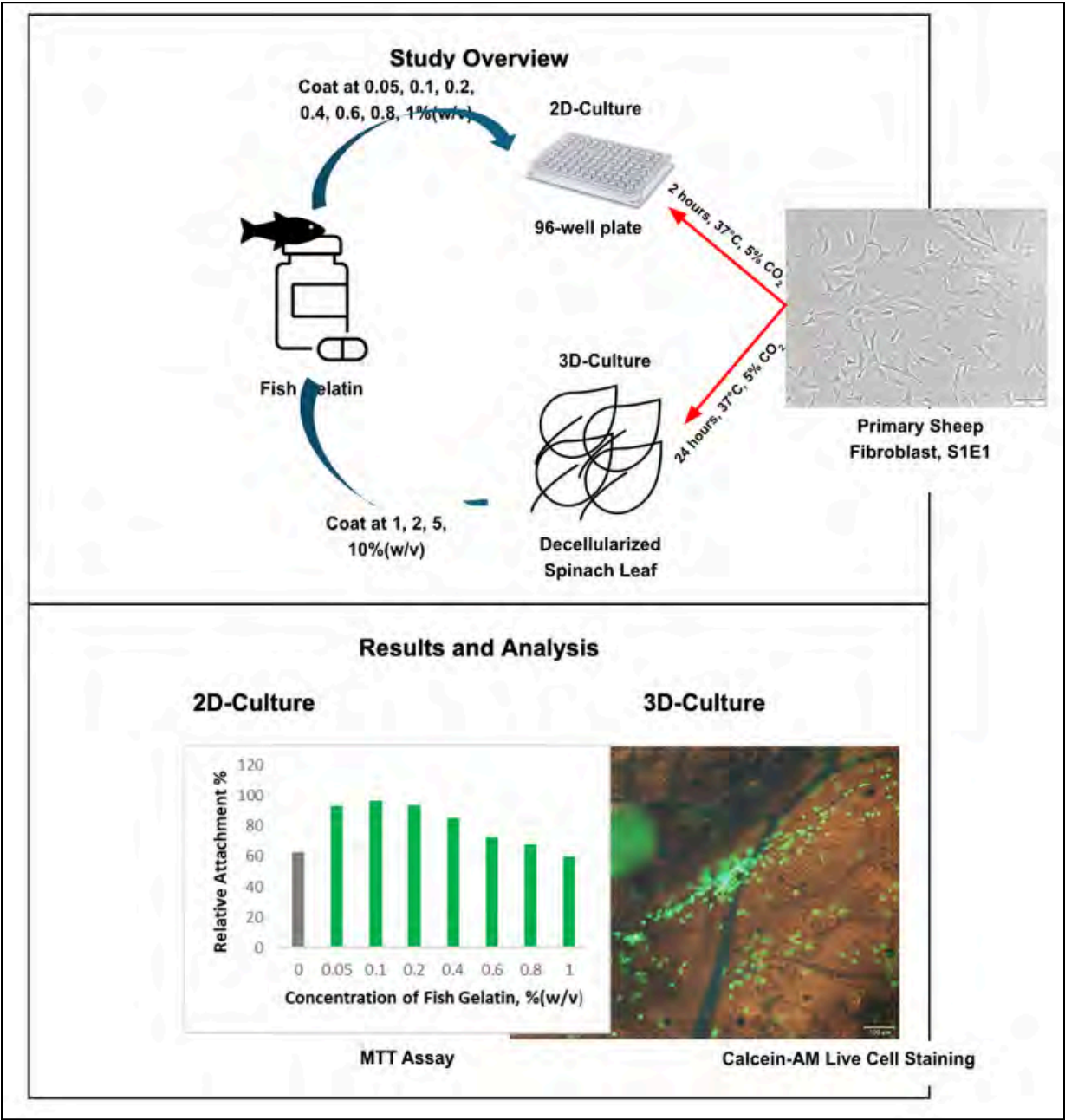


Figure 1: Key findings of the study

# Synthesis and Characterization of PLGA-CQD Fluorescent Nanoprobes for Future Lateral Flow Assay Applications

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Conventional LFAs utilizing gold nanoparticles often exhibit limited sensitivity, particularly for low-abundance biomarkers. To overcome this limitation, the development of advanced nanomaterial-based signal labels has gained increasing attention. Poly(lactic-co-glycolic acid) beads functionalized with carbon quantum dots have emerged as promising fluorescent nanoprobes for enhanced LFA performance. This study aims to develop and optimise the best method to formulate poly(lactic-co-glycolic) acid nanoparticles conjugated with carbon quantum dots to use as fluorescence labels in the lateral flow system. The nanoparticles were synthesized using an emulsion-solvent evaporation method and were further functionalized with carbon quantum dots using carbodiimide chemistry. A statistical software, Design Expert V3 is used to facilitate systematic optimization while minimizing experimental runs. Physicochemical properties of the targeted nanocomposite were evaluated in terms of chemical structure, morphology, particle size, zeta potential, polydispersity index, and fluorescence intensity. Formulated nanobeads exhibited excellent fluorescence intensity show blue to green fluorescence with emission wavelengths around 450–480 nm under UV excitation, with particle size ranges from 50 to 200 nm. The nanoparticles also exhibited a zeta potential of  $-30$  mV, indicating good colloidal stability which is suitable for LFA applications. These findings highlight the potential of the nanocomposite to enhance the performance of lateral flow assays.

**Keywords:** Poly(lactic-co-glycolic) acid nanoparticles, Carbon quantum dots, Fluorescence labels, Emulsion-solvent evaporation, Carbodiimide chemistry

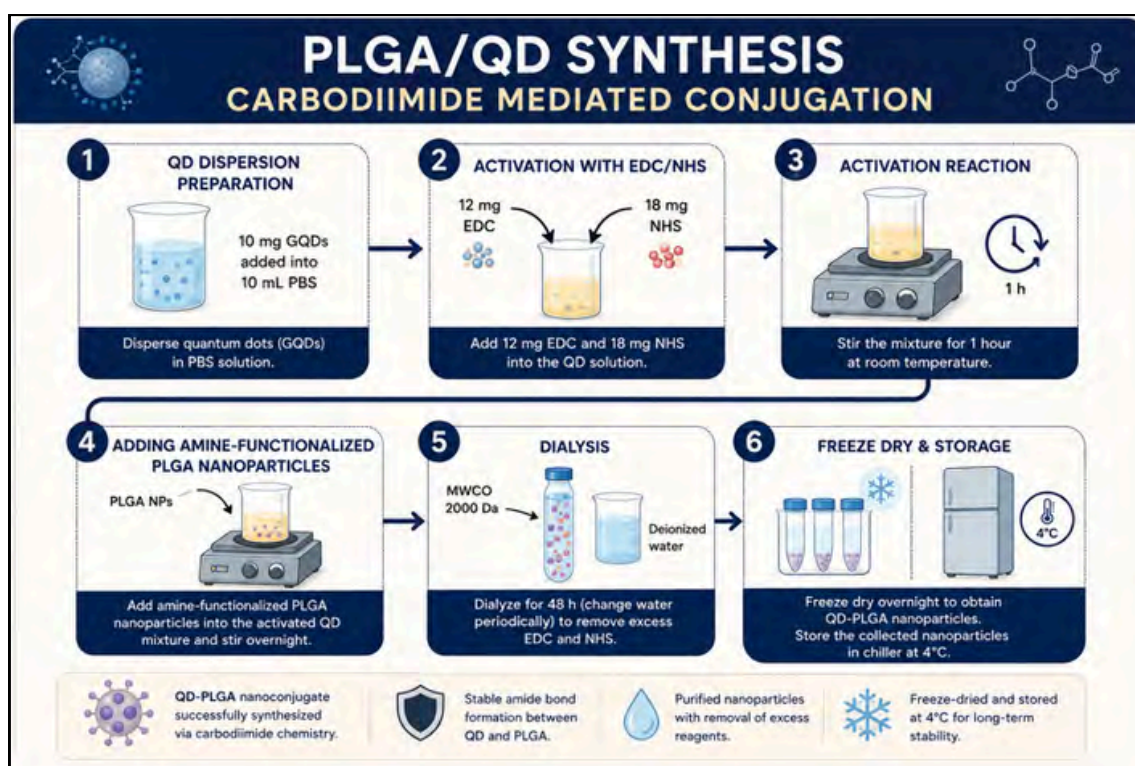


Figure 1: Carbodiimide chemistry



# Bioprinted Gelatin-SIKVAV and Gelatin-Palmitoyl-GDPH Bioinks Promote Skin Regeneration: In Vivo and In Vitro Assessment

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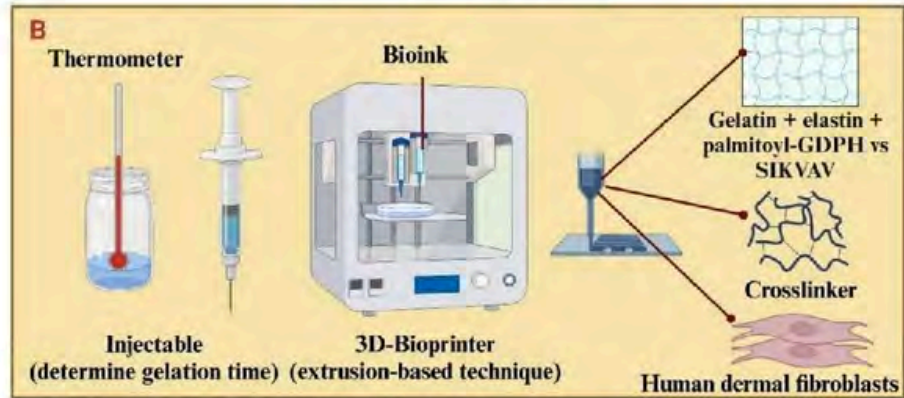
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The advancement of regenerative medicine increasingly relies on 3D bioprinted scaffolds that replicate native extracellular matrix architecture for effective skin repair. Gelatin-based bioinks functionalized with SIKVAV and palmitoyl-GDPH peptides offer enhanced bioactivity, biocompatibility, and cell-matrix interactions. However, their combined in vitro and in vivo regenerative efficacy remains underexplored. Objective: This study evaluates the regenerative performance of extrusion-based 3D bioprinted gelatin-SIKVAV and gelatin-palmitoyl-GDPH bioinks for cutaneous wound healing. Methods: Bioinks were fabricated at different concentrations via extrusion-based 3D bioprinting at 24 °C. Physicochemical characterization and cytocompatibility of the hydrogels were assessed. Human dermal fibroblasts were cultured to evaluate viability and cellular response. For in vivo evaluation, full-thickness excisional wounds were created in mice and treated with bioprinted scaffolds. Gross wound morphology was monitored, followed by histological analysis using hematoxylin and eosin staining. Results: The bioprinted hydrogels demonstrated excellent printability, structural integrity, and ideal physicochemical properties. In vitro analysis confirmed >90% fibroblast viability, indicating strong cytocompatibility and support for cell adhesion and proliferation. In vivo, after 17 days post-operation, GE\_GNP\_ELS\_PAL25 and GE\_GNP\_ELS\_SIK7 significantly accelerated wound closure (>90%) compared to the control group, with visibly improved wound contraction and surface regeneration. Histological assessment revealed a well-organized dermal structure with intact and thickened epidermis, along with complete re-epithelialization and enhanced collagen-rich tissue formation at day 17, particularly in the SIKVAV-functionalized scaffold. Conclusion: The findings suggest that Gelatin-SIKVAV and gelatin-palmitoyl-GDPH bioinks demonstrate strong translational potential as bioactive 3D bioprinting scaffolds for accelerated skin regeneration and wound healing applications.

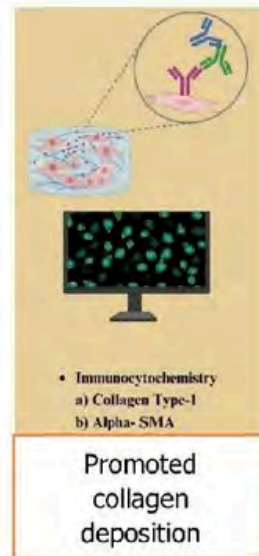
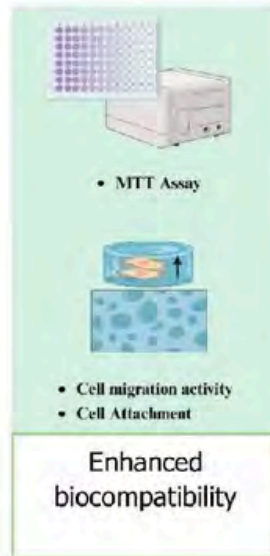
**Keywords:** Bioinks, 3D bioprinting, Gelatin, SIKVAV, Palmitoyl-GDPH



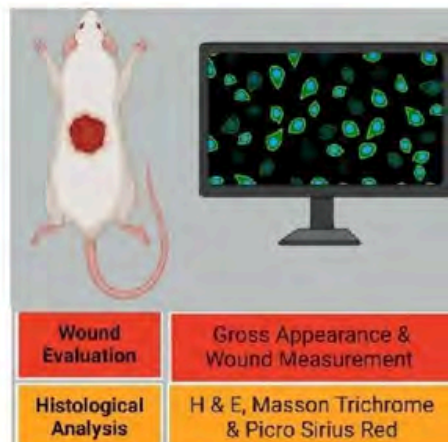
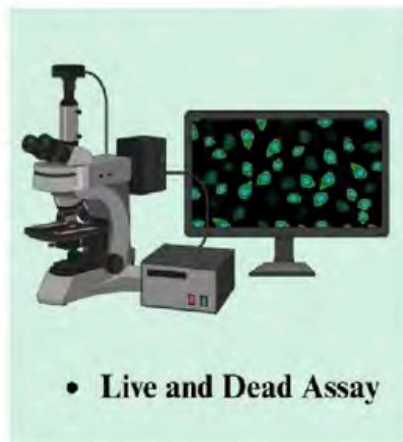
## Study Overview



## Results



## Analysis





## ***e-POSTER* PRESENTER ABSTRACTS**

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### ***e-Poster* Themes:**

**Theme 1 (T1): Advances in Preclinical Evaluation of Cell and Gene Therapy Products**

**Theme 2 (T2): Innovations in Biomaterials for Tissue Engineering and Regenerative Medicine**

**Theme 3 (T3): Emerging Cell-Free Strategies in Tissue Engineering and Regenerative Medicine**

# Therapeutic Potential of Hydroxytyrosol in Modulating Regenerative and Catabolic Activities of Human Osteoarthritic Chondrocytes

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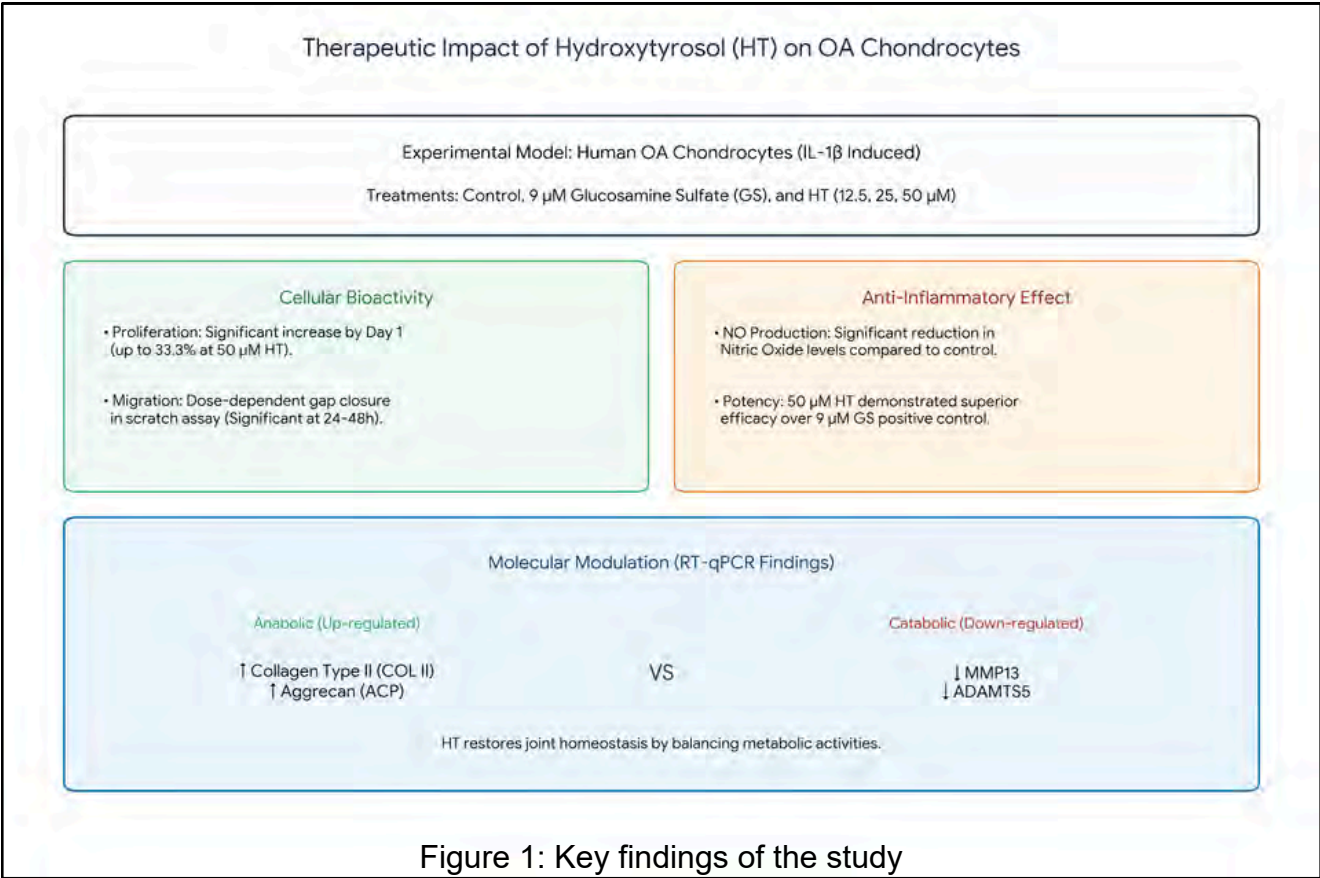
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**Introduction:** Osteoarthritis (OA) is a progressive degenerative joint condition characterized by articular cartilage destruction and remains a primary cause of global disability. Hydroxytyrosol (HT), a potent antioxidant phenolic compound derived from olive oil production, has emerged as a potential therapeutic agent due to its anti-inflammatory and free-radical scavenging properties. **Objective:** This study aimed to evaluate the therapeutic potential of HT in promoting regeneration and mitigating catabolic activity in human OA chondrocytes through in vitro analysis. **Methods:** Human OA chondrocytes were treated with various HT concentrations (0–400  $\mu$ M). Cell viability was assessed via MTT assay over 5 days. Functional effects were evaluated through scratch assays for cell migration, ELISA for nitric oxide (NO) production, and RT-qPCR for anabolic (Collagen II: COL II, Aggrecan: ACP) and catabolic (MMP13, ADAMTS5) gene expression. **Results:** HT supplementation at 12.5, 25, and 50  $\mu$ M significantly promoted chondrocyte proliferation on Day 1 (up to 33.3%,  $p < 0.0001$ ). Scratch assays confirmed that 50  $\mu$ M HT significantly enhanced cell migration after 24 hours ( $p < 0.05$ ), with all concentrations showing significance by 48 hours. Furthermore, HT significantly reduced oxidative stress by decreasing NO production in a dose-dependent manner. Molecular analysis revealed a significant up-regulation of anabolic genes COL II and ACP, while simultaneously down-regulating catabolic markers MMP13 and ADAMTS5 ( $p < 0.05$ ). **Conclusion:** HT effectively restores chondrocyte metabolic activity, enhances migration, and balances joint homeostasis by modulating gene expression. These findings position HT as a promising localized therapy for cartilage regeneration in OA.

**Keywords:** Osteoarthritis, Hydroxytyrosol, Chondrocytes, anti-inflammatory, Gene expression



# Effect of Nutrient Restriction on Mesenchymal Stem Cells: Morphology, Viability, and Surface Markers

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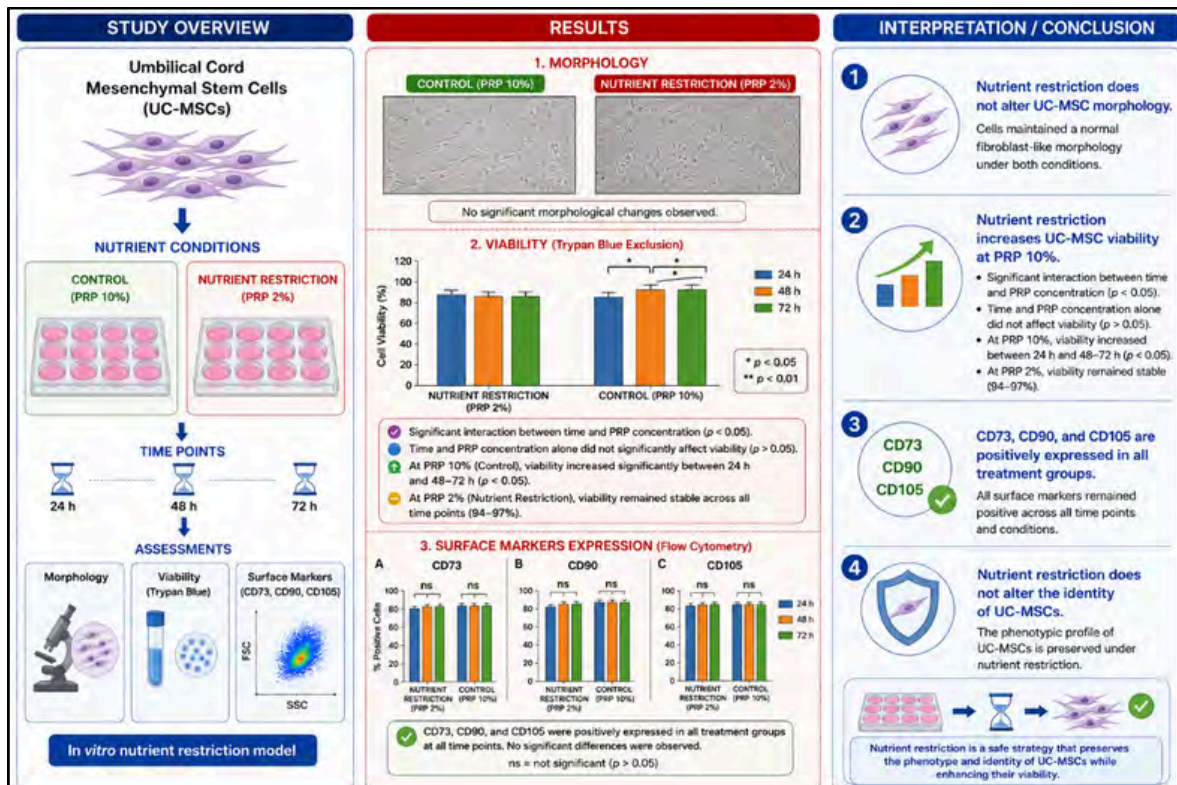
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Mesenchymal stem cells derived from umbilical cord tissue have emerged as a promising cell source in regenerative medicine due to their high proliferative capacity and immunomodulatory properties. Objectives: This study aimed to evaluate the effect of nutrient restriction as a model of intermittent fasting on cell morphology, viability, and surface marker expression. Methods: An in vitro experimental study was conducted using umbilical cord-derived mesenchymal stem cells divided into control and nutrient-restricted groups, observed at 24, 48, and 72 hours. Cell morphology was assessed using an inverted microscope, viability was measured using Trypan Blue exclusion, and surface marker expression (CD73, CD90, and CD105) was analyzed using flow cytometry. Statistical analysis was performed using two-way ANOVA. Results: A significant difference in cell viability was observed between control and nutrient-restricted groups ( $p < 0.05$ ), while no significant differences were found in the expression of CD73, CD90, and CD105 ( $p > 0.05$ ). Morphological changes were minimal and did not indicate loss of cell characteristics. Conclusions: These findings suggest that nutrient restriction affects cell viability but preserves the essential characteristics of mesenchymal stem cells, supporting its potential use as a preconditioning strategy in regenerative therapy.

**Keywords:** Mesenchymal stem cells, Morphology, Nutrient restriction, Surface markers, Viability



# In Vitro Adipogenic Differentiation of Sheep iPSC-Derived iMSCs: Cellular and Molecular Characterization

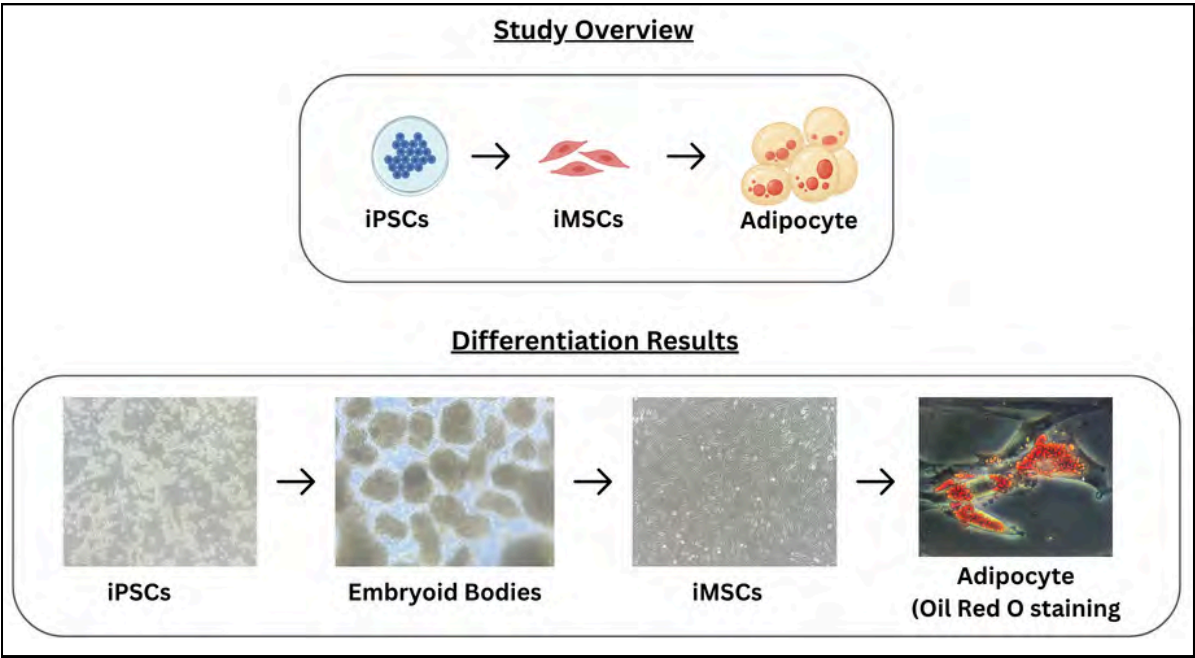
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Induced pluripotent stem cells (iPSCs) occupy the highest position in the stem cell hierarchy and hold significant potential in regenerative medicine. Their pluripotent state enables indefinite self-renewal and differentiation into any cell type except the extraembryonic tissues. Recently, iPSCs have been widely utilized as a powerful platform to generate induced mesenchymal stem cells (iMSCs) as an alternative to primary MSCs derived from sources such as bone marrow, dental pulp, and placenta. iMSCs have been shown to overcome limitations associated with primary MSCs, including limited proliferative capacity, batch-to-batch variability, and invasive procurement procedures. Despite retaining multipotent characteristics similar to MSCs, iMSCs are capable of differentiating into various mesodermal lineages, including adipocytes. Objectives: This study aims to differentiate sheep iPSCs into adipocyte population through an intermediate induced mesenchymal stem cell (iMSC) lineage. Methods: In this study, sheep iPSC-derived embryoid bodies were differentiated into iMSCs using retinoic acid treatment at defined concentrations and durations. The resulting iMSCs were subsequently subjected to adipogenic induction using a defined adipogenic cocktail over 24 days. Results: Differentiated cells exhibited red-stained lipid droplets following Oil Red O staining. Furthermore, molecular assessment revealed upregulation of adipogenic markers, including FABP4, Adiponectin, and PPAR- $\gamma$  at both gene and protein levels, as confirmed by RT-PCR and immunocytochemistry. Conclusion: These findings demonstrate successful adipogenic differentiation of sheep iPSC-derived iMSCs and highlight their potential as a scalable in vitro model for applications in cultured meat production, disease modeling, and drug screening.

**Keywords:** Induced pluripotent stem cells (iPSCs), Induced mesenchymal stem cells (iMSCs), Adipocyte, retinoic acid, Adipogenic cocktail





# Generation of hiPSC-Derived Neurons and Schwann Cells for Establishing Biologically Relevant Human Peripheral Nerve Model

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Peripheral nerve disorders and injuries remain a significant clinical challenge due to the limited regenerative capacity and complexity of the human peripheral nervous system. Current in vitro and animal models often fail to accurately recapitulate human peripheral nerve physiology, highlighting the need for more biologically relevant human-based model systems. Objective: This study aims to systematically differentiate hiPSCs into neurons and Schwann cells by characterizing their lineage-specific markers and morphological maturity for human peripheral nerve modelling. Methods: HiPSCs were differentiated into neuron through chemical differentiation protocol using Gibco PSC Neural Induction Supplement and B27 Neuronal Maturation Medium. HiPSCs were also differentiated into Schwann cells through chemical differentiation protocol using a custom schwann cell differentiation media. Protocol efficacy was evaluated using phase-contrast microscopy for morphological assessment and immunofluorescence for lineage-specific biomarkers. Results: Based on phase contrast microscopy analysis, neurite extension was observed following the neuronal differentiation protocol. The phenotype of the differentiated cell populations were further validated through immunofluorescence, showing positive expression of neuronal markers (TUBB3, and NeuN). Under Schwann cell differentiation protocol, cells were observed under phase contrast microscopy to adopt spindle-shaped morphology with phase-dark cell bodies, typical characteristics of Schwann cells, with positive expression of Schwann cell markers (Egr2 and Nestin) under immunofluorescence analysis. Conclusion: This study confirms the successful differentiation of neurons and Schwann cells from human iPSCs via optimized, chemical differentiation protocols. These findings advance the development of physiologically relevant, human-based models essential for investigating nerve trauma, neurodegeneration, and novel therapeutic interventions.

**Keywords:** Peripheral nerve, Human induced pluripotent stem cells, Neuron, Schwann cell, In Vitro Model

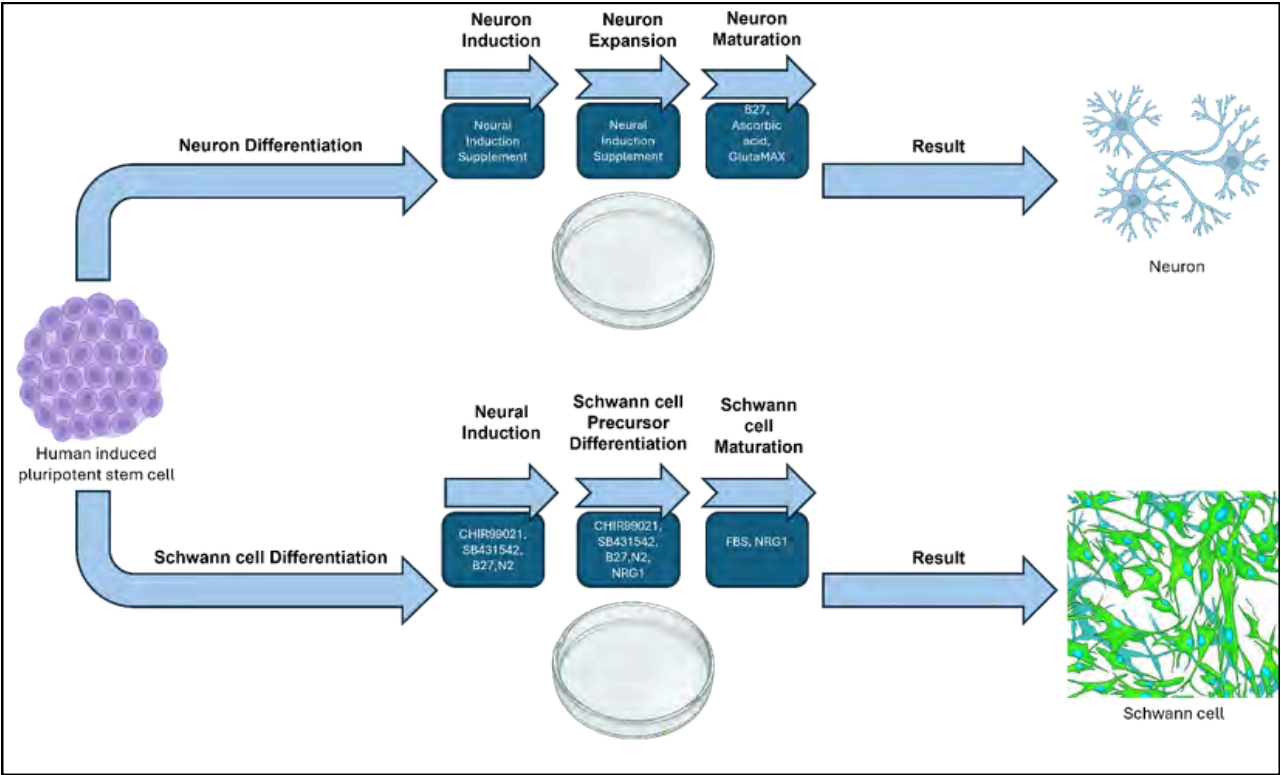


Figure 1: Graphical abstract illustrating the dual differentiation pathways from Human induced pluripotent stem cells (hiPSCs) into neuron and Schwann cells.

# From Screening to Regeneration: Optimization of Kelulut Honey Exposure in Zebrafish Larvae for Fin Regeneration Studies

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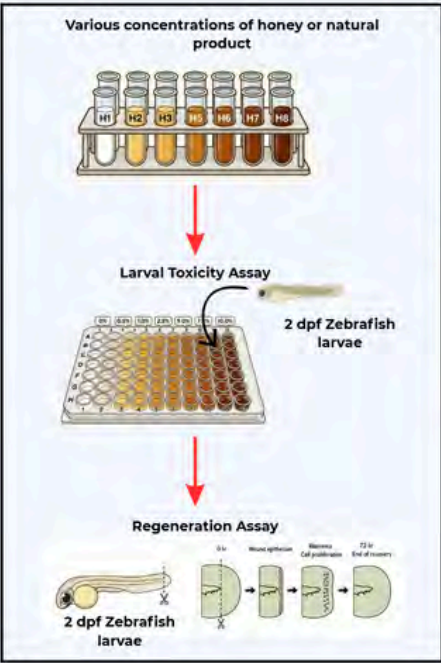
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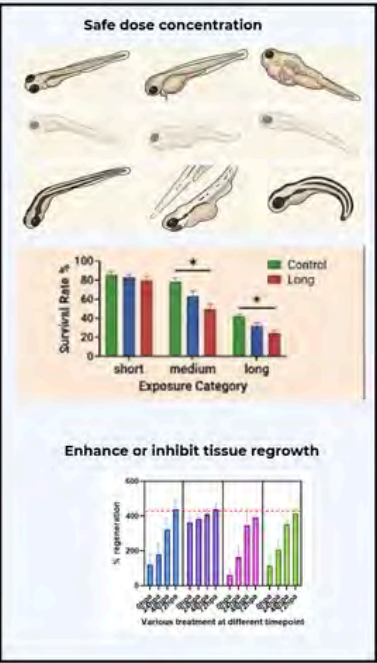
Tissue injury remains a substantial global healthcare challenges with delayed wound healing, emphasizing the need for rapid and effective in vivo regenerative screening platforms. Natural products such as kelulut honey (KH) has been widely reported to possess antimicrobial, antioxidant and wound healing properties in various animal models. However, limited studies have explored KH applications using promising in vivo, zebrafish larval systems particularly in regenerative research. Objectives: This study aimed to optimize KH exposure parameters in zebrafish larvae as a preliminary rapid screening platform for future regeneration studies. Methods: Early-stage zebrafish larvae at 2 days post-fertilization (2dpf) were exposed to varying KH concentrations ranging from 0.025% - 10% under different exposure category duration of 10, 20, 30 minutes and continuous exposure. Larval mortality and morphological abnormalities were observed at 24, 48 and 72 hours following exposure. Results: The findings demonstrated concentration and time dependent effects on larval viability. Lower KH concentrations with shorter exposure durations showed reduced mortality and normal morphology, while higher concentrations with prolonged exposure resulted in acute mortality. The increased mortality observed at high KH concentrations could potentially be associated with osmotic imbalance and physiological stress induced by high sugar content of honey exposure. Based on KH toxicity profiles, selected non toxic concentrations were subsequently chosen for regeneration assessment using zebrafish larval caudal fin amputation technique. Conclusions: This study establishes preliminary exposure parameters of KH that support the use of zebrafish larvae as a rapid and high-throughput screening platform for natural product-based regeneration studies and future regenerative research applications.

**Keywords:** Tissue regeneration, Kelulut honey, Zebrafish larvae, Rapid Screening, Toxicity

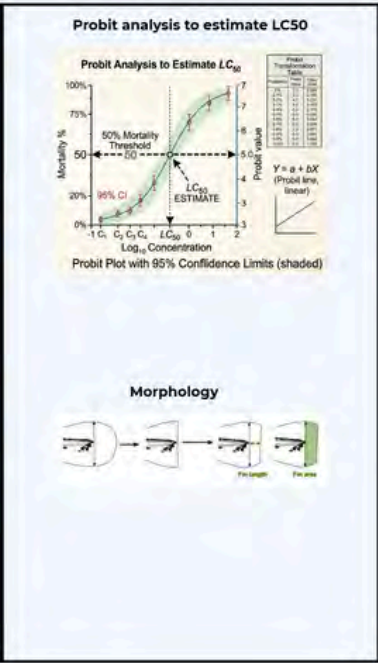
Study Overview



Results



Analysis



# Development and Characterization of a Genipin-Crosslinked Gelatin–Carrageenan Sponge as a Potential Nasal Secretome Delivery System for Nasal Wound Healing

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Impaired healing of the nasal mucosa following surgery or injury can result in delayed re-epithelialization, inflammation, adhesion formation, and disease recurrence. Nasal cell-derived secretome contains cytokines, growth factors, and extracellular matrix-related proteins that may promote epithelial attachment, migration, and tissue repair; however, its therapeutic application is limited by rapid degradation, poor retention, and uncontrolled distribution at the wound site. This preliminary study aimed to fabricate and characterize a gelatin–carrageenan sponge as a potential delivery platform for nasal secretome-mediated wound healing. A gelatin–carrageenan polymer blend (7:3) was prepared and crosslinked using 0.1% genipin. The fabricated sponge was evaluated macroscopically for sponge formation and crosslinking-associated colour development, while surface wettability and fluid-absorption capacity were assessed by contact-angle measurement and swelling studies, respectively. The sponge exhibited a stable porous structure with a characteristic blue colour, confirming successful genipin-mediated crosslinking. It demonstrated an average water contact angle of approximately 40°, indicating a hydrophilic surface that may facilitate fluid interaction and cellular attachment, and a high average swelling ratio of approximately 2400%, reflecting its excellent capacity to absorb and retain aqueous fluid. These properties may support secretome incorporation, maintenance of a moist wound environment, and localized delivery of bioactive factors to the injured nasal mucosa. Overall, the 0.1% genipin-crosslinked gelatin–carrageenan sponge demonstrated promising structural, wettability, and swelling characteristics as a potential nasal wound-healing scaffold. Further studies are warranted to evaluate secretome-loading efficiency, release kinetics, mechanical properties, degradation, cytocompatibility, cell attachment, migration, and wound-healing efficacy in vitro and in vivo.

**Keywords:** Gelatin, Carrageenan, Genipin, Nasal secretome, Sponge scaffold, Swelling, Nasal wound healing, Conditioned medium

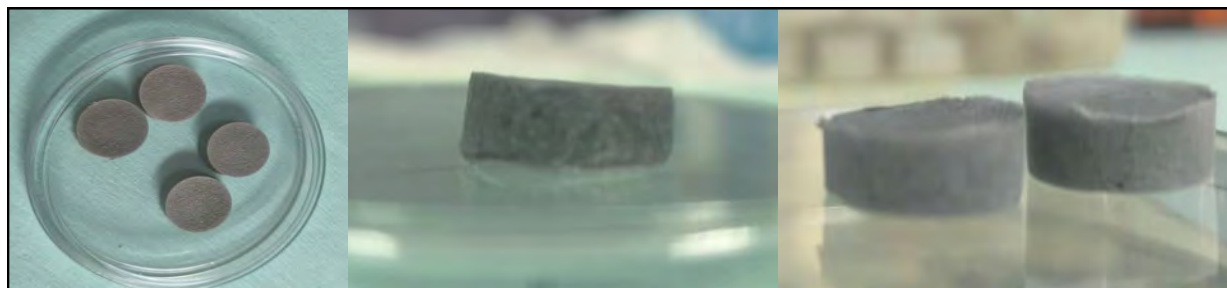


Figure 1: Gross appearance of the fabricated sponges



# Efficacy of Virgin Coconut Oil-Based Emulsion for Oral Applications

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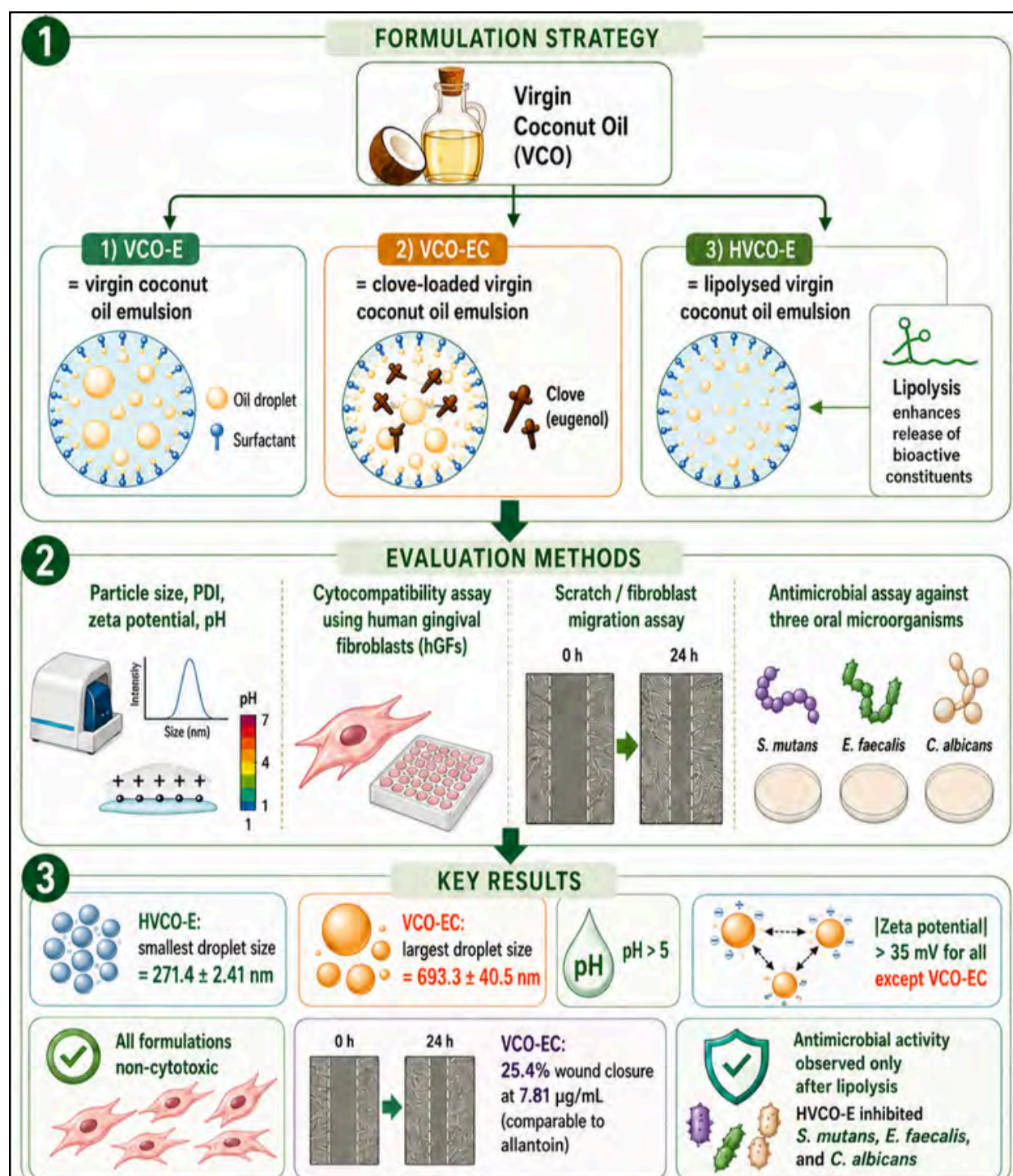
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**Introduction:** Virgin coconut oil (VCO) possesses antimicrobial and wound-healing properties, but its poor aqueous dispersibility and dependence on lipolysis limit its therapeutic efficacy in oromucosal applications. This study aimed to develop and evaluate the physicochemical and therapeutic properties of VCO-based emulsions. **Methods:** Optimised formulations comprising VCO emulsion (VCO-E), clove-loaded VCO emulsion (VCO-EC), and hydrolysed VCO emulsion (hVCO-E) were characterised for particle size, polydispersity index (PDI), zeta potential, and pH. Therapeutic efficacy was evaluated through cytocompatibility and fibroblast migration assays using hGFs, while antimicrobial activity was assessed against *S. mutans*, *E. faecalis*, and *C. albicans*. **Results:** The hVCO-E exhibited the smallest droplet size ( $271.4 \pm 2.41$  nm), whereas VCO-EC showed the largest ( $693.3 \pm 40.5$  nm). All formulations were mildly acidic and exhibited zeta potential exceeding -30 mV, except VCO-EC. All formulations were non-cytotoxic and promoted fibroblast migration, with 7.81 µg/mL VCO-EC achieving 25.4% wound closure comparable to the allantoin- group. Antimicrobial activity was observed only after hydrolysis, with hVCO-E demonstrating inhibitory effects against the tested oral microorganisms. **Conclusion:** Hydrolysis enhanced the physicochemical characterisation and also the release of bioactive constituents, enabling antimicrobial activity while maintaining wound-healing efficacy. These findings support the potential application of hydrolysed VCO emulsions in the formulation of antimicrobial therapy and oral wound healing in the oral environment.

**Keywords:** Virgin Coconut Oil, Emulsion, Clove, Cytotoxicity, Scratch Assay



# Cellular Mineralization Potential of Hydrothermally Prepared Oyster Shell Hydroxyapatite (Oys-HAP) in Human Osteoblast-like Model

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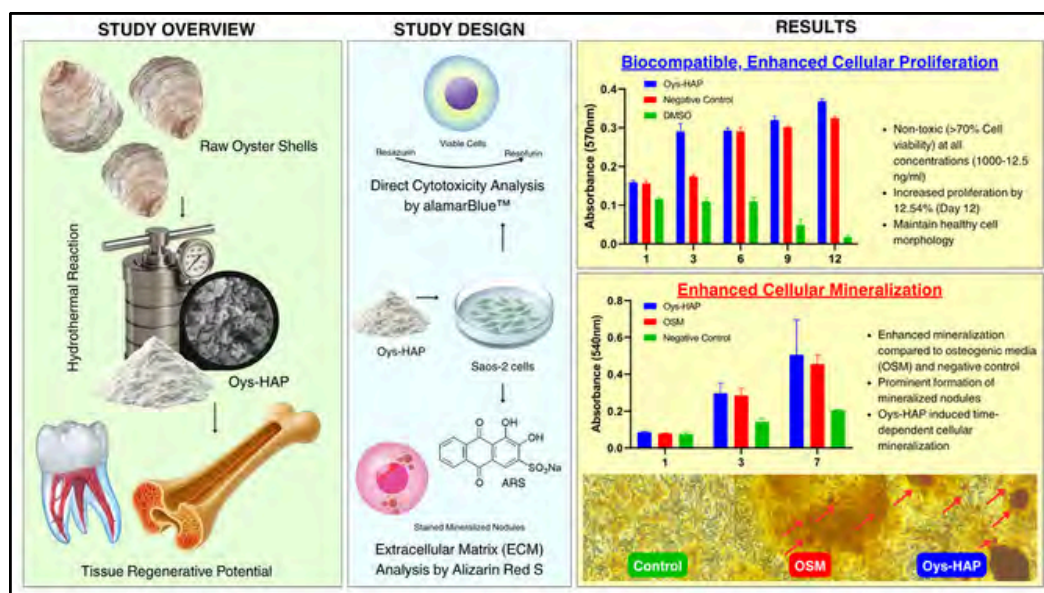
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Calcium carbonate ( $\text{CaCO}_3$ ) from natural sources such as oyster shells had created a way for sustainable hydroxyapatite (HAP) production. HAP has been widely recognized for its tissue regenerative potential owing to its excellent biocompatibility. Objectives: This study aimed to evaluate the cytocompatibility and mineralization potential of Oys-HAP in vitro using the Saos-2 human osteoblast-like cell as a model owing to its closest resemblance to mature primary human osteoblasts and the ability to induce intracellular mineralization. Methods: Oyster shell hydroxyapatite (Oys-HAP) was produced via a 24-hour hydrothermal reaction at  $150^\circ\text{C}$ . The cytocompatibility and mineralization activities were quantified using AlamarBlue™ and Alizarin Red S assays, respectively. Results: Oys-HAP (1000-31.25 ng/ml) was cytocompatible against Saos-2 after 24 hours of exposure. Cellular proliferation was induced by 31.25 ng/ml of Oys-HAP by 12.54% after 12 days of direct exposure compared to the negative control. On the other hand, the formation of extracellular matrix (ECM) was observed to be higher in the Oys-HAP treated group at day 7 compared to osteogenic media (OSM) and negative control. Total calcium increased 25.5% in the Oys-HAP (1.791 mM) compared to the negative (1.423 mM) and 11.66% against OSM (1.604 mM). Conclusions: These findings suggest the promising potential of the Oys-HAP biomaterial as a candidate in tissue engineering applications due to its ability to differentiate and mineralize the Saos-2 human osteoblast-like cell line in vitro.

**Keywords:** Mineralization, Hydroxyapatite, Oyster Shells, Human Osteoblast, Saos-2



# Surface Modifications of PCL Microcarrier for Scalable Human Induced Pluripotent Stem Cell Expansion Towards Stem Cell Therapy Biomanufacturing

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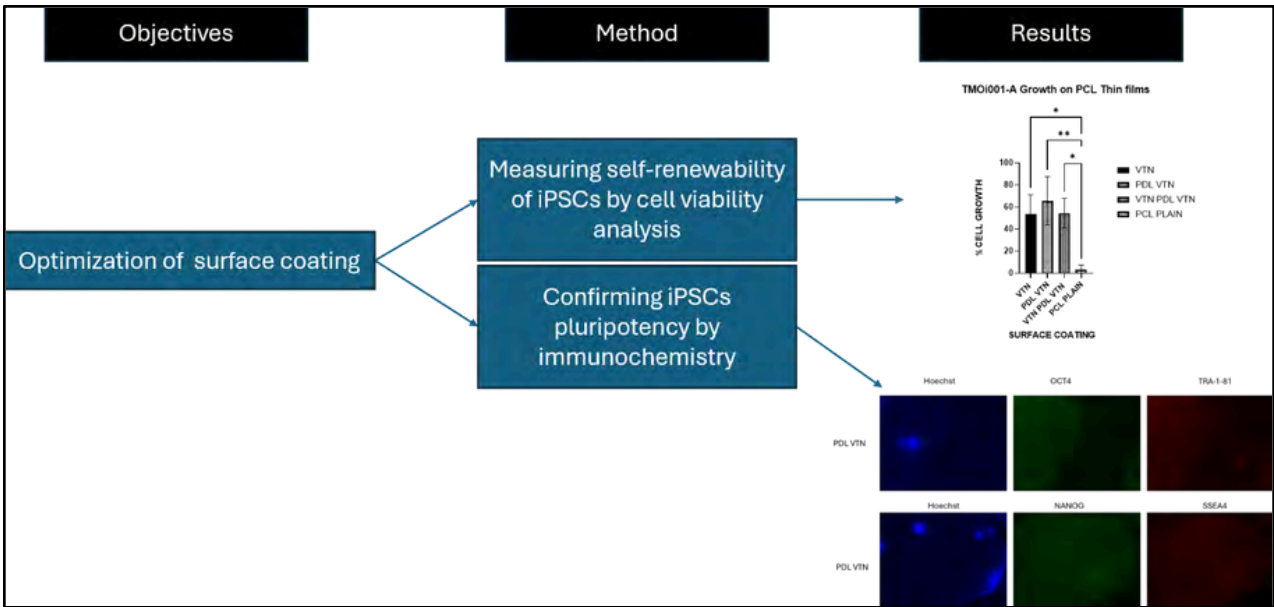
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The clinical translation of human induced pluripotent stem cells (hiPSCs) requires large-scale cell production; however, conventional two-dimensional (2D) culture systems are inadequate to meet therapeutic demands. Three-dimensional (3D) microcarrier systems offer a promising approach, yet optimal parameters for microcarrier design and surface engineering to support hiPSC expansion and differentiation remain undefined. Objective: This study aims to optimize the surface modification protocol for scalable hiPSC expansion using polycaprolactone (PCL) microcarriers towards stem cell therapy biomanufacturing. Method: Firstly, PCL films were fabricated using spin-coating and were modified with different surface coatings: (1) poly-D-lysine (PDL) - vitronectin (VTN) - PDL, (2) PDL - VTN, and (3) VTN alone. Uncoated PCL films were used as control groups. hiPSC growth on the PCL films were assessed by the CCK-8 cell proliferation assay, while pluripotency was confirmed via immunocytochemistry, detecting for Nanog, Oct4, SSEA4, TRA-1-81, and Sox2. Results: PCL films of all surface coating groups were observed to significantly increase hiPSC growth compared to uncoated films. hiPSC growth was slightly higher in PDL + VTN coated PCL films in comparison to PDL + VTN + PDL and VTN only groups, although, the difference is not statistically significant. Conclusion: In conclusion, all types of surface coating support hiPSCs self-renewal and maintain the pluripotency of hiPSCs. This optimised 3D culture system presents a robust and scalable platform, bridging the gap between cell biology, scaffold engineering, and biomanufacturing systems for regenerative medicine applications.

**Keywords:** Stem cells, polycaprolactone, 3D microcarriers, stem cell expansion, surface coating





# Optimization of Tripeptide and Its Incorporation into Various Gelatin Scaffolds for Wound Healing

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Chronic and non-healing wounds remains a significant clinical challenge, where multifactorial impairments hinder tissue regeneration. In response, therapeutic peptides have emerged as promising bioactive agents owing to their bioactivity and capacity to regulate key regenerative processes. Nevertheless, their clinical translation is often restricted by stability and inadequate bioavailability at the wound site. To compensate these shortfalls, biomaterial-based scaffolds therefore offer a promising strategy for sustained and localized peptide delivery. In this study, a novel synthetic Lysine-Arginine-Isoleucine (Lys-Arg-Ile; KRI) tripeptide, rationally derived from antimicrobial peptide fragments, was investigated for its regenerative potential. This study aims to establish the therapeutic applicability of the tripeptide through the optimisation of its concentration and evaluation of biocompatibility to be incorporated into various 3D-bioprinted gelatin scaffolds (hydrogel, sponge, film) for wound healing applications. The physicochemical characteristics of the tripeptide were determined using HPLC, ESI-MS, and CD to respectively assess the purity, molecular weight and secondary structures. Dose-dependent evaluations were subsequently performed on human dermal fibroblasts (HDFs) using a series of cellular assays to evaluate the proliferative, cytotoxicity, and migratory effects of the tripeptide. The tripeptide exhibited concentration-dependency, where lower concentrations enhanced fibroblast proliferation, concentrations exceeding 100 µg/mL induced cytotoxic effects. Following optimisation, the selected peptide concentration was incorporated into 3D-bioprinted gelatin-based hydrogel, film and sponge scaffolds, which were further characterized using FTIR and SEM analysis to evaluate chemical interactions and alterations in microporous morphology upon peptide incorporation. Collectively, the findings support the potential of tripeptide-incorporated scaffolds as promising platforms for translatable next-generation regenerative wound healing.

**Keywords:** Wound healing, Short peptides, Gelatin scaffolds, 3D-Bioprinting



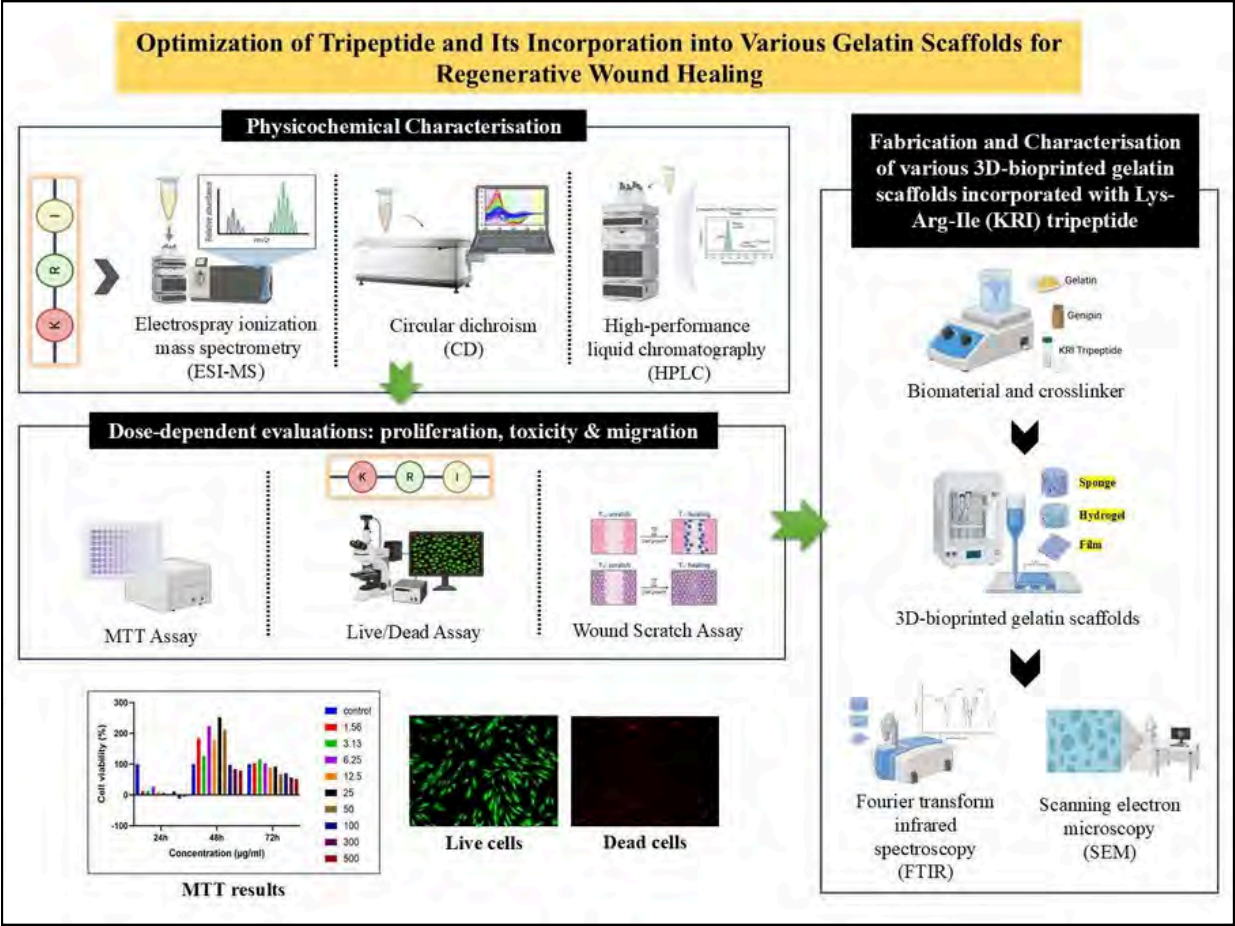


Figure 1: Key findings of the study

# Low-Dose Mesenchymal Stem Cells-Derived Exosome Eye Drops Promote Superior Corneal Epithelial Recovery in a Rabbit Model

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Corneal epithelial injury remains a major clinical concern, as rapid and complete wound closure is essential to prevent complications such as infection, scarring, and vision loss. Objectives: This study aimed to assess the therapeutic potential of human mesenchymal stem cell (MSC)-derived exosome eye drops in enhancing corneal wound healing using a rabbit corneal defect model. Methods: Twenty adult New Zealand White rabbits were subjected to corneal epithelial injury via trephination and randomly divided into four groups: saline control and treatment groups receiving exosome eye drops at 10, 25, and 50 million functional units (MFU). Treatments were administered topically for 14 days. Wound healing was monitored using slit lamp microscope. Histopathological analysis was performed after euthanasia for corneal structural recovery and inflammatory response. Results: Initial analysis showed no significant differences among groups. However, secondary analysis accounting for batch variability and outliers revealed a dose-dependent but non-linear response. The lowest dose (10 MFU) significantly accelerated wound closure compared to control and higher-dose groups, achieving complete epithelial repair by 142 hours and showing reduced cumulative wound area. Histological findings demonstrated improved epithelial structure, absent ulceration, and lower inflammation in treated groups. Higher doses did not yield recovery benefits, suggesting a potential saturation or inhibitory effect at increased concentrations. Conclusions: MSC-derived exosome eye drops enhance corneal epithelial recovery in a dose-dependent but non-linear manner, where low dose demonstrated superior efficacy. These findings emphasize the importance of dose optimization and support continued exploration of exosome-based therapies for corneal wound healing.

**Keywords:** Corneal epithelial injury, Mesenchymal stem cell (MSC) exosomes, Regenerative medicine, Dose optimization, Ocular drug delivery

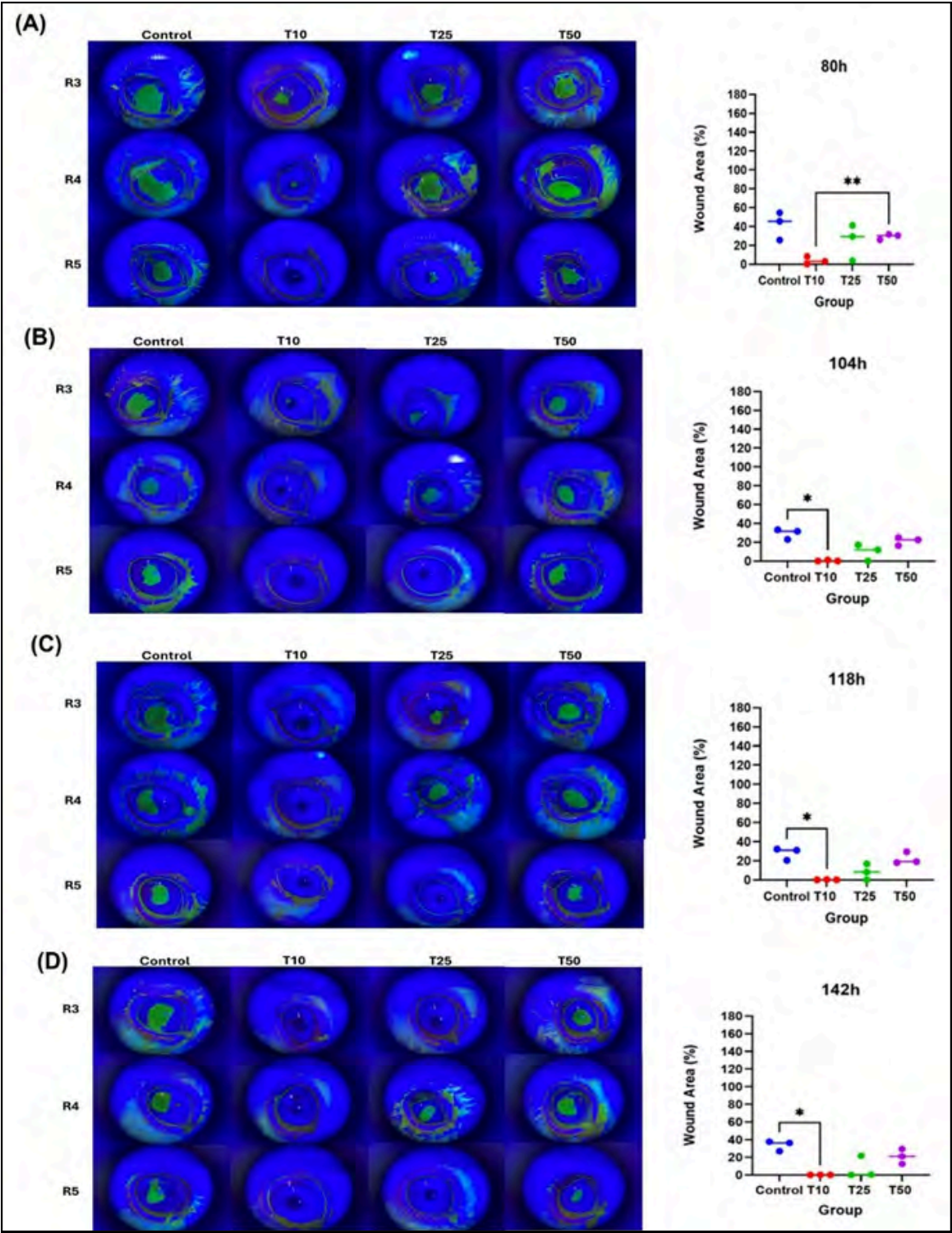


Figure 1: Key findings of the study

# Source-Dependent Extracellular Vesicle Therapy Improves Spermatogenesis in Azoospermia Rats

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**Objectives:** This study compared amniotic fluid-derived extracellular vesicle-enriched preparations (EV-AF) with adult serum-derived extracellular vesicle preparations (EV-Serum) for restoring spermatogenesis in busulfan-induced non-obstructive azoospermia. **Materials and Methods:** Sixteen male Wistar rats were randomized to intact control, azoospermia plus vehicle, azoospermia plus EV-AF, or azoospermia plus EV-Serum (n=4/group). Azoospermia was induced by intratesticular injection of busulfan (5 mg/kg/testis). After eight weeks, 10 µg total vesicular protein in 10 µL phosphate-buffered saline was administered intratesticularly, followed by eight weeks of recovery. Vesicle morphology and size were assessed by transmission electron microscopy. Outcomes included sperm concentration, motility, morphology, spermatogenic cell counts, Johnsen score, seminiferous tubule diameter, and germinal epithelial thickness. **Results:** Both preparations increased spermatocytes and elongated spermatids relative to vehicle. Johnsen scores were 9.62±0.18 in controls, 3.00±0.27 in untreated azoospermia, 7.62±0.18 with EV-AF, and 6.25±0.25 with EV-Serum. EV-AF produced greater recovery of tubule diameter (170.11±3.02 µm) and epithelial thickness (51.67±1.60 µm) and reduced abnormal sperm morphology to near-control levels. Motility and epididymal sperm number improved numerically but remained variable. **Conclusions:** Local EV therapy partially restored spermatogenesis, with EV-AF showing the strongest histological and morphological recovery. Larger studies using particle-normalized dosing and full EV characterization are warranted.

**Keywords:** Azoospermia, Extracellular vesicles, Rat model, Spermatogenesis, Testicular regeneration

# Source-Dependent Extracellular Vesicles Attenuate Ovarian Follicular Loss in POI Rats

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**Objectives:** This study compared amniotic fluid-derived extracellular vesicle-enriched preparations (EV-AF) with serum-derived extracellular vesicle preparations (EV-Serum) for preserving ovarian follicles in cyclophosphamide-induced premature ovarian insufficiency. **Materials and Methods:** Eighteen female Wistar rats were used; six confirmed disease induction, and the remaining animals were allocated to Control, untreated premature ovarian insufficiency, EV-Serum, or EV-AF groups. Cyclophosphamide was administered intraperitoneally at 200 mg/kg on day 1 and 8 mg/kg/day on days 2–14. A single intra-ovarian injection of 10 µL preparation, containing approximately 10–15 µg total protein, was administered. Ovarian histology quantified normal and atretic follicles, normal-follicle proportion, corpus luteum count, and developmental stages. After data-quality control, 11 animal-level observations were analyzed using permutation Kruskal–Wallis tests with false-discovery-rate correction. **Results:** Untreated disease produced severe follicular loss: median normal follicles 3.0, atretic follicles 29.0, and normal-follicle proportion 0.094. EV-AF showed the strongest descriptive recovery, with corresponding medians of 23.0, 5.0, and 0.815, plus a corpus luteum count of 10.0. EV-Serum also improved the histological profile. Global differences remained significant for total normal follicles ( $q=0.0034$ ), total atretic follicles ( $q=0.0129$ ), normal-follicle proportion ( $q=0.0073$ ), and corpus luteum count ( $q=0.0034$ ). **Conclusions:** Both EV-enriched preparations attenuated ovarian follicular injury, with EV-AF showing the strongest descriptive signal.

**Funding:** The study was supported by the program-targeted financing on scientific programs of the Ministry of Healthcare of the Republic of Kazakhstan «Development of an exosome isolation kit from umbilical cord mesenchymal stem cell culture for therapeutic and research application» (2024–2026) (IRN BR25593457).

**Keywords:** Cyclophosphamide, Extracellular vesicles, Ovarian follicles, Premature ovarian insufficiency, Regenerative medicine



# Hypoxia Priming Enhances MSC-Derived Exosome Yield with Preliminary Evidence of Cytocompatibility in MT-ND6–Mutated Fibroblasts

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Hypoxic priming has emerged as a promising strategy to enhance the paracrine activity and regenerative potential of mesenchymal stem cells (MSCs). However, the effects of different hypoxic conditions on the MSC-derived exosome yield, cellular uptake, and cytocompatibility remain insufficiently understood. Objectives: This study aims to investigate the effects of hypoxia priming on particle concentration, size distribution, protein concentration, cellular uptake, and cell viability of MT-ND6 mutated fibroblasts. Methods: MSCs were cultured under normoxia (21% O<sub>2</sub>) or hypoxia (1% or 3% O<sub>2</sub>). Cell viability was assessed before conditioned media collection. Exosomes were isolated and characterized for particle concentration and size by nanoparticle tracking analysis, for protein concentration by protein quantification, and morphology by TEM. The exosomes uptake and effect on cell viability were evaluated in ND6-mutated fibroblasts treated with exosomes from all three conditions. Results: MSC viability was significantly higher under 3% hypoxia than normoxia. Hypoxia increased exosome particle concentration, with 3% hypoxia yielding significantly more particles than 1% hypoxia. Particle sizes remained below 100 nm, with no significant differences among groups. Protein concentration was highest in the 3% hypoxia group, although differences were not statistically significant. Preliminary findings showed that exosomes from all culture conditions were internalized by MT-ND6–mutated fibroblasts, with no apparent adverse effects on cell viability at the tested concentrations. Conclusion: Moderate hypoxia may improve the efficiency of MSC-derived exosome production. Preliminary findings also indicate successful uptake by MT-ND6–mutated fibroblasts without apparent adverse effects on cell viability, supporting further investigation of hypoxia-primed exosomes for mitochondrial dysfunction.

**Keywords:** Hypoxia priming, Mesenchymal stem cells, Exosomes, Particle concentration, Exosomes Uptake

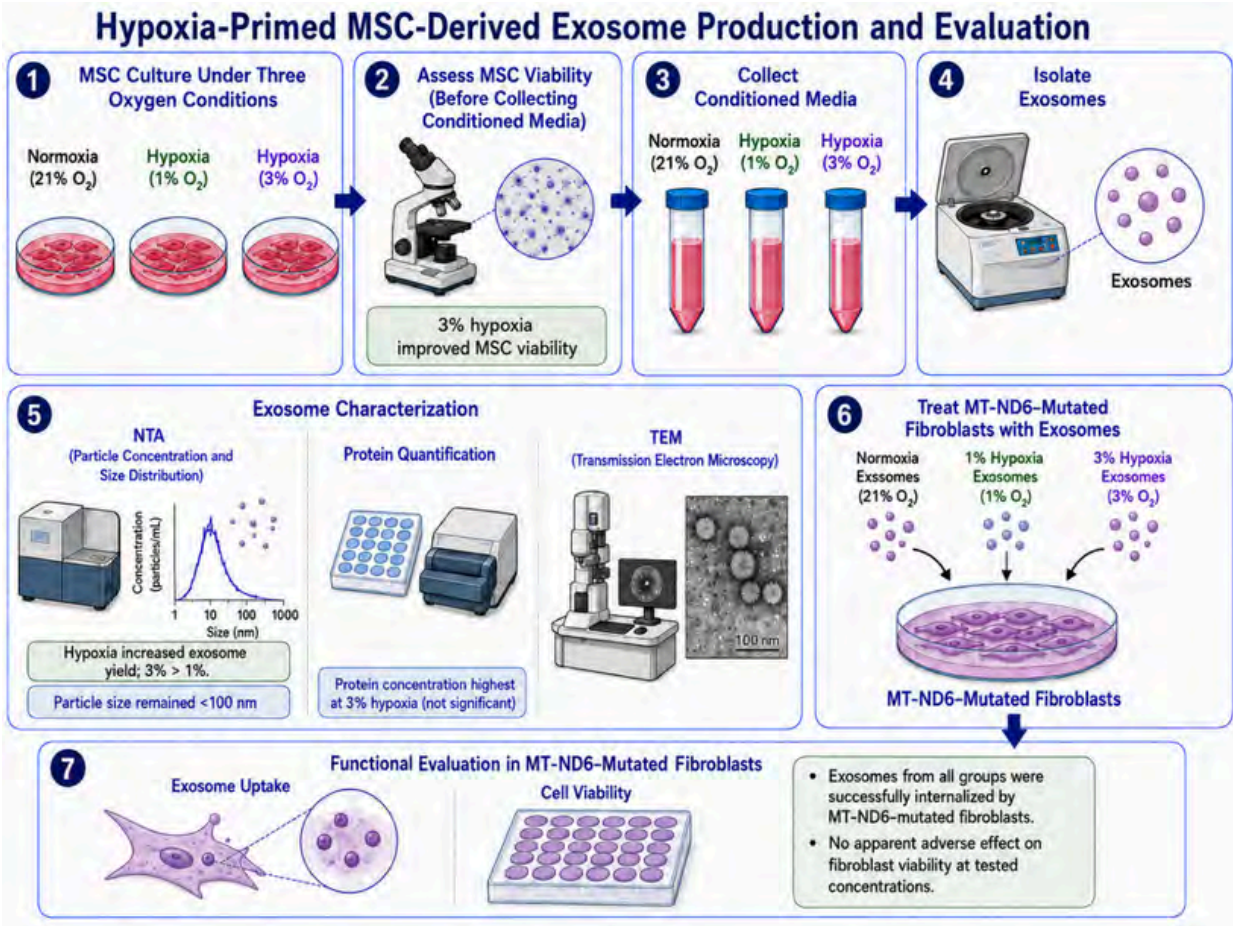


Figure 1: Illustration of the study findings

# Establishing a Replicative Aging Model for Evaluating Exosome-Mediated Telomere Preservation in Wharton's Jelly Mesenchymal Stem Cells

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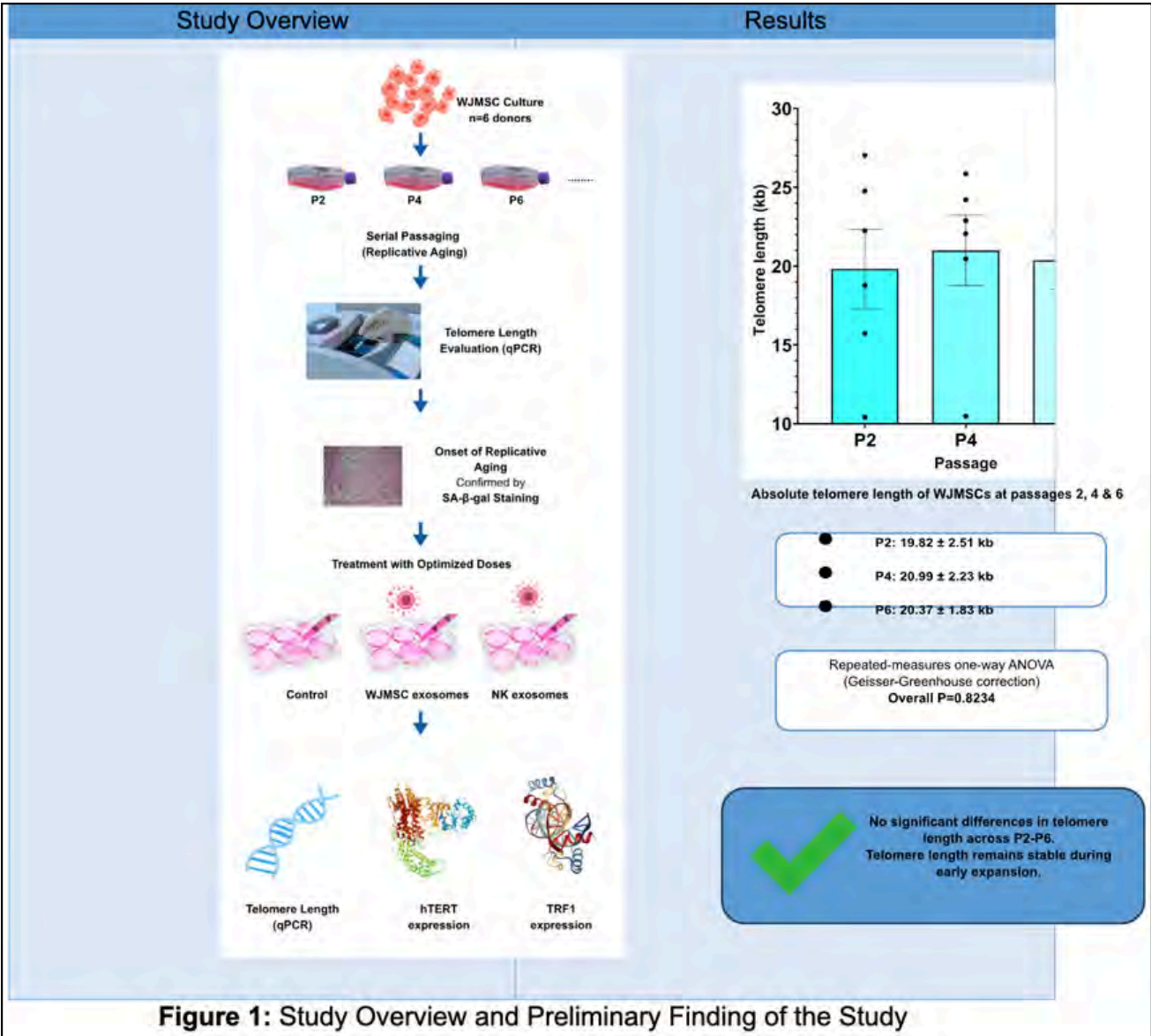
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Replicative senescence limits the clinical application of mesenchymal stem cells (MSCs) because prolonged in vitro expansion results in telomere shortening and reduced regenerative capacity. Identifying the passage at which significant telomere attrition occurs is essential for establishing a replicative aging model to evaluate rejuvenation strategies. This study aimed to determine telomere length (TL) dynamics during serial passaging of Wharton's jelly-derived MSCs (WJMSCs) and establish an in vitro model for evaluation of WJMSC- and natural killer (NK) cell-derived exosomes. WJMSCs from six biological donors were serially passaged, and absolute TL was quantified at passages 2 (P2), 4 (P4), and 6 (P6) using quantitative polymerase chain reaction. Data were analyzed using repeated-measures one-way analysis of variance with the Geisser–Greenhouse correction followed by Šídák's multiple comparisons test. Serial passaging will continue until significant telomere shortening is detected. The corresponding passage will then undergo senescence-associated  $\beta$ -galactosidase staining before treatment with optimized doses of WJMSC- and NK cell-derived exosomes. Post-treatment analyses will evaluate TL, human telomerase reverse transcriptase (hTERT), and telomeric repeat-binding factor 1 (TRF1) expression. Mean TLs were  $19.82 \pm 2.51$  kb,  $20.99 \pm 2.23$  kb, and  $20.37 \pm 1.83$  kb at P2, P4, and P6, respectively. No significant differences were observed across passages (overall  $P = 0.8234$ ), indicating that TL remained stable during early expansion, suggesting that replicative senescence had not yet been established. Continued serial passaging will identify the onset of telomere attrition and provide an in vitro aging model for evaluating the telomere-preserving potential of WJMSC- and NK cell-derived exosomes.

**Keywords:** Exosomes, Mesenchymal stem cells, Natural Killer cells, Replicative aging, Telomere length



# Optimizing Lentiviral Promoters for Efficient Transcription Factor Expression in Human Fibroblasts for Direct Reprogramming

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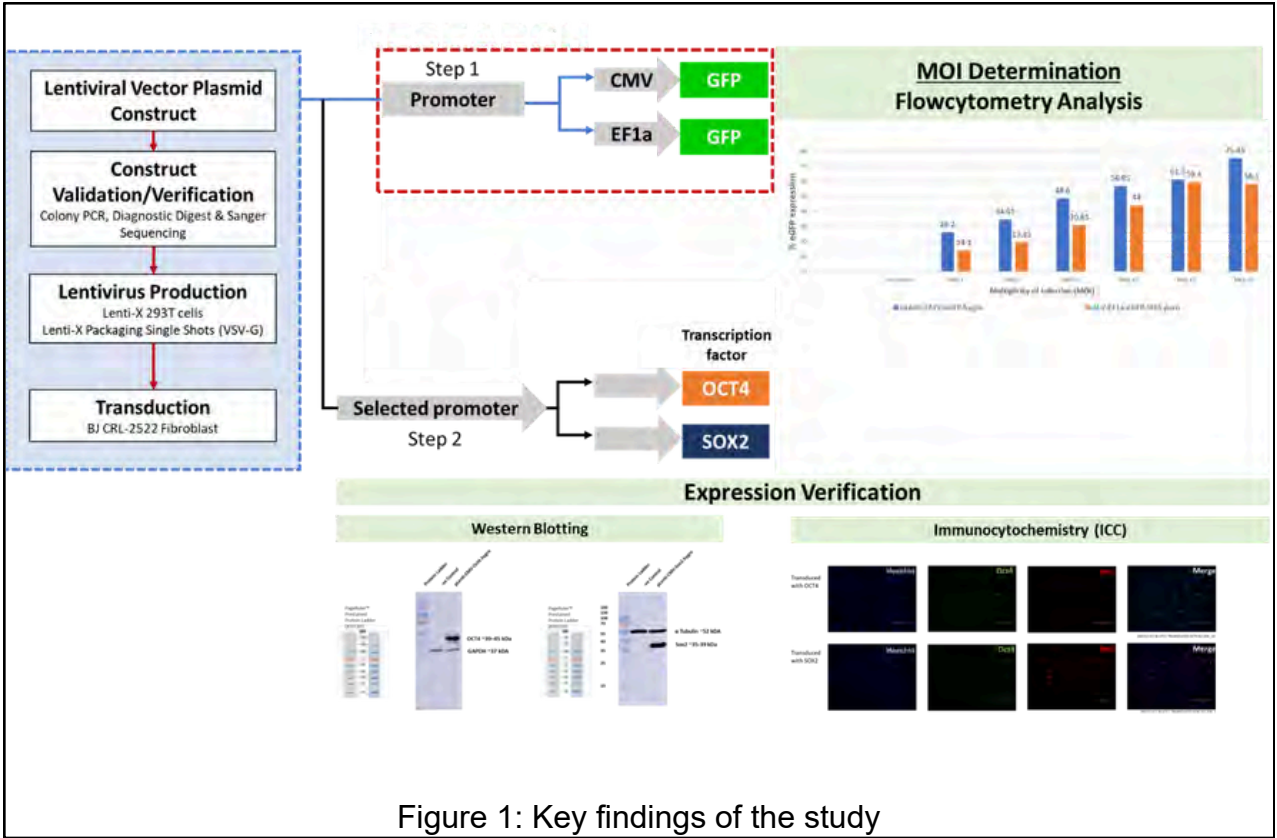
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Direct reprogramming of human fibroblast has potential for in-vitro disease modelling. Efficient delivery and stable expression of transcription factors (TFs) are critical for direct reprogramming. Promoter selection plays a key role in determining the expression levels and overall success. Objective: This study aimed to establish highly efficient and stable lentiviral transduction protocol for targeted transcription factor expression by evaluating the performance of two lentiviral promoters which are cytomegalovirus (CMV) promoter and elongation factor-1 alpha promoter (EF1 $\alpha$ ) promoter. Methods: Lentiviral vectors expressing GFP reporter under the control of either a CMV or EF1 $\alpha$  promoter were constructed. Lentivirus carrying the constructs were produced using fourth-generation lentiviral packaging systems. Flow cytometry used to determine the optimal multiplicity of infection (MOI) and evaluate promoter efficiency in human fibroblasts. The highest-performing promoter were selected to drive SOX2 and OCT4 expression. Protein expression and nuclear localization were confirmed by Western blotting and immunocytochemistry (ICC) 72 hours post transduction. Results: Flow cytometry demonstrated that CMV promoter significantly outperformed the EF1 $\alpha$  promoter in human fibroblasts across all viral MOI. An optimal MOI of 10–15 was identified for the CMV constructs, yielding robust transduction efficiencies (~57–61%) without the toxicity observed even at MOI of 20. Using these parameters, CMV-driven SOX2 and OCT4 resulted in sustained protein expression and correct nuclear localization for both TF, achieving the primary requirement for initiating direct reprogramming. Conclusion: Establishing this optimized lentiviral platform provided a reliable, standardized protocol for the stable overexpression of essential transcription factors for a successful direct reprogramming in human fibroblast.

**Keywords:** Direct Reprogramming, Lentiviral, Promoters, Transcription Factor





# SPR Biosensing for Quality Assessment of Wharton's Jelly Mesenchymal Stem Cells (WJMSCs) via Secretome Biomarkers

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Stem cell-based therapy has shown significant potential in regenerative medicine due to its ability to promote tissue repair and regeneration. However, current stem cell quality assessment methods remain invasive, time-consuming, and destructive to the cells, highlighting the need for rapid and non-invasive monitoring approaches. Objectives: This study aims to develop a label-free Surface Plasmon Resonance (SPR)-based biosensing platform for non-invasive quality assessment of Wharton's jelly-derived mesenchymal stem cells (WJMSCs) using secretome biomarkers, vascular endothelial growth factor (VEGF) and interleukin-6 (IL-6). Methods: WJMSCs were successfully isolated from human umbilical cord tissue and expanded until the required passage stage. Cell morphology analysis and characterisation studies were performed to confirm mesenchymal stem cell properties. Conditioned media containing secreted biomarkers were collected under serum-free conditions. The SPR biosensor platform was fabricated using a gold-coated BK7 prism functionalised with self-assembled monolayers and carboxylated molybdenum disulphide ( $\text{MoS}_2\text{-COOH}$ ) to enhance sensitivity. VEGF and IL-6 interactions were monitored in real time through refractive index changes on the sensor surface. Results: WJMSCs were successfully isolated and maintained stable spindle-shaped morphology throughout passaging. Secretome samples were successfully collected for future VEGF and IL-6 detection studies. UV-Visible analysis demonstrated optical responses from both silver- and gold-coated surfaces. In addition, the bare gold-coated SPR surface demonstrated detectable resonance angle formation and observable SPR angle shifts, indicating successful plasmonic activity and sensor response. Conclusions: These preliminary findings demonstrate the potential integration of WJMSC technology with SPR biosensing for future non-invasive stem cell quality assessment in regenerative medicine.

**Keywords:** Biosensor, Interleukin-6 (IL-6), Surface Plasmon Resonance (SPR), Vascular Endothelial Growth Factor (VEGF), Wharton's Jelly Mesenchymal Stem Cells (WJMSCs)

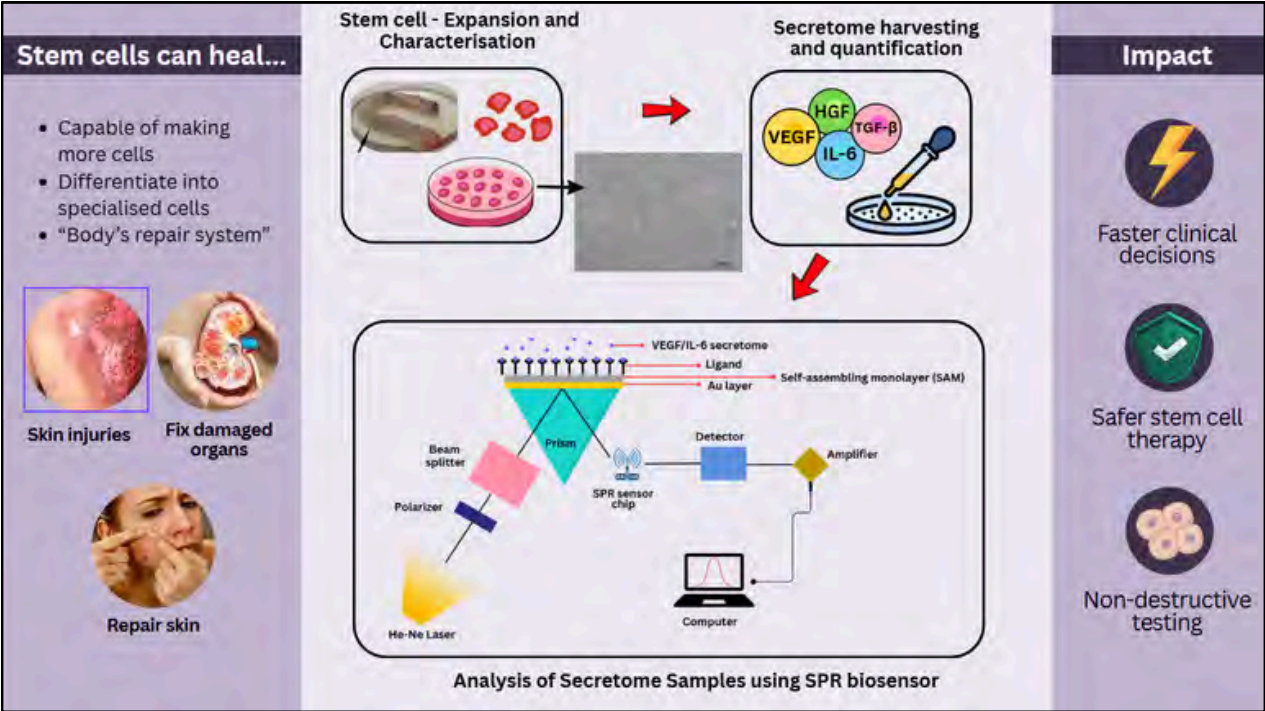


Figure 1: Overview of the SPR-based non-invasive stem cell quality assessment via secretome biomarkers, VEGF and IL-6

# In Vitro Modeling of Atopic Dermatitis Using Cytokine-Induced Human Keratinocytes: Inflammatory Cytokine Profiling and Protein Expression Analysis Dental Tissue Engineering in the 21st Century

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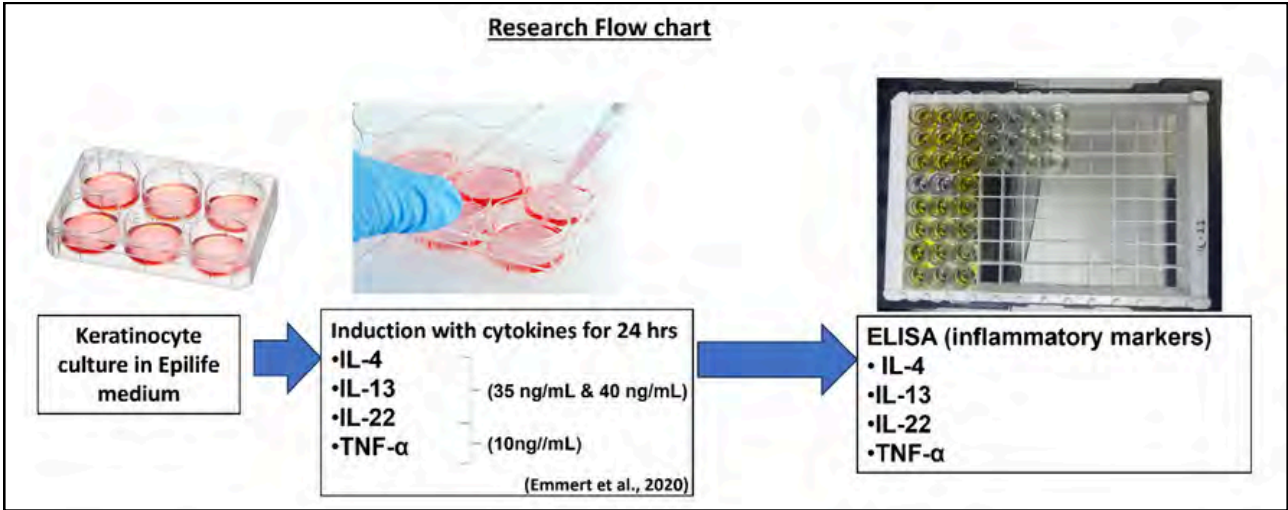
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Atopic dermatitis (AD) is a chronic inflammatory skin disorder characterized by immune dysregulation, increased pro-inflammatory cytokine production, and impaired epidermal barrier function. **Objective:** This study aimed to establish and characterize an in vitro AD-induced keratinocyte model by evaluating the expression of pro-inflammatory mediators following cytokine stimulation. **Methods:** Human keratinocytes were cultured in Epilife medium and induced using a cytokine cocktail consisting of interleukin (IL)-4, IL-13, and IL-22 at concentrations of 35 ng/mL and 40 ng/mL, together with tumor necrosis factor-alpha (TNF- $\alpha$ ) at 10 ng/mL for 24 hours to mimic the inflammatory microenvironment of AD. **Results:** Morphological assessment showed that untreated keratinocytes exhibited the typical cobblestone-like appearance with normal attachment. After induction, no significant morphological changes were observed, suggesting that cytokine exposure primarily activated inflammatory signaling pathways without immediate structural alterations. Bicinchoninic Acid (BCA) assay demonstrated significantly increased total protein concentrations ( $p < 0.05$ ) in both cell lysate and supernatant samples following induction. Cell lysates from both induction groups also showed significantly higher protein concentrations than corresponding supernatants ( $p < 0.05$ ). Enzyme-linked immunosorbent assay (ELISA) revealed significantly elevated levels of IL-4, IL-13, IL-22, and TNF- $\alpha$  in induced keratinocyte supernatants ( $p < 0.05$ ). Increased IL-4 and IL-13 expression confirmed enhanced Th2-mediated inflammatory responses, while elevated TNF- $\alpha$  levels supported active inflammatory signaling. **Conclusion:** Overall, the findings confirm the successful establishment of an AD-like keratinocyte inflammatory model for future evaluation of secretome-based therapeutic approaches. Further analyses involving MTT, transepithelial electrical resistance (TEER), reactive oxygen species (ROS), and immunocytochemistry assays will be conducted to further evaluate the inflammatory response, cellular viability, oxidative stress, and barrier integrity of the established model.

**Keywords:** Atopic Dermatitis, Human keratinocytes, In vitro model, Pro-inflammatory cytokines, Cytokine induction





# The Effects of MyCelest UC-MSC Therapy and Electroacupuncture on Memory Functions and BDNF Expression

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Brain-derived neurotrophic factor (BDNF) is a crucial neurotrophic factor involved in synaptic plasticity, neuronal survival, learning and memory. Alzheimer's disease (AD), a progressive neurodegenerative disease, affects memory functions and neurotrophin signalling in AD patients. While stem cell therapy and electroacupuncture have been individually proven to mitigate AD, the effects of this combined treatments on AD models remain to be explored. Objectives: In this study, we investigated the effects of MyCelest, an umbilicord mesenchymal stem cells (UC-MSCs) product, combined with electroacupuncture on BDNF levels and memory enhancement. Methods: Sprague-dawley rats were divided into Control, AD, AD+MyCelest, AD+MyCelest+EA, AD+EA groups. AD groups were administered aluminium chloride daily to induce Alzheimer's-like model. For treatments, MyCelest UC-MSCs were administered via tail vein injections into the MyCelest groups, and electroacupuncture was performed on the EA groups. A novel object recognition test was conducted to assess recognition memory. Western blotting is currently being performed to determine the expression of BDNF in hippocampus. Results: A significant reduction of recognition index was observed in AD group compared with Control group with  $p < 0.05$ . Marginal increase in recognition index was observed in treatment groups compared with AD group, without statistical significance. Preliminary results of BDNF expression suggest reduction in AD group compared with Control group, and an increase in treatment groups compared with AD group. Conclusions: These findings elucidate the therapeutic effects of MyCelest UC-MSC therapy combined with electroacupuncture in mitigating AD, which may be attributed to improved recognition memory and enhanced neurotrophic factor.

**Keywords:** Stem cell therapy, Electroacupuncture, Alzheimer's disease, Recognition memory, Brain-derived neurotrophic factor

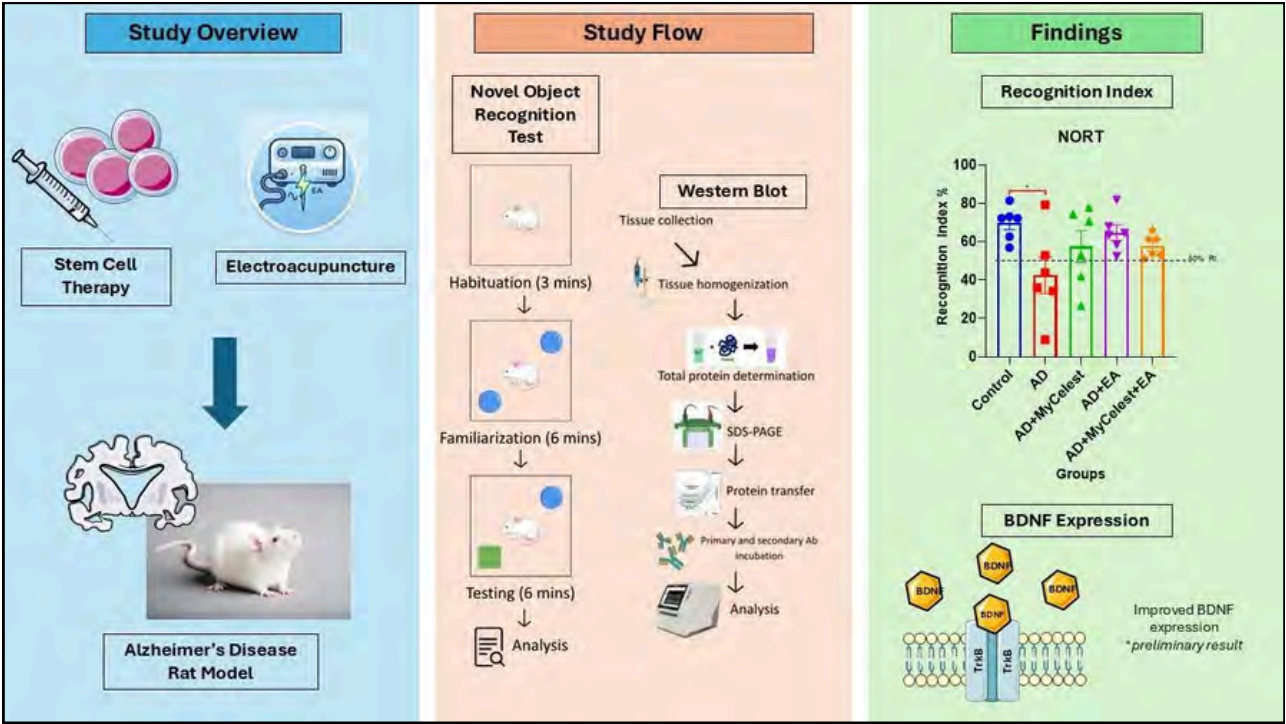


Figure 1: Study overview

# Detection of Cell Damage Following Exposure to Free Radicals in Mesenchymal Stem Cell Culture

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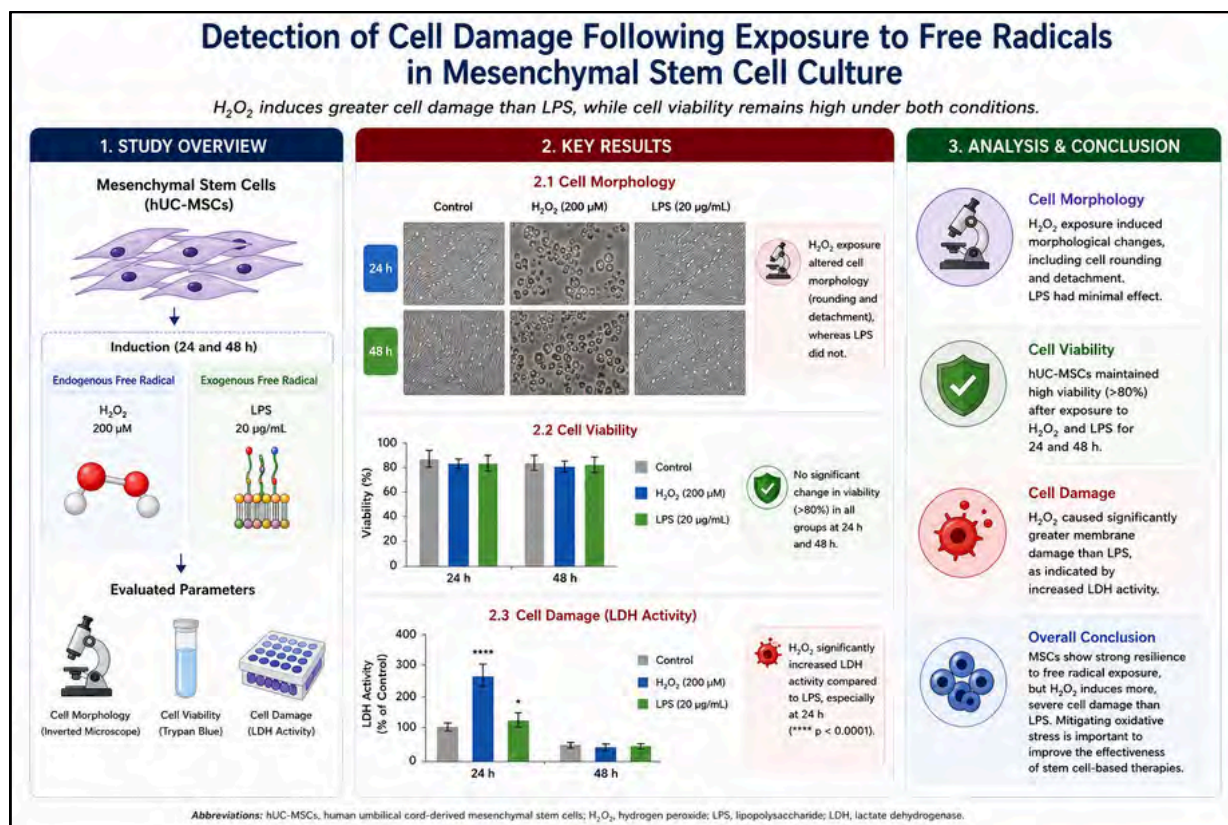
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Mesenchymal stem cells (MSCs) are widely used in regenerative therapy due to their high proliferative and differentiation potential. However, oxidative and inflammatory stress caused by endogenous and exogenous free radicals may impair their therapeutic function. Objectives: This study aimed to evaluate the effects of hydrogen peroxide (H<sub>2</sub>O<sub>2</sub>) and lipopolysaccharide (LPS) exposure on MSC morphology, viability, and cell damage in vitro. Methods: Human umbilical cord-derived MSCs were induced with 200 µM H<sub>2</sub>O<sub>2</sub> and 20 µg/mL LPS for 24 and 48 hours at 37°C and 5% CO<sub>2</sub>. Cell morphology was observed using an inverted microscope, viability was assessed using Trypan Blue exclusion, and lactate dehydrogenase (LDH) activity was measured using a microplate reader. Results: H<sub>2</sub>O<sub>2</sub> exposure caused significant morphological changes, including cell rounding and detachment, whereas LPS-treated cells maintained morphology similar to controls. Cell viability remained high (>80%) in all groups, indicating resistance to oxidative and inflammatory stress. However, LDH activity significantly increased in the H<sub>2</sub>O<sub>2</sub> group compared to the LPS group, especially after 24 hours, suggesting greater membrane damage induced by oxidative stress. Conclusions: MSCs demonstrate adaptive resistance to free radical exposure, although H<sub>2</sub>O<sub>2</sub> induces more severe cellular damage than LPS. These findings may support strategies to improve the effectiveness of regenerative therapy through oxidative stress mitigation.

**Keywords:** Hydrogen peroxide, Cell damage, Lipopolysaccharide, Mesenchymal stem cells, Oxidative stress



# Metabolic Microenvironment Enhances Proliferation and COL2A1 Expression in Umbilical Cord-Derived Mesenchymal Stem Cells

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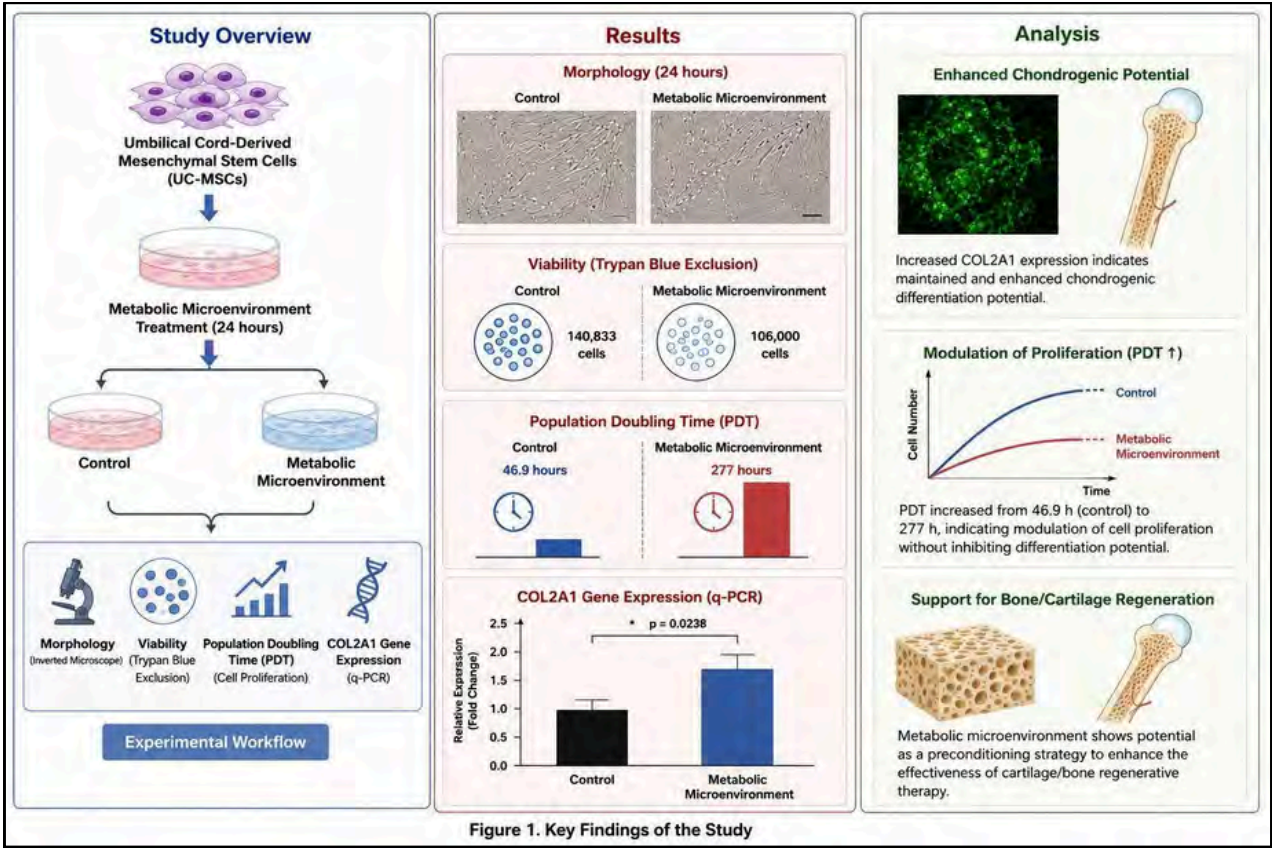
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Umbilical cord-derived mesenchymal stem cells (UC-MSCs) possess high regenerative, immunomodulatory, and chondrogenic differentiation potential; however, their response to metabolic microenvironmental conditions remains poorly understood. Objective: This study aimed to evaluate the effects of the metabolic microenvironment on morphology, viability, Population Doubling Time (PDT), and COL2A1 gene expression as a marker of chondrogenic differentiation in UC-MSCs. Metods: This in vitro experimental study used UC-MSCs divided into a control group and a metabolic microenvironment treatment group for 24 hours. Morphology was observed using an inverted microscope, viability was assessed using the Trypan Blue exclusion method, PDT was calculated to evaluate cell proliferation, and COL2A1 gene expression was analyzed using quantitative Polymerase Chain Reaction (q-PCR). Results: The treatment group showed a longer PDT (277 hours) compared with the control group (46.9 hours), indicating slower cell proliferation under metabolic microenvironmental conditions. Statistical analysis using the Mann–Whitney U test demonstrated a significant difference in COL2A1 gene expression between groups. Conclusions: These findings indicate that the metabolic microenvironment modulated cell proliferation without reducing the chondrogenic differentiation potential of UC-MSCs, as evidenced by maintained COL2A1 gene expression. Furthermore, increased COL2A1 expression suggests that metabolic microenvironmental conditions may support chondrogenic differentiation and could potentially be applied as a preconditioning strategy for cartilage-based regenerative therapy.

**Keywords:** COL2A1 expression, Mesenchymal stem cells, Metabolic microenvironment, Proliferation





## Global Comparison of Cardiac and Vascular Regeneration-Related SNPs in a Malaysian Population

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**Objective:** To characterise allele frequencies of selected variants implicated in post-ischaemic neovascularisation, angiogenesis, endothelial function, and vascular repair in a Malaysian cohort and compare their distribution with global reference populations. **Methods:** Twenty-nine candidate single-nucleotide polymorphisms (SNPs) were submitted for screening against the Prima Nexus Illumina Global Screening Array dataset. Variants with callable genotypes were analysed overall and by self-reported Malay, Chinese, and Indian ethnicity. Minor-allele frequencies were aligned to gnomAD v4.1 reference populations (GRCh38). Malaysian Chinese were compared with East Asian, Indians with South Asian, and Malays with East Asian as a proxy because gnomAD lacks a Malay/Southeast Asian reference. Frequency differences were tested using chi-square or Fisher's exact tests, with Benjamini-Hochberg false-discovery-rate correction. **Results:** Nineteen of 29 SNPs were represented and carried forward. Across these variants, Malaysian Chinese closely mirrored the East Asian reference (Pearson  $r=0.924$ ), yet 11 loci remained significantly different after false-discovery-rate correction. Three unambiguous loci showed both  $q<0.05$  and an absolute frequency difference  $\geq 5\%$ : PLAU rs2227564 was more frequent in Malaysian Chinese than East Asians (0.401 vs 0.295;  $\Delta=+0.106$ ; OR=1.60;  $q=4.0\times 10^{-89}$ ), whereas ICAM1 rs5498 (0.248 vs 0.336;  $\Delta=-0.088$ ; OR=0.65;  $q=6.0\times 10^{-61}$ ) and rs3093030 (0.204 vs 0.290;  $\Delta=-0.085$ ; OR=0.63;  $q=1.5\times 10^{-56}$ ) were less frequent. In the Indian subgroup, PLAU rs2227564 also differed from the South Asian reference (0.415 vs 0.300;  $\Delta=+0.115$ ; OR=1.66;  $q=0.002$ ). Multiple differences were observed in Malays versus the East Asian proxy, although interpretation is limited by the absence of a Southeast Asian reference. **Conclusion:** The Malaysian cohort broadly reflects East Asian ancestry while retaining locus-specific differences in variants related to vascular growth and repair, particularly PLAU and ICAM1. These findings describe population-level genomic variation only and should not be interpreted as disease risk or regenerative capacity. They provide a population-specific genomic basis for future phenotyped studies of cardiac and vascular regeneration.

**Keywords:** Single-nucleotide polymorphisms (SNPs), allele frequencies, Malaysian population genetics, vascular regeneration

## Characterization and Antimicrobial Evaluation of *Ganoderma lucidum*-mediated Silver Nanoparticles

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**Introduction:** Oral infectious diseases and the escalating threat of antimicrobial resistance are major global health concerns, underscoring the need for novel, biocompatible antimicrobial agents. **Objective:** This study aimed to synthesise silver nanoparticles (AgNPs) through a green approach using *Ganoderma lucidum* extract, a medicinal mushroom with diverse pharmacological properties, and to evaluate their antimicrobial activity against selected microorganisms associated with oral infections. **Methods:** The energy-dispersive X-ray (EDX)spectroscopy coupled with scanning electron microscopy (SEM) was used to determine the elemental composition of the *G. lucidum*-mediated silver nanoparticles. Antimicrobial activity against *Escherichia coli*, *Streptococcus mutans*, *Streptococcus pyogenes* and *Candida albicans* was evaluated using minimum inhibitory concentration (MIC) assays. **Results:** EDX-SE analysis confirmed the presence of silver in the synthesized sample, accounting for 14.12 wt.% and 2.04 at.% of the detected elemental composition. The *G. lucidum*-mediated AgNPs demonstrated that *E. coli* was the most susceptible, exhibiting inhibition zones comparable to those produced by chlorhexidine and an MIC of  $\leq 25$   $\mu\text{g/mL}$ . *S. mutans* and *S. pyogenes* showed moderate, concentration-dependent inhibition, with MIC values of  $\leq 25$   $\mu\text{g/mL}$ . In contrast, *C. albicans* exhibited lower susceptibility and required a higher AgNP concentration, with an MIC of approximately 50  $\mu\text{g/mL}$ . **Conclusion:** Overall, the EDX findings support the establishment of silver nanoparticle biosynthesis. The biosynthesized silver nanoparticles showed significant antibacterial and antifungal activities. These findings support the potential application of *G. lucidum*-derived AgNPs as eco-friendly and cost-effective antimicrobial agents in healthcare products, including mouthwashes, biomaterials and other preventive or therapeutic formulations.

**Keywords:** *Ganoderma lucidum* extract, Silver nanoparticles, Biosynthesis, Therapeutic, Antimicrobial

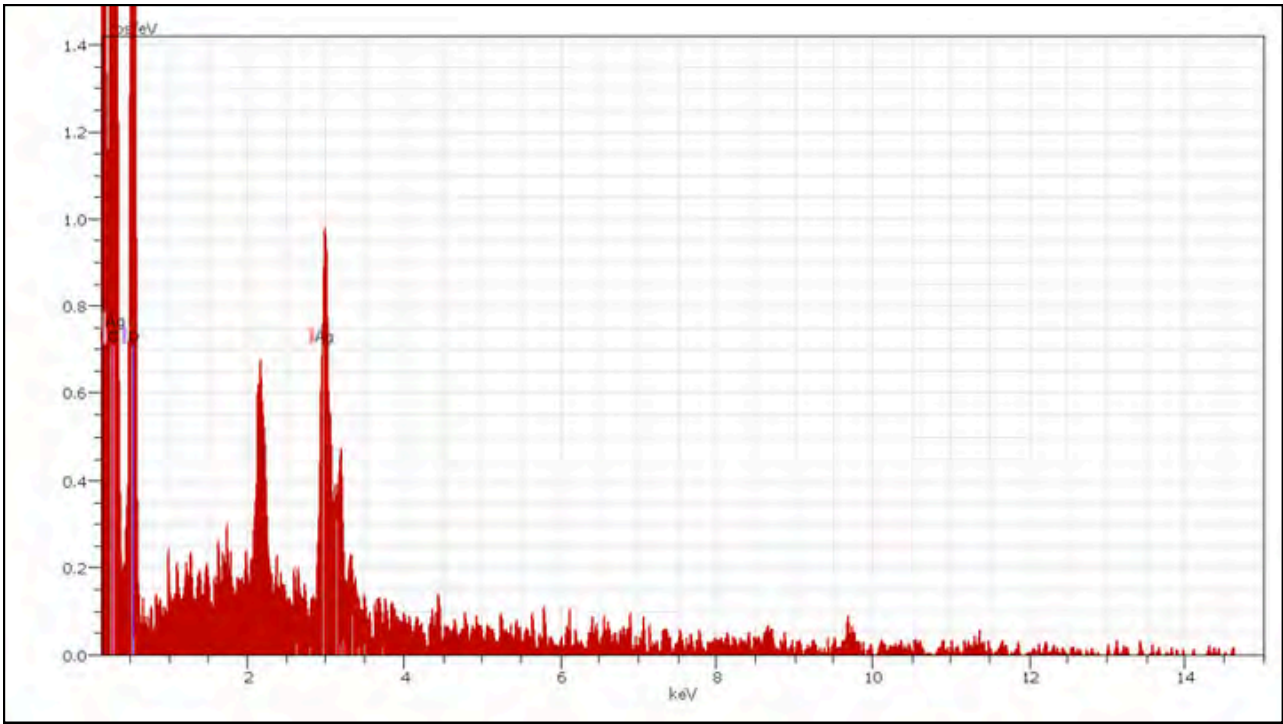


Figure 1: Composition of silver, carbon and oxygen in the samples using EDX machine

# Identification of Central Hub Genes and Biological Pathways in Parkinson's Disease via Transcriptomic Profiling

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Parkinson's disease (PD) remains a debilitating neurodegenerative condition due to its impact on the aging population and the lack of effective treatments. To uncover key molecular mechanisms driving PD, this study identified differentially expressed genes (DEGs), biological pathways, and candidate hub networks using public transcriptomic data. RNA sequencing (RNA-seq) data comparing PD patients and healthy controls were retrieved from the Gene Expression Omnibus and analyzed for differential expression using Tag Count Comparison-Graphical User Interface (TCC-GUI). A total of 25 significant DEGs were identified, comprising 6 upregulated and 19 downregulated genes. Functional enrichment analysis using ShinyGO revealed that these DEGs were predominantly enriched in dopaminergic synaptic transmission, neurotransmitter transport, and neuronal maintenance pathways. Protein-protein interaction (PPI) network topology constructed via STRING and analyzed in Cytoscape identified Dopamine Receptor D2 (DRD2) as the top central hub gene based on its highest degree of connectivity. Additionally, four primary hub genes (PITX3, SLC6A3, SLC18A2, LMX1B) and two secondary bridge genes (NTSR1, ALDH1A1) were prioritized within the core dopaminergic network module. These findings highlight a tightly regulated gene network essential for dopaminergic neuron survival, positioning DRD2 and its co-regulated hub genes as potential targets for therapeutic intervention.

# Physicochemical Characterisation and Dose Optimisation of Germanium- and Green Honey-Loaded Biomaterial Inks for Wound Healing Application

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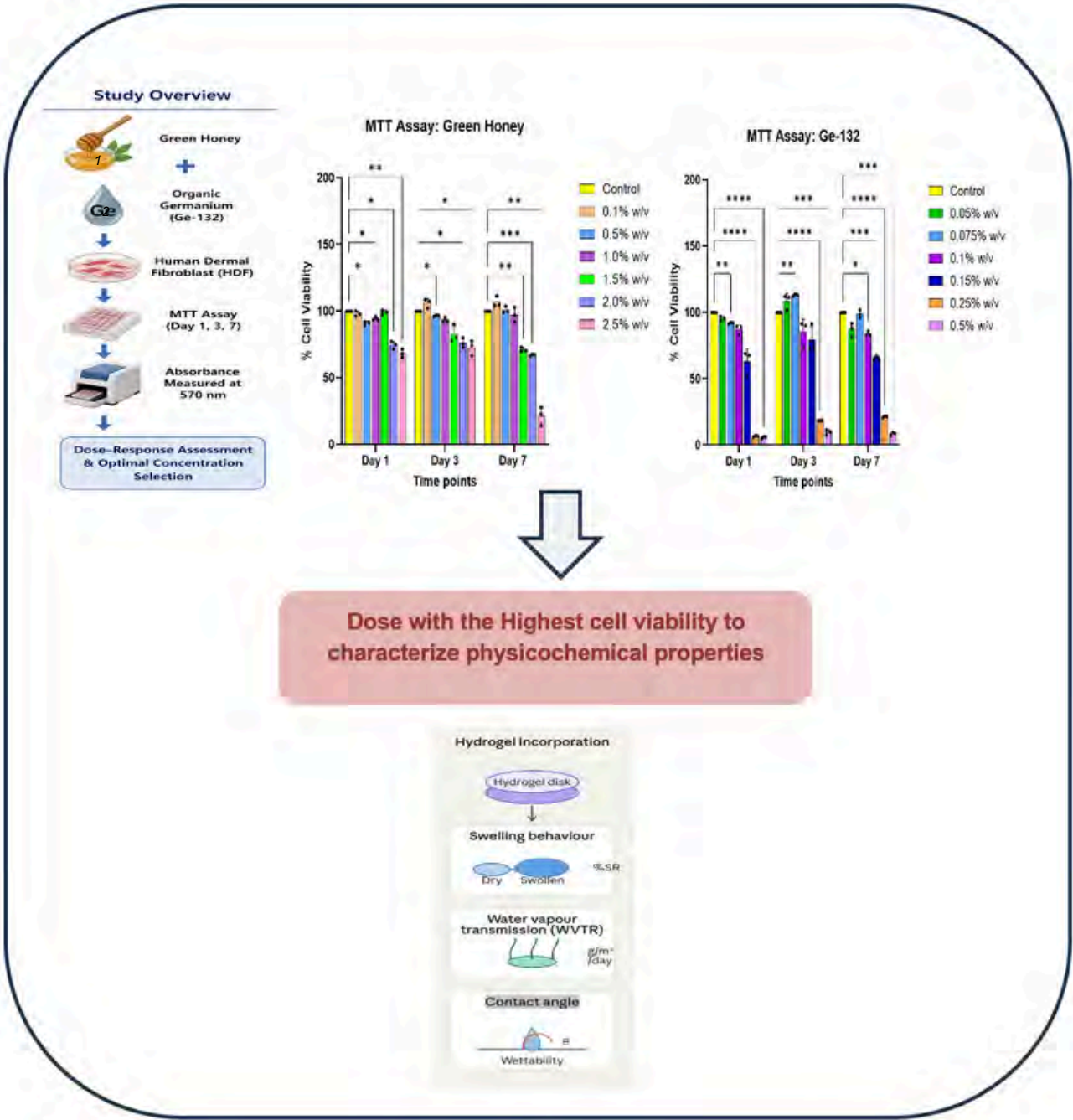
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Natural bioactive compounds are increasingly investigated in tissue engineering for their antioxidant and pro-healing potential. Objective: To determine the optimal therapeutic concentration of organic germanium and green honey for human dermal fibroblasts (HDFs). Methods: HDF cells were seeded in 96-well plates and exposed to six graded concentrations of green honey and organic germanium. After incubation (Day 1, 3, and 7), metabolic activity was assessed using the MTT assay, with absorbance measured at 570 nm following formazan solubilization. Cell viability (%) was calculated relative to untreated controls after blank correction. Concentrations demonstrating the highest viability were shortlisted for further formulation studies. Physicochemical characterization of selected hydrogel formulations includes swelling behaviour, water vapor transmission rate, and contact angle analysis. Results: A range of concentration and time-dependent effect on cell viability was observed. Three concentrations (0.1, 0.5, and 1.0 w/v) for green honey and (0.015, 0.075, 0.1 v/v) for organic germanium consistently showed better cytocompatibility across Days 1, 3, and 7 compared to the higher doses. For green honey, cell viability on Day 1 increased to  $98 \pm 1.71\%$ ,  $91 \pm 0.60\%$ , and  $94 \pm 3.50\%$  at 0.1, 0.5, and 1.0% (v/v), respectively. On Day 3, viability further increased to  $108 \pm 5.90\%$ ,  $113 \pm 0.69\%$ , and  $85 \pm 8.06\%$ , while Day 7 values were  $87 \pm 5.03\%$ ,  $85 \pm 2.99\%$ , and  $84 \pm 1.08\%$ , respectively. For organic germanium, Day 1 viability was  $95.47 \pm 1.71\%$ ,  $92.11 \pm 0.60\%$ , and  $87.40 \pm 3.50\%$  at 0.1, 0.5, and 1.0, respectively. On Day 3, values reached  $108 \pm 5.90\%$ ,  $113 \pm 0.60\%$ , and  $85.28 \pm 8.06\%$ , followed by  $87.40 \pm 5.03\%$ ,  $98.9 \pm 2.09\%$ , and  $84.2 \pm 1.80\%$  on Day 7. Similar to green honey, the strongest response was seen on Day 3 at lower concentrations. In contrast, higher concentrations showed noticeably reduced metabolic activity across all time points, suggesting possible cytotoxic effects at elevated doses. Conclusion: The MTT dose-response screening identified the three ideal dose as cytocompatible concentrations suitable for hydrogel incorporation. Further physicochemical and functional evaluations are ongoing to validate formulation stability and dermal applicability.

**Keywords:** Organo-Germanium (Ge-132), Honey based hydrogel, Angiogenesis and Oxidative Stress, Hyperglycaemic Wound Model, Diabetic Foot Ulcer (DFU)





# A Review of Autologous Platelet Concentrates as Bioactive Scaffolds for Periorbital Tissue Regeneration and Dermal Augmentation

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Periorbital soft tissue volume loss, commonly perceived as “sunken eyes”, is a multifactorial condition involving structural anatomy, genetic predisposition, and age-related changes. Current treatments such as dermal fillers provide immediate volume restoration but do not improve tissue quality and may be associated with complications in delicate periorbital regions, including procedure-related complications, limited longevity, and lack of actual tissue regeneration. Objectives: To evaluate the potential of autologous platelet concentrates as regenerative biomaterials for periorbital soft tissue restoration, and whether platelet-rich plasma (PRP) and platelet-rich fibrin (PRF) offer advantages over conventional volumising treatments. Methods: A narrative synthesis of peer-reviewed articles from 2016–2026 was conducted using PubMed. Search terms included (“platelet-rich plasma” OR PRP OR “platelet-rich fibrin” OR PRF OR “platelet concentrates”) AND (periorbital OR infraorbital OR “under eye” OR “facial rejuvenation”) AND (regeneration OR regenerative OR collagen). Clinical and experimental studies assessing regenerative outcomes, including dermal quality and tissue remodelling, were analysed. Results: Eleven studies were included. Evidence from these showed improvements in skin quality following PRP and PRF interventions, such as enhanced texture, elasticity, and overall appearance. PRF illustrated scaffold-like properties supporting remodeling, while PRP acted via growth factor-mediated mechanisms. Adjunctive treatments provided additional benefit in selected studies, particularly for pigmentation, erythema, and skin texture. Conclusion: Autologous platelet concentrates represent promising bioactive scaffolds that combine structural support with regenerative activity, offering a potential advantage over conventional dermal fillers in periorbital tissue restoration. Standardisation of protocols remains essential for clinical translation.

**Keywords:** Autologous platelet concentrates, Periorbital rejuvenation, Platelet-rich fibrin, Platelet-rich plasma, Soft tissue regeneration

# Characterisation & Release Kinetics of SHED-EVs & WJMSC-EVs in Chitosan Hydrogel for Oral Wound Healing

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Recalcitrant oral wounds associated with systemic conditions such as poorly controlled diabetes or radiation induced mucositis present a significant clinical challenge due to their failure towards normal healing process. Current treatments primarily focus on symptomatic management and infection control but lack of bioactive signals that can stimulate tissue regeneration. Extracellular vesicles (EVs) derived from Mesenchymal Stem Cells (MSC) are a potent cell-free regenerative therapy, but their clinical application is limited to poor retention and rapid clearance by salivary flow. Objectives: This study investigates the characterization and release kinetics of EVs derived from Stem cells from Human Exfoliated Deciduous Teeth (SHED) and Wharton's Jelly MSC (WJMSC) loaded in chitosan hydrogel. Methods: EVs were isolated via Tangential Flow Filtration (TFF). Characterisation was performed for morphology, size and marker expression. Chitosan hydrogel was synthesized and loaded with SHED-EVs or WJMSC-EVs. Field Emission Scanning Electron Microscopy (FESEM) was done to analyse the scaffold's microstructure and EVs integration. Release kinetics were monitored over 7 days in a simulated physiological environment. Results: Morphology of the EVs showed cup-shaped and double-membrane morphology. Nanoparticle tracking analysis (NTA) revealed polydisperse population with a SPAN of 1.74. FESEM showed highly porous, interconnected chitosan network with EVs present within the hydrogel. Both groups showed steady controlled release of the EVs within the 7-day period. Conclusions: SHED-EVs and WJMSC-EVs were successfully characterized and showed sustained release from chitosan hydrogel matrix. This EVs-loaded chitosan system will be able to provide promising regenerative platform for the management of recalcitrant oral wounds.

**Keywords:** WJMSC-EVs, SHED-EVs, Chitosan hydrogel, Oral wound healing

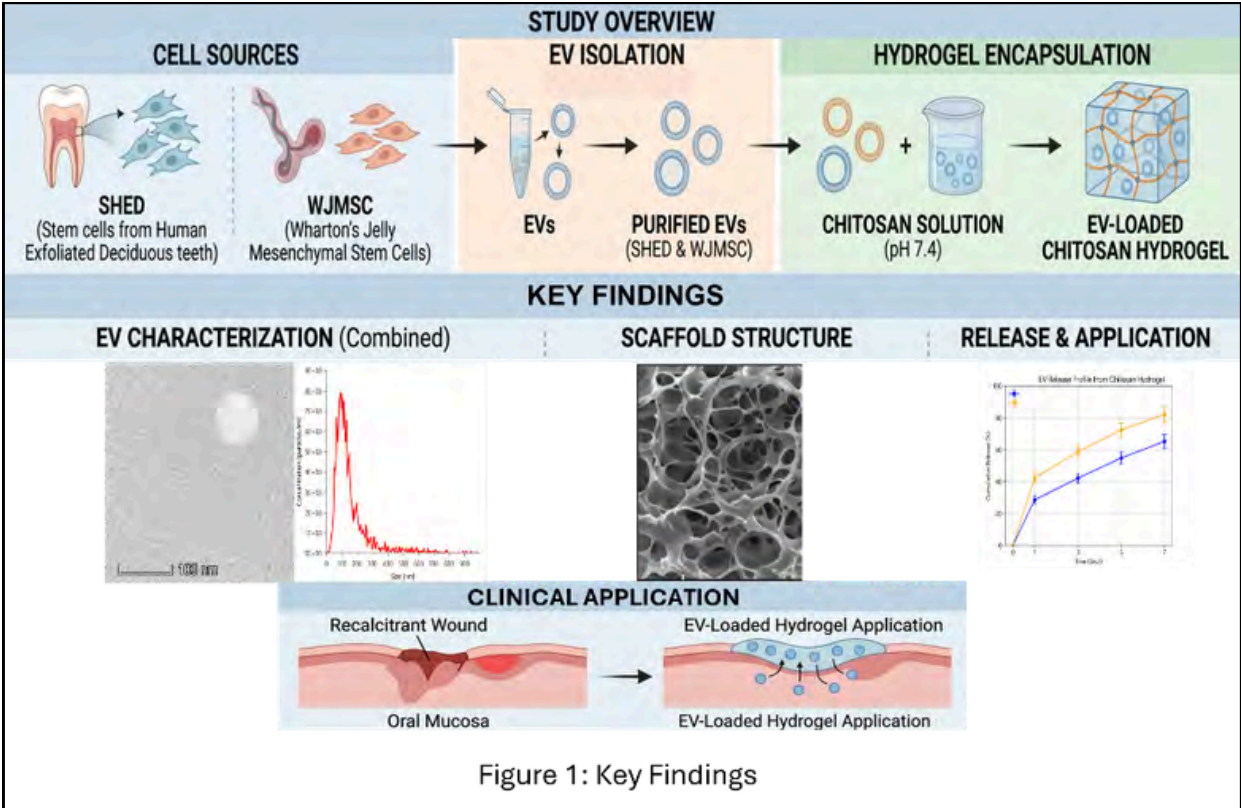


Figure 1: Key Findings

# Topical Extracellular Vesicle-Containing Formulation Accelerates Thermal Burn Wound Repair in Mice

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**Objectives:** This study evaluated the wound-healing efficacy and short-term tolerability of a topical Extracellular Vesicle-enriched formulation in a standardized murine thermal burn model. **Materials and Methods:** Twelve male BALB/c mice were randomized to untreated control, blank formulation or exosome-containing formulation groups (n=4/group). Under ketamine-xylazine anesthesia, a 1-cm metal rod heated to 96±2°C was applied to shaved dorsal skin for 10 seconds. Treatments were applied every other day and covered with adhesive dressings. Wound dimensions and body weight were recorded over 14 days. Repair quality was assessed by hematoxylin and eosin histology. One-way analysis of variance and simple linear regression were used. **Results:** Wound-area regression slopes were -4.169 for control, -4.434 for blank formulation and -5.804 for Extracellular Vesicle based treatment, indicating the fastest closure trajectory in the treated group. By day 14, treated mice had the smallest residual wound area. Histology showed a more uniform epidermis, better-organized dermis, reduced inflammatory cellularity and earlier progression towards remodeling. Body-weight changes did not differ significantly among groups. **Conclusions:** Topical Extracellular Vesicle-containing treatment accelerated burn-wound closure and improved repair quality without an evident body-weight safety signal. Larger, blinded studies should confirm dose-response effects and quantitative histomorphometry.

**Funding:** The study was supported by the program-targeted financing on scientific programs of the Ministry of Healthcare of the Republic of Kazakhstan «Development of an exosome isolation kit from umbilical cord mesenchymal stem cell culture for therapeutic and research application» (2024–2026) (IRN BR25593457)

**Keywords:** Burns, Extracellular vesicles, Skin regeneration, Topical therapy, Wound healing

# Small Extracellular Vesicles Modulate Orthotopic 4T1 Breast Tumors in Mice

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**Objectives:** This study evaluated local 4T1 mammary tumor modulation and major-organ safety after treatment with human serum-derived exosome-enriched small extracellular vesicles (sEVs), alone or combined with doxorubicin in an alginate-gelatin hydrogel. **Materials and Methods:** sEVs were isolated by sequential centrifugation, 0.2-µm filtration and ultracentrifugation at 100,000 × g, then characterized by Western blotting, scanning electron microscopy, dynamic light scattering and zeta-potential analysis. Twelve female BALB/c mice were randomized to hydrogel alone, 4T1-loaded hydrogel, 4T1-loaded hydrogel plus sEVs, or 4T1-loaded hydrogel plus sEVs and doxorubicin (n=3/group). Formulations were implanted into the mammary fat pad. Mammary gland, liver, spleen and kidney were examined by haematoxylin and eosin staining after 10 days. **Results:** Vesicles showed nanoscale morphology and Alix, CD63 and TSG101 signals. The 4T1-loaded hydrogel generated dense tumor-like cellular aggregates. sEV-treated tissue showed lower apparent tumor-like cellular density, with the strongest local reduction after sEV-doxorubicin treatment. Liver and kidney architecture was broadly preserved with sEVs alone, whereas the combination was associated with mild hepatocellular vacuolation and renal tubular epithelial injury. **Conclusions:** The integrated sEV-hydrogel platform was feasible and showed preliminary local tumor modulation. Doxorubicin may strengthen the local effect but requires optimization because of the renal safety signal.

**Funding:** The study was supported by the program-targeted financing on scientific programs of the Ministry of Healthcare of the Republic of Kazakhstan «Development of an exosome isolation kit from umbilical cord mesenchymal stem cell culture for therapeutic and research application» (2024–2026) (IRN BR25593457).

**Keywords:** Breast cancer, Doxorubicin, Hydrogel, Small extracellular vesicles, Tumor model



## ***Lactobacillus* Secretome and Bioactive Metabolites: Mechanistic Insights and Multi-Targeted Strategies Against MASLD**

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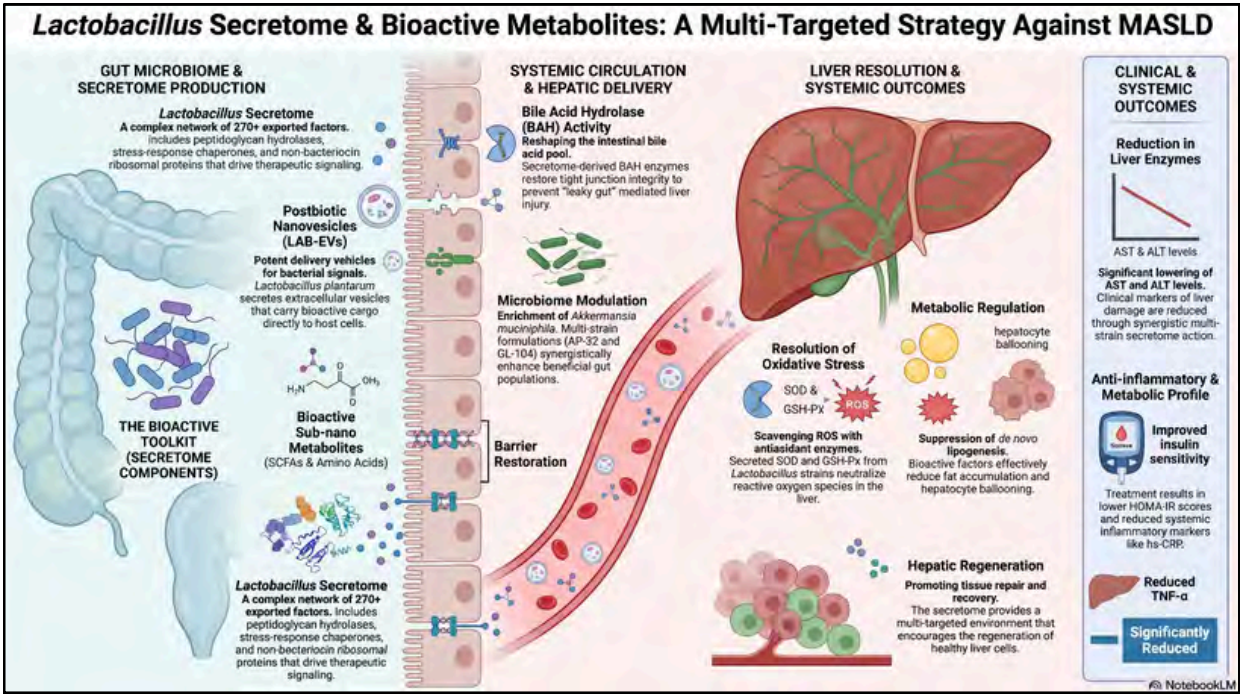
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**Objectives:** This critical review evaluates the therapeutic potential of *Lactobacillus* species, their cell-free secretomes, and postbiotic fractions in modulating the gut-liver axis to alleviate Metabolic Dysfunction-Associated Steatotic Liver Disease (MASLD). **Methods:** Appraisals were conducted on recent genomic, exoproteomic, and surfaceome profiling studies alongside preclinical and clinical trials assessing strain-specific *Lactobacillus* secretomes, bacterial extracellular vesicles (LAB-EVs), bioactive metabolites, and functional surface proteins. **Results:** The molecular profiling showed that the *Lactobacillus* secretome comprises a complex functional network of over 270 exported factors, including peptidoglycan hydrolases, stress-response chaperones, and non-bacteriocin ribosomal proteins. *Lactobacillus murinus* secretome components deliver basic amino acid precursors like arginine to enhance systemic bioavailability, thereby alleviating endoplasmic reticulum stress, lobular inflammation, and hepatocyte ballooning. *Lactobacillus plantarum* secretes LAB-EVs, short-chain fatty acids (SCFAs), and antioxidant enzymes (SOD, GSH-Px) that suppress de novo lipogenesis and scavenge reactive oxygen species. Multi-strain formulations (*L. salivarius* AP-32 and *L. reuteri* GL-104) synergistically reduce transaminases (AST, ALT), lower systemic inflammatory markers (TNF- $\alpha$ , hs-CRP), improve insulin sensitivity (HOMA-IR), and enrich *Akkermansia muciniphila*. Furthermore, secretome-derived bile acid hydrolase (BAH) enzymes reshape the intestinal bile acid pool to restore tight junction integrity and prevent leaky gut-mediated liver injury. **Conclusion:** The dual-action *Lactobacillus* secretome combines the functional surfaceome proteins, postbiotic nanovesicles, and sub-nano metabolites provides a multi-targeted, cell-free biotherapeutic strategy to restore intestinal barrier function, resolve oxidative inflammation, and promote hepatic regeneration in MASLD, a possible multi-target biotherapeutic.

**Keywords:** Hepatic regeneration, LAB-secretome, MASLD



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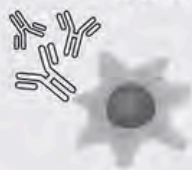


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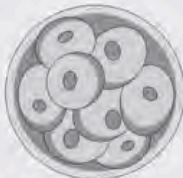


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